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(54) MUSCLE-TARGETING COMPLEXES AND USES THEREOF

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(73) Assignee: **Dyne Therapeutics, Inc.**, Waltham, MA (US)

(21) Appl. No.: **19/211,660**

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Related U.S. Application Data

(63) Continuation of application No. 19/066,035, filed on Feb. 27, 2025, which is a continuation of application No. 18/939,894, filed on Nov. 7, 2024, now Pat. No. 12,319,743, which is a continuation of application No. 18/656,654, filed on May 7, 2024, now Pat. No. 12,173,078, which is a continuation of application No. 18/468,580, filed on Sep. 15, 2023, now Pat. No. 12,018,087, which is a continuation-in-part of application No. 18/184,741, filed on Mar. 16, 2023, now Pat. No. 11,795,233, which is a continuation of application No. 17/936,483, filed on Sep. 29, 2022, now Pat. No. 11,787,869, which is a continuation of application No. 17/846,738, filed on Jun. 22, 2022, now Pat. No. 11,518,816, which is a continuation of application No. 17/671,707, filed on Feb. 15, 2022, now Pat. No. 11,390,682, which is a continuation of application No. 17/400,295, filed on Aug. 12, 2021, now Pat. No. 11,286,305, which is a continuation of application No. 17/205,123, filed on Mar. 18, 2021, now Pat. No. 11,111,309, which is a continuation of application No. 17/264,948, filed on Feb. 1, 2021, filed as application No. PCT/US2019/044990 on Aug. 2, 2019, now abandoned, said application No. 18/468,580 is a continuation-in-part of application No. 17/264,905, filed on Feb. 1, 2021, filed as application No. PCT/US2019/044987 on Aug. 2, 2019, now abandoned, said application No. 18/468,580 is a continuation-in-part of application No. 17/265,016, filed on Feb. 1, 2021, filed as application No. PCT/US2019/044960 on Aug. 2, 2019, now abandoned, said application No. 18/468,580 is a continuation-in-part of application No. 17/264,998, filed on Feb. 1, 2021, filed as application No. PCT/US2019/044982 on Aug. 2, 2019, now abandoned, said application No. 18/468,580 is a continuation-in-part of applica-

(Continued)

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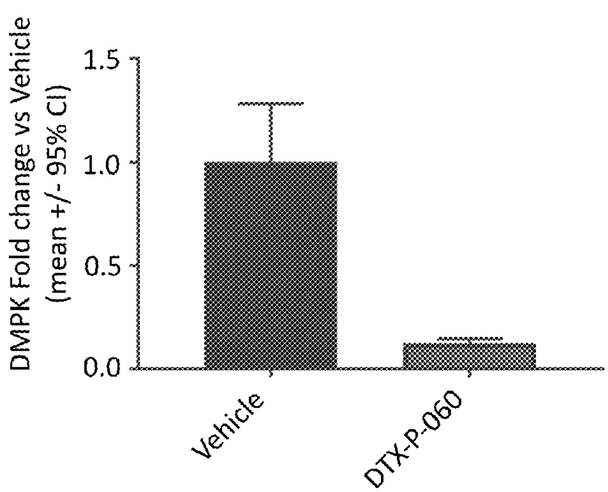
(52) **U.S. Cl.**

CPC ***A61K 47/6849*** (2017.08); ***A61P 21/00*** (2018.01); ***C07K 16/2881*** (2013.01); ***C12N 15/113*** (2013.01); *C07K 2317/24* (2013.01); *C07K 2317/524* (2013.01); *C07K 2317/526* (2013.01); *C07K 2317/565* (2013.01); *C07K 2317/567* (2013.01); *C07K 2317/71* (2013.01); *C12N 2310/11* (2013.01); *C12N 2310/315* (2013.01); *C12N 2310/321* (2013.01); *C12N 2310/322* (2013.01); *C12N 2320/32* (2013.01)

(57) **ABSTRACT**

Aspects of the disclosure relate to complexes comprising a muscle-targeting agent covalently linked to a molecular payload. In some embodiments, the muscle-targeting agent specifically binds to an internalizing cell surface receptor on muscle cells. In some embodiments, the molecular payload inhibits activity of a disease allele associated with muscle disease. In some embodiments, the molecular payload is an oligonucleotide, such as an antisense oligonucleotide or RNAi oligonucleotide.

Specification includes a Sequence Listing.



Related U.S. Application Data

tion No. 17/265,044, filed on Feb. 1, 2021, filed as application No. PCT/US2019/044955 on Aug. 2, 2019, now abandoned, said application No. 18/468,580 is a continuation-in-part of application No. 17/265,024, filed on Feb. 1, 2021, filed as application No. PCT/US2019/044949 on Aug. 2, 2019, now abandoned, said application No. 18/468,580 is a continuation-in-part of application No. 17/265,019, filed on Feb. 1, 2021, filed as application No. PCT/US2019/044959 on Aug. 2, 2019, now abandoned, said application No. 18/468,580 is a continuation-in-part of application No. 17/264,972, filed on Feb. 1, 2021, filed as application No. PCT/US2019/044961 on Aug. 2, 2019, now abandoned.

(60) Provisional application No. 62/713,933, filed on Aug. 2, 2018, provisional application No. 62/859,672, filed on Jun. 10, 2019, provisional application No. 62/858,888, filed on Jun. 7, 2019, provisional application No. 62/855,761, filed on May 31, 2019, provisional application No. 62/779,161, filed on Dec. 13, 2018, provisional application No. 62/713,914, filed on Aug. 2, 2018, provisional application No. 62/713,959, filed on Aug. 2, 2018, provisional application No. 62/859,694, filed on Jun. 10, 2019, provisional application No. 62/858,925, filed on Jun. 7, 2019, provisional application No. 62/855,781, filed on May 31, 2019, provisional application No. 62/779,173, filed on Dec. 13, 2018, provisional application No. 62/714,010, filed on Aug. 2, 2018, provisional application No. 62/714,025, filed on Aug. 2, 2018, provisional application No. 62/855,766, filed on May 31, 2019, provisional application No. 62/714,031, filed on Aug. 2, 2018, provisional application No. 62/714,035, filed on Aug. 2, 2018, provisional application No. 62/714,034, filed on Aug. 2, 2018.

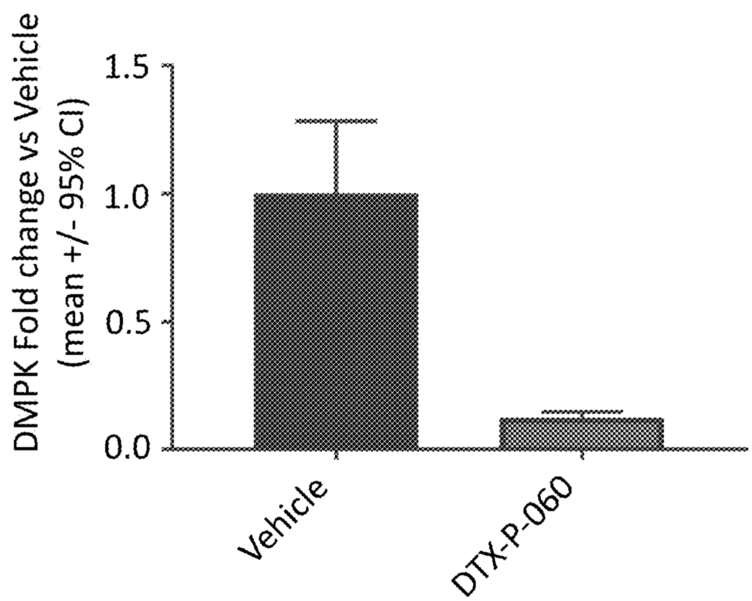

DMPK Fold change vs Vehicle
(mean +/- 95% CI)
1.5
1.0
0.5
0.0
Vehicle
DTX-P-060

FIG. 1

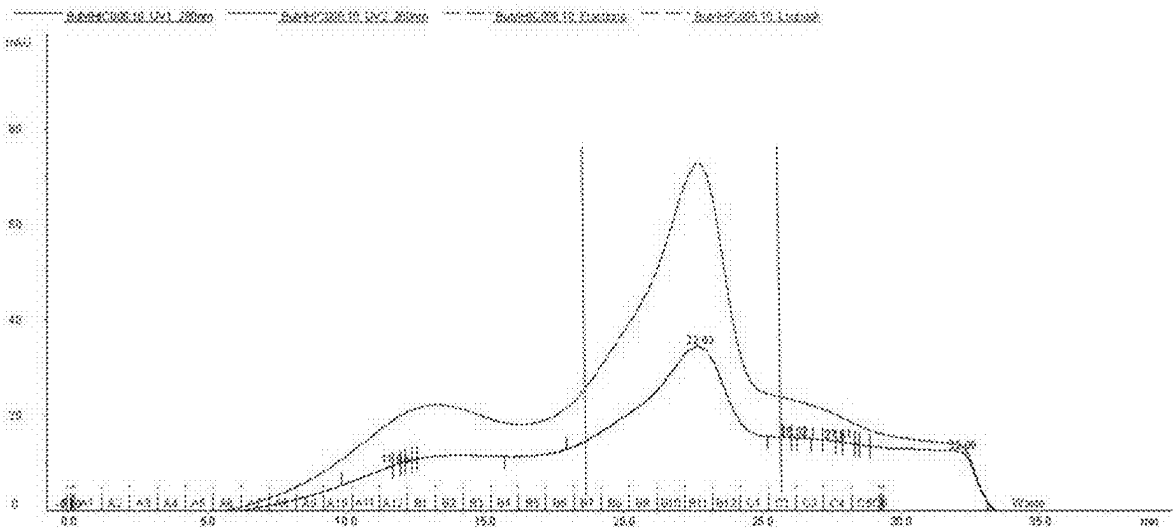
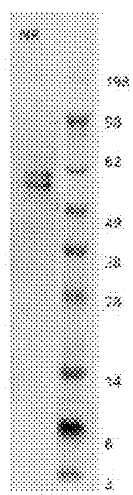


FIG. 2A



NuPage 4-12% 1mm SDS-PAGE
MES running buffer, 150v 50min

FIG. 2B

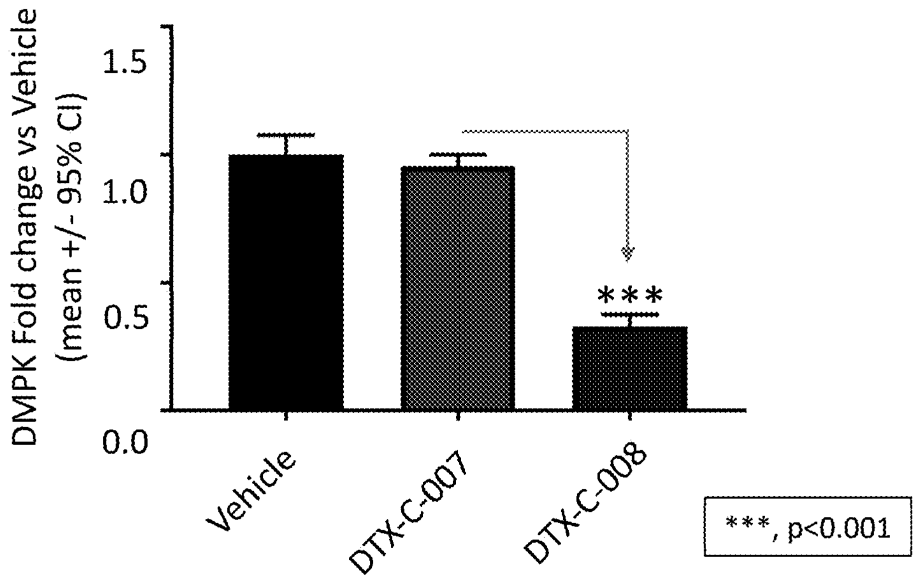

DMPK Fold change vs Vehicle
(mean +/- 95% CI)
1.5
1.0
0.5
0.0
***
Vehicle
DTX-C-007
DTX-C-008
***, p<0.001

FIG. 3

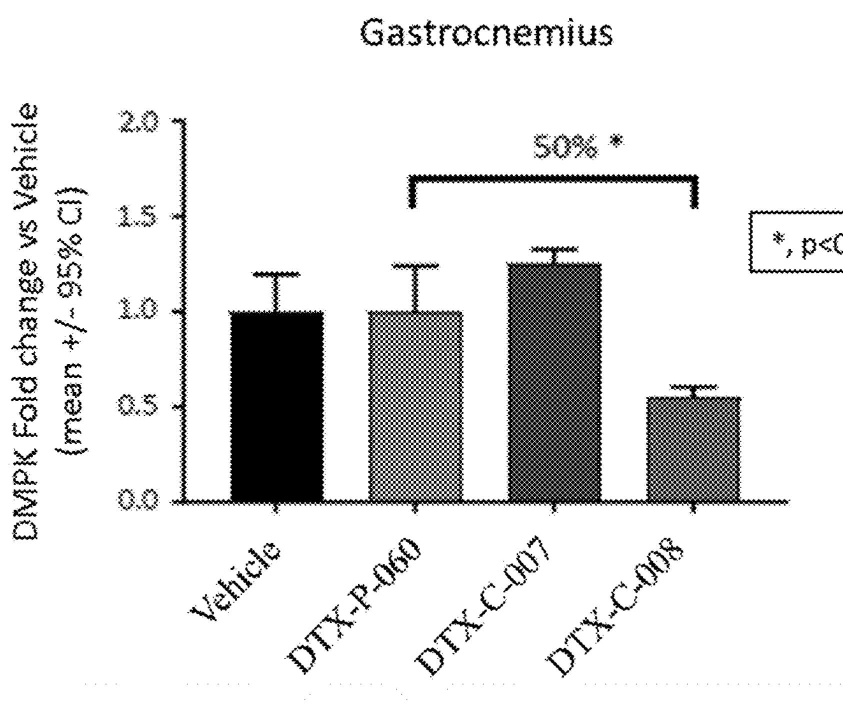

Gastrocnemius
DMPK Fold change vs Vehicle
(mean +/- 95% CI)
2.0
1.5
1.0
0.5
0.0
50% *
*, p<0.05
Vehicle
DTX-P-060
DTX-C-007
DTX-C-008

FIG. 4A

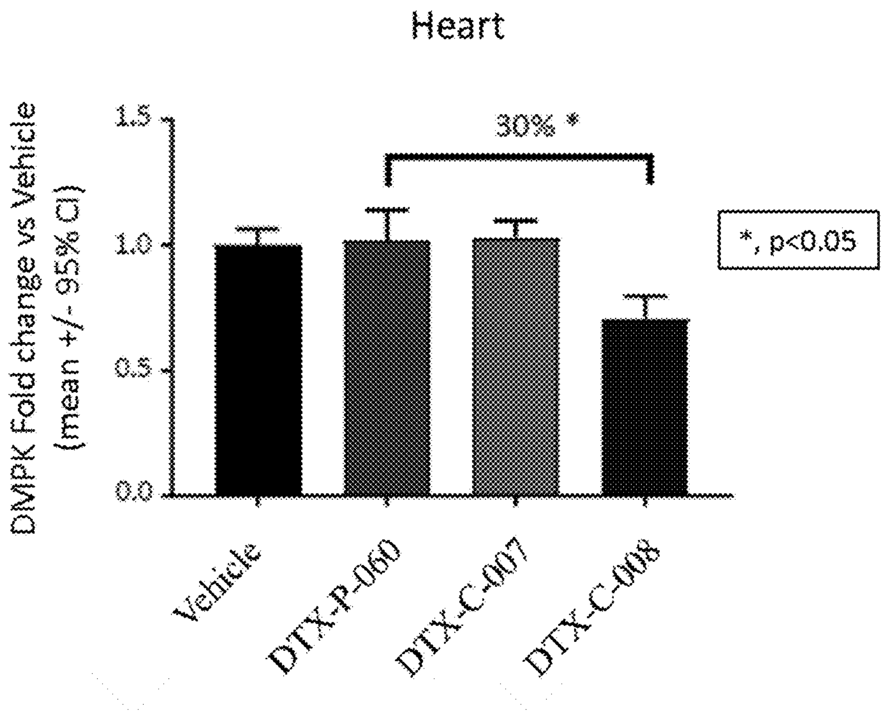

Heart
DMPK Fold change vs Vehicle (mean +/- 95% CI)
1.5
1.0
0.5
0.0
30% *
*, p<0.05
Vehicle
DTX-P-060
DTX-C-007
DTX-C-008

FIG. 4B

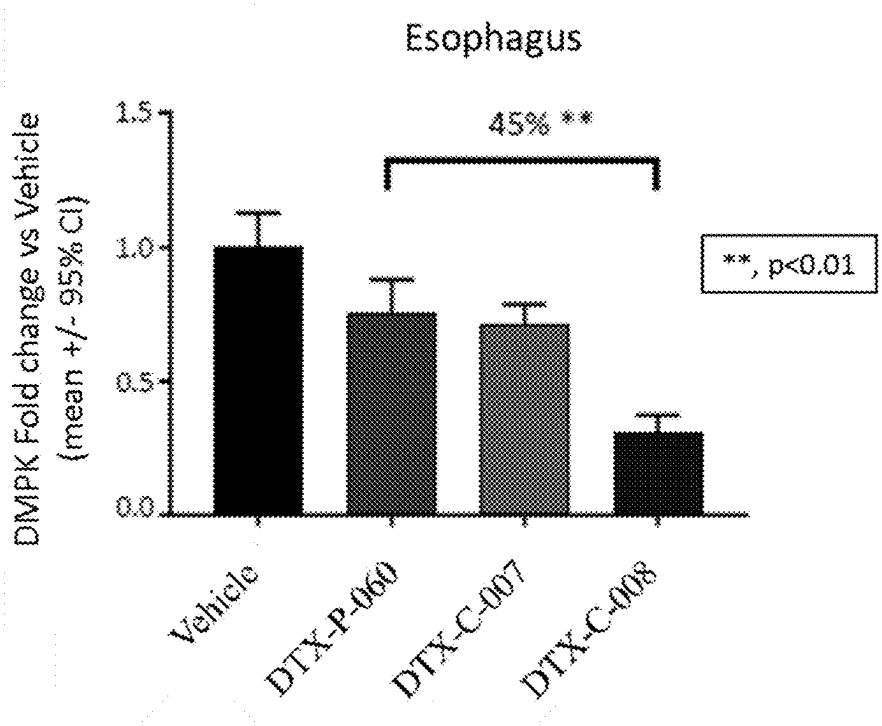

Esophagus
DMPK Fold change vs Vehicle (mean +/- 95% CI)
1.5
1.0
0.5
0.0
45% **
**, p<0.01
Vehicle
DTX-P-060
DTX-C-007
DTX-C-008

FIG. 4C

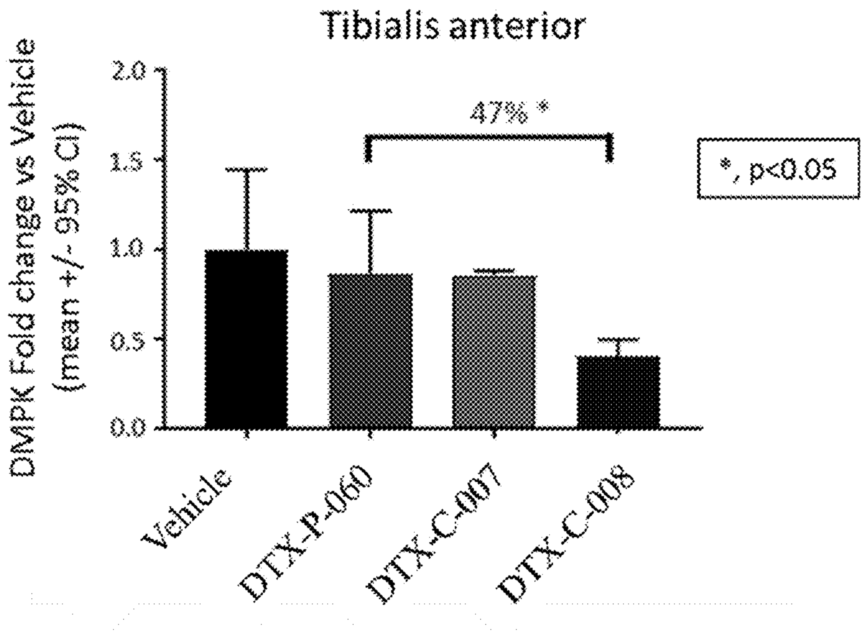

Tibialis anterior
DMPK Fold change vs Vehicle (mean +/- 95% CI)
2.0
1.5
1.0
0.5
0.0
47% *
*, p<0.05
Vehicle
DTX-P-060
DTX-C-007
DTX-C-008

FIG. 4D

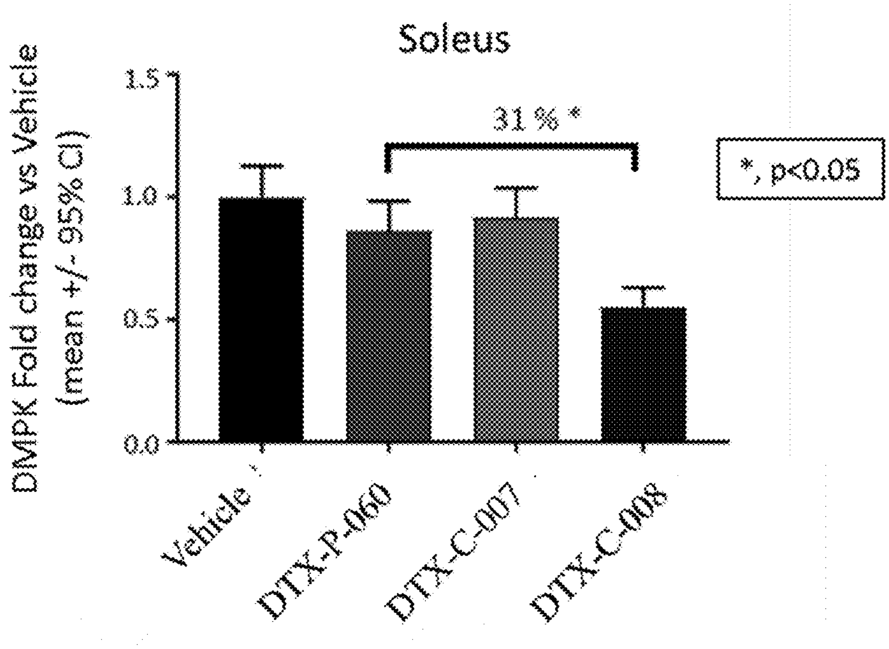

Soleus
DMPK Fold change vs Vehicle (mean +/- 95% CI)
1.5
1.0
0.5
0.0
31 % *
*, p<0.05
Vehicle
DTX-P-060
DTX-C-007
DTX-C-008

FIG. 4E

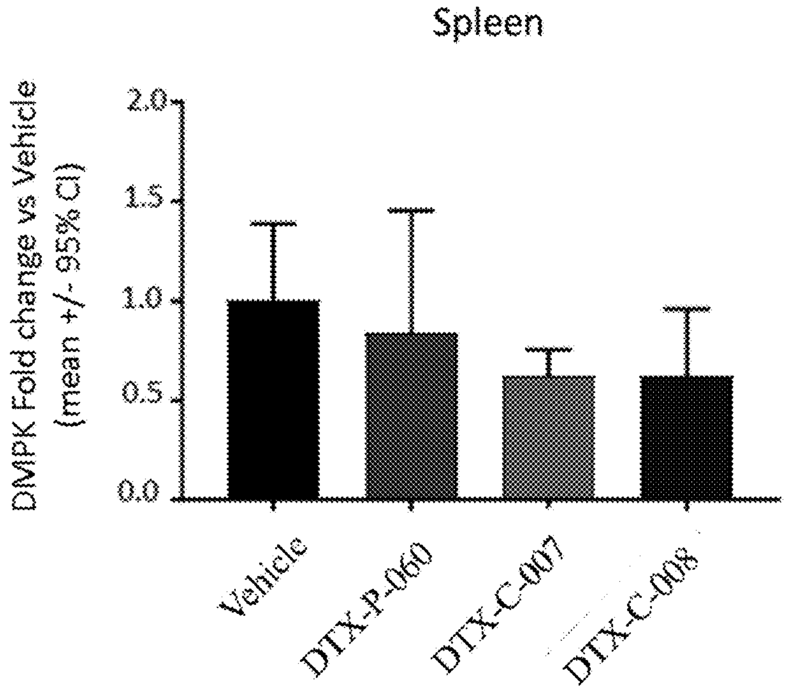

Spleen
DMPK Fold change vs Vehicle (mean +/- 95% CI)
2.0
1.5
1.0
0.5
0.0
Vehicle
DTX-P-060
DTX-C-007
DTX-C-008

FIG. 5A

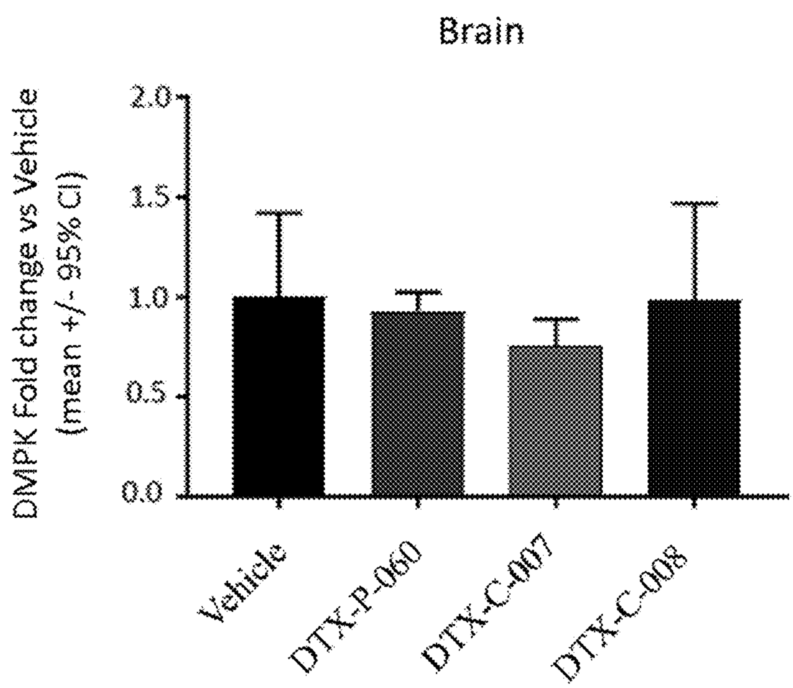

Brain
DMPK Fold change vs Vehicle (mean +/- 95% CI)
2.0
1.5
1.0
0.5
0.0
Vehicle
DTX-P-060
DTX-C-007
DTX-C-008

FIG. 5B

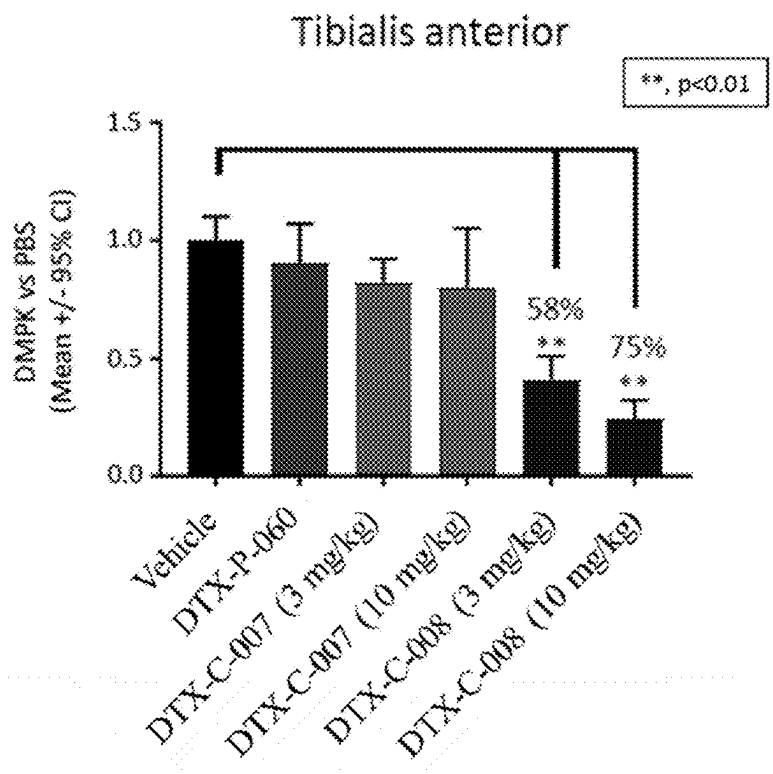

Tibialis anterior
**, p<0.01
DMPK vs PBS
(Mean +/- 95% CI)
1.5
1.0
0.5
0.0
58%
**
75%
**
Vehicle
DTX-P-060
DTX-C-007 (3 mg/kg)
DTX-C-007 (10 mg/kg)
DTX-C-008 (3 mg/kg)
DTX-C-008 (10 mg/kg)

FIG. 6A

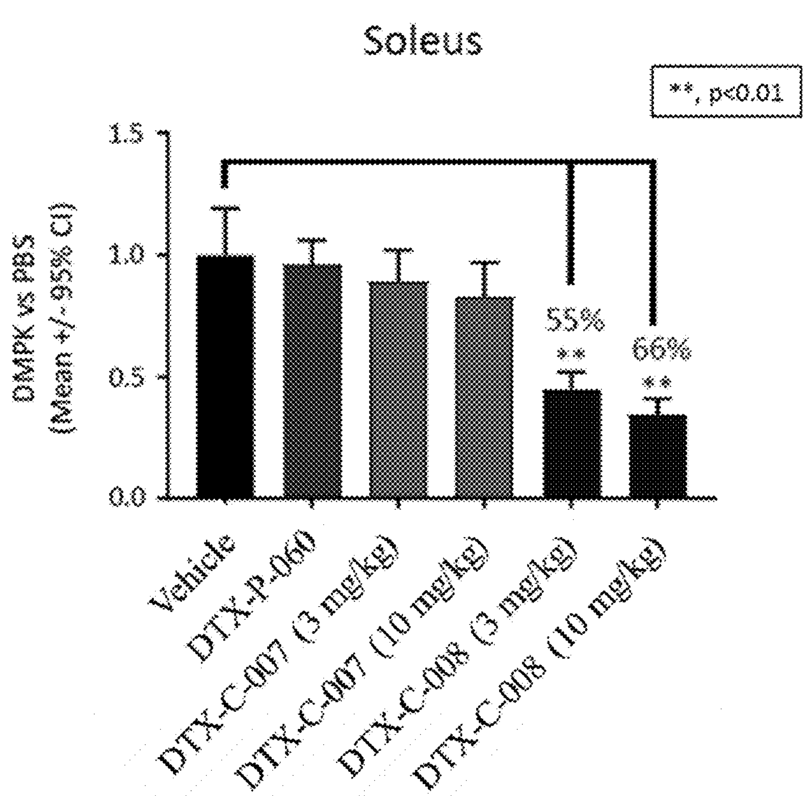

Soleus
**, p<0.01
DMPK vs PBS
(Mean +/- 95% CI)
1.5
1.0
0.5
0.0
55%
**
66%
**
Vehicle
DTX-P-060
DTX-C-007 (3 mg/kg)
DTX-C-007 (10 mg/kg)
DTX-C-008 (3 mg/kg)
DTX-C-008 (10 mg/kg)

FIG. 6B

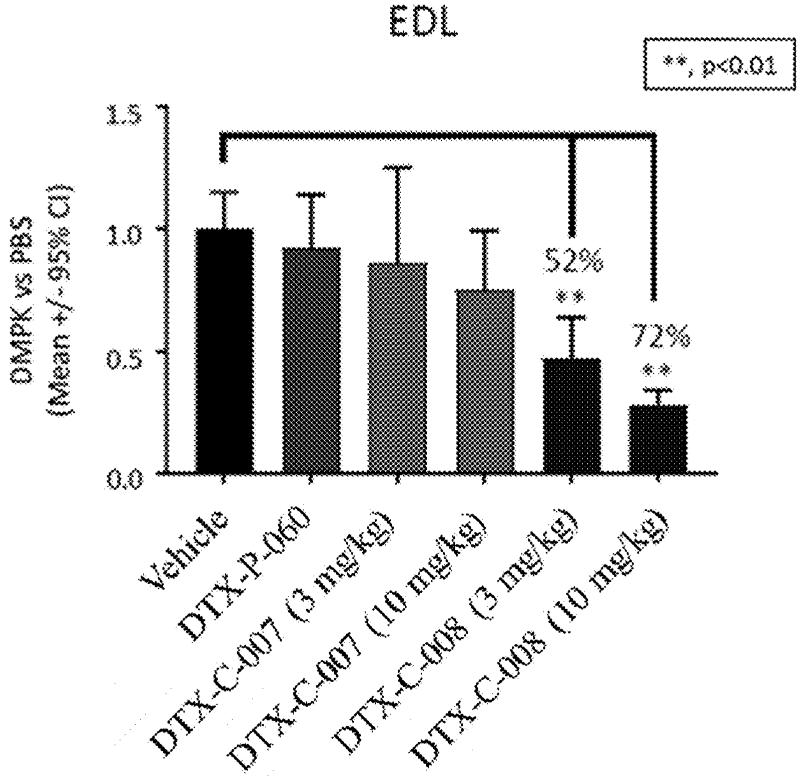

EDL
**, p<0.01
DMPK vs PBS
(Mean +/- 95% CI)
1.5
1.0
0.5
0.0
52%
**
72%
**
Vehicle
DTX-P-060
DTX-C-007 (3 mg/kg)
DTX-C-007 (10 mg/kg)
DTX-C-008 (3 mg/kg)
DTX-C-008 (10 mg/kg)

FIG. 6C

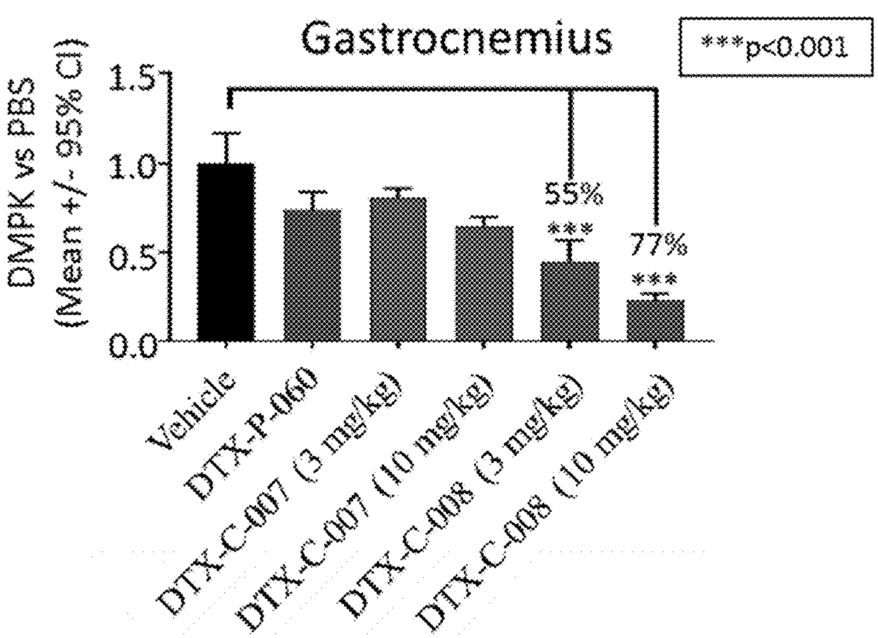
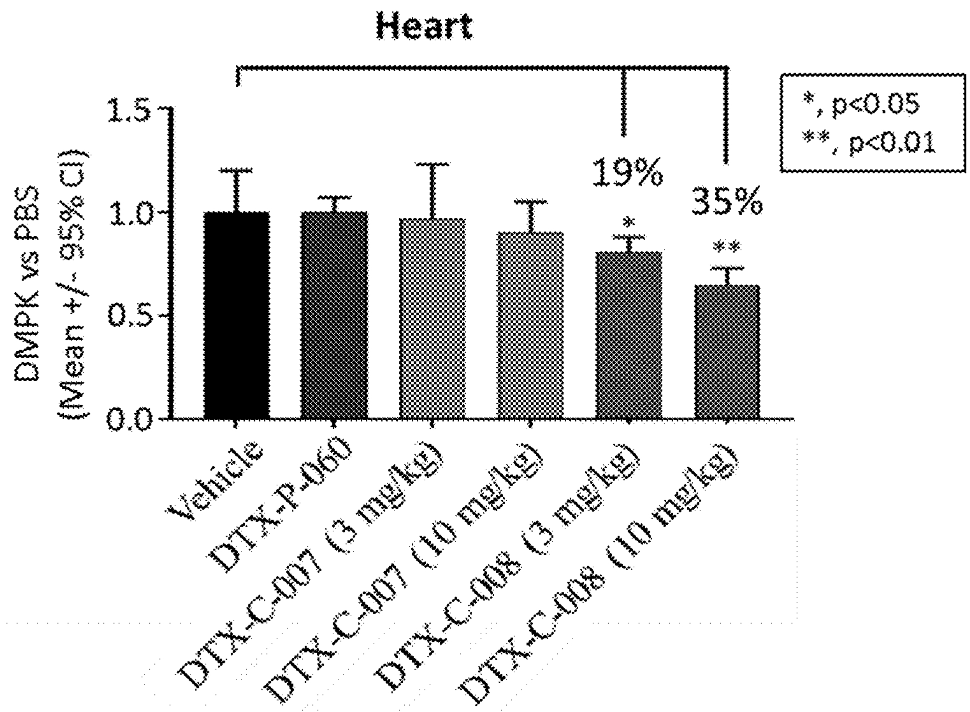
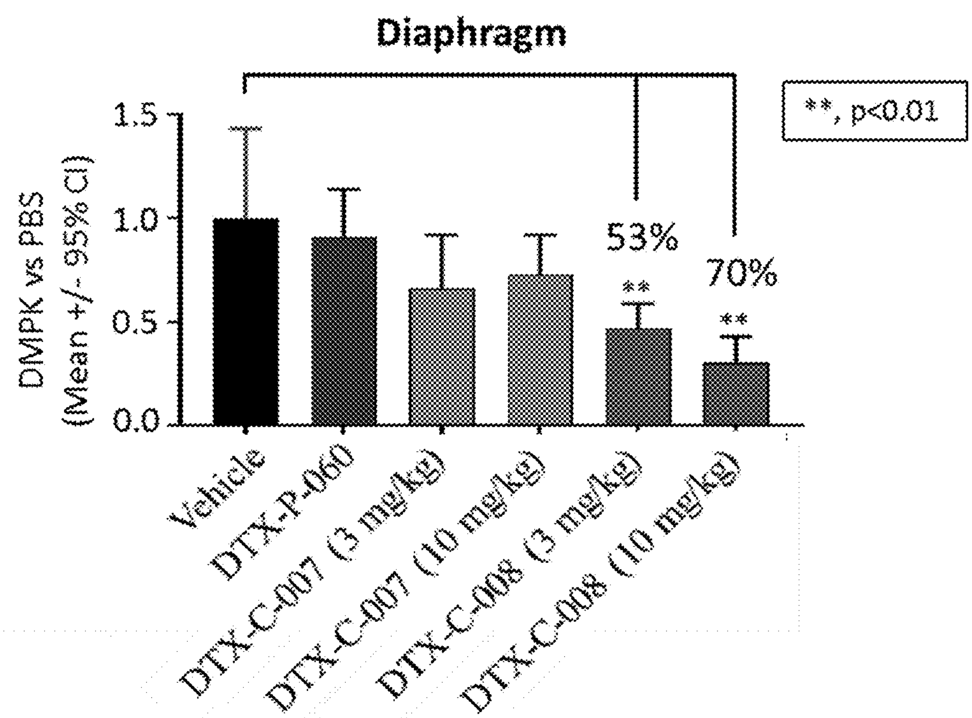

Gastrocnemius
***p<0.001
DMPK vs PBS
(Mean +/- 95% CI)
1.5
1.0
0.5
0.0
55%
***
77%
***
Vehicle
DTX-P-060
DTX-C-007 (3 mg/kg)
DTX-C-007 (10 mg/kg)
DTX-C-008 (3 mg/kg)
DTX-C-008 (10 mg/kg)

FIG. 6D



Heart
*, p<0.05
**, p<0.01
1.5
1.0
0.5
0.0
DMPK vs PBS
(Mean +/- 95% CI)
19%
*
35%
**
Vehicle
DTX-P-060
DTX-C-007 (3 mg/kg)
DTX-C-007 (10 mg/kg)
DTX-C-008 (3 mg/kg)
DTX-C-008 (10 mg/kg)

FIG. 6E



Diaphragm
**, p<0.01
1.5
1.0
0.5
0.0
DMPK vs PBS
(Mean +/- 95% CI)
53%
**
70%
**
Vehicle
DTX-P-060
DTX-C-007 (3 mg/kg)
DTX-C-007 (10 mg/kg)
DTX-C-008 (3 mg/kg)
DTX-C-008 (10 mg/kg)

FIG. 6F

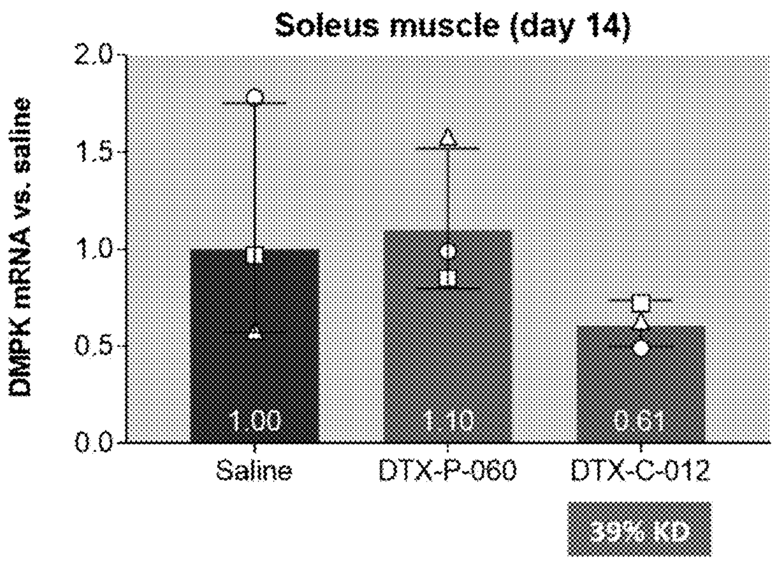

Soleus muscle (day 14)
DMPK mRNA vs. saline
2.0
1.5
1.0
0.5
0.0
1.00
1.10
0.61
Saline
DTX-P-060
DTX-C-012
39% KD

FIG. 7A

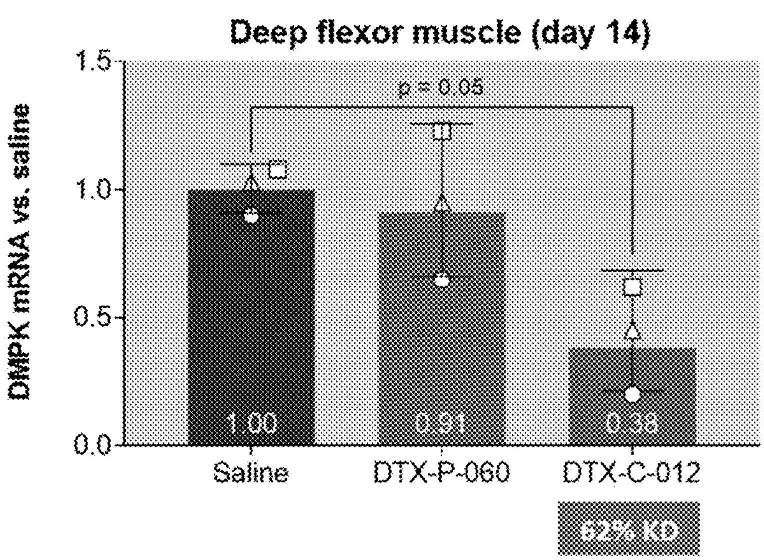

Deep flexor muscle (day 14)
DMPK mRNA vs. saline
1.5
1.0
0.5
0.0
p = 0.05
1.00
0.91
0.38
Saline
DTX-P-060
DTX-C-012
62% KD

FIG. 7B

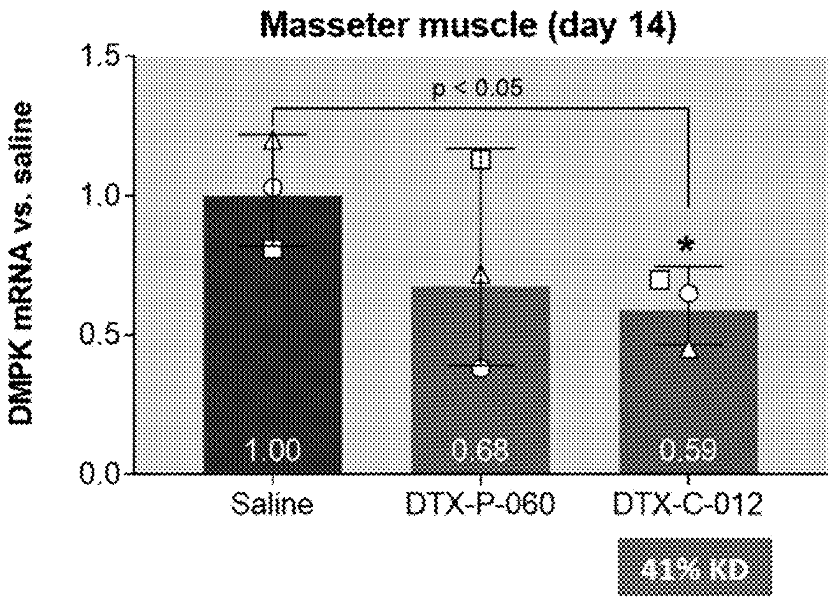

Masseter muscle (day 14)
DMPK mRNA vs. saline
1.5
1.0
0.5
0.0
p < 0.05
*
1.00
0.68
0.59
Saline
DTX-P-060
DTX-C-012
41% KD

FIG. 7C

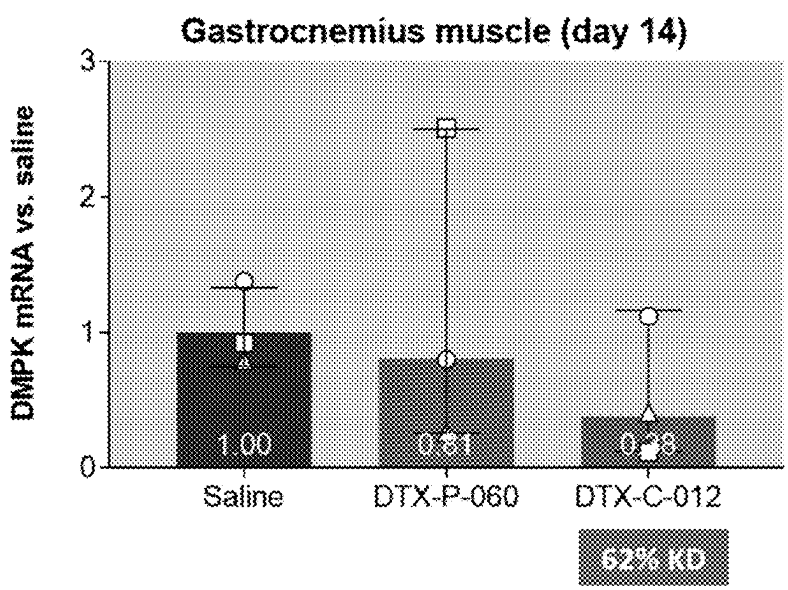

Gastrocnemius muscle (day 14)
DMPK mRNA vs. saline
3
2
1
0
1.00
0.81
Saline
DTX-P-060
DTX-C-012
62% KD

FIG. 7D

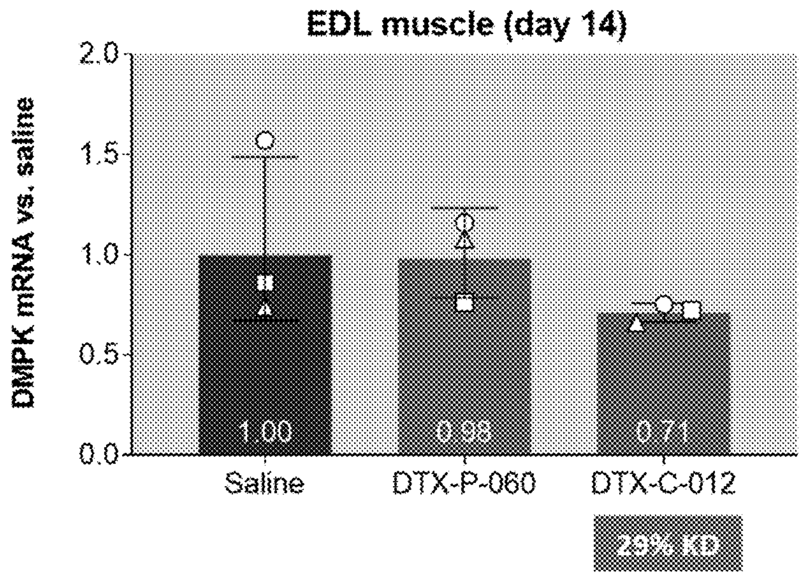

EDL muscle (day 14)
DMPK mRNA vs. saline
2.0
1.5
1.0
0.5
0.0
1.00
0.98
0.71
Saline
DTX-P-060
DTX-C-012
29% KD

FIG. 7E

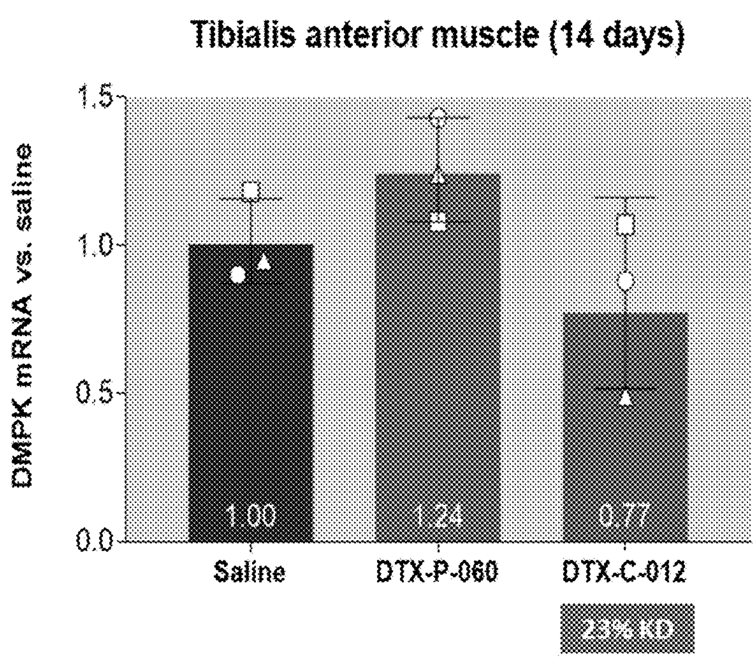

Tibialis anterior muscle (14 days)
DMPK mRNA vs. saline
1.5
1.0
0.5
0.0
1.00
1.24
0.77
Saline
DTX-P-060
DTX-C-012
23% KD

FIG. 7F

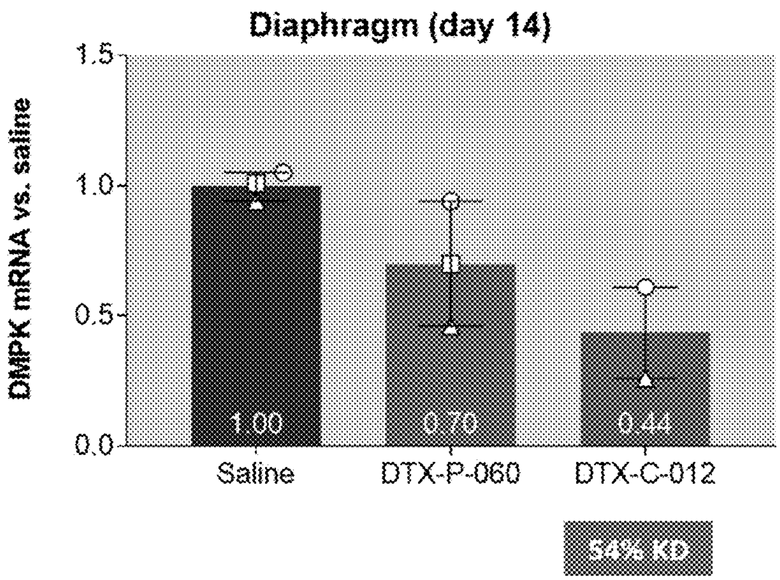

Diaphragm (day 14)
DMPK mRNA vs. saline
1.5
1.0
0.5
0.0
1.00
0.70
0.44
Saline
DTX-P-060
DTX-C-012
54% KD

FIG. 7G

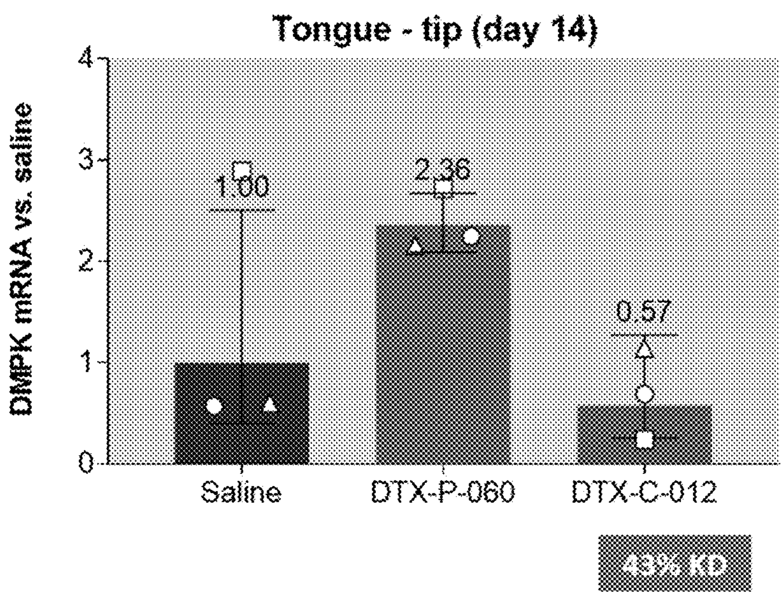

Tongue - tip (day 14)
DMPK mRNA vs. saline
4
3
2
1
0
1.00
2.36
0.57
Saline
DTX-P-060
DTX-C-012
43% KD

FIG. 7H

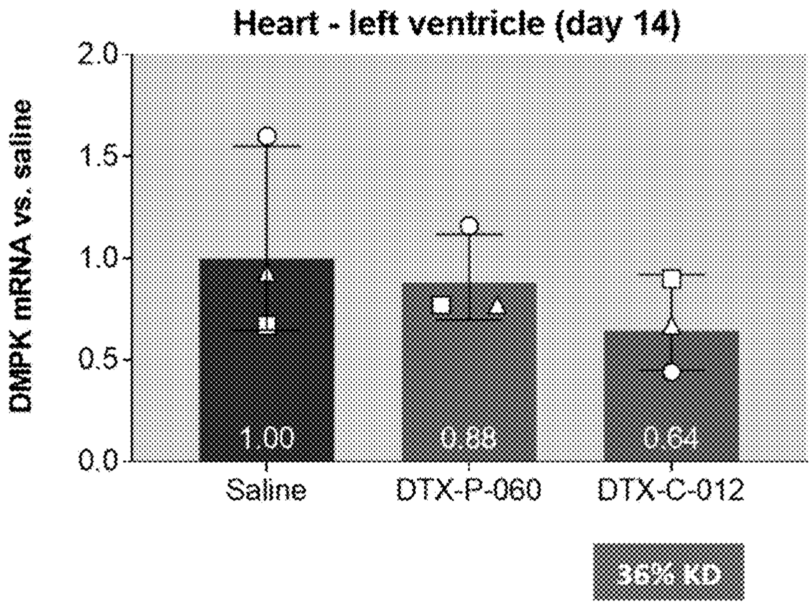

Heart - left ventricle (day 14)
DMPK mRNA vs. saline
2.0
1.5
1.0
0.5
0.0
1.00
0.88
0.64
Saline
DTX-P-060
DTX-C-012
36% KD

FIG. 7I

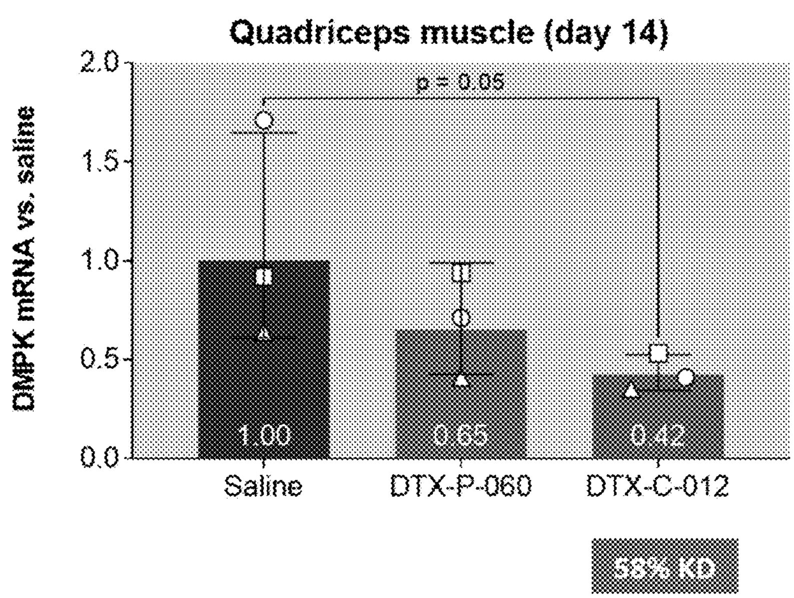

Quadriceps muscle (day 14)
DMPK mRNA vs. saline
2.0
1.5
1.0
0.5
0.0
p = 0.05
1.00
0.65
0.42
Saline
DTX-P-060
DTX-C-012
58% KD

FIG. 7J

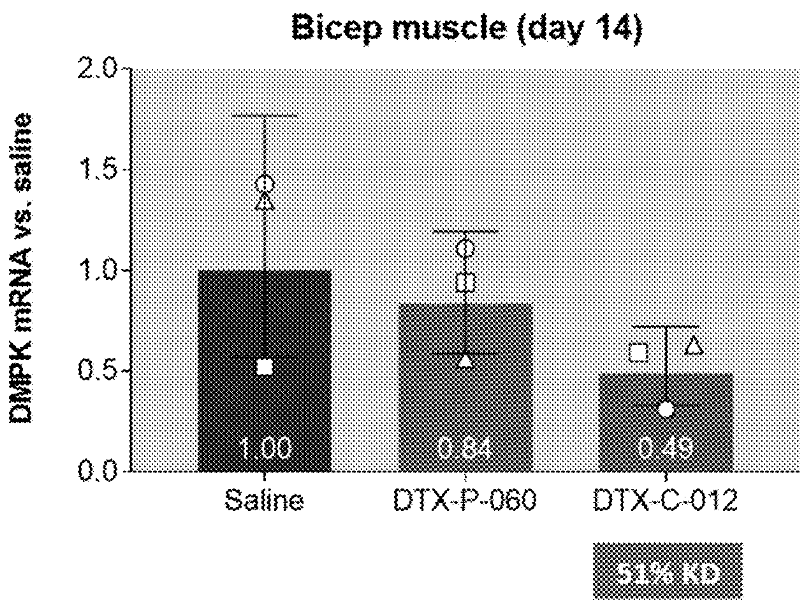

Bicep muscle (day 14)
DMPK mRNA vs. saline
2.0
1.5
1.0
0.5
0.0
1.00
0.84
0.49
Saline
DTX-P-060
DTX-C-012
51% KD

FIG. 7K

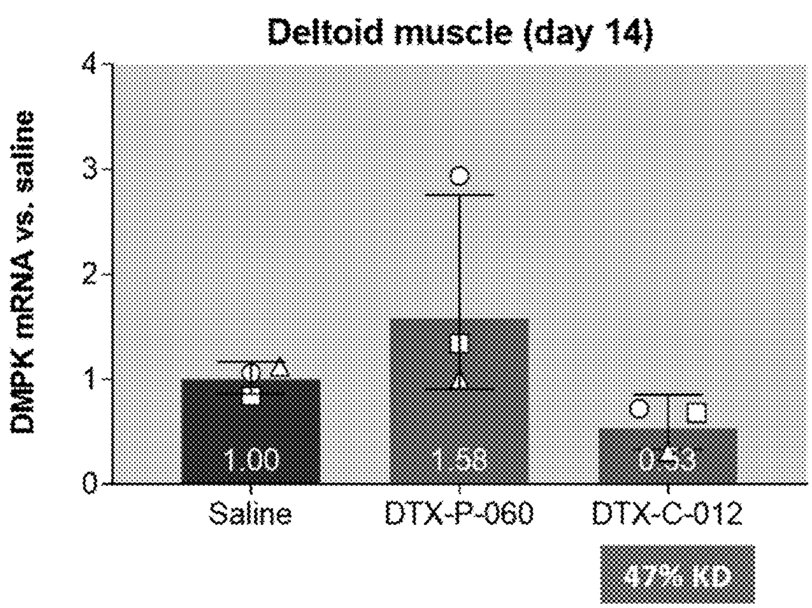

Deltoid muscle (day 14)
DMPK mRNA vs. saline
4
3
2
1
0
1.00
1.58
0.53
Saline
DTX-P-060
DTX-C-012
47% KD

FIG. 7L

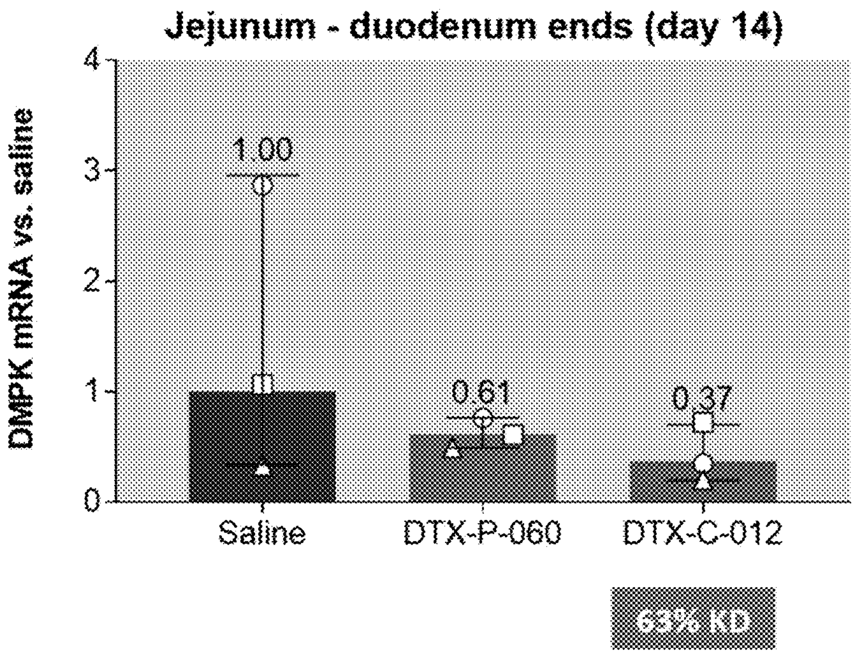

Jejunum - duodenum ends (day 14)
DMPK mRNA vs. saline
4
3
2
1
0
1.00
0.61
0.37
Saline
DTX-P-060
DTX-C-012
63% KD

FIG. 8A

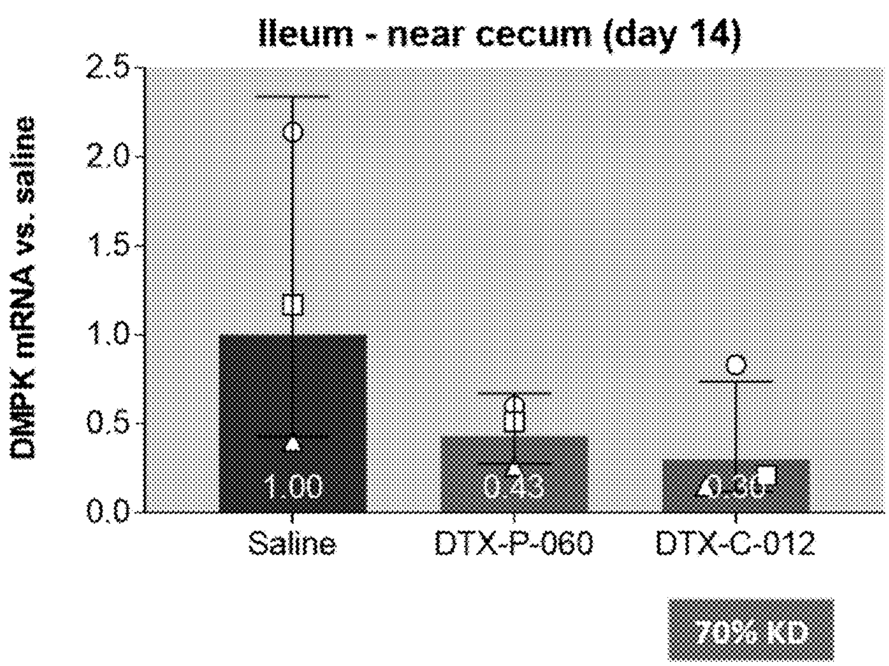

Ileum - near cecum (day 14)
DMPK mRNA vs. saline
2.5
2.0
1.5
1.0
0.5
0.0
1.00
0.43
0.30
Saline
DTX-P-060
DTX-C-012
70% KD

FIG. 8B

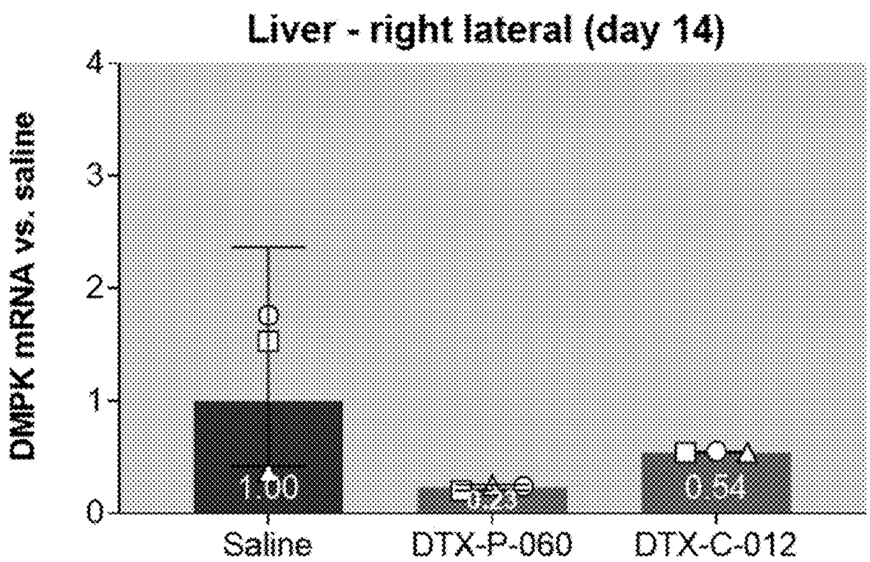

Liver - right lateral (day 14)
DMPK mRNA vs. saline
4
3
2
1
0
1.00
0.23
0.54
Saline
DTX-P-060
DTX-C-012

FIG. 9A

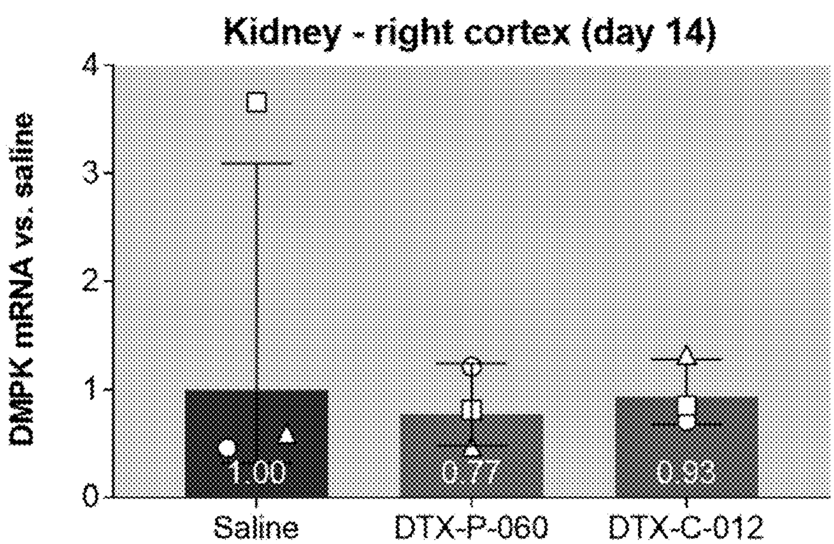
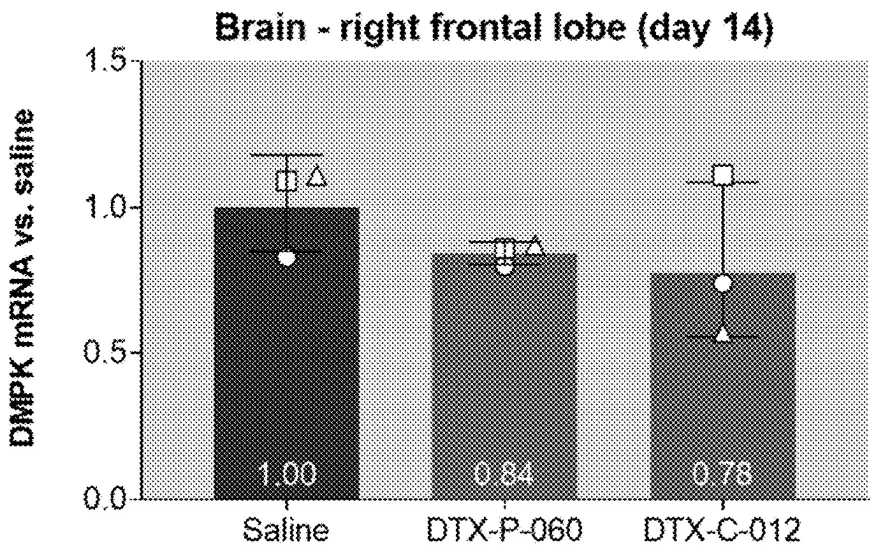
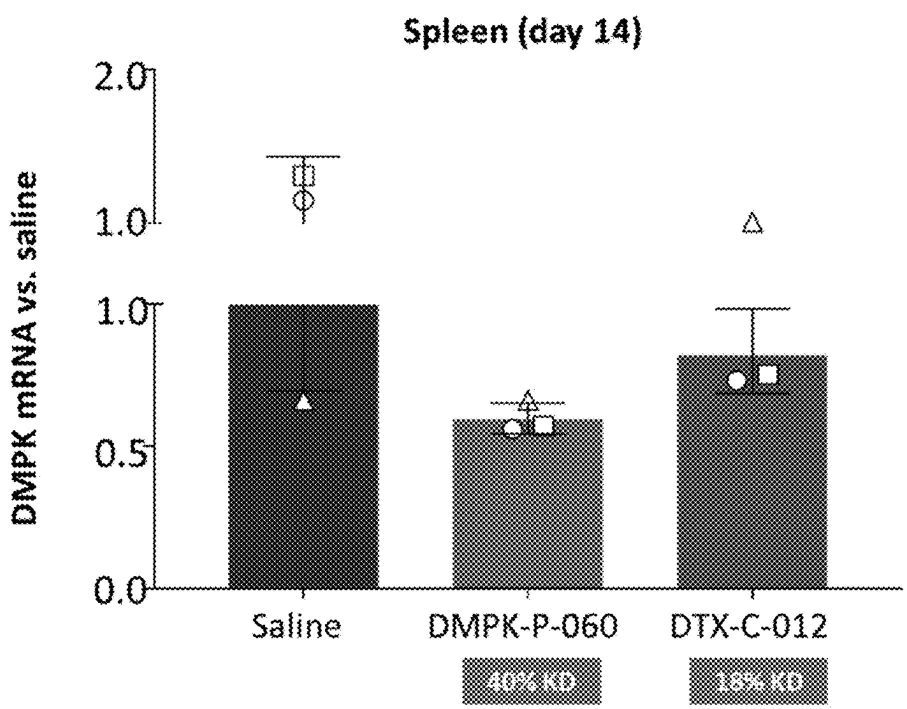

Kidney - right cortex (day 14)
DMPK mRNA vs. saline
4
3
2
1
0
1.00
0.77
0.93
Saline
DTX-P-060
DTX-C-012

FIG. 9B



Brain - right frontal lobe (day 14)
DMPK mRNA vs. saline
1.5
1.0
0.5
0.0
1.00
0.84
0.78
Saline
DTX-P-060
DTX-C-012

FIG. 9C



Spleen (day 14)
DMPK mRNA vs. saline
2.0
1.0
1.0
0.5
0.0
Saline
DMPK-P-060
DTX-C-012
40% KD
18% KD

FIG. 9D

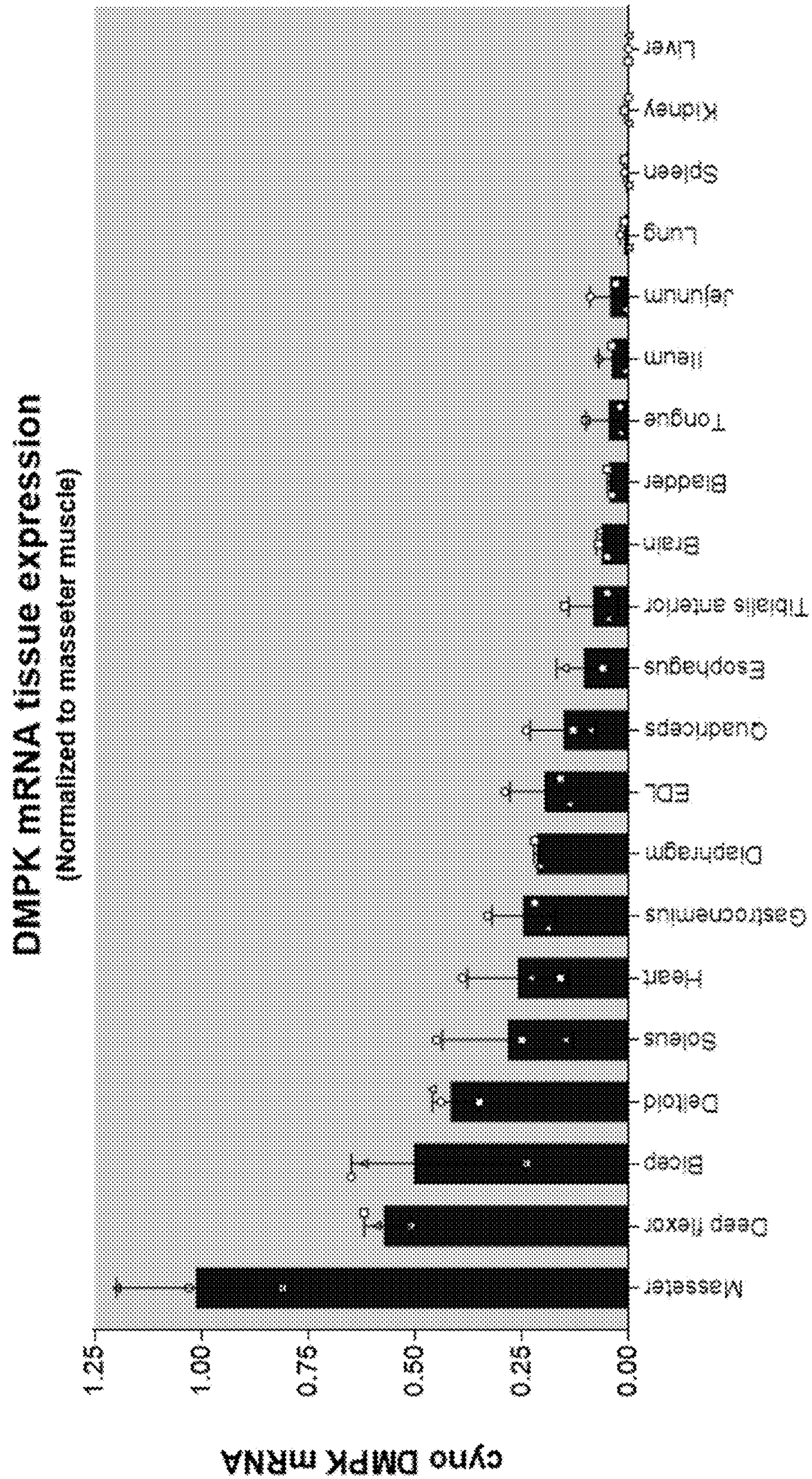

DMPK mRNA tissue expression
(Normalized to masseter muscle)
cyno DMPK mRNA
1.25
1.00
0.75
0.50
0.25
0.00
Masseter
Deep flexor
Bicep
Deltoid
Soleus
Heart
Gastrocnemius
Diaphragm
EDL
Quadriceps
Esophagus
Tibialis anterior
Brain
Bladder
Tongue
Ileum
Jejunum
Lung
Spleen
Kidney
Liver

FIG. 10

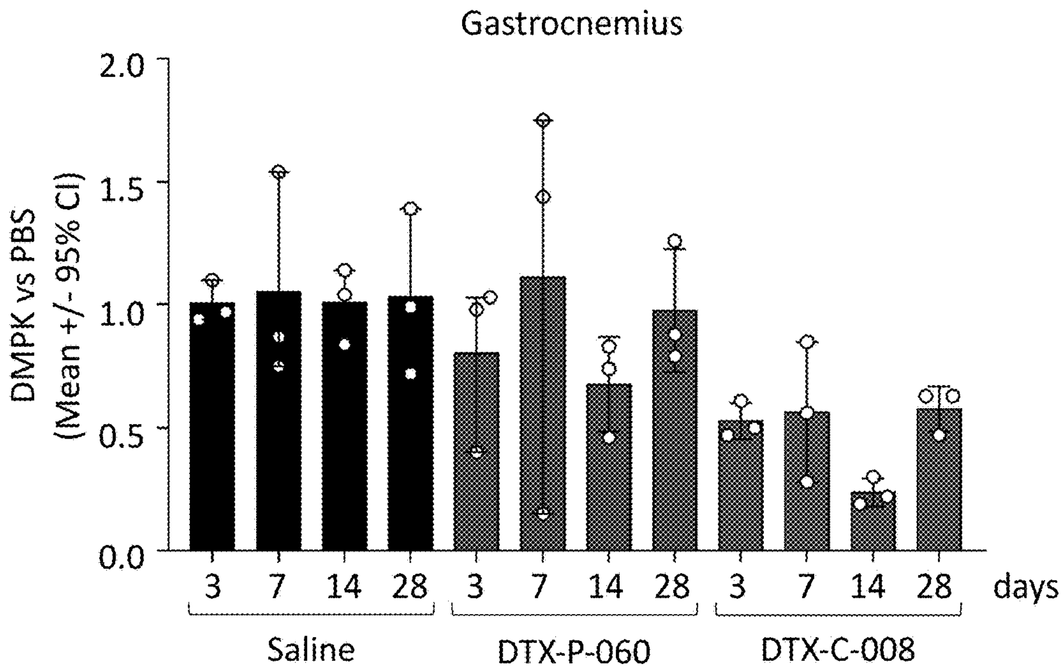


FIG. 11A

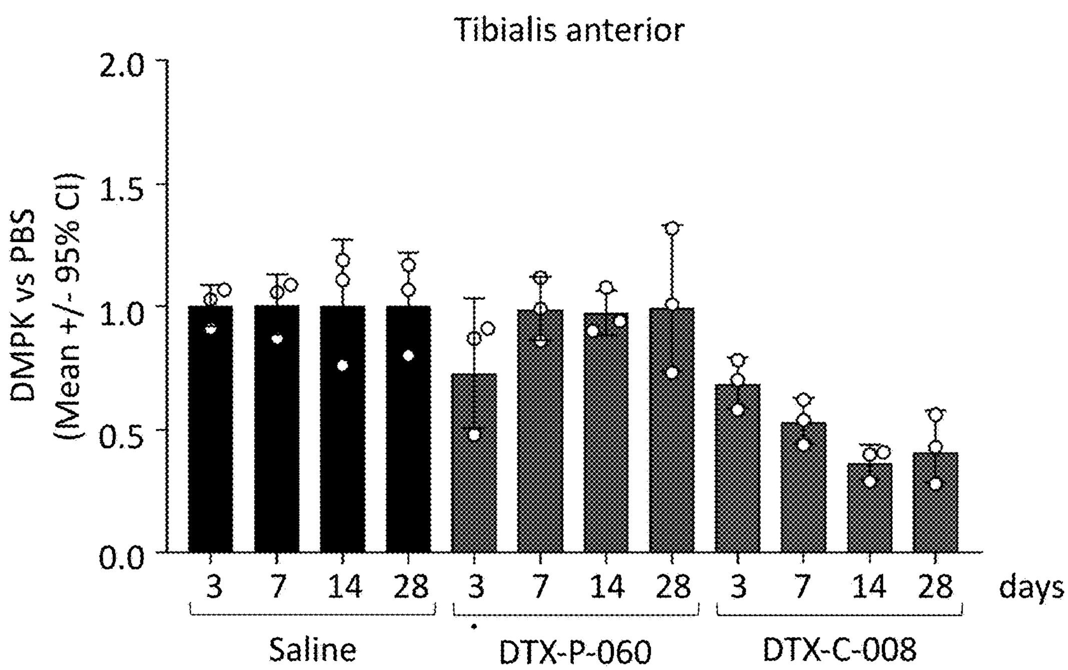


FIG. 11B

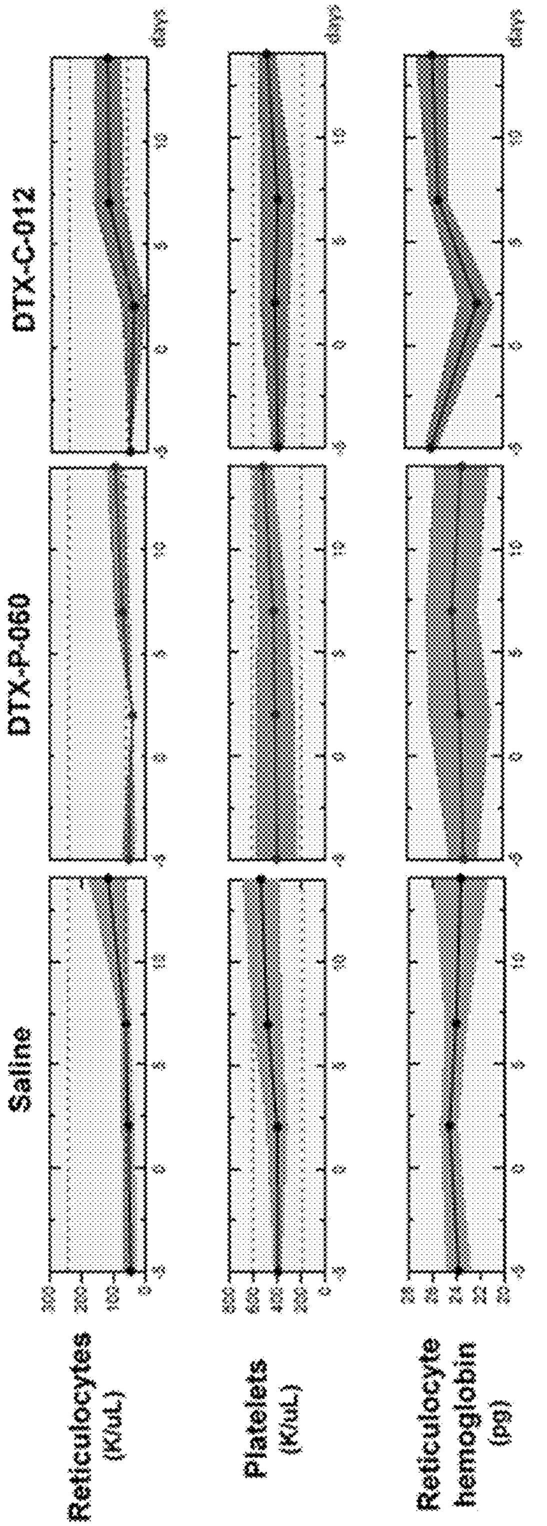

Saline
DTX-P-060
DTX-C-012
Reticulocytes (K/uL)
Platelets (K/uL)
Reticulocyte hemoglobin (pg)
days

FIG. 12

MUSCLE-TARGETING COMPLEXES AND USES THEREOF

FIELD OF THE INVENTION

[0001] The present application relates to targeting complexes for delivering molecular payloads (e.g., oligonucleotides) to cells and uses thereof, particularly uses relating to treatment of disease.

REFERENCE TO THE SEQUENCE LISTING

[0002] The content of the electronic sequence listing (D082470001US19-SEQ-ZJG.xml; Size: 1,174,452 bytes; and Date of Creation: May 16, 2025) is herein incorporated by reference in its entirety.

BACKGROUND OF INVENTION

[0003] Muscle diseases are often associated with muscle weakness and/or muscle dysfunction that lead to life-threatening complications. Many examples of such diseases have been characterized, including various forms of muscular dystrophy (e.g., Duchenne, facioscapulohumeral, myotonic, and oculopharyngeal), Pompe disease, centronuclear myopathy, familial hypertrophic cardiomyopathy, Laing distal myopathy, Fibrodysplasia Ossificans Progressiva, Friedereich's ataxia, myofibrilar myopathy, and others. These conditions are generally hereditary, but can arise spontaneously. These conditions are often congenital but can arise later in life. Many rare muscle disease are single gene disorders associated with gain-of-function or loss-of-function mutations, which may have dominant or recessive phenotypes. For example, activating mutations have been identified in genes encoding ion channels, structural proteins, metabolic proteins, and signaling proteins that contribute to muscle disease. Despite advances in understanding the genetic etiology of muscle disease, effective treatment options remain limited.

SUMMARY OF INVENTION

[0004] According to some aspects, the disclosure provides complexes that target muscle cells for purposes of delivering molecular payloads to those cells. In some embodiments, the complexes of the present disclosure facilitate muscle-specific delivery of molecular payloads that target muscle disease alleles. For example, in some embodiments, complexes provided herein are particularly useful for delivering molecular payloads that modulate the expression or activity of a gene in a subject having or suspected of having a muscle disease associated with the gene (e.g., a gene/disease of Table 1). In some embodiments, complexes provided herein comprise muscle-targeting agents (e.g., muscle targeting antibodies) that specifically bind to receptors on the surface of muscle cells for purposes of delivering molecular payloads to the muscle cells. In some embodiments, the complexes are taken up into the cells via a receptor (e.g., transferrin receptor) mediated internalization, following which the molecular payload may be released to perform a function inside the cells. For example, complexes engineered to deliver oligonucleotides may release the oligonucleotides such that the oligonucleotides can modulate expression or activity of a muscle disease allele. In some embodiments, the oligonucleotides are released by endosomal cleavage of covalent linkers connecting oligonucleotides and muscle-targeting agents of the complexes.

[0005] In some embodiments, methods are provided for treating a subject diagnosed as having a muscle disease associated with a disease allele (e.g., a gain-of-function disease allele). In some embodiments, the methods involve administering to the subject a complex comprising a muscle-targeting agent covalently linked to a molecular payload configured to inhibit expression or activity of the disease allele. In some embodiments, the muscle-targeting agent specifically binds to an internalizing cell surface receptor on muscle cells of the subject. In some embodiments, the muscle disease is hereditary, and may exhibit increased severity in sequential family generations of the subject. In some embodiments, the subject has been diagnosed as having the muscle disease based on a genetic analysis of the disease allele. In some embodiments, the subject exhibits progressive muscle weakness and/or sarcopenia prior to the administration. In some embodiments, the subject exhibits myotonia prior to the administration.

[0006] According to some aspects, a method for treating a subject diagnosed as having a muscle disease (e.g., associated with a gain-of-function disease allele) is provided. In some embodiments, the methods comprise administering to the subject a complex comprising a muscle-targeting agent covalently linked to a molecular payload configured to inhibit expression or activity of the disease allele. In some embodiments, the muscle-targeting agent specifically binds to an internalizing cell surface receptor on muscle cells of the subject.

[0007] In some embodiments, the muscle disease is hereditary. In some embodiments, the muscle disease exhibits increased severity in sequential family generations of the subject. In some embodiments, the subject was diagnosed as having the muscle disease based on a genetic analysis of a disease allele. In some embodiments, the subject exhibits progressive muscle weakness and/or sarcopenia prior to the administration. In some embodiments, the subject exhibits myotonia, e.g., measurable with electromyography, prior to the administration.

[0008] In some embodiments, the muscle-targeting agent is a muscle-targeting antibody. In some embodiments, the muscle-targeting antibody specifically binds to an extracellular epitope of a transferrin receptor. In some embodiments, the extracellular epitope of the transferrin receptor comprises an epitope of the apical domain of the transferrin receptor. In some embodiments, the muscle-targeting antibody specifically binds to an epitope of a sequence in the range of C89 to F760 of SEQ ID NO: 1-3. In some embodiments, the equilibrium dissociation constant (Kd) of binding of the muscle-targeting antibody to the transferrin receptor is in a range from 10^{-11} M to 10^{-6} M. In some embodiments, the muscle-targeting antibody competes for specific binding to an epitope of a transferrin receptor with an antibody listed in Table 2.

[0009] In some embodiments, the muscle-targeting antibody competes for specific binding to an epitope of a transferrin receptor with a Kd of less than or equal to 10^{-6} M. In some embodiments, the Kd is in a range of 10^{-11} M to 10^{-6} M.

[0010] In some embodiments, the muscle-targeting antibody does not specifically bind to the transferrin binding site of the transferrin receptor and/or the muscle-targeting antibody does not inhibit binding of transferrin to the transferrin receptor. In some embodiments, the muscle-targeting antibody is cross-reactive with extracellular epitopes of two or more of a human, non-human primate and rodent transferrin receptor. In some embodiments, the method is configured to promote transferrin receptor mediated internalization of the molecular payload into a muscle cell.

[0011] In some embodiments, the muscle-targeting antibody is a chimeric antibody, optionally wherein the chimeric antibody is a humanized monoclonal antibody. In some embodiments, the muscle-targeting antibody is in the form of a ScFv, a Fab fragment, Fab' fragment, $F(ab')_2$ fragment, or Fv fragment.

[0012] In some embodiments, the molecular payload is an oligonucleotide. In some embodiments, the oligonucleotide comprises a region of complementarity to gene listed in Table 1 or mRNA encoded therefrom. In some embodiments, the oligonucleotide is a gapmer oligonucleotide, a mixmer oligonucleotide, an antisense oligonucleotide, a RNAi oligonucleotide, a messenger RNA (mRNA), or a guide sequence.

[0013] In some embodiments, the complex is administered to the subject by extramuscular parenteral administration. In some embodiments, the complex is administered to the subject by intravenous administration. In some embodiments, the complex is administered to the subject by subcutaneous administration of the complex.

[0014] In some aspects, a complex is provided that comprises a muscle-targeting agent linked to a single-stranded oligonucleotide. In some embodiments, the muscle-targeting agent specifically binds to an internalizing cell surface receptor on muscle cells, and wherein the oligonucleotide comprises a region of complementarity to a muscle disease gene.

[0015] In some embodiments, a composition is provided that comprises a plurality of complexes, each complex comprising a muscle-targeting agent covalently linked to at two, at least three or more (e.g., 2 to 6) oligonucleotides. In some embodiments, the muscle-targeting agent specifically binds to an internalizing cell surface receptor on muscle cells of a subject, and each oligonucleotide comprises a region of complementarity to a muscle disease gene.

[0016] In some aspects, a complex is provided that comprises a muscle-targeting agent covalently linked to a molecular payload configured to modulate expression or activity of a muscle disease gene that encodes a non-secreted product that functions within muscle cells. In some embodiments, the muscle-targeting agent specifically binds to an internalizing cell surface receptor on muscle cells.

[0017] In some embodiments, the muscle-targeting agent is a muscle-targeting antibody. In some embodiments, the muscle-targeting antibody specifically binds to an extracellular epitope of a transferrin receptor. In some embodiments, the extracellular epitope of the transferrin receptor comprises an epitope of the apical domain of the transferrin receptor. In some embodiments, the muscle-targeting antibody specifically binds to an epitope of a sequence within amino acids C89 to F760 of SEQ ID NO: 1-3. In some embodiments, the equilibrium dissociation constant (Kd) of binding of the muscle-targeting antibody to the transferrin receptor is in a range from 10^{-11} M to 10^{-6} M. In some embodiments, the muscle-targeting antibody competes for specific binding to an epitope of a transferrin receptor with an antibody listed in Table 2. In some embodiments, the muscle-targeting antibody competes for specific binding to an epitope of a transferrin receptor with a Kd of less than or equal to 10^{-6} M. In some embodiments, the Kd is in a range of 10^{-11} M to 10^{-6} M.

[0018] In some embodiments, the muscle-targeting antibody does not specifically bind to the transferrin binding site of the transferrin receptor and/or wherein the muscle-targeting antibody does not inhibit binding of transferrin to the transferrin receptor. In some embodiments, the muscle-targeting antibody is cross-reactive with extracellular epitopes of two or more of a human, non-human primate and rodent transferrin receptor.

[0019] In some embodiments, the complex is configured to promote transferrin receptor mediated internalization of the molecular payload into a muscle cell. In some embodiments, the muscle-targeting antibody is a chimeric antibody. In some embodiments, the chimeric antibody is a humanized monoclonal antibody.

[0020] In some embodiments, the muscle-targeting antibody is in the form of a ScFv, a Fab fragment, Fab' fragment, $F(ab')_2$ fragment, or Fv fragment.

[0021] In some embodiments, the molecular payload is an oligonucleotide. In some embodiments, the oligonucleotide comprises a region of complementarity to a muscle disease gene having a gain-of-function disease allele.

[0022] In some embodiments, the molecular payload is an polypeptide. In some embodiments, the polypeptide is an E3 ubiquitin ligase inhibitor peptide.

[0023] In some embodiments, the oligonucleotide comprises at least one modified internucleotide linkage. In some embodiments, the at least one modified internucleotide linkage is a phosphorothioate linkage. In some embodiments, the oligonucleotide comprises phosphorothioate linkages in the Rp stereochemical conformation and/or in the Sp stereochemical conformation. In some embodiments, the oligonucleotide comprises phosphorothioate linkages that are all in the Rp stereochemical conformation or that are all in the Sp stereochemical conformation.

[0024] In some embodiments, the oligonucleotide comprises one or more modified nucleotides. In some embodiments, the one or more modified nucleotides are 2'-modified nucleotides.

[0025] In some embodiments, the oligonucleotide is a gapmer oligonucleotide that directs RNAse H-mediated cleavage of an mRNA transcript encoded by the muscle disease gene in a cell. In some embodiments, the gapmer oligonucleotide comprises a central portion of 5 to 15 deoxyribonucleotides flanked by wings of 2 to 8 modified nucleotides.

[0026] In some embodiments, the modified nucleotides of the wings are 2'-modified nucleotides. In some embodiments, the oligonucleotide is a mixmer oligonucleotide.

[0027] In some embodiments, the mixmer oligonucleotide comprises two or more different 2' modified nucleotides. In some embodiments, the oligonucleotide is an RNAi oligonucleotide that promotes RNAi-mediated cleavage of a mRNA transcript encoded by the muscle disease gene.

[0028] In some embodiments, the oligonucleotide is a double-stranded oligonucleotide of 19 to 25 nucleotides in length. In some embodiments, the RNAi oligonucleotide comprises at least one 2' modified nucleotide. In some embodiments, each 2' modified nucleotide is selected from the group consisting of: 2'-O-methyl, 2'-fluoro (2'-F), 2'-O-methoxyethyl (2'-MOE), and 2', 4'-bridged nucleotides.

[0029] In some embodiments, the one or more modified nucleotides are bridged nucleotides. In some embodiments, at least one 2' modified nucleotide is a 2',4'-bridged nucleotide selected from: 2',4'-constrained 2'-O-ethyl (cEt) and locked nucleic acid (LNA) nucleotides.

[0030] In some embodiments, the oligonucleotide comprises a guide sequence for a genome editing nuclease.

[0031] In some embodiments, the oligonucleotide is phosphorodiamidite morpholino oligomer. In some embodiments, the muscle-targeting agent is covalently linked to the molecular payload via a cleavable linker.

[0032] In some embodiments, the cleavable linker is selected from: a protease-sensitive linker, pH-sensitive linker, and glutathione-sensitive linker. In some embodiments, the cleavable linker is a protease-sensitive linker. In some embodiments, the protease-sensitive linker comprises a sequence cleavable by a lysosomal protease and/or an endosomal protease. In some embodiments, the protease-sensitive linker comprises a valine-citrulline dipeptide sequence. In some embodiments, the linker is a pH-sensitive linker that is cleaved at a pH in a range of 4 to 6.

[0033] In some embodiments, the muscle-targeting agent is covalently linked to the molecular payload via a non-cleavable linker. In some embodiments, the non-cleavable linker is an alkane linker.

[0034] In some embodiments, the muscle-targeting antibody comprises a non-natural amino acid to which the oligonucleotide is covalently linked. In some embodiments, the muscle-targeting antibody is covalently linked to the oligonucleotide via conjugation to a lysine residue or a cysteine residue of the antibody. In some embodiments, the muscle-targeting antibody is conjugated to the cysteine via a maleimide-containing linker, optionally wherein the maleimide-containing linker comprises a maleimidocaproyl or maleimidomethyl cyclohexane-1-carboxylate group.

[0035] In some embodiments, the muscle-targeting antibody is a glycosylated antibody that comprises at least one sugar moiety to which the oligonucleotide is covalently linked. In some embodiments, the sugar moiety is a branched mannose. In some embodiments, the muscle-targeting antibody is a glycosylated antibody that comprises one to four sugar moieties each of which is covalently linked to a separate oligonucleotide.

[0036] In some embodiments, the muscle-targeting antibody is a fully-glycosylated antibody. In some embodiments, the muscle-targeting antibody is a partially-glycosylated antibody. In some embodiments, the partially-glycosylated antibody is produced via chemical or enzymatic means. In some embodiments, the partially-glycosylated antibody is produced in a cell, cell that is deficient for an enzyme in the N- or O-glycosylation pathway.

[0037] According to some aspects, methods of delivering a molecular payload to a cell expressing transferrin receptor are provided. In some embodiments, the methods comprise contacting the cell with a complex provided herein.

[0038] According to some aspects, methods of inhibiting expression or activity of muscle disease gene in a cell are provided. In some embodiments, the methods comprise contacting the cell with a complex provided herein in an amount effective for promoting internalization of the molecular payload to the cell. In some embodiments, the cell is in vitro. In some embodiments, the cell is in a subject. In some embodiments, the subject is a human.

[0039] According to some aspects, methods of treating a subject having a muscle disease are provided. In some embodiments, the methods comprise administering to the subject an effective amount of a complex provided herein. In some embodiments, the muscle disease is a disease listed in Table 1. In some embodiments, the muscle disease is a disease selected from the group consisting of: Adult Pompe Disease, Centronuclear myopathy (CNM), Duchenne Muscular Dystrophy, Facioscapulohumeral Muscular Dystrophy (FSHD), Familial Hypertrophic Cardiomyopathy, Fibrodysplasia Ossificans Progressiva (FOP), Friedreich's Ataxia (FRDA), Inclusion Body Myopathy 2, Laing Distal Myopathy, Myofibrillar Myopathy, Myotonia Congenita (autosomal dominant form, Thomsen Disease), Myotonic Dystrophy Type I, Myotonic Dystrophy Type II, Myotubular Myopathy, Oculopharyngeal Muscular Dystrophy, and Paramyotonia Congenita.

BRIEF DESCRIPTION OF THE DRAWINGS

[0040] FIG. **1** depicts a non-limiting schematic showing the effect of transfecting Hepa 1-6 cells with an antisense oligonucleotide that targets DMPK (DTX-P-060) on expression levels of DMPK relative to a vehicle transfection;

[0041] FIG. **2**A depicts a non-limiting schematic showing an HIL-HPLC trace obtained during purification of a muscle targeting complex comprising an anti-transferrin receptor antibody covalently linked to a DMPK antisense oligonucleotide.

[0042] FIG. **2**B depicts a non-limiting image of an SDS-PAGE analysis of a muscle targeting complex.

[0043] FIG. **3** depicts a non-limiting schematic showing the ability of a muscle targeting complex (DTX-C-008) comprising DTX-P-060 to reduce expression levels of DMPK.

[0044] FIGS. **4**A-**4**E depict non-limiting schematics showing the ability of a muscle targeting complex (DTX-C-008) comprising DTX-P-060 to reduce expression levels of DMPK in mouse muscle tissues in vivo, relative to a vehicle experiment. (N=3 C57Bl/6 WT mice)

[0045] FIGS. **5**A-**5**B depict non-limiting schematics showing the tissue selectivity of a muscle targeting complex (DTX-C-008) comprising DTX-P-060. The muscle targeting complex (DTX-C-008) comprising DTX-P-060 does not reduce expression levels of DMPK in mouse brain or spleen tissues in vivo, relative to a vehicle experiment. (N=3 C57Bl/6 WT mice)

[0046] FIGS. **6**A-**6**F depict non-limiting schematics showing the ability of a muscle targeting complex (DTX-C-008) comprising DTX-P-060 to reduce expression levels of DMPK in mouse muscle tissues in vivo, relative to a vehicle experiment. (N=5 C57Bl/6 WT mice)

[0047] FIGS. **7**A-**7**L depict non-limiting schematics showing the ability of a muscle targeting complex (DTX-C-012) comprising DTX-P-060 to reduce expression levels of DMPK in cynomolgus monkey muscle tissues in vivo, relative to a vehicle experiment and compared to a naked DMPK ASO (DTX-P-060). (N=3 male cynomolgus monkeys)

[0048] FIGS. **8**A-**8**B depict non-limiting schematics showing the ability of a muscle targeting complex (DTX-C-012) comprising DTX-P-060 to reduce expression levels of DMPK in cynomolgus monkey smooth muscle tissues in vivo, relative to a vehicle experiment and compared to a naked DMPK ASO (DTX-P-060). (N=3 male cynomolgus monkeys)

[0049] FIGS. **9**A-**9**D depict non-limiting schematics showing the tissue selectivity of a muscle targeting complex (DTX-C-012) comprising DTX-P-060. The muscle targeting complex comprising DMPK-ASO does not reduce expression levels of DMPK in cynomolgus monkey liver, kidney, brain, or spleen tissues in vivo, relative to a vehicle experiment. (N=3 male cynomolgus monkeys)

[0050] FIG. **10** shows normalized DMPK mRNA tissue expression levels across several tissue types in cynomolgus monkeys. (N=3 male cynomolgus monkeys)

[0051] FIGS. **11**A-**11**B depict non-limiting schematics showing the ability of a muscle targeting complex (DTX-C-008) comprising DTX-P-060 to reduce expression levels of DMPK in mouse muscle tissues in vivo for up to 28 days after dosing with DTX-C-008, relative to a vehicle experiment and compared to a naked DMPK ASO (DTX-P-060).

[0052] FIG. **12** shows that a single dose of a muscle targeting complex (DTX-C-012) comprising DTX-P-060 is safe and tolerated in cynomolgus monkeys. (N=3 male cynomolgus monkeys)

DETAILED DESCRIPTION OF INVENTION

[0053] Aspects of the disclosure relate to a recognition that while certain molecular payloads (e.g., oligonucleotides, peptides, small molecules) can have beneficial effects in muscle cells, it has proven challenging to effectively target such cells. As described herein, the present disclosure provides complexes comprising muscle-targeting agents covalently linked to molecular payloads in order to overcome such challenges. In some embodiments, the complexes are particularly useful for delivering molecular payloads that modulate expression or activity of target genes in muscle cells, e.g., in a subject having or suspected of having a muscle disease. For example, in some embodiments, complexes are useful for treating subjects having rare muscle diseases, including Pompe disease, Centronuclear myopathy, Fibrodysplasia Ossificans Progressiva, Friedreich's ataxia, or Duchenne muscular dystrophy. In some embodiments, depending on the condition to be treated, different molecular payloads may be used in such complexes. For example, if the underlying mutation gives rise to a splicing defect, then an oligonucleotide or other payload may be used to correct the splicing defect (e.g., an oligonucleotide that inhibits exon skipping or promotes alternative splicing). If the underlying mutation results in a gain-of-function allele, then an oligonucleotide (e.g., RNAi, PMO, ASO-gapmer) may be used to inhibit the expression or activity of the allele. In some embodiments, e.g., when the mutation results in a loss-of-function allele, the payload may comprise an expression construct, e.g., for expressing a wild-type version of the allele. In some embodiments, the payload may comprise machinery (e.g., a guide nucleic acid, expression construct encoding a gene editing enzyme) for correcting the underlying defect, e.g., by gene editing.

[0054] Further aspects of the disclosure, including a description of defined terms, are provided below.

I. Definitions

[0055] Administering: As used herein, the terms "administering" or "administration" means to provide a complex to a subject in a manner that is physiologically and/or pharmacologically useful (e.g., to treat a condition in the subject).

[0056] Approximately: As used herein, the term "approximately" or "about," as applied to one or more values of interest, refers to a value that is similar to a stated reference value. In certain embodiments, the term "approximately" or "about" refers to a range of values that fall within 15%, 14%, 13%, 12%, 11%, 10%, 9%, 8%, 7%, 6%, 5%, 4%, 3%, 2%, 1%, or less in either direction (greater than or less than) of the stated reference value unless otherwise stated or otherwise evident from the context (except where such number would exceed 100% of a possible value).

[0057] Antibody: As used herein, the term "antibody" refers to a polypeptide that includes at least one immunoglobulin variable domain or at least one antigenic determinant, e.g., paratope that specifically binds to an antigen. In some embodiments, an antibody is a full-length antibody. In some embodiments, an antibody is a chimeric antibody. In some embodiments, an antibody is a humanized antibody. However, in some embodiments, an antibody is a Fab fragment, a F(ab')2 fragment, a Fv fragment or a scFv fragment. In some embodiments, an antibody is a nanobody derived from a camelid antibody or a nanobody derived from shark antibody. In some embodiments, an antibody is a diabody. In some embodiments, an antibody comprises a framework having a human germline sequence. In another embodiment, an antibody comprises a heavy chain constant domain selected from the group consisting of IgG, IgG1, IgG2, IgG2A, IgG2B, IgG2C, IgG3, IgG4, IgA1, IgA2, IgD, IgM, and IgE constant domains. In some embodiments, an antibody comprises a heavy (H) chain variable region (abbreviated herein as VH), and/or a light (L) chain variable region (abbreviated herein as VL). In some embodiments, an antibody comprises a constant domain, e.g., an Fc region. An immunoglobulin constant domain refers to a heavy or light chain constant domain. Human IgG heavy chain and light chain constant domain amino acid sequences and their functional variations are known. With respect to the heavy chain, in some embodiments, the heavy chain of an antibody described herein can be an alpha (a), delta (A), epsilon (F), gamma (7) or mu (p) heavy chain. In some embodiments, the heavy chain of an antibody described herein can comprise a human alpha (a), delta (A), epsilon (v), gamma (7) or mu (p) heavy chain. In a particular embodiment, an antibody described herein comprises a human gamma 1 CH1, CH2, and/or CH3 domain. In some embodiments, the amino acid sequence of the VH domain comprises the amino acid sequence of a human gamma (7) heavy chain constant region, such as any known in the art. Non-limiting examples of human constant region sequences have been described in the art, e.g., see U.S. Pat. No. 5,693,780 and Kabat E A et al., (1991) supra. In some embodiments, the VH domain comprises an amino acid sequence that is at least 70%, 75%, 80%, 85%, 90%, 95%, 98%, or at least 99% identical to any of the variable chain constant regions provided herein. In some embodiments, an antibody is modified, e.g., modified via glycosylation, phosphorylation, sumoylation, and/or methylation. In some embodiments, an antibody is a glycosylated antibody, which is conjugated to one or more sugar or carbohydrate molecules. In some embodiments, the one or more sugar or carbohydrate molecule are conjugated to the antibody via N-glycosylation, O-glycosylation, C-glycosylation, glypiation (GPI anchor attachment), and/or phosphoglycosylation. In some embodiments, the one or more sugar or carbohydrate molecule are monosaccharides, disaccharides, oligosaccharides, or glycans. In some embodiments, the one or more sugar or carbohydrate molecule is a branched oligosaccharide or a branched glycan. In some embodiments, the one or more sugar or carbohydrate molecule includes a mannose unit, a glucose unit, an N-acetylglucosamine unit, an N-acetylgalactosamine unit, a galactose unit, a fucose unit, or a phospholipid unit. In some embodiments, an antibody is a construct that comprises a polypeptide comprising one or more antigen binding fragments of the disclosure linked to a linker polypeptide or an immunoglobulin constant domain. Linker polypeptides comprise two or more amino acid residues joined by peptide bonds and are used to link one or more antigen binding portions. Examples of linker polypeptides have been reported (see e.g., Holliger, P., et al. (1993) Proc. Natl. Acad. Sci. USA 90:6444-6448; Poljak, R. J., et al. (1994) Structure 2:1121-1123). Still further, an antibody may be part of a larger immunoadhesion molecule, formed by covalent or noncovalent association of the antibody or antibody portion with one or more other proteins or peptides. Examples of such immunoadhesion molecules include use of the streptavidin core region to make a tetrameric scFv molecule (Kipriyanov, S. M., et al. (1995) Human Antibodies and Hybridomas 6:93-101) and use of a cysteine residue, a marker peptide and a C-terminal polyhistidine tag to make bivalent and biotinylated scFv molecules (Kipriyanov, S. M., et al. (1994) Mol. Immunol. 31:1047-1058).

[0058] CDR: As used herein, the term "CDR" refers to the complementarity determining region within antibody variable sequences. There are three CDRs in each of the variable regions of the heavy chain and the light chain, which are designated CDR1, CDR2 and CDR3, for each of the variable regions. The term "CDR set" as used herein refers to a group of three CDRs that occur in a single variable region capable of binding the antigen. The exact boundaries of these CDRs have been defined differently according to different systems. The system described by Kabat (Kabat et al., Sequences of Proteins of Immunological Interest (National Institutes of Health, Bethesda, Md. (1987) and (1991)) not only provides an unambiguous residue numbering system applicable to any variable region of an antibody, but also provides precise residue boundaries defining the three CDRs. These CDRs may be referred to as Kabat CDRs. Sub-portions of CDRs may be designated as L1, L2 and L3 or H1, H2 and H3 where the "L" and the "H" designates the light chain and the heavy chains regions, respectively. These regions may be referred to as Chothia CDRs, which have boundaries that overlap with Kabat CDRs. Other boundaries defining CDRs overlapping with the Kabat CDRs have been described by Padlan (FASEB J. 9:133-139 (1995)) and MacCallum (J Mol Biol 262(5):732-45 (1996)). Still other CDR boundary definitions may not strictly follow one of the above systems, but will nonetheless overlap with the Kabat CDRs, although they may be shortened or lengthened in light of prediction or experimental findings that particular residues or groups of residues or even entire CDRs do not significantly impact antigen binding. The methods used herein may utilize CDRs defined according to any of these systems, although preferred embodiments use Kabat or Chothia defined CDRs.

[0059] CDR-grafted antibody: The term "CDR-grafted antibody" refers to antibodies which comprise heavy and light chain variable region sequences from one species but in which the sequences of one or more of the CDR regions of VH and/or VL are replaced with CDR sequences of another species, such as antibodies having murine heavy and light chain variable regions in which one or more of the murine CDRs (e.g., CDR3) has been replaced with human CDR sequences.

[0060] Chimeric antibody: The term "chimeric antibody" refers to antibodies which comprise heavy and light chain variable region sequences from one species and constant region sequences from another species, such as antibodies having murine heavy and light chain variable regions linked to human constant regions.

[0061] Complementary: As used herein, the term "complementary" refers to the capacity for precise pairing between two nucleotides or two sets of nucleotides. In particular, complementary is a term that characterizes an extent of hydrogen bond pairing that brings about binding between two nucleotides or two sets of nucleotides. For example, if a base at one position of an oligonucleotide is capable of hydrogen bonding with a base at the corresponding position of a target nucleic acid (e.g., an mRNA), then the bases are considered to be complementary to each other at that position. Base pairings may include both canonical Watson-Crick base pairing and non-Watson-Crick base pairing (e.g., Wobble base pairing and Hoogsteen base pairing). For example, in some embodiments, for complementary base pairings, adenosine-type bases (A) are complementary to thymidine-type bases (T) or uracil-type bases (U), that cytosine-type bases (C) are complementary to guanosine-type bases (G), and that universal bases such as 3-nitropyrrole or 5-nitroindole can hybridize to and are considered complementary to any A, C, U, or T. Inosine (I) has also been considered in the art to be a universal base and is considered complementary to any A, C, U or T.

[0062] Conservative amino acid substitution: As used herein, a "conservative amino acid substitution" refers to an amino acid substitution that does not alter the relative charge or size characteristics of the protein in which the amino acid substitution is made. Variants can be prepared according to methods for altering polypeptide sequence known to one of ordinary skill in the art such as are found in references which compile such methods, e.g. Molecular Cloning: A Laboratory Manual, J. Sambrook, et al., eds., Fourth Edition, Cold Spring Harbor Laboratory Press, Cold Spring Harbor, New York, 2012, or Current Protocols in Molecular Biology, F. M. Ausubel, et al., eds., John Wiley & Sons, Inc., New York. Conservative substitutions of amino acids include substitutions made amongst amino acids within the following groups: (a) M, I, L, V; (b) F, Y, W; (c) K, R, H; (d) A, G; (e) S, T; (f) Q, N; and (g) E, D.

[0063] Covalently linked: As used herein, the term "covalently linked" refers to a characteristic of two or more molecules being linked together via at least one covalent bond. In some embodiments, two molecules can be covalently linked together by a single bond, e.g., a disulfide bond or disulfide bridge, that serves as a linker between the molecules. However, in some embodiments, two or more molecules can be covalently linked together via a molecule that serves as a linker that joins the two or more molecules together through multiple covalent bonds. In some embodiments, a linker may be a cleavable linker. However, in some embodiments, a linker may be a non-cleavable linker.

[0064] Cross-reactive: As used herein and in the context of a targeting agent (e.g., antibody), the term "cross-reactive," refers to a property of the agent being capable of specifically binding to more than one antigen of a similar type or class (e.g., antigens of multiple homologs, paralogs, or orthologs) with similar affinity or avidity. For example, in some embodiments, an antibody that is cross-reactive against human and non-human primate antigens of a similar type or class (e.g., a human transferrin receptor and non-human primate transferring receptor) is capable of binding to the human antigen and non-human primate antigens with a similar affinity or avidity. In some embodiments, an antibody is cross-reactive against a human antigen and a rodent antigen of a similar type or class. In some embodiments, an antibody is cross-reactive against a rodent antigen and a non-human primate antigen of a similar type or class. In some embodiments, an antibody is cross-reactive against a human antigen, a non-human primate antigen, and a rodent antigen of a similar type or class.

[0065] Disease allele: As used herein, the term "disease allele" refers to any one of alternative forms (e.g., mutant forms) of a gene for which the allele is correlated with and/or directly or indirectly contributes to, or causes, disease. A disease allele may comprise gene alterations including, but not limited to, insertions (e.g., disease-associated repeats described below), deletions, missense mutations, nonsense mutations and splice-site mutations relative to a wild-type (non-disease) allele. In some embodiments, a disease allele has a loss-of-function mutation. In some embodiments, a disease allele has a gain-of-function mutation. In some embodiments, a disease allele encodes an activating mutation (e.g., encodes a protein that is constitutively active). In some embodiments, a disease allele is a recessive allele having a recessive phenotype. In some embodiments, a disease allele is a dominant allele having a dominant phenotype.

[0066] Disease-associated-repeat: As used herein, the term "disease-associated-repeat" refers to a repeated nucleotide sequence at a genomic location for which the number of units of the repeated nucleotide sequence is correlated with and/or directly or indirectly contributes to, or causes, genetic disease. Each repeating unit of a disease associated repeat may be 2, 3, 4, 5 or more nucleotides in length. For example, in some embodiments, a disease associated repeat is a dinucleotide repeat. In some embodiments, a disease associated repeat is a trinucleotide repeat. In some embodiments, a disease associated repeat is a tetranucleotide repeat. In some embodiments, a disease associated repeat is a pentanucleotide repeat. In some embodiments, embodiments, the disease-associated-repeat comprises CAG repeats, CTG repeats, CUG repeats, CGG repeats, CCTG repeats, or a nucleotide complement of any thereof. In some embodiments, a disease-associated-repeat is in a non-coding portion of a gene. However, in some embodiments, a disease-associated-repeat is in a coding region of a gene. In some embodiments, a disease-associated-repeat is expanded from a normal state to a length that directly or indirectly contributes to, or causes, genetic disease. In some embodiments, a disease-associated-repeat is in RNA (e.g., an RNA transcript). In some embodiments, a disease-associated-repeat is in DNA (e.g., a chromosome, a plasmid). In some embodiments, a disease-associated-repeat is expanded in a chromosome of a germline cell. In some embodiments, a disease-associated-repeat is expanded in a chromosome of a somatic cell. In some embodiments, a disease-associated-repeat is expanded to a number of repeating units that is associated with congenital onset of disease. In some embodiments, a disease-associated-repeat is expanded to a number of repeating units that is associated with childhood onset of disease. In some embodiments, a disease-associated-repeat is expanded to a number of repeating units that is associated with adult onset of disease.

[0067] Framework: As used herein, the term "framework" or "framework sequence" refers to the remaining sequences of a variable region minus the CDRs. Because the exact definition of a CDR sequence can be determined by different systems, the meaning of a framework sequence is subject to correspondingly different interpretations. The six CDRs (CDR-L1, CDR-L2, and CDR-L3 of light chain and CDR-H1, CDR-H2, and CDR-H3 of heavy chain) also divide the framework regions on the light chain and the heavy chain into four sub-regions (FR1, FR2, FR3 and FR4) on each chain, in which CDR1 is positioned between FR1 and FR2, CDR2 between FR2 and FR3, and CDR3 between FR3 and FR4. Without specifying the particular sub-regions as FR1, FR2, FR3 or FR4, a framework region, as referred by others, represents the combined FRs within the variable region of a single, naturally occurring immunoglobulin chain. As used herein, a FR represents one of the four sub-regions, and FRs represents two or more of the four sub-regions constituting a framework region. Human heavy chain and light chain acceptor sequences are known in the art. In one embodiment, the acceptor sequences known in the art may be used in the antibodies disclosed herein.

[0068] Human antibody: The term "human antibody", as used herein, is intended to include antibodies having variable and constant regions derived from human germline immunoglobulin sequences. The human antibodies of the disclosure may include amino acid residues not encoded by human germline immunoglobulin sequences (e.g., mutations introduced by random or site-specific mutagenesis in vitro or by somatic mutation in vivo), for example in the CDRs and in particular CDR3. However, the term "human antibody", as used herein, is not intended to include antibodies in which CDR sequences derived from the germline of another mammalian species, such as a mouse, have been grafted onto human framework sequences.

[0069] Humanized antibody: The term "humanized antibody" refers to antibodies which comprise heavy and light chain variable region sequences from a non-human species (e.g., a mouse) but in which at least a portion of the VH and/or VL sequence has been altered to be more "human-like", i.e., more similar to human germline variable sequences. One type of humanized antibody is a CDR-grafted antibody, in which human CDR sequences are introduced into non-human VH and VL sequences to replace the corresponding nonhuman CDR sequences. In one embodiment, humanized anti-transferrin receptor antibodies and antigen binding portions are provided. Such antibodies may be generated by obtaining murine anti-transferrin receptor monoclonal antibodies using traditional hybridoma technology followed by humanization using in vitro genetic engineering, such as those disclosed in Kasaian et al PCT publication No. WO 2005/123126 A2.

[0070] Internalizing cell surface receptor: As used herein, the term, "internalizing cell surface receptor" refers to a cell surface receptor that is internalized by cells, e.g., upon external stimulation, e.g., ligand binding to the receptor. In some embodiments, an internalizing cell surface receptor is internalized by endocytosis. In some embodiments, an internalizing cell surface receptor is internalized by clathrin-mediated endocytosis. However, in some embodiments, an internalizing cell surface receptor is internalized by a clathrin-independent pathway, such as, for example, phagocytosis, macropinocytosis, caveolae- and raft-mediated uptake or constitutive clathrin-independent endocytosis. In some embodiments, the internalizing cell surface receptor comprises an intracellular domain, a transmembrane domain, and/or an extracellular domain, which may optionally further comprise a ligand-binding domain. In some embodiments, a cell surface receptor becomes internalized by a cell after ligand binding. In some embodiments, a ligand may be a muscle-targeting agent or a muscle-targeting antibody. In some embodiments, an internalizing cell surface receptor is a transferrin receptor.

[0071] Isolated antibody: An "isolated antibody", as used herein, is intended to refer to an antibody that is substantially free of other antibodies having different antigenic specificities (e.g., an isolated antibody that specifically binds transferrin receptor is substantially free of antibodies that specifically bind antigens other than transferrin receptor). An isolated antibody that specifically binds transferrin receptor complex may, however, have cross-reactivity to other antigens, such as transferrin receptor molecules from other species. Moreover, an isolated antibody may be substantially free of other cellular material and/or chemicals.

[0072] Kabat numbering: The terms "Kabat numbering", "Kabat definitions and "Kabat labeling" are used interchangeably herein. These terms, which are recognized in the art, refer to a system of numbering amino acid residues which are more variable (i.e. hypervariable) than other amino acid residues in the heavy and light chain variable regions of an antibody, or an antigen binding portion thereof (Kabat et al. (1971) Ann. NY Acad, Sci. 190:382-391 and, Kabat, E. A., et al. (1991) Sequences of Proteins of Immunological Interest, Fifth Edition, U.S. Department of Health and Human Services, NIH Publication No. 91-3242). For the heavy chain variable region, the hypervariable region ranges from amino acid positions 31 to 35 for CDR1, amino acid positions 50 to 65 for CDR2, and amino acid positions 95 to 102 for CDR3. For the light chain variable region, the hypervariable region ranges from amino acid positions 24 to 34 for CDR1, amino acid positions 50 to 56 for CDR2, and amino acid positions 89 to 97 for CDR3.

[0073] Molecular payload: As used herein, the term "molecular payload" refers to a molecule or species that functions to modulate a biological outcome. In some embodiments, a molecular payload is linked to, or otherwise associated with a muscle-targeting agent. In some embodiments, the molecular payload is a small molecule, a protein, a peptide, a nucleic acid, or an oligonucleotide. In some embodiments, the molecular payload functions to modulate the transcription of a DNA sequence, to modulate the expression of a protein, or to modulate the activity of a protein. In some embodiments, the molecular payload is an oligonucleotide that comprises a strand having a region of complementarity to a target gene.

[0074] Muscle Disease Gene: As used herein, the term "muscle disease gene" refers to a gene having a least one disease allele correlated with and/or directly or indirectly contributing to, or causing, a muscle disease. In some embodiments, the muscle disease is a rare disease, e.g., as defined by the Genetic and Rare Diseases Information Center (GARD), which is a program of the National Center for Advancing Translational Sciences (NCATS). In some embodiments, the muscle disease is a rare disease that is characterized as affecting fewer than 200,000 people. In some embodiments, the muscle disease is a single-gene disease. In some embodiments, a muscle disease gene is a gene listed in Table 1.

[0075] Muscle-targeting agent: As used herein, the term, "muscle-targeting agent," refers to a molecule that specifically binds to an antigen expressed on muscle cells. The antigen in or on muscle cells may be a membrane protein, for example an integral membrane protein or a peripheral membrane protein. Typically, a muscle-targeting agent specifically binds to an antigen on muscle cells that facilitates internalization of the muscle-targeting agent (and any associated molecular payload) into the muscle cells. In some embodiments, a muscle-targeting agent specifically binds to an internalizing, cell surface receptor on muscles and is capable of being internalized into muscle cells through receptor mediated internalization. In some embodiments, the muscle-targeting agent is a small molecule, a protein, a peptide, a nucleic acid (e.g., an aptamer), or an antibody. In some embodiments, the muscle-targeting agent is linked to a molecular payload.

[0076] Muscle-targeting antibody: As used herein, the term, "muscle-targeting antibody," refers to a muscle-targeting agent that is an antibody that specifically binds to an antigen found in or on muscle cells. In some embodiments, a muscle-targeting antibody specifically binds to an antigen on muscle cells that facilitates internalization of the muscle-targeting antibody (and any associated molecular payment) into the muscle cells. In some embodiments, the muscle-targeting antibody specifically binds to an internalizing, cell surface receptor present on muscle cells. In some embodiments, the muscle-targeting antibody is an antibody that specifically binds to a transferrin receptor.

[0077] Oligonucleotide: As used herein, the term "oligonucleotide" refers to an oligomeric nucleic acid compound of up to 200 nucleotides in length. Examples of oligonucleotides include, but are not limited to, RNAi oligonucleotides (e.g., siRNAs, shRNAs), microRNAs, gapmers, mixmers, phosphorodiamidite morpholinos, peptide nucleic acids, aptamers, guide nucleic acids (e.g., Cas9 guide RNAs), etc. Oligonucleotides may be single-stranded or double-stranded. In some embodiments, an oligonucleotide may comprise one or more modified nucleotides (e.g. 2'-O-methyl sugar modifications, purine or pyrimidine modifications). In some embodiments, an oligonucleotide may comprise one or more modified internucleotide linkage. In some embodiments, an oligonucleotide may comprise one or more phosphorothioate linkages, which may be in the Rp or Sp stereochemical conformation.

[0078] Recombinant antibody: The term "recombinant human antibody", as used herein, is intended to include all human antibodies that are prepared, expressed, created or isolated by recombinant means, such as antibodies expressed using a recombinant expression vector transfected into a host cell (described in more details in this disclosure), antibodies isolated from a recombinant, combinatorial human antibody library (Hoogenboom H. R., (1997) TIB

Tech. 15:62-70; Azzazy H., and Highsmith W. E., (2002) Clin. Biochem. 35:425-445; Gavilondo J. V., and Larrick J. W. (2002) BioTechniques 29:128-145; Hoogenboom H., and Chames P. (2000) Immunology Today 21:371-378), antibodies isolated from an animal (e.g., a mouse) that is transgenic for human immunoglobulin genes (see e.g., Taylor, L. D., et al. (1992) Nucl. Acids Res. 20:6287-6295; Kellermann S-A., and Green L. L. (2002) Current Opinion in Biotechnology 13:593-597; Little M. et al (2000) Immunology Today 21:364-370) or antibodies prepared, expressed, created or isolated by any other means that involves splicing of human immunoglobulin gene sequences to other DNA sequences. Such recombinant human antibodies have variable and constant regions derived from human germline immunoglobulin sequences. In certain embodiments, however, such recombinant human antibodies are subjected to in vitro mutagenesis (or, when an animal transgenic for human Ig sequences is used, in vivo somatic mutagenesis) and thus the amino acid sequences of the VH and VL regions of the recombinant antibodies are sequences that, while derived from and related to human germline VH and VL sequences, may not naturally exist within the human antibody germline repertoire in vivo. One embodiment of the disclosure provides fully human antibodies capable of binding human transferrin receptor which can be generated using techniques well known in the art, such as, but not limited to, using human Ig phage libraries such as those disclosed in Jermutus et al., PCT publication No. WO 2005/007699 A2.

[0079] Region of complementarity: As used herein, the term "region of complementarity" refers to a nucleotide sequence, e.g., of a oligonucleotide, that is sufficiently complementary to a cognate nucleotide sequence, e.g., of a target nucleic acid, such that the two nucleotide sequences are capable of annealing to one another under physiological conditions (e.g., in a cell). In some embodiments, a region of complementarity is fully complementary to a cognate nucleotide sequence of target nucleic acid. However, in some embodiments, a region of complementarity is partially complementary to a cognate nucleotide sequence of target nucleic acid (e.g., at least 80%, 90%, 95% or 99% complementarity). In some embodiments, a region of complementarity contains 1, 2, 3, or 4 mismatches compared with a cognate nucleotide sequence of a target nucleic acid.

[0080] Specifically binds: As used herein, the term "specifically binds" refers to the ability of a molecule to bind to a binding partner with a degree of affinity or avidity that enables the molecule to be used to distinguish the binding partner from an appropriate control in a binding assay or other binding context. With respect to an antibody, the term, "specifically binds", refers to the ability of the antibody to bind to a specific antigen with a degree of affinity or avidity, compared with an appropriate reference antigen or antigens, that enables the antibody to be used to distinguish the specific antigen from others, e.g., to an extent that permits preferential targeting to certain cells, e.g., muscle cells, through binding to the antigen, as described herein. In some embodiments, an antibody specifically binds to a target if the antibody has a K_D for binding the target of at least about 10^{-4} M, 10^{-5} M, 10^{-6} M, 10^{-7} M, 10^{-8} M, 10^{-9} M, 10^{-10} M, 10^{-11} M, 10^{-12} M, 10^{-13} M, or less. In some embodiments, an antibody specifically binds to the transferrin receptor, e.g., an epitope of the apical domain of transferrin receptor.

[0081] Subject: As used herein, the term "subject" refers to a mammal. In some embodiments, a subject is non-human primate, or rodent. In some embodiments, a subject is a human. In some embodiments, a subject is a patient, e.g., a human patient that has or is suspected of having a disease. In some embodiments, the subject is a human patient who has or is suspected of having a muscle disease (e.g., any of the diseases provided in Table 1).

[0082] Transferrin receptor: As used herein, the term, "transferrin receptor" (also known as TFRC, CD71, p90, or TFR1) refers to an internalizing cell surface receptor that binds transferrin to facilitate iron uptake by endocytosis. In some embodiments, a transferrin receptor may be of human (NCBI Gene ID 7037), non-human primate (e.g., NCBI Gene ID 711568 or NCBI Gene ID 102136007), or rodent (e.g., NCBI Gene ID 22042) origin. In addition, multiple human transcript variants have been characterized that encoded different isoforms of the receptor (e.g., as annotated under GenBank RefSeq Accession Numbers: NP_001121620.1, NP_003225.2, NP_001300894.1, and NP_001300895.1).

II. Complexes

[0083] Provided herein are complexes that comprise a targeting agent, e.g. an antibody, covalently linked to a molecular payload. In some embodiments, a complex comprises a muscle-targeting antibody covalently linked to an oligonucleotide. A complex may comprise an antibody that specifically binds a single antigenic site or that binds to at least two antigenic sites that may exist on the same or different antigens. A complex may be used to modulate the activity or function of at least one gene, protein, and/or nucleic acid. In some embodiments, the molecular payload present with a complex is responsible for the modulation of a gene, protein, and/or nucleic acids. A molecular payload may be a small molecule, protein, nucleic acid, oligonucleotide, or any molecular entity capable of modulating the activity or function of a gene, protein, and/or nucleic acid in a cell. In some embodiments, a molecular payload is an oligonucleotide that targets a muscle disease allele in muscle cells.

[0084] In some embodiments, a complex comprises a muscle-targeting agent, e.g. an anti-transferrin receptor antibody, covalently linked to a molecular payload, e.g. an antisense oligonucleotide that targets a muscle disease allele.

[0085] In some embodiments, a complex is useful for treating a muscle disease, in which a molecular payload affects the activity of the corresponding gene provided in Table 1. For example, depending on the condition, a molecular payload may modulate (e.g., decrease, increase) transcription or expression of the gene, modulate the expression of a protein encoded by the gene, or to modulate the activity of the encoded protein. In some embodiments, the molecular payload is an oligonucleotide that comprises a strand having a region of complementarity to a target gene provided in Table 1.

TABLE 1

List of muscle diseases and corresponding genes.

Disease	Gene Symbol	GenBank Accession No.
Adult Pompe	GAA	NM_000152; NM_001079803; NM_001079804
Adult Pompe	GYS1	NM_001161587; NM_002103
Centronuclear myopathy (CNM)	DNM2	NM_001190716; NM_004945; NM_001005362; NM_001005360; NM_001005361; NM_007871
Duchenne muscular dystrophy	DMD	NM_004023; NM_004020; NM_004018; NM_004012
Facioscapulohumeral muscular dystrophy (FSHD)	DUX4	NM_001306068; NM_001363820; NM_001205218; NM_001293798
Familial hypertrophic cardiomyopathy	MYBPC3	NM_000256
Familial hypertrophic cardiomyopathy	MYH6	NM_002471; NM_001164171; NM_010856
Familial hypertrophic cardiomyopathy	MYH7	NM_000257; NM_080728
Familial hypertrophic cardiomyopathy	TNNI3	NM_000363
Familial hypertrophic cardiomyopathy	TNNT2	NM_001001432; NM_001001431; NM_000364; NM_001001430; NM_001276347; NM_001276346; NM_001276345
Fibrodysplasia Ossificans Progressiva (FOP)	ACVR1	NM_001105; NM_001347663; NM_001347664; NM_001347665; NM_001347666; NM_001347667; NM_001111067
Friedreich's ataxia (FRDA)	FXN	NM_001161706; NM_181425; NM_000144
Inclusion body myopathy 2	GNE	NM_001190383; NM_001190384; NM_001128227; NM_005476; NM_001190388
Laing distal myopathy	MYH7	NM_000257; NM_080728
Myofibrillar myopathy	BAG3	NM_004281
Myofibrillar myopathy	CRYAB	NM_001885; NM_001330379; NM_001289807; NM_001289808
Myofibrillar myopathy	DES	NM_001927
Myofibrillar myopathy	DNAJB6	NM_005494; NM_058246
Myofibrillar myopathy	FHL1	NM_001159701; NM_001159699; NM_001159702; NM_001159703; NM_001159704;NM_001159700; NM_001167819; NM_001330659; NM_001449; NM_001077362
Myofibrillar myopathy	FLNC	NM_001458; NM_001127487
Myofibrillar myopathy	LDB3	NM_007078; NM_001171611; NM_001171610; NM_001080114; NM_001080115; NM_001080116
Myofibrillar myopathy	MYOT	NM_001300911; NM_006790; NM_001135940
Myofibrillar myopathy	PLEC	NM_201378; NM_201379; NM_201380; NM_201381; NM_201382; NM_201383; NM_201384; NM_000445
Myofibrillar myopathy	TTN	NM_133432; NM_133379; NM_133437; NM_003319; NM_001256850; NM_001267550; NM_133378
Myotonia congenita (autosomal dominant form, Thomsen Disease)	CLCN1	NM_000083; NM_013491
Myotonic dystrophy type I	DMPK	NM_001081563; NM_004409; NM_001081560; NM_001081562; NM_001288764; NM_001288765; NM_001288766

TABLE 1-continued

List of muscle diseases and corresponding genes.		
Disease	Gene Symbol	GenBank Accession No.
Myotonic dystrophy type II	CNBP	NM_001127192; NM_001127193; NM_001127194; NM_001127195; NM_001127196; NM_003418
Myotubular myopathy	MTM1	NM_000252
Oculopharyngeal muscular dystrophy	PABPN1	NM_004643
Paramyotonia congenita	SCN4A	NM_000334

A. Muscle-Targeting Agents

[0086] Some aspects of the disclosure provide muscle-targeting agents, e.g., for delivering a molecular payload to a muscle cell. In some embodiments, such muscle-targeting agents are capable of binding to a muscle cell, e.g., via specifically binding to an antigen on the muscle cell, and delivering an associated molecular payload to the muscle cell. In some embodiments, the molecular payload is bound (e.g., covalently bound) to the muscle targeting agent and is internalized into the muscle cell upon binding of the muscle targeting agent to an antigen on the muscle cell, e.g., via endocytosis. It should be appreciated that various types of muscle-targeting agents may be used in accordance with the disclosure. For example, the muscle-targeting agent may comprise, or consist of, a nucleic acid (e.g., DNA or RNA), a peptide (e.g., an antibody), a lipid (e.g., a microvesicle), or a sugar moiety (e.g., a polysaccharide). Exemplary muscle-targeting agents are described in further detail herein, however, it should be appreciated that the exemplary muscle-targeting agents provided herein are not meant to be limiting.

[0087] Some aspects of the disclosure provide muscle-targeting agents that specifically bind to an antigen on muscle, such as skeletal muscle, smooth muscle, or cardiac muscle. In some embodiments, any of the muscle-targeting agents provided herein bind to (e.g., specifically bind to) an antigen on a skeletal muscle cell, a smooth muscle cell, and/or a cardiac muscle cell.

[0088] By interacting with muscle-specific cell surface recognition elements (e.g., cell membrane proteins), both tissue localization and selective uptake into muscle cells can be achieved. In some embodiments, molecules that are substrates for muscle uptake transporters are useful for delivering a molecular payload into muscle tissue. Binding to muscle surface recognition elements followed by endocytosis can allow even large molecules such as antibodies to enter muscle cells. As another example molecular payloads conjugated to transferrin or anti-transferrin receptor antibodies can be taken up by muscle cells via binding to transferrin receptor, which may then be endocytosed, e.g., via clathrin-mediated endocytosis.

[0089] The use of muscle-targeting agents may be useful for concentrating a molecular payload (e.g., oligonucleotide) in muscle while reducing toxicity associated with effects in other tissues. In some embodiments, the muscle-targeting agent concentrates a bound molecular payload in muscle cells as compared to another cell type within a subject. In some embodiments, the muscle-targeting agent concentrates a bound molecular payload in muscle cells (e.g., skeletal, smooth, or cardiac muscle cells) in an amount that is at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, 30, 40, 50, 60, 70, 80, 90, or 100 times greater than an amount in non-muscle cells (e.g., liver, neuronal, blood, or fat cells). In some embodiments, a toxicity of the molecular payload in a subject is reduced by at least 1%, 2%, 3%, 4%, 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 90%, or 95% when it is delivered to the subject when bound to the muscle-targeting agent.

[0090] In some embodiments, to achieve muscle selectivity, a muscle recognition element (e.g., a muscle cell antigen) may be required. As one example, a muscle-targeting agent may be a small molecule that is a substrate for a muscle-specific uptake transporter. As another example, a muscle-targeting agent may be an antibody that enters a muscle cell via transporter-mediated endocytosis. As another example, a muscle targeting agent may be a ligand that binds to cell surface receptor on a muscle cell. It should be appreciated that while transporter-based approaches provide a direct path for cellular entry, receptor-based targeting may involve stimulated endocytosis to reach the desired site of action.

[0091] Muscle cells encompassed by the present disclosure include, but are not limited to, skeletal muscle cells, smooth muscle cells, cardiac muscle cells, myoblasts and myocytes.

i. Muscle-Targeting Antibodies

[0092] In some embodiments, the muscle-targeting agent is an antibody. Generally, the high specificity of antibodies for their target antigen provides the potential for selectively targeting muscle cells (e.g., skeletal, smooth, and/or cardiac muscle cells). This specificity may also limit off-target toxicity. Examples of antibodies that are capable of targeting a surface antigen of muscle cells have been reported and are within the scope of the disclosure. For example, antibodies that target the surface of muscle cells are described in Arahata K., et al. "Immunostaining of skeletal and cardiac muscle surface membrane with antibody against Duchenne muscular dystrophy peptide" *Nature* 1988; 333: 861-3; Song K. S., et al. "Expression of caveolin-3 in skeletal, cardiac, and smooth muscle cells. Caveolin-3 is a component of the sarcolemma and co-fractionates with dystrophin and dystrophin-associated glycoproteins" *J Biol Chem* 1996; 271: 15160-5; and Weisbart R. H. et al., "Cell type specific targeted intracellular delivery into muscle of a monoclonal antibody that binds myosin IIb" *Mol Immunol.* 2003 March, 39(13):78309; the entire contents of each of which are incorporated herein by reference.

a. Anti-Transferrin Receptor Antibodies

[0093] Some aspects of the disclosure are based on the recognition that agents binding to transferrin receptor, e.g., anti-transferrin-receptor antibodies, are capable of targeting muscle cell. Transferrin receptors are internalizing cell surface receptors that transport transferrin across the cellular membrane and participate in the regulation and homeostasis of intracellular iron levels. Some aspects of the disclosure provide transferrin receptor binding proteins, which are capable of binding to transferrin receptor. Accordingly, aspects of the disclosure provide binding proteins (e.g., antibodies) that bind to transferrin receptor. In some embodiments, binding proteins that bind to transferrin receptor are internalized, along with any bound molecular payload, into a muscle cell. As used herein, an antibody that binds to a transferrin receptor may be referred to as an anti-transferrin receptor antibody. Antibodies that bind, e.g. specifically bind, to a transferrin receptor may be internalized into the cell, e.g. through receptor-mediated endocytosis, upon binding to a transferrin receptor.

[0094] It should be appreciated that anti-transferrin receptor antibodies may be produced, synthesized, and/or derivatized using several known methodologies, e.g. library design using phage display. Exemplary methodologies have been characterized in the art and are incorporated by reference (Diez, P. et al. "High-throughput phage-display screening in array format", Enzyme and microbial technology, 2015, 79, 34-41.; Christoph M. H. and Stanley, J. R. "Antibody Phage Display: Technique and Applications" J Invest Dermatol. 2014, 134:2.; Engleman, Edgar (Ed.) "Human Hybridomas and Monoclonal Antibodies." 1985, Springer.). In other embodiments, an anti-transferrin antibody has been previously characterized or disclosed. Antibodies that specifically bind to transferrin receptor are known in the art (see, e.g. U.S. Pat. No. 4,364,934, filed Dec. 4, 1979, "Monoclonal antibody to a human early thymocyte antigen and methods for preparing same"; U.S. Pat. No. 8,409,573, filed Jun. 14, 2006, "Anti-CD71 monoclonal antibodies and uses thereof for treating malignant tumor cells"; U.S. Pat. No. 9,708,406, filed May 20, 2014, "Anti-transferrin receptor antibodies and methods of use"; U.S. Pat. No. 9,611,323, filed Dec. 19, 2014, "Low affinity blood brain barrier receptor antibodies and uses therefor"; WO 2015/098989, filed Dec. 24, 2014, "Novel anti-Transferrin receptor antibody that passes through blood-brain barrier"; Schneider C. et al. "Structural features of the cell surface receptor for transferrin that is recognized by the monoclonal antibody OKT9." J Biol Chem. 1982, 257:14, 8516-8522.; Lee et al. "Targeting Rat Anti-Mouse Transferrin Receptor Monoclonal Antibodies through Blood-Brain Barrier in Mouse" 2000, J Pharmacol. Exp. Ther., 292: 1048-1052.).

[0095] Any appropriate anti-transferrin receptor antibodies may be used in the complexes disclosed herein. Examples of anti-transferrin receptor antibodies, including associated references and binding epitopes, are listed in Table 2. In some embodiments, the anti-transferrin receptor antibody comprises the complementarity determining regions (CDR-H1, CDR-H2, CDR-H3, CDR-L1, CDR-L2, and CDR-L3) of any of the anti-transferrin receptor antibodies provided herein, e.g., anti-transferrin receptor antibodies listed in Table 2.

TABLE 1

List of anti-transferrin receptor antibody clones, including associated references and binding epitope information.

Antibody Clone Name	Reference(s)	Epitope/Notes
OKT9	U.S. Pat. No. 4,364,934, filed Dec. 4, 1979, entitled "MONOCLONAL ANTIBODY TO A HUMAN EARLY THYMOCYTE ANTIGEN AND METHODS FOR PREPARING SAME" Schneider C. et al. "Structural features of the cell surface receptor for transferrin that is recognized by the monoclonal antibody OKT9." J Biol Chem. 1982, 257:14, 8516-8522.	Apical domain of TfR (residues 305-366 of human TfR sequence XM_052730.3, available in GenBank)
(From JCR) Clone M11 Clone M23 Clone M27 Clone B84	WO 2015/098989, filed Dec. 24, 2014, "Novel anti-Transferrin receptor antibody that passes through blood-brain barrier" U.S. Pat. No. 9,994,641, filed Dec. 24, 2014, "Novel anti-Transferrin receptor antibody that passes through blood-brain barrier"	Apical domain (residues 230-244 and 326-347 of TfR) and protease-like domain (residues 461-473)
(From Genentech) 7A4, 8A2, 15D2, 10D11, 7B10, 15G11, 16G5, 13C3, 16G4, 16F6, 7G7, 4C2, 1B12, and 13D4	WO 2016/081643, filed May 26, 2016, entitled "ANTI-TRANSFERRIN RECEPTOR ANTIBODIES AND METHODS OF USE" U.S. Pat. No. 9,708,406, filed May 20, 2014, "Anti-transferrin receptor antibodies and methods of use"	Apical domain and non-apical regions
(From Armagen) 8D3	Lee et al. "Targeting Rat Anti-Mouse Transferrin Receptor Monoclonal Antibodies through Blood-Brain Barrier in Mouse" 2000, J Pharmacol. Exp. Ther., 292: 1048-1052.	

TABLE 1-continued

List of anti-transferrin receptor antibody clones, including associated references and binding epitope information.

Antibody Clone Name	Reference(s)	Epitope/Notes
	U.S. Pat. App. 2010/077498, filed Sep. 11, 2008, entitled "COMPOSITIONS AND METHODS FOR BLOOD-BRAIN BARRIER DELIVERY IN THE MOUSE"	
OX26	Haobam, B. et al. 2014. Rab17-mediated recycling endosomes contribute to autophagosome formation in response to Group A *Streptococcus* invasion. Cellular microbiology. 16: 1806-21.	
DF1513	Ortiz-Zapater E et al. Trafficking of the human transferrin receptor in plant cells: effects of tyrphostin A23 and brefeldin A. Plant J 48:757-70 (2006).	
1A1B2, 66IG10, MEM-189, JF0956, 29806, 1A1B2, TFRC/1818, 1E6, 66Ig10, TFRC/1059, Q1/71, 23D10, 13E4, TFRC/1149, ER-MP21, YTA74.4, BU54, 2B6, RI7 217	Commercially available anti-transferrin receptor antibodies.	Novus Biologicals 8100 Southpark Way, A-8 Littleton CO 80120
(From INSERM) BA120g	U.S. Pat. App. 2011/0311544A1, filed Jun. 15, 2005, entitled "ANTI-CD71 MONOCLONAL ANTIBODIES AND USES THEREOF FOR TREATING MALIGNANT TUMOR CELLS"	Does not compete with OKT9
LUCA31	U.S. Pat. No. 7,572,895, filed Jun. 7, 2004, entitled "TRANSFERRIN RECEPTOR ANTIBODIES"	"LUCA31 epitope"
(Salk Institute) B3/25 T58/30	Trowbridge, I.S. et al. "Anti-transferrin receptor monoclonal antibody and toxin-antibody conjugates affect growth of human tumour cells." Nature, 1981, volume 294, pages 171-173	
R17 217.1.3, 5E9C11, OKT9 BE0023 clone)	Commercially available anti-transferrin receptor antibodies.	BioXcell 10 Technology Dr., Suite 2B West Lebanon, NH 03784-1671 USA
BK19.9, B3/25, T56/14 and T58/1	Gatter, K.C. et al. "Transferrin receptors in human tissues: their distribution and possible clinical relevance." J Clin Pathol. 1983 May; 36(5):539-45.	

[0096] In some embodiments, the muscle-targeting agent is an anti-transferrin receptor antibody. In some embodiment, an anti-transferrin receptor antibody specifically binds to a transferrin protein having an amino acid sequence as disclosed herein. In some embodiments, an anti-transferrin receptor antibody may specifically bind to any extracellular epitope of a transferrin receptor or an epitope that becomes exposed to an antibody, including the apical domain, the transferrin binding domain, and the protease-like domain. In some embodiments, an anti-transferrin receptor antibody binds to an amino acid segment of a human or non-human primate transferrin receptor, as provided in SEQ ID Nos. 1-3 in the range of amino acids C89 to F760. In some embodiments, an anti-transferrin receptor antibody specifically binds with binding affinity of at least about 10^{-4} M, 10^{-5} M, 10^{-6} M, 10^{-7} M, 10^{-8} M, 10^{-9} M, 10^{-10} M, 10^{-11} M, 10^{-12} M, 10^{-13} M, or less. Anti-transferrin receptor antibodies used herein may be capable of competing for binding with other anti-transferrin receptor antibodies, e.g. OKT9, 8D3, that bind to transferrin receptor with 10^{-3} M, 10^{-4} M, 10^{-5} M, 10^{-6} M, 10^{-7} M, or less.

[0097] An example human transferrin receptor amino acid sequence, corresponding to NCBI sequence NP_003225.2 (transferrin receptor protein 1 isoform 1, *Homo sapiens*) is as follows:

(SEQ ID NO: 1)
MMDQARSAFSNLFGGEPLSYTRFSLARQVDGDNSHVEMKLAVDEEENADN
NTKANVTKPKRCSGSICYGTIAVIVFFLIGFMIGYLGYCKGVEPKTECER
LAGTESPVREEPGEDFPAARRLYWDDLKRKLSEKLDSTDFTGTIKLLNEN
SYVPREAGSQKDENLALYVENQFREFKLSKVWRDQHFVKIQVKDSAQNSV
IIVDKNGRLVYLVENPGGYVAYSKAATVTGKLVHANFGTKKDFEDLYTPV
NGSIVIVRAGKITFAEKVANAESLNAIGVLIYMDQTKFPIVNAELSFFGH
AHLGTGDPYTPGFPSFNHTQFPPSRSSGLPNIPVQTISRAAAEKLFGNME
GDCPSDWKTDSTCRMVTSESKNVKLTVSNVLKEIKILNIFGVIKGFVEPD
HYVVVGAQRDAWGPGAAKSGVGTALLLKLAQMFSDMVLKDGFQPSRSIIF
ASWSAGDFGSVGATEWLEGYLSSLHLKAFTYINLDKAVLGTSNFKVSASP
LLYTLIEKTMQNVKHPVTGQFLYQDSNWASKVEKLTLDNAAFPFLAYSGI
PAVSFCFCEDTDYPYLGTTMDTYKELIERIPELNKVARAAAEVAGQFVIK
LTHDVELNLDYERYNSQLLSFVRDLNQYRADIKEMGLSLQWLYSARGDFF
RATSRLTTDFGNAEKTDRFVMKKLNDRVMRVEYHFLSPYVSPKESPFRHV
FWGSGSHTLPALLENLKLRKQNNGAFNETLFRNQLALATWTIQGAANALS
GDVWDIDNEF

[0098] An example non-human primate transferrin receptor amino acid sequence, corresponding to NCBI sequence NP_001244232.1(transferrin receptor protein 1, *Macaca mulatta*) is as follows:

(SEQ ID NO: 2)
MMDQARSAFSNLFGGEPLSYTRFSLARQVDGDNSHVEMKLGVDEEENTDN
NTKPNGTKPKRCGGNICYGTIAVIIFFLIGFMIGYLGYCKGVEPKTECER
LAGTESPAREEPEEDFPAAPRLYWDDLKRKLSEKLDTTDFTSTIKLLNEN
LYVPREAGSQKDENLALYIENQFREFKLSKVWRDQHFVKIQVKDSAQNSV
IIVDKNGGLVYLVENPGGYVAYSKAATVTGKLVHANFGTKKDFEDLDSPV
NGSIVIVRAGKITFAEKVANAESLNAIGVLIYMDQTKFPIVKADLSFFGH
AHLGTGDPYTPGFPSFNHTQFPPSQSSGLPNIPVQTISRAAAEKLFGNME
GDCPSDWKTDSTCKMVTSENKSVKLTVSNVLKETKILNIFGVIKGFVEPD
HYVVVGAQRDAWGPGAAKSSVGTALLLKLAQMFSDMVLKDGFQPSRSIIF
ASWSAGDFGSVGATEWLEGYLSSLHLKAFTYINLDKAVLGTSNFKVSASP
LLYTLIEKTMQDVKHPVTGRSLYQDSNWASKVEKLTLDNAAFPFLAYSGI
PAVSFCFCEDTDYPYLGTTMDTYKELVERIPELNKVARAAAEVAGQFVIK
LTHDTELNLDYERYNSQLLLFLRDLNQYRADVKEMGLSLQWLYSARGDFF
RATSRLTTDFRNAEKRDKFVMKKLNDRVMRVEYYFLSPYVSPKESPFRHV
FWGSGSHTLSALLESLKLRRQNNSAFNETLFRNQLALATWTIQGAANALS
GDVWDIDNEF

[0099] An example non-human primate transferrin receptor amino acid sequence, corresponding to NCBI sequence XP_005545315.1 (transferrin receptor protein 1, *Macaca fascicularis*) is as follows:

(SEQ ID NO: 3)
MMDQARSAFSNLFGGEPLSYTRFSLARQVDGDNSHVEMKLGVDEEENTDN
NTKANGTKPKRCGGNICYGTIAVIIFFLIGFMIGYLGYCKGVEPKTECER
LAGTESPAREEPEEDFPAAPRLYWDDLKRKLSEKLDTTDFTSTIKLLNEN
LYVPREAGSQKDENLALYIENQFREFKLSKVWRDQHFVKIQVKDSAQNSV
IIVDKNGGLVYLVENPGGYVAYSKAATVTGKLVHANFGTKKDFEDLDSPV
NGSIVIVRAGKITFAEKVANAESLNAIGVLIYMDQTKFPIVKADLSFFGH
AHLGTGDPYTPGFPSFNHTQFPPSQSSGLPNIPVQTISRAAAEKLFGNME
GDCPSDWKTDSTCKMVTSENKSVKLTVSNVLKETKILNIFGVIKGFVEPD
HYVVVGAQRDAWGPGAAKSSVGTALLLKLAQMFSDMVLKDGFQPSRSIIF
ASWSAGDFGSVGATEWLEGYLSSLHLKAFTYINLDKAVLGTSNFKVSASP
LLYTLIEKTMQDVKHPVTGRSLYQDSNWASKVEKLTLDNAAFPFLAYSGI
PAVSFCFCEDTDYPYLGTTMDTYKELVERIPELNKVARAAAEVAGQFVIK
LTHDTELNLDYERYNSQLLLFLRDLNQYRADVKEMGLSLQWLYSARGDFF
RATSRLTTDFRNAEKRDKFVMKKLNDRVMRVEYYFLSPYVSPKESPFRHV
FWGSGSHTLSALLESLKLRRQNNSAFNETLFRNQLALATWTIQGAANALS
GDVWDIDNEF.

[0100] An example mouse transferrin receptor amino acid sequence, corresponding to NCBI sequence NP_001344227.1 (transferrin receptor protein 1, *Mus musculus*) is as follows:

(SEQ ID NO: 4)
MMDQARSAFSNLFGGEPLSYTRFSLARQVDGDNSHVEMKLAADEEENADN
NMKASVRKPKRFNGRLCFAAIALVIFFLIGFMSGYLGYCKRVEQKEECVK
LAETEETDKSETMETEDVPTSSRLYWADLKTLLSEKLNSIEFADTIKQLS
QNTYTPREAGSQKDESLAYYIENQFHEFKFSKVWRDEHYVKIQVKSSIGQ
NMVTIVQSNGNLDPVESPEGYVAFSKPTEVSGKLVHANFGTKKDFEELSY
SVNGSLVIVRAGEITFAEKVANAQSFNAIGVLIYMDKNKFPVVEADLALF
GHAHLGTGDPYTPGFPSFNHTQFPPSQSSGLPNIPVQTISRAAAEKLFGK
MEGSCPARWNIDSSCKLELSQNQNVKLIVKNVLKERRILNIFGVIKGYEE
PDRYVVVGAQRDALGAGVAAKSSVGTGLLLKLAQVFSDMISKDGFRPSRS
IIFASWTAGDFGAVGATEWLEGYLSSLHLKAFTYINLDKVVLGTSNFKVS
ASPLLYTLMGKIMQDVKHPVDGKSLYRDSNWISKVEKLSFDNAAYPFLAY
SGIPAVSFCFCEDADYPYLGTRLDTYEALTQKVPQLNQMVRTAAEVAGQL
IIKLTHDVELNLDYEMYNSKLLSFMKDLNQFKTDIRDMGLSLQWLYSARG
DYFRATSRLTTDFHNAEKTNRFVMREINDRIMKVEYHFLSPYVSPRESPF
RHIFWGSGSHTLSALVENLKLRQKNITAFNETLFRNQLALATWTIQGVAN
ALSGDIWNIDNEF

[0101] In some embodiments, an anti-transferrin receptor antibody binds to an amino acid segment of the receptor as follows: FVKIQVKDSAQNSVIIVDKNGRLVYLVENPG-GYVAYSKAATVTGKLVHANFGTKKDFE DLYTPVNG-SIVIVRAGKITFAEKVANAESLN-

AIGVLIYMDQTKFPIVNAELSFFGHAHLG TGDPYTPGFPSFNHTQFPPSRSSGLPNIPVQTIS-RAAAEKLFGNMEGDCPSDWKTDSTCR MVTS-ESKNVKLTVSNVLKE (SEQ ID NO: 5) and does not inhibit the binding interactions between transferrin receptors and transferrin and/or human hemochromatosis protein (also known as HFE).

[0102] Appropriate methodologies may be used to obtain and/or produce antibodies, antibody fragments, or antigen-binding agents, e.g., through the use of recombinant DNA protocols. In some embodiments, an antibody may also be produced through the generation of hybridomas (see, e.g., Kohler, G and Milstein, C. "Continuous cultures of fused cells secreting antibody of predefined specificity" Nature, 1975, 256: 495-497). The antigen-of-interest may be used as the immunogen in any form or entity, e.g., recombinant or a naturally occurring form or entity. Hybridomas are screened using standard methods, e.g. ELISA screening, to find at least one hybridoma that produces an antibody that targets a particular antigen. Antibodies may also be produced through screening of protein expression libraries that express antibodies, e.g., phage display libraries. Phage display library design may also be used, in some embodiments, (see, e.g. U.S. Pat. No. 5,223,409, filed Mar. 1, 1991, "Directed evolution of novel binding proteins"; WO 1992/18619, filed Apr. 10, 1992, "Heterodimeric receptor libraries using phagemids"; WO 1991/17271, filed May 1, 1991, "Recombinant library screening methods"; WO 1992/20791, filed May 15, 1992, "Methods for producing members of specific binding pairs"; WO 1992/15679, filed Feb. 28, 1992, and "Improved epitope displaying phage"). In some embodiments, an antigen-of-interest may be used to immunize a non-human animal, e.g., a rodent or a goat. In some embodiments, an antibody is then obtained from the non-human animal, and may be optionally modified using a number of methodologies, e.g., using recombinant DNA techniques. Additional examples of antibody production and methodologies are known in the art (see, e.g. Harlow et al. "Antibodies: A Laboratory Manual", Cold Spring Harbor Laboratory, 1988.).

[0103] In some embodiments, an antibody is modified, e.g., modified via glycosylation, phosphorylation, sumoylation, and/or methylation. In some embodiments, an antibody is a glycosylated antibody, which is conjugated to one or more sugar or carbohydrate molecules. In some embodiments, the one or more sugar or carbohydrate molecule are conjugated to the antibody via N-glycosylation, O-glycosylation, C-glycosylation, glypiation (GPI anchor attachment), and/or phosphoglycosylation. In some embodiments, the one or more sugar or carbohydrate molecules are monosaccharides, disaccharides, oligosaccharides, or glycans. In some embodiments, the one or more sugar or carbohydrate molecule is a branched oligosaccharide or a branched glycan. In some embodiments, the one or more sugar or carbohydrate molecule includes a mannose unit, a glucose unit, an N-acetylglucosamine unit, an N-acetylgalactosamine unit, a galactose unit, a fucose unit, or a phospholipid unit. In some embodiments, there are about 1-10, about 1-5, about 5-10, about 1-4, about 1-3, or about 2 sugar molecules. In some embodiments, a glycosylated antibody is fully or partially glycosylated. In some embodiments, an antibody is glycosylated by chemical reactions or by enzymatic means. In some embodiments, an antibody is glycosylated in vitro or inside a cell, which may optionally be deficient in an enzyme in the N- or O-glycosylation pathway, e.g. a glycosyltransferase. In some embodiments, an antibody is functionalized with sugar or carbohydrate molecules as described in International Patent Application Publication WO2014065661, published on May 1, 2014, entitled, "Modified antibody, antibody-conjugate and process for the preparation thereof".

[0104] Some aspects of the disclosure provide proteins that bind to transferrin receptor (e.g., an extracellular portion of the transferrin receptor). In some embodiments, transferrin receptor antibodies provided herein bind specifically to transferrin receptor (e.g., human transferrin receptor). Transferrin receptors are internalizing cell surface receptors that transport transferrin across the cellular membrane and participate in the regulation and homeostasis of intracellular iron levels. In some embodiments, transferrin receptor antibodies provided herein bind specifically to transferrin receptor from human, non-human primates, mouse, rat, etc. In some embodiments, transferrin receptor antibodies provided herein bind to human transferrin receptor. In some embodiments, transferrin receptor antibodies provided herein specifically bind to human transferrin receptor. In some embodiments, transferrin receptor antibodies provided herein bind to an apical domain of human transferrin receptor. In some embodiments, transferrin receptor antibodies provided herein specifically bind to an apical domain of human transferrin receptor.

[0105] In some embodiments, transferrin receptor antibodies of the present disclosure include one or more of the CDR-H (e.g., CDR-H1, CDR-H2, and CDR-H3) amino acid sequences from any one of the anti-transferrin receptor antibodies selected from Table 2. In some embodiments, transferrin receptor antibodies include the CDR-H1, CDR-H2, and CDR-H3 as provided for any one of the anti-transferrin receptor antibodies selected from Table 2. In some embodiments, anti-transferrin receptor antibodies include the CDR-L1, CDR-L2, and CDR-L3 as provided for any one of the anti-transferrin receptor antibodies selected from Table 2. In some embodiments, anti-transferrin antibodies include the CDR-H1, CDR-H2, CDR-H3, CDR-L1, CDR-L2, and CDR-L3 as provided for any one of the anti-transferrin receptor antibodies selected from Table 2. The disclosure also includes any nucleic acid sequence that encodes a molecule comprising a CDR-H1, CDR-H2, CDR-H3, CDR-L1, CDR-L2, or CDR-L3 as provided for any one of the anti-transferrin receptor antibodies selected from Table 2. In some embodiments, antibody heavy and light chain CDR3 domains may play a particularly important role in the binding specificity/affinity of an antibody for an antigen. Accordingly, anti-transferrin receptor antibodies of the disclosure may include at least the heavy and/or light chain CDR3s of any one of the anti-transferrin receptor antibodies selected from Table 2.

[0106] In some examples, any of the anti-transferrin receptor antibodies of the disclosure have one or more CDR (e.g., CDR-H or CDR-L) sequences substantially similar to any of the CDR-H1, CDR-H2, CDR-H3, CDR-L1, CDR-L2, and/or CDR-L3 sequences from one of the anti-transferrin receptor antibodies selected from Table 2. In some embodiments, the position of one or more CDRs along the VH (e.g., CDR-H1, CDR-H2, or CDR-H3) and/or VL (e.g., CDR-L1, CDR-L2, or CDR-L3) region of an antibody described herein can vary by one, two, three, four, five, or six amino acid positions so long as immunospecific binding to transferrin receptor (e.g., human transferrin receptor) is maintained (e.g., substantially maintained, for example, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95% of the binding of the original antibody from which it is derived). For example, in some embodiments, the position defining a CDR of any antibody described herein can vary by shifting the N-terminal and/or C-terminal boundary of the CDR by one, two, three, four, five, or six amino acids, relative to the CDR position of any one of the antibodies described herein, so long as immunospecific binding to transferrin receptor (e.g., human transferrin receptor) is maintained (e.g., substantially maintained, for example, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95% of the binding of the original antibody from which it is derived). In another embodiment, the length of one or more CDRs along the VH (e.g., CDR-H1, CDR-H2, or CDR-H3) and/or VL (e.g., CDR-L1, CDR-L2, or CDR-L3) region of an antibody described herein can vary (e.g., be shorter or longer) by one, two, three, four, five, or more amino acids, so long as immunospecific binding to transferrin receptor (e.g., human transferrin receptor) is maintained (e.g., substantially maintained, for example, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95% of the binding of the original antibody from which it is derived).

[0107] Accordingly, in some embodiments, a CDR-L1, CDR-L2, CDR-L3, CDR-H1, CDR-H2, and/or CDR-H3 described herein may be one, two, three, four, five or more amino acids shorter than one or more of the CDRs described herein (e.g., CDRS from any of the anti-transferrin receptor antibodies selected from Table 2) so long as immunospecific binding to transferrin receptor (e.g., human transferrin receptor) is maintained (e.g., substantially maintained, for example, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95% relative to the binding of the original antibody from which it is derived). In some embodiments, a CDR-L1, CDR-L2, CDR-L3, CDR-H1, CDR-H2, and/or CDR-H3 described herein may be one, two, three, four, five or more amino acids longer than one or more of the CDRs described herein (e.g., CDRS from any of the anti-transferrin receptor antibodies selected from Table 2) so long as immunospecific binding to transferrin receptor (e.g., human transferrin receptor) is maintained (e.g., substantially maintained, for example, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95% relative to the binding of the original antibody from which it is derived). In some embodiments, the amino portion of a CDR-L1, CDR-L2, CDR-L3, CDR-H1, CDR-H2, and/or CDR-H3 described herein can be extended by one, two, three, four, five or more amino acids compared to one or more of the CDRs described herein (e.g., CDRS from any of the anti-transferrin receptor antibodies selected from Table 2) so long as immunospecific binding to transferrin receptor (e.g., human transferrin receptor is maintained (e.g., substantially maintained, for example, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95% relative to the binding of the original antibody from which it is derived). In some embodiments, the carboxy portion of a CDR-L1, CDR-L2, CDR-L3, CDR-H1, CDR-H2, and/or CDR-H3 described herein can be extended by one, two, three, four, five or more amino acids compared to one or more of the CDRs described herein (e.g., CDRS from any of the anti-transferrin receptor antibodies selected from Table 2) so long as immunospecific binding to transferrin receptor (e.g., human transferrin receptor) is maintained (e.g., substantially maintained, for example, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95% relative to the binding of the original antibody from which it is derived). In some embodiments, the amino portion of a CDR-L1, CDR-L2, CDR-L3, CDR-H1, CDR-H2, and/or CDR-H3 described herein can be shortened by one, two, three, four, five or more amino acids compared to one or more of the CDRs described herein (e.g., CDRS from any of the anti-transferrin receptor antibodies selected from Table 2) so long as immunospecific binding to transferrin receptor (e.g., human transferrin receptor) is maintained (e.g., substantially maintained, for example, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95% relative to the binding of the original antibody from which it is derived). In some embodiments, the carboxy portion of a CDR-L1, CDR-L2, CDR-L3, CDR-H1, CDR-H2, and/or CDR-H3 described herein can be shortened by one, two, three, four, five or more amino acids compared to one or more of the CDRs described herein (e.g., CDRS from any of the anti-transferrin receptor antibodies selected from Table 2) so long as immunospecific binding to transferrin receptor (e.g., human transferrin receptor) is maintained (e.g., substantially maintained, for example, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95% relative to the binding of the original antibody from which it is derived). Any method can be used to ascertain whether immunospecific binding to transferrin receptor (e.g., human transferrin receptor) is maintained, for example, using binding assays and conditions described in the art.

[0108] In some examples, any of the anti-transferrin receptor antibodies of the disclosure have one or more CDR (e.g., CDR-H or CDR-L) sequences substantially similar to any one of the anti-transferrin receptor antibodies selected from Table 2. For example, the antibodies may include one or more CDR sequence(s) from any of the anti-transferrin receptor antibodies selected from Table 2 containing up to 5, 4, 3, 2, or 1 amino acid residue variations as compared to the corresponding CDR region in any one of the CDRs provided herein (e.g., CDRs from any of the anti-transferrin receptor antibodies selected from Table 2) so long as immunospecific binding to transferrin receptor (e.g., human transferrin receptor) is maintained (e.g., substantially maintained, for example, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95% relative to the binding of the original antibody from which it is derived). In some embodiments, any of the amino acid variations in any of the CDRs provided herein may be conservative variations. Conservative variations can be introduced into the CDRs at positions where the residues are not likely to be involved in interacting with a transferrin receptor protein (e.g., a human transferrin receptor protein), for example, as determined based on a crystal structure. Some aspects of the disclosure provide transferrin receptor antibodies that comprise one or more of the heavy chain variable (VH) and/or light chain variable (VL) domains provided herein. In some embodiments, any of the VH domains provided herein include one or more of the CDR-H sequences (e.g., CDR-H1, CDR-H2, and CDR-H3) provided herein, for example, any of the CDR-H sequences provided in any one of the anti-transferrin receptor antibodies selected from Table 2. In some embodiments, any of the VL domains provided herein include one or more of the CDR-L sequences (e.g., CDR-L1, CDR-L2, and CDR-L3) provided herein, for example, any of the CDR-L sequences provided in any one of the anti-transferrin receptor antibodies selected from Table 2.

[0109] In some embodiments, anti-transferrin receptor antibodies of the disclosure include any antibody that includes a heavy chain variable domain and/or a light chain variable domain of any anti-transferrin receptor antibody, such as any one of the anti-transferrin receptor antibodies selected from Table 2. In some embodiments, anti-transferrin receptor antibodies of the disclosure include any antibody that includes the heavy chain variable and light chain variable pairs of any anti-transferrin receptor antibody, such as any one of the anti-transferrin receptor antibodies selected from Table 2.

[0110] Aspects of the disclosure provide anti-transferrin receptor antibodies having a heavy chain variable (VH) and/or a light chain variable (VL) domain amino acid sequence homologous to any of those described herein. In some embodiments, the anti-transferrin receptor antibody comprises a heavy chain variable sequence or a light chain variable sequence that is at least 75% (e.g., 80%, 85%, 90%, 95%, 98%, or 99%) identical to the heavy chain variable sequence and/or any light chain variable sequence of any anti-transferrin receptor antibody, such as any one of the anti-transferrin receptor antibodies selected from Table 2. In some embodiments, the homologous heavy chain variable and/or a light chain variable amino acid sequences do not vary within any of the CDR sequences provided herein. For example, in some embodiments, the degree of sequence variation (e.g., 75%, 80%, 85%, 90%, 95%, 98%, or 99%) may occur within a heavy chain variable and/or a light chain variable sequence excluding any of the CDR sequences provided herein. In some embodiments, any of the anti-transferrin receptor antibodies provided herein comprise a heavy chain variable sequence and a light chain variable sequence that comprises a framework sequence that is at least 75%, 80%, 85%, 90%, 95%, 98%, or 99% identical to the framework sequence of any anti-transferrin receptor antibody, such as any one of the anti-transferrin receptor antibodies selected from Table 2.

[0111] In some embodiments, an anti-transferrin receptor antibody, which specifically binds to transferrin receptor (e.g., human transferrin receptor), comprises a light chain variable VL domain comprising any of the CDR-L domains (CDR-L1, CDR-L2, and CDR-L3), or CDR-L domain variants provided herein, of any of the anti-transferrin receptor antibodies selected from Table 2. In some embodiments, an anti-transferrin receptor antibody, which specifically binds to transferrin receptor (e.g., human transferrin receptor), comprises a light chain variable VL domain comprising the CDR-L1, the CDR-L2, and the CDR-L3 of any anti-transferrin receptor antibody, such as any one of the anti-transferrin receptor antibodies selected from Table 2. In some embodiments, the anti-transferrin receptor antibody comprises a light chain variable (VL) region sequence comprising one, two, three or four of the framework regions of the light chain variable region sequence of any anti-transferrin receptor antibody, such as any one of the anti-transferrin receptor antibodies selected from Table 2. In some embodiments, the anti-transferrin receptor antibody comprises one, two, three or four of the framework regions of a light chain variable region sequence which is at least 75%, 80%, 85%, 90%, 95%, or 100% identical to one, two, three or four of the framework regions of the light chain variable region sequence of any anti-transferrin receptor antibody, such as any one of the anti-transferrin receptor antibodies selected from Table 2. In some embodiments, the light chain variable framework region that is derived from said amino acid sequence consists of said amino acid sequence but for the presence of up to 10 amino acid substitutions, deletions, and/or insertions, preferably up to 10 amino acid substitutions. In some embodiments, the light chain variable framework region that is derived from said amino acid sequence consists of said amino acid sequence with 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid residues being substituted for an amino acid found in an analogous position in a corresponding non-human, primate, or human light chain variable framework region.

[0112] In some embodiments, an anti-transferrin receptor antibody that specifically binds to transferrin receptor comprises the CDR-L1, the CDR-L2, and the CDR-L3 of any anti-transferrin receptor antibody, such as any one of the anti-transferrin receptor antibodies selected from Table 2. In some embodiments, the antibody further comprises one, two, three or all four VL framework regions derived from the VL of a human or primate antibody. The primate or human light chain framework region of the antibody selected for use with the light chain CDR sequences described herein, can have, for example, at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, or at least 99%) identity with a light chain framework region of a non-human parent antibody. The primate or human antibody selected can have the same or substantially the same number of amino acids in its light chain complementarity determining regions to that of the light chain complementarity determining regions of any of the antibodies provided herein, e.g., any of the anti-transferrin receptor antibodies selected from Table 2. In some embodiments, the primate or human light chain framework region amino acid residues are from a natural primate or human antibody light chain framework region having at least 75% identity, at least 80% identity, at least 85% identity, at least 90% identity, at least 95% identity, at least 98% identity, at least 99% (or more) identity with the light chain framework regions of any anti-transferrin receptor antibody, such as any one of the anti-transferrin receptor antibodies selected from Table 2. In some embodiments, an anti-transferrin receptor antibody further comprises one, two, three or all four VL framework regions derived from a human light chain variable kappa subfamily. In some embodiments, an anti-transferrin receptor antibody further comprises one, two, three or all four VL framework regions derived from a human light chain variable lambda subfamily.

[0113] In some embodiments, any of the anti-transferrin receptor antibodies provided herein comprise a light chain variable domain that further comprises a light chain constant region. In some embodiments, the light chain constant region is a kappa, or a lambda light chain constant region. In some embodiments, the kappa or lambda light chain constant region is from a mammal, e.g., from a human, monkey, rat, or mouse. In some embodiments, the light chain constant region is a human kappa light chain constant region. In some embodiments, the light chain constant region is a human lambda light chain constant region. It should be appreciated that any of the light chain constant regions provided herein may be variants of any of the light chain constant regions provided herein. In some embodiments, the light chain constant region comprises an amino acid sequence that is at least 75%, 80%, 85%, 90%, 95%, 98%, or 99% identical to any of the light chain constant regions of any anti-transferrin receptor antibody, such as any one of the anti-transferrin receptor antibodies selected from Table 2.

[0114] In some embodiments, the anti-transferrin receptor antibody is any anti-transferrin receptor antibody, such as any one of the anti-transferrin receptor antibodies selected from Table 2.

[0115] In some embodiments, an anti-transferrin receptor antibody comprises a VL domain comprising the amino acid sequence of any anti-transferrin receptor antibody, such as any one of the anti-transferrin receptor antibodies selected from Table 2, and wherein the constant regions comprise the amino acid sequences of the constant regions of an IgG, IgE, IgM, IgD, IgA or IgY immunoglobulin molecule, or a human IgG, IgE, IgM, IgD, IgA or IgY immunoglobulin molecule. In some embodiments, an anti-transferrin receptor antibody comprises any of the VL domains, or VL domain variants, and any of the VH domains, or VH domain variants, wherein the VL and VH domains, or variants thereof, are from the same antibody clone, and wherein the constant regions comprise the amino acid sequences of the constant regions of an IgG, IgE, IgM, IgD, IgA or IgY immunoglobulin molecule, any class (e.g., IgG1, IgG2, IgG3, IgG4, IgA1 and IgA2), or any subclass (e.g., IgG2a and IgG2b) of immunoglobulin molecule. Non-limiting examples of human constant regions are described in the art, e.g., see Kabat E A et al., (1991) supra.

[0116] In some embodiments, an antibody of the disclosure can bind to a target antigen (e.g., transferrin receptor) with relatively high affinity, e.g., with a K_D less than 10^{-6} M, 10^{-7} M, 10^{-8} M, 10^{-9} M, 10^{-10} M, 10^{-11} M or lower. For example, anti-transferrin receptor antibodies can bind to a transferrin receptor protein (e.g., human transferrin receptor) with an affinity between 5 pM and 500 nM, e.g., between 50 pM and 100 nM, e.g., between 500 pM and 50 nM. The disclosure also includes antibodies that compete with any of the antibodies described herein for binding to a transferrin receptor protein (e.g., human transferrin receptor) and that have an affinity of 50 nM or lower (e.g., 20 nM or lower, 10 nM or lower, 500 pM or lower, 50 pM or lower, or 5 pM or lower). The affinity and binding kinetics of the anti-transferrin receptor antibody can be tested using any suitable method including but not limited to biosensor technology (e.g., OCTET or BIACORE).

[0117] In some embodiments, an antibody of the disclosure can bind to a target antigen (e.g., transferrin receptor) with relatively high affinity, e.g., with a K_D less than 10^{-6} M, 10^{-7} M, 10^{-8} M, 10^{-9} M, 10^{-10} M, 10^{-11} M or lower. For example, anti-transferrin receptor antibodies can bind to a transferrin receptor protein (e.g., human transferrin receptor) with an affinity between 5 pM and 500 nM, e.g., between 50 pM and 100 nM, e.g., between 500 pM and 50 nM. The disclosure also includes antibodies that compete with any of the antibodies described herein for binding to a transferrin receptor protein (e.g., human transferrin receptor) and that have an affinity of 50 nM or lower (e.g., 20 nM or lower, 10 nM or lower, 500 pM or lower, 50 pM or lower, or 5 pM or lower). The affinity and binding kinetics of the anti-transferrin receptor antibody can be tested using any suitable method including but not limited to biosensor technology (e.g., OCTET or BIACORE).

[0118] In some embodiments, the muscle-targeting agent is a transferrin receptor antibody (e.g., the antibody and variants thereof as described in International Application Publication WO 2016/081643, incorporated herein by reference).

[0119] The heavy chain and light chain CDRs of the antibody according to different definition systems are provided in Table 1.1. The different definition systems, e.g., the Kabat definition, the Chothia definition, and/or the contact definition have been described. See, e.g., (e.g., Kabat, E. A., et al. (1991) Sequences of Proteins of Immunological Interest, Fifth Edition, U.S. Department of Health and Human Services, NIH Publication No. 91-3242, Chothia et al., (1989) Nature 342:877; Chothia, C. et al. (1987) J. Mol. Biol. 196:901-917, Al-lazikani et al (1997) J. Molec. Biol. 273:927-948; and Almagro, J. Mol. Recognit. 17:132-143 (2004). See also hgmp.mrc.ac.uk and bioinf.org.uk/abs).

TABLE 1.1

Heavy chain and light chain CDRs of a transferrin receptor antibody

CDRs	Kabat	Chothia	Contact
CDR-H1	SYWMH (SEQ ID NO: 17)	GYTFTSY (SEQ ID NO: 23)	TSYWMH (SEQ ID NO: 25)
CDR-H2	EINPTNGRTNYIEKFKS (SEQ ID NO: 18)	NPTNGR (SEQ ID NO: 24)	WIGEINPTNGRTN (SEQ ID NO: 26)
CDR-H3	GTRAYHY (SEQ ID NO: 19)	GTRAYHY (SEQ ID NO: 19)	ARGTRA (SEQ ID NO: 27)
CDR-L1	RASDNLYSNLA (SEQ ID NO: 20)	RASDNLYSNLA (SEQ ID NO: 20)	YSNLAWY (SEQ ID NO: 28)
CDR-L2	DATNLAD (SEQ ID NO: 21)	DATNLAD (SEQ ID NO: 21)	LLVYDATNLA (SEQ ID NO: 29)
CDR-L3	QHFWGTPLT (SEQ ID NO: 22)	QHFWGTPLT (SEQ ID NO: 22)	QHFWGTPL (SEQ ID NO: 30)

[0120] The heavy chain variable domain (VH) and light chain variable domain sequences are also provided:

VH
(SEQ ID NO: 33)
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGE
INPTNGRTNYIEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARGT
RAYHYWGQGTSVTVSS

VL
(SEQ ID NO: 34)
DIQMTQSPASLSVSVGETVTITCRASDNLYSNLAWYQQKQGKSPQLLVYD
ATNLADGVPSRFSGSGSGTQYSLKINSLQSEDFGTYYCQHFWGTPLTFGA
GTKLELK

[0121] In some embodiments, the transferrin receptor antibody of the present disclosure comprises a CDR-H1, a CDR-H2, and a CDR-H3 that are the same as the CDR-H1, CDR-H2, and CDR-H3 shown in Table 1.1. Alternatively or in addition, the transferrin receptor antibody of the present disclosure comprises a CDR-L1, a CDR-L2, and a CDR-L3 that are the same as the CDR-L1, CDR-L2, and CDR-L3 shown in Table 1.1.

[0122] In some embodiments, the transferrin receptor antibody of the present disclosure comprises a CDR-H1, a CDR-H2, and a CDR-H3, which collectively contains no more than 5 amino acid variations (e.g., no more than 5, 4, 3, 2, or 1 amino acid variation) as compared with the CDR-H1, CDR-H2, and CDR-H3 as shown in Table 1.1. "Collectively" means that the total number of amino acid variations in all of the three heavy chain CDRs is within the defined range. Alternatively or in addition, the transferrin receptor antibody of the present disclosure may comprise a CDR-L1, a CDR-L2, and a CDR-L3, which collectively contains no more than 5 amino acid variations (e.g., no more than 5, 4, 3, 2 or 1 amino acid variation) as compared with the CDR-L1, CDR-L2, and CDR-L3 as shown in Table 1.1.

[0123] In some embodiments, the transferrin receptor antibody of the present disclosure comprises a CDR-H1, a CDR-H2, and a CDR-H3, at least one of which contains no more than 3 amino acid variations (e.g., no more than 3, 2, or 1 amino acid variation) as compared with the counterpart heavy chain CDR as shown in Table 1.1. Alternatively or in addition, the transferrin receptor antibody of the present disclosure may comprise CDR-L1, a CDR-L2, and a CDR-L3, at least one of which contains no more than 3 amino acid variations (e.g., no more than 3, 2, or 1 amino acid variation) as compared with the counterpart light chain CDR as shown in Table 1.1.

[0124] In some embodiments, the transferrin receptor antibody of the present disclosure comprises a CDR-L3, which contains no more than 3 amino acid variations (e.g., no more than 3, 2, or 1 amino acid variation) as compared with the CDR-L3 as shown in Table 1.1. In some embodiments, the transferrin receptor antibody of the present disclosure comprises a CDR-L3 containing one amino acid variation as compared with the CDR-L3 as shown in Table 1.1. In some embodiments, the transferrin receptor antibody of the present disclosure comprises a CDR-L3 of QHFAGTPLT (SEQ ID NO: 31 according to the Kabat and Chothia definition system) or QHFAGTPL (SEQ ID NO: 32 according to the Contact definition system). In some embodiments, the transferrin receptor antibody of the present disclosure comprises a CDR-H1, a CDR-H2, a CDR-H3, a CDR-L1 and a CDR-L2 that are the same as the CDR-H1, CDR-H2, and CDR-H3 shown in Table 1.1, and comprises a CDR-L3 of QHFAGTPLT (SEQ ID NO: 31 according to the Kabat and Chothia definition system) or QHFAGTPL (SEQ ID NO: 32 according to the Contact definition system).

[0125] In some embodiments, the transferrin receptor antibody of the present disclosure comprises heavy chain CDRs that collectively are at least 80% (e.g., 80%, 85%, 90%, 95%, or 98%) identical to the heavy chain CDRs as shown in Table 1.1. Alternatively or in addition, the transferrin receptor antibody of the present disclosure comprises light chain CDRs that collectively are at least 80% (e.g., 80%, 85%, 90%, 95%, or 98%) identical to the light chain CDRs as shown in Table 1.1.

[0126] In some embodiments, the transferrin receptor antibody of the present disclosure comprises a VH comprising the amino acid sequence of SEQ ID NO: 33. Alternatively or in addition, the transferrin receptor antibody of the present disclosure comprises a VL comprising the amino acid sequence of SEQ ID NO: 34.

[0127] In some embodiments, the transferrin receptor antibody of the present disclosure comprises a VH containing no more than 20 amino acid variations (e.g., no more than 20, 19, 18, 17, 16, 15, 14, 13, 12, 11, 10, 9, 8, 7, 6, 5, 4, 3, 2, or 1 amino acid variation) as compared with the VH as set forth in SEQ ID NO: 33. Alternatively or in addition, the transferrin receptor antibody of the present disclosure comprises a VL containing no more than 15 amino acid variations (e.g., no more than 20, 19, 18, 17, 16, 15, 14, 13, 12, 11, 9, 8, 7, 6, 5, 4, 3, 2, or 1 amino acid variation) as compared with the VL as set forth in SEQ ID NO: 34.

[0128] In some embodiments, the transferrin receptor antibody of the present disclosure comprises a VH comprising an amino acid sequence that is at least 80% (e.g., 80%, 85%, 90%, 95%, or 98%) identical to the VH as set forth in SEQ ID NO: 33. Alternatively or in addition, the transferrin receptor antibody of the present disclosure comprises a VL comprising an amino acid sequence that is at least 80% (e.g., 80%, 85%, 90%, 95%, or 98%) identical to the VL as set forth in SEQ ID NO: 34.

[0129] In some embodiments, the transferrin receptor antibody of the present disclosure is a humanized antibody (e.g., a humanized variant of an antibody). In some embodiments, the transferrin receptor antibody of the present disclosure comprises a CDR-H1, a CDR-H2, a CDR-H3, a CDR-L1, a CDR-L2, and a CDR-L3 that are the same as the CDR-H1, CDR-H2, and CDR-H3 shown in Table 1.1, and comprises a humanized heavy chain variable region and/or a humanized light chain variable region.

[0130] Humanized antibodies are human immunoglobulins (recipient antibody) in which residues from a complementary determining region (CDR) of the recipient are replaced by residues from a CDR of a non-human species (donor antibody) such as mouse, rat, or rabbit having the desired specificity, affinity, and capacity. In some embodiments, Fv framework region (FR) residues of the human immunoglobulin are replaced by corresponding non-human residues. Furthermore, the humanized antibody may comprise residues that are found neither in the recipient antibody nor in the imported CDR or framework sequences, but are included to further refine and optimize antibody performance. In general, the humanized antibody will comprise substantially all of at least one, and typically two, variable domains, in which all or substantially all of the CDR regions correspond to those of a non-human immunoglobulin and all or substantially all of the FR regions are those of a human immunoglobulin consensus sequence. The humanized antibody optimally also will comprise at least a portion of an immunoglobulin constant region or domain (Fc), typically that of a human immunoglobulin. Antibodies may have Fc regions modified as described in WO 99/58572. Other forms of humanized antibodies have one or more CDRs (one, two, three, four, five, six) which are altered with respect to the original antibody, which are also termed one or more CDRs derived from one or more CDRs from the original antibody. Humanized antibodies may also involve affinity maturation.

[0131] In some embodiments, humanization is achieved by grafting the CDRs (e.g., as shown in Table 1.1) into the IGKV1-NL1*01 and IGHV1-3*01 human variable domains. In some embodiments, the transferrin receptor antibody of the present disclosure is a humanized variant comprising one or more amino acid substitutions at positions 9, 13, 17, 18, 40, 45, and 70 as compared with the VL as set forth in SEQ ID NO: 34, and/or one or more amino acid substitutions at positions 1, 5, 7, 11, 12, 20, 38, 40, 44, 66, 75, 81, 83, 87, and 108 as compared with the VH as set forth in SEQ ID NO: 33. In some embodiments, the transferrin receptor antibody of the present disclosure is a humanized variant comprising amino acid substitutions at all of positions 9, 13, 17, 18, 40, 45, and 70 as compared with the VL as set forth in SEQ ID NO: 34, and/or amino acid substitutions at all of positions 1, 5, 7, 11, 12, 20, 38, 40, 44, 66, 75, 81, 83, 87, and 108 as compared with the VH as set forth in SEQ ID NO: 33.

[0132] In some embodiments, the transferrin receptor antibody of the present disclosure is a humanized antibody and contains the residues at positions 43 and 48 of the VL as set forth in SEQ ID NO: 34. Alternatively or in addition, the transferrin receptor antibody of the present disclosure is a humanized antibody and contains the residues at positions 48, 67, 69, 71, and 73 of the VH as set forth in SEQ ID NO: 33.

[0133] The VH and VL amino acid sequences of an example humanized antibody that may be used in accordance with the present disclosure are provided:

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Humanized VH
                                          (SEQ ID NO: 35)
EVQLVQSGAEVKKPGASVKVSCKASGYTFTSYWMHWVRQAPGQRLEWIGE
INPTNGRTNYIEKFKSRATLTVDKSASTAYMELSSLRSEDTAVYYCARGT
RAYHYWGQGTMVTVSS

Humanized VL
                                          (SEQ ID NO: 36)
DIQMTQSPSSLSASVGDRVTITCRASDNLYSNLAWYQQKPGKSPKLLVYD
ATNLADGVPSRFSGSGSGTDYSLKINSLQSEDFGTYYCQHFWGTPLTFGA
GTKLELK
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[0134] In some embodiments, the transferrin receptor antibody of the present disclosure comprises a VH comprising the amino acid sequence of SEQ ID NO: 35. Alternatively or in addition, the transferrin receptor antibody of the present disclosure comprises a VL comprising the amino acid sequence of SEQ ID NO: 36.

[0135] In some embodiments, the transferrin receptor antibody of the present disclosure comprises a VH containing no more than 20 amino acid variations (e.g., no more than 20, 19, 18, 17, 16, 15, 14, 13, 12, 11, 10, 9, 8, 7, 6, 5, 4, 3, 2, or 1 amino acid variation) as compared with the VH as set forth in SEQ ID NO: 35. Alternatively or in addition, the transferrin receptor antibody of the present disclosure comprises a VL containing no more than 15 amino acid variations (e.g., no more than 20, 19, 18, 17, 16, 15, 14, 13, 12, 11, 9, 8, 7, 6, 5, 4, 3, 2, or 1 amino acid variation) as compared with the VL as set forth in SEQ ID NO: 36.

[0136] In some embodiments, the transferrin receptor antibody of the present disclosure comprises a VH comprising an amino acid sequence that is at least 80% (e.g., 80%, 85%, 90%, 95%, or 98%) identical to the VH as set forth in SEQ ID NO: 35. Alternatively or in addition, the transferrin receptor antibody of the present disclosure comprises a VL comprising an amino acid sequence that is at least 80% (e.g., 80%, 85%, 90%, 95%, or 98%) identical to the VL as set forth in SEQ ID NO: 36.

[0137] In some embodiments, the transferrin receptor antibody of the present disclosure is a humanized variant comprising amino acid substitutions at one or more of positions 43 and 48 as compared with the VL as set forth in SEQ ID NO: 34, and/or amino acid substitutions at one or more of positions 48, 67, 69, 71, and 73 as compared with the VH as set forth in SEQ ID NO: 33. In some embodiments, the transferrin receptor antibody of the present disclosure is a humanized variant comprising a S43A and/or a V48L mutation as compared with the VL as set forth in SEQ ID NO: 34, and/or one or more of A67V, L69I, V71R, and K73T mutations as compared with the VH as set forth in SEQ ID NO: 33

[0138] In some embodiments, the transferrin receptor antibody of the present disclosure is a humanized variant comprising amino acid substitutions at one or more of positions 9, 13, 17, 18, 40, 43, 48, 45, and 70 as compared with the VL as set forth in SEQ ID NO: 34, and/or amino acid substitutions at one or more of positions 1, 5, 7, 11, 12, 20, 38, 40, 44, 48, 66, 67, 69, 71, 73, 75, 81, 83, 87, and 108 as compared with the VH as set forth in SEQ ID NO: 33.

[0139] In some embodiments, the transferrin receptor antibody of the present disclosure is a chimeric antibody, which can include a heavy constant region and a light constant region from a human antibody. Chimeric antibodies refer to antibodies having a variable region or part of variable region from a first species and a constant region from a second species. Typically, in these chimeric antibodies, the variable region of both light and heavy chains mimics the variable regions of antibodies derived from one species of mammals (e.g., a non-human mammal such as mouse, rabbit, and rat), while the constant portions are homologous to the sequences in antibodies derived from another mammal such as human. In some embodiments, amino acid modifications can be made in the variable region and/or the constant region.

[0140] In some embodiments, the transferrin receptor antibody described herein is a chimeric antibody, which can include a heavy constant region and a light constant region from a human antibody. Chimeric antibodies refer to antibodies having a variable region or part of variable region from a first species and a constant region from a second species. Typically, in these chimeric antibodies, the variable region of both light and heavy chains mimics the variable regions of antibodies derived from one species of mammals (e.g., a non-human mammal such as mouse, rabbit, and rat), while the constant portions are homologous to the sequences in antibodies derived from another mammal such as human. In some embodiments, amino acid modifications can be made in the variable region and/or the constant region.

[0141] In some embodiments, the heavy chain of any of the transferrin receptor antibodies as described herein may comprises a heavy chain constant region (CH) or a portion thereof (e.g., CH1, CH2, CH3, or a combination thereof). The heavy chain constant region can of any suitable origin, e.g., human, mouse, rat, or rabbit. In one specific example, the heavy chain constant region is from a human IgG (a gamma heavy chain), e.g., IgG1, IgG2, or IgG4. An exemplary human IgG1 constant region is given below:

```
                                          (SEQ ID NO: 37)
ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGV

HTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEP

KSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVS

HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGK

EYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTC

LVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRW

QQGNVFSCSVMHEALHNHYTQKSLSLSPGK
```

[0142] In some embodiments, the light chain of any of the transferrin receptor antibodies described herein may further comprise a light chain constant region (CL), which can be any CL known in the art. In some examples, the CL is a kappa light chain. In other examples, the CL is a lambda light chain. In some embodiments, the CL is a kappa light chain, the sequence of which is provided below:

```
                                        (SEQ ID NO: 38)
ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGV

HTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEP

KSCDKTHTCP
```

[0143] Other antibody heavy and light chain constant regions are well known in the art, e.g., those provided in the IMGT database (www.imgt.org) or at www.vbase2.org/vbstat.php., both of which are incorporated by reference herein.

[0144] Exemplary heavy chain and light chain amino acid sequences of the transferrin receptor antibodies described are provided below:

```
Heavy Chain (VH + human IgG1 constant region)
                                        (SEQ ID NO: 39)
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGE
INPTNGRTNYIEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARGT
RAYHYWGQGTSVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFP
EPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICN
VNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTL
MISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYR
VVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTL
PPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSD
GSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK

Light Chain (VL + kappa light chain)
                                        (SEQ ID NO: 40)
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGE
INPTNGRTNYIEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARGT
RAYHYWGQGTSVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFP
EPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICN
VNHKPSNTKVDKKVEPKSCDKTHTCP

Heavy Chain (humanized VH + human IgG1 constant
region)
                                        (SEQ ID NO: 41)
EVQLVQSGAEVKKPGASVKVSCKASGYTFTSYWMHWVRQAPGQRLEWIGE
INPTNGRTNYIEKFKSRATLTVDKSASTAYMELSSLRSEDTAVYYCARGT
RAYHYWGQGTMVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFP
EPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICN
VNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTL
MISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYR
VVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTL
PPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSD
GSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK

Light Chain (humanized VL + kappa light chain)
                                        (SEQ ID NO: 42)
DIQMTQSPSSLSASVGDRVTITCRASDNLYSNLAWYQQKPGKSPKLLVYD
ATNLADGVPSRFSGSGSGTDYSLKINSLQSEDFGTYYCQHFWGTPLTFGA
GTKLELKASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNS
GALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTK
VDKKVEPKSCDKTHTCP
```

[0145] In some embodiments, the transferrin receptor antibody described herein comprises a heavy chain comprising an amino acid sequence that is at least 80% (e.g., 80%, 85%, 90%, 95%, or 98%) identical to SEQ ID NO: 39. Alternatively or in addition, the transferrin receptor antibody described herein comprises a light chain comprising an amino acid sequence that is at least 80% (e.g., 80%, 85%, 90%, 95%, or 98%) identical to SEQ ID NO: 40. In some embodiments, the transferrin receptor antibody described herein comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 39. Alternatively or in addition, the transferrin receptor antibody described herein comprises a light chain comprising the amino acid sequence of SEQ ID NO: 40.

[0146] In some embodiments, the transferrin receptor antibody of the present disclosure comprises a heavy chain containing no more than 20 amino acid variations (e.g., no more than 20, 19, 18, 17, 16, 15, 14, 13, 12, 11, 10, 9, 8, 7, 6, 5, 4, 3, 2, or 1 amino acid variation) as compared with the heavy chain as set forth in SEQ ID NO: 39. Alternatively or in addition, the transferrin receptor antibody of the present disclosure comprises a light chain containing no more than 15 amino acid variations (e.g., no more than 20, 19, 18, 17, 16, 15, 14, 13, 12, 11, 9, 8, 7, 6, 5, 4, 3, 2, or 1 amino acid variation) as compared with the light chain as set forth in SEQ ID NO: 40.

[0147] In some embodiments, the transferrin receptor antibody described herein comprises a heavy chain comprising an amino acid sequence that is at least 80% (e.g., 80%, 85%, 90%, 95%, or 98%) identical to SEQ ID NO: 41. Alternatively or in addition, the transferrin receptor antibody described herein comprises a light chain comprising an amino acid sequence that is at least 80% (e.g., 80%, 85%, 90%, 95%, or 98%) identical to SEQ ID NO: 42. In some embodiments, the transferrin receptor antibody described herein comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 41. Alternatively or in addition, the transferrin receptor antibody described herein comprises a light chain comprising the amino acid sequence of SEQ ID NO: 42.

[0148] In some embodiments, the transferrin receptor antibody of the present disclosure comprises a heavy chain containing no more than 20 amino acid variations (e.g., no more than 20, 19, 18, 17, 16, 15, 14, 13, 12, 11, 10, 9, 8, 7, 6, 5, 4, 3, 2, or 1 amino acid variation) as compared with the heavy chain of humanized antibody as set forth in SEQ ID NO: 39. Alternatively or in addition, the transferrin receptor antibody of the present disclosure comprises a light chain containing no more than 15 amino acid variations (e.g., no more than 20, 19, 18, 17, 16, 15, 14, 13, 12, 11, 9, 8, 7, 6, 5, 4, 3, 2, or 1 amino acid variation) as compared with the light chain of humanized antibody as set forth in SEQ ID NO: 40.

[0149] In some embodiments, the transferrin receptor antibody is an antigen binding fragment (FAB) of an intact antibody (full-length antibody). Antigen binding fragment of an intact antibody (full-length antibody) can be prepared via routine methods. For example, F(ab')2 fragments can be produced by pepsin digestion of an antibody molecule, and Fab fragments that can be generated by reducing the disulfide bridges of F(ab')2 fragments. Exemplary FABs amino acid sequences of the transferrin receptor antibodies described herein are provided below:

```
Heavy Chain FAB (VH + a portion of human IgG1
constant region)
                                        (SEQ ID NO: 43)
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQRPGQGLEWIGE
INPTNGRTNYIEKFKSKATLTVDKSSSTAYMQLSSLTSEDSAVYYCARGT
RAYHYWGQGTSVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFP
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-continued

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EPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICN
VNHKPSNTKVDKKVEPKSCDKTHTCP
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Heavy Chain FAB (humanized VH + a portion of human IgG1 constant region)

(SEQ ID NO: 44)

```
EVQLVQSGAEVKKPGASVKVSCKASGYTFTSYWMHWVRQAPGQRLEWIGE
INPTNGRTNYIEKFKSRATLTVDKSASTAYMELSSLRSEDTAVYYCARGT
RAYHYWGQGTMVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFP
EPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICN
VNHKPSNTKVDKKVEPKSCDKTHTCP
```

[0150] The transferrin receptor antibodies described herein can be in any antibody form, including, but not limited to, intact (i.e., full-length) antibodies, antigen-binding fragments thereof (such as Fab, Fab', F(ab')2, Fv), single chain antibodies, bi-specific antibodies, or nanobodies. In some embodiments, the transferrin receptor antibody described herein is a scFv. In some embodiments, the transferrin receptor antibody described herein is a scFv-Fab (e.g., scFv fused to a portion of a constant region). In some embodiments, the transferrin receptor antibody described herein is a scFv fused to a constant region (e.g., human IgG1 constant region as set forth in SEQ ID NO: 39).

b. Other Muscle-Targeting Antibodies

[0151] In some embodiments, the muscle-targeting antibody is an antibody that specifically binds hemojuvelin, caveolin-3, Duchenne muscular dystrophy peptide, myosin Jib or CD63. In some embodiments, the muscle-targeting antibody is an antibody that specifically binds a myogenic precursor protein. Exemplary myogenic precursor proteins include, without limitation, ABCG2, M-Cadherin/Cadherin-15, Caveolin-1, CD34, FoxK1, Integrin alpha 7, Integrin alpha 7 beta 1, MYF-5, MyoD, Myogenin, NCAM-1/CD56, Pax3, Pax7, and Pax9. In some embodiments, the muscle-targeting antibody is an antibody that specifically binds a skeletal muscle protein. Exemplary skeletal muscle proteins include, without limitation, alpha-Sarcoglycan, beta-Sarcoglycan, Calpain Inhibitors, Creatine Kinase MM/CKMM, eIF5A, Enolase 2/Neuron-specific Enolase, epsilon-Sarcoglycan, FABP3/H-FABP, GDF-8/Myostatin, GDF-11/GDF-8, Integrin alpha 7, Integrin alpha 7 beta 1, Integrin beta 1/CD29, MCAM/CD146, MyoD, Myogenin, Myosin Light Chain Kinase Inhibitors, NCAM-1/CD56, and Troponin I. In some embodiments, the muscle-targeting antibody is an antibody that specifically binds a smooth muscle protein. Exemplary smooth muscle proteins include, without limitation, alpha-Smooth Muscle Actin, VE-Cadherin, Caldesmon/CALD1, Calponin 1, Desmin, Histamine H2 R, Motilin R/GPR38, Transgelin/TAGLN, and Vimentin. However, it should be appreciated that antibodies to additional targets are within the scope of this disclosure and the exemplary lists of targets provided herein are not meant to be limiting.

c. Antibody Features/Alterations

[0152] In some embodiments, conservative mutations can be introduced into antibody sequences (e.g., CDRs or framework sequences) at positions where the residues are not likely to be involved in interacting with a target antigen (e.g., transferrin receptor), for example, as determined based on a crystal structure. In some embodiments, one, two or more mutations (e.g., amino acid substitutions) are introduced into the Fc region of a muscle-targeting antibody described herein (e.g., in a CH2 domain (residues 231-340 of human IgG1) and/or CH3 domain (residues 341-447 of human IgG1) and/or the hinge region, with numbering according to the Kabat numbering system (e.g., the EU index in Kabat)) to alter one or more functional properties of the antibody, such as serum half-life, complement fixation, Fc receptor binding and/or antigen-dependent cellular cytotoxicity.

[0153] In some embodiments, one, two or more mutations (e.g., amino acid substitutions) are introduced into the hinge region of the Fc region (CH1 domain) such that the number of cysteine residues in the hinge region are altered (e.g., increased or decreased) as described in, e.g., U.S. Pat. No. 5,677,425. The number of cysteine residues in the hinge region of the CH1 domain can be altered to, e.g., facilitate assembly of the light and heavy chains, or to alter (e.g., increase or decrease) the stability of the antibody or to facilitate linker conjugation.

[0154] In some embodiments, one, two or more mutations (e.g., amino acid substitutions) are introduced into the Fc region of a muscle-targeting antibody described herein (e.g., in a CH2 domain (residues 231-340 of human IgG1) and/or CH3 domain (residues 341-447 of human IgG1) and/or the hinge region, with numbering according to the Kabat numbering system (e.g., the EU index in Kabat)) to increase or decrease the affinity of the antibody for an Fc receptor (e.g., an activated Fc receptor) on the surface of an effector cell. Mutations in the Fc region of an antibody that decrease or increase the affinity of an antibody for an Fc receptor and techniques for introducing such mutations into the Fc receptor or fragment thereof are known to one of skill in the art. Examples of mutations in the Fc receptor of an antibody that can be made to alter the affinity of the antibody for an Fc receptor are described in, e.g., Smith P et al., (2012) PNAS 109: 6181-6186, U.S. Pat. No. 6,737,056, and International Publication Nos. WO 02/060919; WO 98/23289; and WO 97/34631, which are incorporated herein by reference.

[0155] In some embodiments, one, two or more amino acid mutations (i.e., substitutions, insertions or deletions) are introduced into an IgG constant domain, or FcRn-binding fragment thereof (preferably an Fc or hinge-Fc domain fragment) to alter (e.g., decrease or increase) half-life of the antibody in vivo. See, e.g., International Publication Nos. WO 02/060919; WO 98/23289; and WO 97/34631; and U.S. Pat. Nos. 5,869,046, 6,121,022, 6,277,375 and 6,165,745 for examples of mutations that will alter (e.g., decrease or increase) the half-life of an antibody in vivo.

[0156] In some embodiments, one, two or more amino acid mutations (i.e., substitutions, insertions or deletions) are introduced into an IgG constant domain, or FcRn-binding fragment thereof (preferably an Fc or hinge-Fc domain fragment) to decrease the half-life of the anti-transferrin receptor antibody in vivo. In some embodiments, one, two or more amino acid mutations (i.e., substitutions, insertions or deletions) are introduced into an IgG constant domain, or FcRn-binding fragment thereof (preferably an Fc or hinge-Fc domain fragment) to increase the half-life of the antibody in vivo. In some embodiments, the antibodies can have one or more amino acid mutations (e.g., substitutions) in the second constant (CH2) domain (residues 231-340 of human IgG1) and/or the third constant (CH3) domain (residues 341-447 of human IgG1), with numbering according to the EU index in Kabat (Kabat E A et al., (1991) supra). In some embodiments, the constant region of the IgG1 of an antibody described herein comprises a methionine (M) to tyrosine (Y) substitution in position 252, a serine (S) to threonine (T) substitution in position 254, and a threonine (T) to glutamic acid (E) substitution in position 256, numbered according to the EU index as in Kabat. See U.S. Pat. No. 7,658,921, which is incorporated herein by reference. This type of mutant IgG, referred to as "YTE mutant" has been shown to display fourfold increased half-life as compared to wild-type versions of the same antibody (see Dall'Acqua W F et al., (2006) J Biol Chem 281: 23514-24). In some embodiments, an antibody comprises an IgG constant domain comprising one, two, three or more amino acid substitutions of amino acid residues at positions 251-257, 285-290, 308-314, 385-389, and 428-436, numbered according to the EU index as in Kabat.

[0157] In some embodiments, one, two or more amino acid substitutions are introduced into an IgG constant domain Fc region to alter the effector function(s) of the anti-transferrin receptor antibody. The effector ligand to which affinity is altered can be, for example, an Fc receptor or the C1 component of complement. This approach is described in further detail in U.S. Pat. Nos. 5,624,821 and 5,648,260. In some embodiments, the deletion or inactivation (through point mutations or other means) of a constant region domain can reduce Fc receptor binding of the circulating antibody thereby increasing tumor localization. See, e.g., U.S. Pat. Nos. 5,585,097 and 8,591,886 for a description of mutations that delete or inactivate the constant domain and thereby increase tumor localization. In some embodiments, one or more amino acid substitutions may be introduced into the Fc region of an antibody described herein to remove potential glycosylation sites on Fc region, which may reduce Fc receptor binding (see, e.g., Shields R L et al., (2001) J Biol Chem 276: 6591-604).

[0158] In some embodiments, one or more amino in the constant region of a muscle-targeting antibody described herein can be replaced with a different amino acid residue such that the antibody has altered Clq binding and/or reduced or abolished complement dependent cytotoxicity (CDC). This approach is described in further detail in U.S. Pat. No. 6,194,551 (Idusogie et al). In some embodiments, one or more amino acid residues in the N-terminal region of the CH2 domain of an antibody described herein are altered to thereby alter the ability of the antibody to fix complement. This approach is described further in International Publication No. WO 94/29351. In some embodiments, the Fc region of an antibody described herein is modified to increase the ability of the antibody to mediate antibody dependent cellular cytotoxicity (ADCC) and/or to increase the affinity of the antibody for an Fc7 receptor. This approach is described further in International Publication No. WO 00/42072.

[0159] In some embodiments, the heavy and/or light chain variable domain(s) sequence(s) of the antibodies provided herein can be used to generate, for example, CDR-grafted, chimeric, humanized, or composite human antibodies or antigen-binding fragments, as described elsewhere herein. As understood by one of ordinary skill in the art, any variant, CDR-grafted, chimeric, humanized, or composite antibodies derived from any of the antibodies provided herein may be useful in the compositions and methods described herein and will maintain the ability to specifically bind transferrin receptor, such that the variant, CDR-grafted, chimeric, humanized, or composite antibody has at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95% or more binding to transferrin receptor relative to the original antibody from which it is derived.

[0160] In some embodiments, the antibodies provided herein comprise mutations that confer desirable properties to the antibodies. For example, to avoid potential complications due to Fab-arm exchange, which is known to occur with native IgG4 mAbs, the antibodies provided herein may comprise a stabilizing 'Adair' mutation (Angal S., et al., "A single amino acid substitution abolishes the heterogeneity of chimeric mouse/human (IgG4) antibody," Mol Immunol 30, 105-108; 1993), where serine 228 (EU numbering; residue 241 Kabat numbering) is converted to proline resulting in an IgG1-like hinge sequence. Accordingly, any of the antibodies may include a stabilizing 'Adair' mutation.

[0161] As provided herein, antibodies of this disclosure may optionally comprise constant regions or parts thereof. For example, a VL domain may be attached at its C-terminal end to a light chain constant domain like Cx or Ck. Similarly, a VH domain or portion thereof may be attached to all or part of a heavy chain like IgA, IgD, IgE, IgG, and IgM, and any isotype subclass. Antibodies may include suitable constant regions (see, for example, Kabat et al., Sequences of Proteins of Immunological Interest, No. 91-3242, National Institutes of Health Publications, Bethesda, Md. (1991)). Therefore, antibodies within the scope of this may disclosure include VH and VL domains, or an antigen binding portion thereof, combined with any suitable constant regions.

ii. Muscle-Targeting Peptides

[0162] Some aspects of the disclosure provide muscle-targeting peptides as muscle-targeting agents. Short peptide sequences (e.g., peptide sequences of 5-20 amino acids in length) that bind to specific cell types have been described. For example, cell-targeting peptides have been described in Vines e., et al., A. "Cell-penetrating and cell-targeting peptides in drug delivery" Biochim Biophys Acta 2008, 1786: 126-38; Jarver P., et al., "In vivo biodistribution and efficacy of peptide mediated delivery" Trends Pharmacol Sci 2010; 31: 528-35; Samoylova T. I., et al., "Elucidation of muscle-binding peptides by phage display screening" Muscle Nerve 1999; 22: 460-6; U.S. Pat. No. 6,329,501, issued on Dec. 11, 2001, entitled "METHODS AND COMPOSITIONS FOR TARGETING COMPOUNDS TO MUSCLE"; and Samoylov A. M., et al., "Recognition of cell-specific binding of phage display derived peptides using an acoustic wave sensor." Biomol Eng 2002; 18: 269-72; the entire contents of each of which are incorporated herein by reference. By designing peptides to interact with specific cell surface antigens (e.g., receptors), selectivity for a desired tissue, e.g., muscle, can be achieved. Skeletal muscle-targeting has been investigated and a range of molecular payloads are able to be delivered. These approaches may have high selectivity for muscle tissue without many of the practical disadvantages of a large antibody or viral particle. Accordingly, in some embodiments, the muscle-targeting agent is a muscle-targeting peptide that is from 4 to 50 amino acids in length. In some embodiments, the muscle-targeting peptide is 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, or 50 amino acids in length. Muscle-targeting peptides can be generated using any of several methods, such as phage display.

[0163] In some embodiments, a muscle-targeting peptide may bind to an internalizing cell surface receptor that is overexpressed or relatively highly expressed in muscle cells, e.g. a transferrin receptor, compared with certain other cells. In some embodiments, a muscle-targeting peptide may target, e.g., bind to, a transferrin receptor. In some embodiments, a peptide that targets a transferrin receptor may comprise a segment of a naturally occurring ligand, e.g., transferrin. In some embodiments, a peptide that targets a transferrin receptor is as described in U.S. Pat. No. 6,743,893, filed Nov. 30, 2000, "RECEPTOR-MEDIATED UPTAKE OF PEPTIDES THAT BIND THE HUMAN TRANSFERRIN RECEPTOR". In some embodiments, a peptide that targets a transferrin receptor is as described in Kawamoto, M. et al, "A novel transferrin receptor-targeted hybrid peptide disintegrates cancer cell membrane to induce rapid killing of cancer cells." BMC Cancer. 2011 Aug. 18; 11:359. In some embodiments, a peptide that targets a transferrin receptor is as described in U.S. Pat. No. 8,399,653, filed May 20, 2011, "TRANSFERRIN/TRANSFERRIN RECEPTOR-MEDIATED SIRNA DELIVERY".

[0164] As discussed above, examples of muscle targeting peptides have been reported. For example, muscle-specific peptides were identified using phage display library presenting surface heptapeptides. As one example a peptide having the amino acid sequence ASSLNIA (SEQ ID NO: 6) bound to C2C12 murine myotubes in vitro, and bound to mouse muscle tissue in vivo. Accordingly, in some embodiments, the muscle-targeting agent comprises the amino acid sequence ASSLNIA (SEQ ID NO: 6). This peptide displayed improved specificity for binding to heart and skeletal muscle tissue after intravenous injection in mice with reduced binding to liver, kidney, and brain. Additional muscle-specific peptides have been identified using phage display. For example, a 12 amino acid peptide was identified by phage display library for muscle targeting in the context of treatment for DMD. See, Yoshida D., et al., "Targeting of salicylate to skin and muscle following topical injections in rats." Int J Pharm 2002; 231: 177-84; the entire contents of which are hereby incorporated by reference. Here, a 12 amino acid peptide having the sequence SKTFNTHPQSTP (SEQ ID NO: 7) was identified and this muscle-targeting peptide showed improved binding to C2C12 cells relative to the ASSLNIA (SEQ ID NO: 6) peptide.

[0165] An additional method for identifying peptides selective for muscle (e.g., skeletal muscle) over other cell types includes in vitro selection, which has been described in Ghosh D., et al., "Selection of muscle-binding peptides from context-specific peptide-presenting phage libraries for adenoviral vector targeting" J Virol 2005; 79: 13667-72; the entire contents of which are incorporated herein by reference. By pre-incubating a random 12-mer peptide phage display library with a mixture of non-muscle cell types, non-specific cell binders were selected out. Following rounds of selection the 12 amino acid peptide TARGEHKEEELI (SEQ ID NO: 8) appeared most frequently. Accordingly, in some embodiments, the muscle-targeting agent comprises the amino acid sequence TARGEHKEEELI (SEQ ID NO: 8).

[0166] A muscle-targeting agent may an amino acid-containing molecule or peptide. A muscle-targeting peptide may correspond to a sequence of a protein that preferentially binds to a protein receptor found in muscle cells. In some embodiments, a muscle-targeting peptide contains a high propensity of hydrophobic amino acids, e.g. valine, such that the peptide preferentially targets muscle cells. In some embodiments, a muscle-targeting peptide has not been previously characterized or disclosed. These peptides may be conceived of, produced, synthesized, and/or derivatized using any of several methodologies, e.g. phage displayed peptide libraries, one-bead one-compound peptide libraries, or positional scanning synthetic peptide combinatorial libraries. Exemplary methodologies have been characterized in the art and are incorporated by reference (Gray, B. P. and Brown, K. C. "Combinatorial Peptide Libraries: Mining for Cell-Binding Peptides" Chem Rev. 2014, 114:2, 1020-1081.; Samoylova, T. I. and Smith, B. F. "Elucidation of muscle-binding peptides by phage display screening." Muscle Nerve, 1999, 22:4. 460-6.). In some embodiments, a muscle-targeting peptide has been previously disclosed (see, e.g. Writer M. J. et al. "Targeted gene delivery to human airway epithelial cells with synthetic vectors incorporating novel targeting peptides selected by phage display." J. Drug Targeting. 2004; 12:185; Cai, D. "BDNF-mediated enhancement of inflammation and injury in the aging heart." Physiol Genomics. 2006, 24:3, 191-7.; Zhang, L. "Molecular profiling of heart endothelial cells." Circulation, 2005, 112:11, 1601-11.; McGuire, M. J. et al. "In vitro selection of a peptide with high selectivity for cardiomyocytes in vivo." J Mol Biol. 2004, 342:1, 171-82.). Exemplary muscle-targeting peptides comprise an amino acid sequence of the following group: CQAQGQLVC (SEQ ID NO: 9), CSERSMNFC (SEQ ID NO: 10), CPKTRRVPC (SEQ ID NO: 11), WLSEAGPVVTVRALRGTGSW (SEQ ID NO: 12), ASSLNIA (SEQ ID NO: 6), CMQHSMRVC (SEQ ID NO: 13), and DDTRHWG (SEQ ID NO: 14). In some embodiments, a muscle-targeting peptide may comprise about 2-25 amino acids, about 2-20 amino acids, about 2-15 amino acids, about 2-10 amino acids, or about 2-5 amino acids. Muscle-targeting peptides may comprise naturally-occurring amino acids, e.g. cysteine, alanine, or non-naturally-occurring or modified amino acids. Non-naturally occurring amino acids include 3-amino acids, homo-amino acids, proline derivatives, 3-substituted alanine derivatives, linear core amino acids, N-methyl amino acids, and others known in the art. In some embodiments, a muscle-targeting peptide may be linear; in other embodiments, a muscle-targeting peptide may be cyclic, e.g. bicyclic (see, e.g. Silvana, M. G. et al. Mol. Therapy, 2018, 26:1, 132-147.).

iii. Muscle-Targeting Receptor Ligands

[0167] A muscle-targeting agent may be a ligand, e.g. a ligand that binds to a receptor protein. A muscle-targeting ligand may be a protein, e.g. transferrin, which binds to an internalizing cell surface receptor expressed by a muscle cell. Accordingly, in some embodiments, the muscle-targeting agent is transferrin, or a derivative thereof that binds to a transferrin receptor. A muscle-targeting ligand may alternatively be a small molecule, e.g. a lipophilic small molecule that preferentially targets muscle cells relative to other cell types. Exemplary lipophilic small molecules that may target muscle cells include compounds comprising cholesterol, cholesteryl, stearic acid, palmitic acid, oleic acid, oleyl, linolene, linoleic acid, myristic acid, sterols, dihydrotestosterone, testosterone derivatives, glycerine, alkyl chains, trityl groups, and alkoxy acids.

iv. Muscle-Targeting Aptamers

[0168] A muscle-targeting agent may be an aptamer, e.g. an RNA aptamer, which preferentially targets muscle cells relative to other cell types. In some embodiments, a muscle-targeting aptamer has not been previously characterized or disclosed. These aptamers may be conceived of, produced, synthesized, and/or derivatized using any of several methodologies, e.g. Systematic Evolution of Ligands by Exponential Enrichment. Exemplary methodologies have been characterized in the art and are incorporated by reference

(Yan, A. C. and Levy, M. "Aptamers and aptamer targeted delivery" RNA biology, 2009, 6:3, 316-20.; Germer, K. et al. "RNA aptamers and their therapeutic and diagnostic applications." Int. J. Biochem. Mol. Biol. 2013; 4: 27-40.). In some embodiments, a muscle-targeting aptamer has been previously disclosed (see, e.g. Phillippou, S. et al. "Selection and Identification of Skeletal-Muscle-Targeted RNA Aptamers." Mol Ther Nucleic Acids. 2018, 10:199-214.; Thiel, W. H. et al. "Smooth Muscle Cell-targeted RNA Aptamer Inhibits Neointimal Formation." Mol Ther. 2016, 24:4, 779-87.). Exemplary muscle-targeting aptamers include the A01B RNA aptamer and RNA Apt 14. In some embodiments, an aptamer is a nucleic acid-based aptamer, an oligonucleotide aptamer or a peptide aptamer. In some embodiments, an aptamer may be about 5-15 kDa, about 5-10 kDa, about 10-15 kDa, about 1-5 Da, about 1-3 kDa, or smaller.

v. Other Muscle-Targeting Agents

[0169] One strategy for targeting a muscle cell (e.g., a skeletal muscle cell) is to use a substrate of a muscle transporter protein, such as a transporter protein expressed on the sarcolemma. In some embodiments, the muscle-targeting agent is a substrate of an influx transporter that is specific to muscle tissue. In some embodiments, the influx transporter is specific to skeletal muscle tissue. Two main classes of transporters are expressed on the skeletal muscle sarcolemma, (1) the adenosine triphosphate (ATP) binding cassette (ABC) superfamily, which facilitate efflux from skeletal muscle tissue and (2) the solute carrier (SLC) superfamily, which can facilitate the influx of substrates into skeletal muscle. In some embodiments, the muscle-targeting agent is a substrate that binds to an ABC superfamily or an SLC superfamily of transporters. In some embodiments, the substrate that binds to the ABC or SLC superfamily of transporters is a naturally-occurring substrate. In some embodiments, the substrate that binds to the ABC or SLC superfamily of transporters is a non-naturally occurring substrate, for example, a synthetic derivative thereof that binds to the ABC or SLC superfamily of transporters.

[0170] In some embodiments, the muscle-targeting agent is a substrate of an SLC superfamily of transporters. SLC transporters are either equilibrative or use proton or sodium ion gradients created across the membrane to drive transport of substrates. Exemplary SLC transporters that have high skeletal muscle expression include, without limitation, the SATT transporter (ASCT1; SLC1A4), GLUT4 transporter (SLC2A4), GLUT7 transporter (GLUT7; SLC2A7), ATRC2 transporter (CAT-2; SLC7A2), LAT3 transporter (KIAA0245; SLC7A6), PHT1 transporter (PTR4; SLC15A4), OATP-J transporter (OATP5A1; SLC21A15), OCT3 transporter (EMT; SLC22A3), OCTN2 transporter (FLJ46769; SLC22A5), ENT transporters (ENT1; SLC29A1 and ENT2; SLC29A2), PAT2 transporter (SLC36A2), and SAT2 transporter (KIAA1382; SLC38A2). These transporters can facilitate the influx of substrates into skeletal muscle, providing opportunities for muscle targeting.

[0171] In some embodiments, the muscle-targeting agent is a substrate of an equilibrative nucleoside transporter 2 (ENT2) transporter. Relative to other transporters, ENT2 has one of the highest mRNA expressions in skeletal muscle. While human ENT2 (hENT2) is expressed in most body organs such as brain, heart, placenta, thymus, pancreas, prostate, and kidney, it is especially abundant in skeletal muscle. Human ENT2 facilitates the uptake of its substrates depending on their concentration gradient. ENT2 plays a role in maintaining nucleoside homeostasis by transporting a wide range of purine and pyrimidine nucleobases. The hENT2 transporter has a low affinity for all nucleosides (adenosine, guanosine, uridine, thymidine, and cytidine) except for inosine. Accordingly, in some embodiments, the muscle-targeting agent is an ENT2 substrate. Exemplary ENT2 substrates include, without limitation, inosine, 2',3'-dideoxyinosine, and calofarabine. In some embodiments, any of the muscle-targeting agents provided herein are associated with a molecular payload (e.g., oligonucleotide payload). In some embodiments, the muscle-targeting agent is covalently linked to the molecular payload. In some embodiments, the muscle-targeting agent is non-covalently linked to the molecular payload.

[0172] In some embodiments, the muscle-targeting agent is a substrate of an organic cation/carnitine transporter (OCTN2), which is a sodium ion-dependent, high affinity carnitine transporter. In some embodiments, the muscle-targeting agent is carnitine, mildronate, acetylcarnitine, or any derivative thereof that binds to OCTN2. In some embodiments, the carnitine, mildronate, acetylcarnitine, or derivative thereof is covalently linked to the molecular payload (e.g., oligonucleotide payload).

[0173] A muscle-targeting agent may be a protein that is protein that exists in at least one soluble form that targets muscle cells. In some embodiments, a muscle-targeting protein may be hemojuvelin (also known as repulsive guidance molecule C or hemochromatosis type 2 protein), a protein involved in iron overload and homeostasis. In some embodiments, hemojuvelin may be full length or a fragment, or a mutant with at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 98% or at least 99% sequence identity to a functional hemojuvelin protein. In some embodiments, a hemojuvelin mutant may be a soluble fragment, may lack a N-terminal signaling, and/or lack a C-terminal anchoring domain. In some embodiments, hemojuvelin may be annotated under GenBank RefSeq Accession Numbers NM_001316767.1, NM_145277.4, NM_202004.3, NM_213652.3, or NM_213653.3. It should be appreciated that a hemojuvelin may be of human, non-human primate, or rodent origin.

B. Molecular Payloads

[0174] Some aspects of the disclosure provide molecular payloads, e.g., for modulating a biological outcome, e.g., the transcription of a DNA sequence, the expression of a protein, or the activity of a protein. In some embodiments, a molecular payload is linked to, or otherwise associated with a muscle-targeting agent. In some embodiments, such molecular payloads are capable of targeting to a muscle cell, e.g., via specifically binding to a nucleic acid or protein in the muscle cell following delivery to the muscle cell by an associated muscle-targeting agent. It should be appreciated that various types of muscle-targeting agents may be used in accordance with the disclosure. For example, the molecular payload may comprise, or consist of, an oligonucleotide (e.g., antisense oligonucleotide), a peptide (e.g., a peptide that binds a nucleic acid or protein associated with disease in a muscle cell), a protein (e.g., a protein that binds a nucleic acid or protein associated with disease in a muscle cell), or a small molecule (e.g., a small molecule that modulates the function of a nucleic acid or protein associated with disease in a muscle cell). In some embodiments, the molecular payload is an oligonucleotide that comprises a strand having a region of complementarity to a gene provided in Table 1. Exemplary molecular payloads are described in further detail herein, however, it should be appreciated that the exemplary molecular payloads provided herein are not meant to be limiting.

[0175] In some embodiments at least one (e.g., at least 2, at least 3, at least 4, at least 5, at least 10) molecular payload (e.g., oligonucleotides) is linked to a muscle-targeting agent. In some embodiments, all molecular payloads attached to a muscle-targeting agent are the same, e.g. target the same gene. In some embodiments, all molecular payloads attached to a muscle-targeting agent are different, for example the molecular payloads may target different portions of the same target gene, or the molecular payloads may target at least two different target genes. In some embodiments, a muscle-targeting agent may be attached to some molecular payloads that are the same and some molecular payloads that are different.

[0176] The present disclosure also provides a composition comprising a plurality of complexes, for which at least 80% (e.g., at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%) of the complexes comprise a muscle-targeting agent linked to the same number of molecular payloads (e.g., oligonucleotides).

i. Oligonucleotides

[0177] Any suitable oligonucleotide may be used as a molecular payload, as described herein. In some embodiments, the oligonucleotide may be designed to cause degradation of an mRNA (e.g., the oligonucleotide may be a gapmer, an siRNA, a ribozyme or an aptamer that causes degradation). In some embodiments, the oligonucleotide may be designed to block translation of an mRNA (e.g., the oligonucleotide may be a mixmer, an siRNA or an aptamer that blocks translation). In some embodiments, an oligonucleotide may be designed to caused degradation and block translation of an mRNA. In some embodiments, an oligonucleotide may be a guide nucleic acid (e.g., guide RNA) for directing activity of an enzyme (e.g., a gene editing enzyme). Other examples of oligonucleotides are provided herein. It should be appreciated that, in some embodiments, oligonucleotides in one format (e.g., antisense oligonucleotides) may be suitably adapted to another format (e.g., siRNA oligonucleotides) by incorporating functional sequences (e.g., antisense strand sequences) from one format to the other format.

[0178] In some embodiments, an oligonucleotide may comprise a region of complementarity to a target gene provided in Table 1. Further non-limiting examples are provided below for selected genes of Table 1.

DMPK/DM1

[0179] In some embodiments, examples of oligonucleotides useful for targeting DMPK, e.g., for the treatment of DM1, are provided in US Patent Application Publication 20100016215A1, published on Jan. 1, 2010, entitled Compound And Method For Treating Myotonic Dystrophy; US Patent Application Publication 20130237585A1, published Jul. 19, 2010, Modulation Of Dystrophia Myotonica-Protein Kinase (DMPK) Expression; US Patent Application Publication 20150064181A1, published on Mar. 5, 2015, entitled "Antisense Conjugates For Decreasing Expression Of Dmpk"; US Patent Application Publication 20150238627A1, published on Aug. 27, 2015, entitled "Peptide-Linked Morpholino Antisense Oligonucleotides For Treatment Of Myotonic Dystrophy"; Pandey, S. K. et al. "Identification and Characterization of Modified Antisense Oligonucleotides Targeting DMPK in Mice and Nonhuman Primates for the Treatment of Myotonic Dystrophy Type 1" J. of Pharmacol Exp Ther, 2015, 355:329-340.; Langlois, M. et al. "Cytoplasmic and Nuclear Retained DMPK mRNAs Are Targets for RNA Interference in Myotonic Dystrophy Cells" J. Biological Chemistry, 2005, 280:17, 16949-16954.; Jauvin, D. et al. "Targeting DMPK with Antisense Oligonucleotide Improves Muscle Strength in Myotonic Dystrophy Type 1 Mice", Mol. Ther: Nucleic Acids, 2017, 7:465-474.; Mulders, S. A. et al. "Triplet-repeat oligonucleotide-mediated reversal of RNA toxicity in myotonic dystrophy" PNAS, 2009, 106:33, 13915-13920.; Wheeler, T. M. et al., "Targeting nuclear RNA for in vivo correction of myotonic dystrophy" Nature, 2012, 488(7409):111-115.; and US Patent Application Publication 20160304877A1, published on Oct. 20, 2016, entitled "Compounds And Methods For Modulation Of Dystrophia Myotonica-Protein Kinase (Dmpk) Expression," the contents of each of which are incorporated herein by reference in their entireties.

[0180] Examples of oligonucleotides for promoting DMPK gene editing include US Patent Application Publication 20170088819A1, published on Mar. 3, 2017, entitled "Genetic Correction Of Myotonic Dystrophy Type 1"; and International Patent Application Publication WO18002812A1, published on Apr. 1, 2018, entitled "Materials And Methods For Treatment Of Myotonic Dystrophy Type 1 (DM1) And Other Related Disorders," the contents of each of which are incorporated herein by reference in their entireties.

[0181] In some embodiments, the oligonucleotide may have region of complementarity to a mutant form of DMPK, for example, a mutant form as reported in Botta A. et al. "The CTG repeat expansion size correlates with the splicing defects observed in muscles from myotonic dystrophy type 1 patients." J Med Genet. 2008 October; 45(10):639-46.; and Machuca-Tzili L. et al. "Clinical and molecular aspects of the myotonic dystrophies: a review." Muscle Nerve. 2005 July; 32(1):1-18.; the contents of each of which are incorporated herein by reference in their entireties.

[0182] In some embodiments, an oligonucleotide provided herein is an antisense oligonucleotide targeting DMPK. In some embodiments, the oligonucleotide targeting is any one of the antisense oligonucleotides (e.g., a Gapmer) targeting DMPK as described in US Patent Application Publication US20160304877A1, published on Oct. 20, 2016, entitled "Compounds And Methods For Modulation Of Dystrophia Myotonica-Protein Kinase (DMPK) Expression," incorporated herein by reference. In some embodiments, the DMPK targeting oligonucleotide targets a region of the DMPK gene sequence as set forth in Genbank accession No. NM_001081560.2 or as set forth in Genbank accession No. NG_009784.1.

[0183] In some embodiments, the DMPK targeting oligonucleotide comprises a nucleotide sequence comprising a region complementary to a target region that is at least 10 continuous nucleotides (e.g., at least 10, at least 12, at least 14, at least 16, or more continuous nucleotides) in Genbank accession No. NM_001081560.2.

[0184] In some embodiments, the DMPK targeting oligonucleotide comprise a gapmer motif. "Gapmer" means a chimeric antisense compound in which an internal region having a plurality of nucleotides that support RNase H cleavage is positioned between external regions having one or more nucleotides, wherein the nucleotides comprising the internal region are chemically distinct from the nucleotide or nucleotides comprising the external regions. The internal region can be referred to as a "gap segment" and the external regions can be referred to as "wing segments." In some embodiments, the DMPK targeting oligonucleotide comprises one or more modified nucleotides, and/or one or more modified internucleotide linkages. In some embodiments, the internucleotide linkage is a phosphorothioate linkage. In some embodiments, the oligonucleotide comprises a full phosphorothioate backbone. In some embodiments, the oligonucleotide is a DNA gapmer with cET ends (e.g., 3-10^{-3}; cET-DNA-cET). In some embodiments, the DMPK targeting oligonucleotide comprises one or more 6'-(S)-CH_3 biocyclic nucleotides, one or more P-D-2'-deoxyribonucleotides, and/or one or more 5-methylcytosine nucleotides.

DUX4/FSHD

[0185] In some embodiments, examples of oligonucleotides useful for targeting DUX4, e.g., for the treatment of FSHD, are provided in U.S. Pat. No. 9,988,628, published on Feb. 2, 2017, entitled "AGENTS USEFUL IN TREATING FACIOSCAPULOHUMERAL MUSCULAR DYSTROPHY"; U.S. Pat. No. 9,469,851, published Oct. 30, 2014, entitled "RECOMBINANT VIRUS PRODUCTS AND METHODS FOR INHIBITING EXPRESSION OF DUX4"; US Patent Application Publication 20120225034, published on Sep. 6, 2012, entitled "AGENTS USEFUL IN TREATING FACIOSCAPULOHUMERAL MUSCULAR DYSTROPHY"; PCT Patent Application Publication Number WO 2013/120038, published on Aug. 15, 2013, entitled "MORPHOLINO TARGETING DUX4 FOR TREATING FSHD"; Chen et al., "Morpholino-mediated Knockdown of DUX4 Toward Facioscapulohumeral Muscular Dystrophy Therapeutics," Molecular Therapy, 2016, 24:8, 1405-1411.; and Ansseau et al., "Antisense Oligonucleotides Used to Target the DUX4 mRNA as Therapeutic Approaches in Facioscapulohumeral Muscular Dystrophy (FSHD)," Genes, 2017, 8, 93.; the contents of each of which are incorporated herein in their entireties. In some embodiments, the oligonucleotide is an antisense oligonucleotide, a morpholino, a siRNA, a shRNA, or another nucleotide which hybridizes with the target DUX4 gene or mRNA.

[0186] In some embodiments, e.g., for the treatment of FSHD, oligonucleotides may have a region of complementarity to a hypomethylated, contracted D4Z4 repeat, as in Daxinger, et al., "Genetic and Epigenetic Contributors to FSHD," published in Curr Opin Genet Dev in 2015, Lim J-W, et al., DICER/AGO-dependent epigenetic silencing of D4Z4 repeats enhanced by exogenous siRNA suggests mechanisms and therapies for FSHD Hum Mol Genet. 2015 Sep. 1; 24(17): 4817-4828, the contents of each of which are incorporated in their entireties.

DNM2/CNM

[0187] In some embodiments, examples of oligonucleotides useful for targeting DNM2, e.g., for the treatment of CNM, are provided in US Patent Application Publication Number 20180142008, published on May 24, 2018, entitled "DYNAMIN 2 INHIBITOR FOR THE TREATMENT OF DUCHENNE'S MUSCULAR DYSTROPHY", and in PCT Application Publication Number WO 2018/100010A1, published on Jun. 7, 2018, entitled "ALLELE-SPECIFIC SILENCING THERAPY FOR DYNAMIN 2-RELATED DISEASES". For example, in some embodiments, the oligonucleotide is a RNAi, an antisense nucleic acid, a siRNA, or a ribozyme that interferes specifically with DNM2 expression. Other examples of oligonucleotides useful for targeting DNM2 are provided in Tasfaout, et al., "Single Intramuscular Injection of AAV-shRNA Reduces DNM2 and Prevents Myotubular Myopathy in Mice," published in Mol. Ther. on Apr. 4, 2018, and in Tasfaout, et al., "Antisense oligonucleotide-mediated Dnm2 knockdown prevents and reverts myotubular myopathy in mice," Nature Communications volume 8, Article number: 15661 (2017). In some embodiments, the oligonucleotide is a shRNA or a morpholino that efficiently targets DNM2 mRNA. In some embodiments, the oligonucleotide encodes wild-type DNM2 which is resistant to miR-133 activity, as in Todaka, et al. "Overexpression of NF90-NF45 Represses Myogenic MicroRNA Biogenesis, Resulting in Development of Skeletal Muscle Atrophy and Centronuclear Muscle Fibers," published in Mol. Cell Biol. in July 2015 Further examples of oligonucleotides useful for targeting DNM2 are provided in Gibbs, et al., "Two Dynamin-2 Genes are Required for Normal Zebrafish Development" published in PLoS One in 2013, the contents of each of which are incorporated herein in their entirety.

[0188] In some embodiments, e.g., for the treatment of CNM, the oligonucleotide may have a region of complementarity to a mutant in DNM2 associated with CNM, as in Bohm et al, "Mutation Spectrum in the Large GTPase Dynamin 2, and Genotype-Phenotype Correlation in Autosomal Dominant Centronuclear Myopathy," as published in Hum. Mutat. in 2012, the contents of which are incorporated herein in its entirety.

Pompe Disease

[0189] In some embodiments, e.g., for the treatment of Pompe disease, an oligonucleotide mediates exon 2 inclusion in a GAA disease allele as in van der Wal, et al., "GAA Deficiency in Pompe Disease is Alleviated by Exon Inclusion in iPSC-Derived Skeletal Muscle Cells," Mol Ther Nucleic Acids. 2017 Jun. 16; 7: 101-115, the contents of which are incorporated herein by reference. Accordingly, in some embodiments, the oligonucleotide may have a region of complementarity to a GAA disease allele.

[0190] In some embodiments, e.g., for the treatment of Pompe disease, an oligonucleotide, such as an RNAi or antisense oligonucleotide, is utilized to suppress expression of wild-type GYS1 in muscle cells, as reported, for example, in Clayton, et al., "Antisense Oligonucleotide-mediated Suppression of Muscle Glycogen Synthase 1 Synthesis as an Approach for Substrate Reduction Therapy of Pompe Disease," published in Mol Ther Nucleic Acids in 2017, or US Patent Application Publication Number 2017182189, published on Jun. 29, 2017, entitled "INHIBITING OR DOWN-REGULATING GLYCOGEN SYNTHASE BY CREATING PREMATURE STOP CODONS USING ANTISENSE OLIGONUCLEOTIDES", the contents of which are incorporated herein by reference. Accordingly, in some embodiments, oligonucleotides may have an antisense strand having a region of complementarity to a sequence a human GYS1 sequence, corresponding to RefSeq number NM_002103.4 and/or a mouse GYS1 sequence, corresponding to RefSeq number NM_030678.3.

ACVR1/FOP

[0191] In some embodiments, examples of oligonucleotides useful for targeting ACVR1, e.g., for the treatment of FOP, are provided in US Patent Application 2009/0253132, published Oct. 8, 2009, "Mutated ACVR1 for diagnosis and treatment of fibrodysplasia ossificans progressiva (FOP)"; WO 2015/152183, published Oct. 8, 2015, "Prophylactic agent and therapeutic agent for fibrodysplasia ossificans progressive"; Lowery, J. W. et al, "Allele-specific RNA Interference in FOP—Silencing the FOP gene", GENE THERAPY, vol. 19, 2012, pages 701-702; Takahashi, M. et al. "Disease-causing allele-specific silencing against the ALK2 mutants, R206H and G356D, in fibrodysplasia ossificans progressiva" Gene Therapy (2012) 19, 781-785; Shi, S. et al. "Antisense-Oligonucleotide Mediated Exon Skipping in Activin-Receptor-Like Kinase 2: Inhibiting the Receptor That Is Overactive in Fibrodysplasia Ossificans Progressiva" Plos One, July 2013, Vol 8:7, e69096.; US Patent Application 2017/0159056, published Jun. 8, 2017, "Antisense oligonucleotides and methods of use thereof"; U.S. Pat. No. 8,859,752, issued Oct. 4, 2014, "SIRNA-based therapy of Fibrodyplasia Ossificans Progressiva (FOP)"; WO 2004/094636, published Nov. 4, 2004, "Effective sirna knock-down constructs", the contents of each of which are incorporated herein in their entireties.

FXN/Friedreich's Ataxia

[0192] In some embodiments, examples of oligonucleotides useful for targeting FXN and/or otherwise compensating for frataxin deficiency, e.g., for the treatment of Freidrich Ataxia, are provided in Li, L. et al "Activating frataxin expression by repeat-targeted nucleic acids" Nat. Comm. 2016, 7:10606.; WO 2016/094374, published Jun. 16, 2016, "Compositions and methods for treatment of friedreich's ataxia."; WO 2015/020993, published Feb. 12, 2015, "RNAi COMPOSITIONS AND METHODS FOR TREATMENT OF FRIEDREICH'S ATAXIA"; WO 2017/186815, published Nov. 2, 2017, "Antisense oligonucleotides for enhanced expression of frataxin"; WO 2008/018795, published Feb. 14, 2008, "Methods and means for treating dna repeat instability associated genetic disorders"; US Patent Application 2018/0028557, published Feb. 1, 2018, "Hybrid oligonucleotides and uses thereof"; WO 2015/023975, published Feb. 19, 2015, "Compositions and methods for modulating RNA"; WO 2015/023939, published Feb. 19, 2015, "Compositions and methods for modulating expression of frataxin"; US Patent Application 2017/0281643, published Oct. 5, 2017, "Compounds and methods for modulating frataxin expression"; Li L. et al., "Activating frataxin expression by repeat-targeted nucleic acids" Nature Communications, Published 4 Feb. 2016; and Li L. et al. "Activation of Frataxin Protein Expression by Antisense Oligonucleotides Targeting the Mutant Expanded Repeat" Nucleic Acid Ther. 2018 Feb. 28(1):23-33., the contents of each of which are incorporated herein in their entireties.

[0193] In some embodiments, an oligonucleotide payload is configured (e.g., as a gapmer or RNAi oligonucleotide) for inhibiting expression of a natural antisense transcript that inhibits FXN expression, e.g., as disclosed in U.S. Pat. No. 9,593,330, filed Jun. 9, 2011, "Treatment of frataxin (FXN) related diseases by inhibition of natural antisense transcript to FXN", the contents of which are incorporated herein by reference in its entirety.

[0194] Examples of oligonucleotides for promoting FXN gene editing include WO 2016/094845, published Jun. 16, 2016, "Compositions and methods for editing nucleic acids in cells utilizing oligonucleotides"; WO 2015/089354, published Jun. 18, 2015, "Compositions and methods of use of CRISPR-Cas systems in nucleotide repeat disorders"; WO 2015/139139, published Sep. 24, 2015, "CRISPR-based methods and products for increasing frataxin levels and uses thereof"; and WO 2018/002783, published Jan. 4, 2018, "Materials and methods for treatment of Friedreich ataxia and other related disorders", the contents of each of which are incorporated herein in their entireties.

[0195] Examples of oligonucleotides for promoting FXN gene expression through targeting of non-FXN genes, e.g. epigenetic regulators of FXN, include WO 2015/023938, published Feb. 19, 2015, "Epigenetic regulators of frataxin", the contents of which are incorporated herein in its entirety.

[0196] In some embodiments, oligonucleotides may have a region of complementarity to a sequence set forth as: a FXN gene from humans (Gene ID 2395; NC_000009.12) and/or a FXN gene from mice (Gene ID 14297; NC_000085.6). In some embodiments, the oligonucleotide may have region of complementarity to a mutant form of FXN, for example as reported in e.g., Montermini, L. et al. "The Friedreich ataxia GAA triplet repeat: premutation and normal alleles." Hum. Molec. Genet., 1997, 6: 1261-1266.; Filla, A. et al. "The relationship between trinucleotide (GAA) repeat length and clinical features in Friedreich ataxia." Am. J. Hum. Genet. 1996, 59: 554-560.; Pandolfo, M. Friedreich ataxia: the clinical picture. J. Neurol. 2009, 256, 3-8.; the contents of each of which are incorporated herein by reference in their entireties.

DMD/Dystrophinopathies

[0197] Examples of oligonucleotides useful for targeting DMD are provided in U.S. Patent Application Publication US20100130591A1, published on May 27, 2010, entitled "MULTIPLE EXON SKIPPING COMPOSITIONS FOR DMD"; U.S. Pat. No. 8,361,979, issued Jan. 29, 2013, entitled "MEANS AND METHOD FOR INDUCING EXON-SKIPPING"; U.S. Patent Application Publication 20120059042, published Mar. 8, 2012, entitled "METHOD FOR EFFICIENT EXON (44) SKIPPING IN DUCHENNE MUSCULAR DYSTROPHY AND ASSOCIATED MEANS; U.S. Patent Application Publication 20140329881, published Nov. 6, 2014, entitled "EXON SKIPPING COMPOSITIONS FOR TREATING MUSCULAR DYSTROPHY"; U.S. Pat. No. 8,232,384, issued Jul. 31, 2012, entitled "ANTISENSE OLIGONUCLEOTIDES FOR INDUCING EXON SKIPPING AND METHODS OF USE THEREOF"; U.S. Patent Application Publication 20120022134A1, published Jan. 26, 2012, entitled "METHODS AND MEANS FOR EFFICIENT SKIPPING OF EXON 45 IN DUCHENNE MUSCULAR DYSTROPHY PRE-MRNA; U.S. Patent Application Publication 20120077860, published Mar. 29, 2012, entitled "ADENO-ASSOCIATED VIRAL VECTOR FOR EXON SKIPPING IN A GENE ENCODING A DISPENSABLE DOMAN PROTEIN"; U.S. Pat. No. 8,324,371, issued Dec. 4, 2012, entitled "OLIGOMERS"; U.S. Pat. No. 9,078,911, issued Jul. 14, 2015, entitled "ANTISENSE OLIGONUCLEOTIDES"; U.S. Pat. No. 9,079,934, issued Jul. 14, 2015, entitled "ANTISENSE NUCLEIC ACIDS"; U.S. Pat. No. 9,034,838, issued May 19, 2015, entitled "MIR-31 IN DUCHENNE MUSCULAR DYSTROPHY THERAPY"; and International Patent Publication WO2017062862A3, published Apr. 13, 2017, entitled "OLIGONUCLEOTIDE COMPOSITIONS AND METHODS THEREOF"; the contents of each of which are incorporated herein in their entireties.

[0198] Examples of oligonucleotides for promoting DMD gene editing include International Patent Publication WO2018053632A1, published Mar. 29, 2018, entitled "METHODS OF MODIFYING THE DYSTROPHIN GENE AND RESTORING DYSTROPHIN EXPRESSION AND USES THEREOF"; International Patent Publication WO2017049407A1, published Mar. 30, 2017, entitled "MODIFICATION OF THE DYSTROPHIN GENE AND USES THEREOF"; International Patent Publication WO2016161380A1, published Oct. 6, 2016, entitled "CRISPR/CAS-RELATED METHODS AND COMPOSITIONS FOR TREATING DUCHENNE MUSCULAR DYSTROPHY AND BECKER MUSCULAR DYSTROPHY"; International Patent Publication WO2017095967, published Jun. 8, 2017, entitled "THERAPEUTIC TARGETS FOR THE CORRECTION OF THE HUMAN DYSTROPHIN GENE BY GENE EDITING AND METHODS OF USE"; International Patent Publication WO2017072590A1, published May 4, 2017, entitled "MATERIALS AND METHODS FOR TREATMENT OF DUCHENNE MUSCULAR DYSTROPHY"; International Patent Publication WO2018098480A1, published May 31, 2018, entitled "PREVENTION OF MUSCULAR DYSTROPHY BY CRISPR/CPF1-MEDIATED GENE EDITING"; US Patent Application Publication US20170266320A1, published Sep. 21, 2017, entitled "RNA-Guided Systems for In Vivo Gene Editing"; International Patent Publication WO2016025469A1, published Feb. 18, 2016, entitled "PREVENTION OF MUSCULAR DYSTROPHY BY CRISPR/CAS9-MEDIATED GENE EDITING"; U.S. Patent Application Publication 2016/0201089, published Jul. 14, 2016, entitled "RNA-GUIDED GENE EDITING AND GENE REGULATION"; and U.S. Patent Application Publication 2013/0145487, published Jun. 6, 2013, entitled "MEGANUCLEASE VARIANTS CLEAVING A DNA TARGET SEQUENCE FROM THE DYSTROPHN GENE AND USES THEREOF", the contents of each of which are incorporated herein in their entireties. In some embodiments, an oligonucleotide may have a region of complementarity to DMD gene sequences of multiple species, e.g., selected from human, mouse and non-human species.

[0199] In some embodiments, the oligonucleotide may have region of complementarity to a mutant DMD allele, for example, a DMD allele with at least one mutation in any of exons 1-79 of DMD in humans that leads to a frameshift and improper RNA splicing/processing.

MYH7/Hypertrophic Cardiomyopathy

[0200] Examples of oligonucleotides useful as payloads, e.g., for targeting MYH7, are provided in US Patent Application Publication 20180094262, published on Apr. 5, 2018, entitled Inhibitors of MYH7B and Uses Thereof; US Patent Application Publication 20160348103, published on Dec. 1, 2016, entitled Oligonucleotides and Methods for Treatment of Cardiomyopathy Using RNA Interference; US Patent Application Publication 20160237430, published on Aug. 18, 2016, entitled "Allele-specific RNA Silencing for the Treatment of Hypertrophic Cardiomyopathy"; US Patent Application Publication 20160032286, published on Feb. 4, 2016, entitled "Inhibitors of MYH7B and Uses Thereof"; US Patent Application Publication 20140187603, published on Jul. 3, 2014, entitled "MicroRNA Inhibitors Comprising Locked Nucleotides"; US Patent Application Publication 20140179764, published on Jun. 26, 2014, entitled "Dual Targeting of miR-208 and miR-499 in the Treatment of Cardiac Disorders"; US Patent Application Publication 20120114744, published on May 10, 2012, entitled "Compositions and Methods to Treat Muscular and Cardiovascular Disorders"; the contents of each of which are incorporated herein in their entireties.

[0201] In some embodiments, the oligonucleotide may target lncRNA or mRNA, e.g., for degradation. In some embodiments, the oligonucleotide may target, e.g., for degradation, a nucleic acid encoding a protein involved in a mismatch repair pathway, e.g., MSH2, MutLalpha, MutSbeta, MutLalpha. Non-limiting examples of proteins involved in mismatch repair pathways, for which mRNAs encoding such proteins may be targeted by oligonucleotides described herein, are described in Iyer, R. R. et al., "DNA triplet repeat expansion and mismatch repair" Annu Rev Biochem. 2015; 84:199-226.; and Schmidt M. H. and Pearson C. E., "Disease-associated repeat instability and mismatch repair" DNA Repair (Amst). 2016 February; 38:117-26.

a. Oligonucleotide Size/Sequence

[0202] Oligonucleotides may be of a variety of different lengths, e.g., depending on the format. In some embodiments, an oligonucleotide is 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 35, 40, 45, 50, 75, or more nucleotides in length. In a some embodiments, the oligonucleotide is 8 to 50 nucleotides in length, 8 to 40 nucleotides in length, 8 to 30 nucleotides in length, 10 to 15 nucleotides in length, 10 to 20 nucleotides in length, 15 to 25 nucleotides in length, 21 to 23 nucleotides in lengths, etc.

[0203] In some embodiments, a complementary nucleic acid sequence of an oligonucleotide for purposes of the present disclosure is specifically hybridizable or specific for the target nucleic acid when binding of the sequence to the target molecule (e.g., mRNA) interferes with the normal function of the target (e.g., mRNA) to cause a loss of activity (e.g., inhibiting translation) or expression (e.g., degrading a target mRNA) and there is a sufficient degree of complementarity to avoid non-specific binding of the sequence to non-target sequences under conditions in which avoidance of non-specific binding is desired, e.g., under physiological conditions in the case of in vivo assays or therapeutic treatment, and in the case of in vitro assays, under conditions in which the assays are performed under suitable conditions of stringency. Thus, in some embodiments, an oligonucleotide may be at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% or 100% complementary to the consecutive nucleotides of an target nucleic acid. In some embodiments a complementary nucleotide sequence need not be 100% complementary to that of its target to be specifically hybridizable or specific for a target nucleic acid.

[0204] In some embodiments, an oligonucleotide comprises region of complementarity to a target nucleic acid that is in the range of 8 to 15, 8 to 30, 8 to 40, or 10 to 50, or 5 to 50, or 5 to 40 nucleotides in length. In some embodiments, a region of complementarity of an oligonucleotide to a target nucleic acid is 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, or 50 nucleotides in length. In some embodiments, the region of complementarity is complementary with at least 8 consecutive nucleotides of a target nucleic acid. In some embodiments, an oligonucleotide may contain 1, 2 or 3 base mismatches compared to the portion of the consecutive nucleotides of target nucleic acid. In some embodiments the oligonucleotide may have up to 3 mismatches over 15 bases, or up to 2 mismatches over 10 bases.

b. Oligonucleotide Modifications:

[0205] The oligonucleotides described herein may be modified, e.g., comprise a modified sugar moiety, a modified internucleoside linkage, a modified nucleotide and/or combinations thereof. In addition, in some embodiments, oligonucleotides may exhibit one or more of the following properties: do not mediate alternative splicing; are not immune stimulatory; are nuclease resistant; have improved cell uptake compared to unmodified oligonucleotides; are not toxic to cells or mammals; have improved endosomal exit internally in a cell; minimizes TLR stimulation; or avoid pattern recognition receptors. Any of the modified chemistries or formats of oligonucleotides described herein can be combined with each other. For example, one, two, three, four, five, or more different types of modifications can be included within the same oligonucleotide.

[0206] In some embodiments, certain nucleotide modifications may be used that make an oligonucleotide into which they are incorporated more resistant to nuclease digestion than the native oligodeoxynucleotide or oligoribonucleotide molecules; these modified oligonucleotides survive intact for a longer time than unmodified oligonucleotides. Specific examples of modified oligonucleotides include those comprising modified backbones, for example, modified internucleoside linkages such as phosphorothioates, phosphotriesters, methyl phosphonates, short chain alkyl or cycloalkyl intersugar linkages or short chain heteroatomic or heterocyclic intersugar linkages. Accordingly, oligonucleotides of the disclosure can be stabilized against nucleolytic degradation such as by the incorporation of a modification, e.g., a nucleotide modification.

[0207] In some embodiments, an oligonucleotide may be of up to 50 or up to 100 nucleotides in length in which 2 to 10, 2 to 15, 2 to 16, 2 to 17, 2 to 18, 2 to 19, 2 to 20, 2 to 25, 2 to 30, 2 to 40, 2 to 45, or more nucleotides of the oligonucleotide are modified nucleotides. The oligonucleotide may be of 8 to 30 nucleotides in length in which 2 to 10, 2 to 15, 2 to 16, 2 to 17, 2 to 18, 2 to 19, 2 to 20, 2 to 25, 2 to 30 nucleotides of the oligonucleotide are modified nucleotides. The oligonucleotide may be of 8 to 15 nucleotides in length in which 2 to 4, 2 to 5, 2 to 6, 2 to 7, 2 to 8, 2 to 9, 2 to 10, 2 to 11, 2 to 12, 2 to 13, 2 to 14 nucleotides of the oligonucleotide are modified nucleotides. Optionally, the oligonucleotides may have every nucleotide except 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 nucleotides modified. Oligonucleotide modifications are described further herein.

c. Modified Nucleotides

[0208] In some embodiments, an oligonucleotide include a 2'-modified nucleotide, e.g., a 2'-deoxy, 2'-deoxy-2'-fluoro, 2'-O-methyl, 2'-O-methoxyethyl (2'-O-MOE), 2'-O-aminopropyl (2'-O-AP), 2'-O-dimethylaminoethyl (2'-O-DMAOE), 2'-O-dimethylaminopropyl (2'-O-DMAP), 2'-O-dimethylaminoethyloxyethyl (2'-O-DMAEOE), or 2'-O-N-methylacetamido (2'-O-NMA).

[0209] In some embodiments, an oligonucleotide can include at least one 2'-O-methyl-modified nucleotide, and in some embodiments, all of the nucleotides include a 2'-O-methyl modification. In some embodiments, an oligonucleotide comprises modified nucleotides in which the ribose ring comprises a bridge moiety connecting two atoms in the ring, e.g., connecting the 2'-O atom to the 4'-C atom. In some embodiments, the oligonucleotides are "locked," e.g., comprise modified nucleotides in which the ribose ring is "locked" by a methylene bridge connecting the 2'-O atom and the 4'-C atom. Examples of LNAs are described in International Patent Application Publication WO/2008/043753, published on Apr. 17, 2008, and entitled "RNA Antagonist Compounds For The Modulation Of PCSK9", the contents of which are incorporated herein by reference in its entirety.

[0210] Other modifications that may be used in the oligonucleotides disclosed herein include ethylene-bridged nucleic acids (ENAs). ENAs include, but are not limited to, 2'-0,4'-C-ethylene-bridged nucleic acids. Examples of ENAs are provided in International Patent Publication No. WO 2005/042777, published on May 12, 2005, and entitled "APP/ENA Antisense"; Morita et al., Nucleic Acid Res., Suppl 1:241-242, 2001; Surono et al., Hum. Gene Ther., 15:749-757, 2004; Koizumi, Curr. Opin. Mol. Ther., 8:144-149, 2006 and Horie et al., Nucleic Acids Symp. Ser (Oxf), 49:171-172, 2005; the disclosures of which are incorporated herein by reference in their entireties.

[0211] In some embodiments, the oligonucleotide may comprise a bridged nucleotide, such as a locked nucleic acid (LNA) nucleotide, a constrained ethyl (cEt) nucleotide, or an ethylene bridged nucleic acid (ENA) nucleotide. In some embodiments, the oligonucleotide comprises a modified nucleotide disclosed in one of the following United States Patent or Patent Application Publications: U.S. Pat. No. 7,399,845, issued on Jul. 15, 2008, and entitled "6-Modified Bicyclic Nucleic Acid Analogs"; U.S. Pat. No. 7,741,457, issued on Jun. 22, 2010, and entitled "6-Modified Bicyclic Nucleic Acid Analogs"; U.S. Pat. No. 8,022,193, issued on Sep. 20, 2011, and entitled "6-Modified Bicyclic Nucleic Acid Analogs"; U.S. Pat. No. 7,569,686, issued on Aug. 4, 2009, and entitled "Compounds And Methods For Synthesis Of Bicyclic Nucleic Acid Analogs"; U.S. Pat. No. 7,335,765, issued on Feb. 26, 2008, and entitled "Novel Nucleoside And Oligonucleotide Analogues"; U.S. Pat. No. 7,314,923, issued on Jan. 1, 2008, and entitled "Novel Nucleoside And Oligonucleotide Analogues"; U.S. Pat. No. 7,816,333, issued on Oct. 19, 2010, and entitled "Oligonucleotide Analogues And Methods Utilizing The Same" and US Publication Number 2011/0009471 now U.S. Pat. No. 8,957,201, issued on Feb. 17, 2015, and entitled "Oligonucleotide Analogues And Methods Utilizing The Same", the entire contents of each of which are incorporated herein by reference for all purposes.

[0212] In some embodiments, the oligonucleotide comprises at least one nucleotide modified at the 2' position of the sugar, preferably a 2'-O-alkyl, 2'-O-alkyl-O-alkyl or 2'-fluoro-modified nucleotide. In other preferred embodiments, RNA modifications include 2'-fluoro, 2'-amino and 2' O-methyl modifications on the ribose of pyrimidines, abasic residues or an inverted base at the 3' end of the RNA.

[0213] In some embodiments, the oligonucleotide may have at least one modified nucleotide that results in an increase in Tm of the oligonucleotide in a range of 1° C., 2° C., 3° C., 4° C., or 5° C. compared with an oligonucleotide that does not have the at least one modified nucleotide. The oligonucleotide may have a plurality of modified nucleotides that result in a total increase in Tm of the oligonucleotide in a range of 2° C., 3° C., 4° C., 5° C., 6° C., 7° C., 8° C., 9° C., 10° C., 15° C., 20° C., 25° C., 30° C., 35° C., 40° C., 45° C. or more compared with an oligonucleotide that does not have the modified nucleotide.

[0214] The oligonucleotide may comprise alternating nucleotides of different kinds. For example, an oligonucleotide may comprise alternating deoxyribonucleotides or ribonucleotides and 2'-fluoro-deoxyribonucleotides. An oligonucleotide may comprise alternating deoxyribonucleotides or ribonucleotides and 2'-O-methyl nucleotides. An oligonucleotide may comprise alternating 2'-fluoro nucleotides and 2'-O-methyl nucleotides. An oligonucleotide may comprise alternating bridged nucleotides and 2'-fluoro or 2'-O-methyl nucleotides.

d. Internucleotide Linkages/Backbones

[0215] In some embodiments, oligonucleotide may contain a phosphorothioate or other modified internucleotide linkage. In some embodiments, the oligonucleotide comprises phosphorothioate internucleoside linkages. In some embodiments, the oligonucleotide comprises phosphorothioate internucleoside linkages between at least two nucleotides. In some embodiments, the oligonucleotide comprises phosphorothioate internucleoside linkages between all nucleotides. For example, in some embodiments, oligonucleotides comprise modified internucleotide linkages at the first, second, and/or third internucleoside linkage at the 5' or 3' end of the nucleotide sequence.

[0216] Phosphorus-containing linkages that may be used include, but are not limited to, phosphorothioates, chiral phosphorothioates, phosphorodithioates, phosphotriesters, aminoalkylphosphotriesters, methyl and other alkyl phosphonates comprising 3'alkylene phosphonates and chiral phosphonates, phosphinates, phosphoramidates comprising 3'-amino phosphoramidate and aminoalkylphosphoramidates, thionophosphoramidates, thionoalkylphosphonates, thionoalkylphosphotriesters, and boranophosphates having normal 3'-5' linkages, 2'-5' linked analogs of these, and those having inverted polarity wherein the adjacent pairs of nucleoside units are linked 3'-5' to 5'-3' or 2'-5' to 5'-2'; see U.S. Pat. Nos. 3,687,808; 4,469,863; 4,476,301; 5,023,243; 5,177,196; 5,188,897; 5,264,423; 5,276,019; 5,278,302; 5,286,717; 5,321,131; 5,399,676; 5,405,939; 5,453,496; 5,455, 233; 5,466,677; 5,476,925; 5,519,126; 5,536,821; 5,541,306; 5,550,111; 5,563,253; 5,571,799; 5,587,361; and 5,625,050.

[0217] In some embodiments, oligonucleotides may have heteroatom backbones, such as methylene(methylimino) or MMI backbones; amide backbones (see De Mesmaeker et al. Ace. Chem. Res. 1995, 28:366-374); morpholino backbones (see Summerton and Weller, U.S. Pat. No. 5,034,506); or peptide nucleic acid (PNA) backbones (wherein the phosphodiester backbone of the oligonucleotide is replaced with a polyamide backbone, the nucleotides being bound directly or indirectly to the aza nitrogen atoms of the polyamide backbone, see Nielsen et al., Science 1991, 254, 1497).

e. Stereospecific Oligonucleotides

[0218] In some embodiments, internucleotidic phosphorus atoms of oligonucleotides are chiral, and the properties of the oligonucleotides are adjusted based on the configuration of the chiral phosphorus atoms. In some embodiments, appropriate methods may be used to synthesize P-chiral oligonucleotide analogs in a stereocontrolled manner (e.g., as described in Oka N, Wada T, Stereocontrolled synthesis of oligonucleotide analogs containing chiral internucleotidic phosphorus atoms. Chem Soc Rev. 2011 December; 40(12): 5829-43.) In some embodiments, phosphorothioate containing oligonucleotides are provided that comprise nucleoside units that are joined together by either substantially all Sp or substantially all Rp phosphorothioate intersugar linkages. In some embodiments, such phosphorothioate oligonucleotides having substantially chirally pure intersugar linkages are prepared by enzymatic or chemical synthesis, as described, for example, in U.S. Pat. No. 5,587,261, issued on Dec. 12, 1996, the contents of which are incorporated herein by reference in their entirety. In some embodiments, chirally controlled oligonucleotides provide selective cleavage patterns of a target nucleic acid. For example, in some embodiments, a chirally controlled oligonucleotide provides single site cleavage within a complementary sequence of a nucleic acid, as described, for example, in US Patent Application Publication 20170037399 A1, published on Feb. 2, 2017, entitled "CHIRAL DESIGN", the contents of which are incorporated herein by reference in their entirety.

f. Morpholinos

[0219] In some embodiments, the oligonucleotide may be a morpholino-based compounds. Morpholino-based oligomeric compounds are described in Dwaine A. Braasch and David R. Corey, Biochemistry, 2002, 41(14), 4503-4510); Genesis, volume 30, issue 3, 2001; Heasman, J., Dev. Biol., 2002, 243, 209-214; Nasevicius et al., Nat. Genet., 2000, 26, 216-220; Lacerra et al., Proc. Natl. Acad. Sci., 2000, 97, 9591-9596; and U.S. Pat. No. 5,034,506, issued Jul. 23, 1991. In some embodiments, the morpholino-based oligomeric compound is a phosphorodiamidate morpholino oligomer (PMO) (e.g., as described in Iverson, Curr. Opin. Mol. Ther., 3:235-238, 2001; and Wang et al., J. Gene Med., 12:354-364, 2010; the disclosures of which are incorporated herein by reference in their entireties).

g. Peptide Nucleic Acids (PNAs)

[0220] In some embodiments, both a sugar and an internucleoside linkage (the backbone) of the nucleotide units of an oligonucleotide are replaced with novel groups. In some embodiments, the base units are maintained for hybridization with an appropriate nucleic acid target compound. One such oligomeric compound, an oligonucleotide mimetic that has been shown to have excellent hybridization properties, is referred to as a peptide nucleic acid (PNA). In PNA compounds, the sugar-backbone of an oligonucleotide is replaced with an amide containing backbone, for example, an aminoethylglycine backbone. The nucleobases are retained and are bound directly or indirectly to aza nitrogen atoms of the amide portion of the backbone. Representative publication that report the preparation of PNA compounds include, but are not limited to, U.S. Pat. Nos. 5,539,082;

5,714,331; and 5,719,262, each of which is herein incorporated by reference. Further teaching of PNA compounds can be found in Nielsen et al., Science, 1991, 254, 1497-1500.

h. Gapmers

[0221] In some embodiments, the oligonucleotide is a gapmer. A gapmer oligonucleotide generally has the formula 5'-X-Y-Z-3', with X and Z as flanking regions around a gap region Y. In some embodiments, the Y region is a contiguous stretch of nucleotides, e.g., a region of at least 6 DNA nucleotides, which are capable of recruiting an RNAse, such as RNAse H. In some embodiments, the gapmer binds to the target nucleic acid, at which point an RNAse is recruited and can then cleave the target nucleic acid. In some embodiments, the Y region is flanked both 5' and 3' by regions X and Z comprising high-affinity modified nucleotides, e.g., one to six modified nucleotides. Examples of modified nucleotides include, but are not limited to, 2' MOE or 2'OMe or Locked Nucleic Acid bases (LNA). The flanking sequences X and Z may be of one to twenty nucleotides, one to eight nucleotides or one to five nucleotides in length, in some embodiments. The flanking sequences X and Z may be of similar length or of dissimilar lengths. The gap-segment Y may be a nucleotide sequence of five to twenty nucleotides, size to twelve nucleotides or six to ten nucleotides in length, in some embodiments.

[0222] In some embodiments, the gap region of the gapmer oligonucleotides may contain modified nucleotides known to be acceptable for efficient RNase H action in addition to DNA nucleotides, such as C4'-substituted nucleotides, acyclic nucleotides, and arabino-configured nucleotides. In some embodiments, the gap region comprises one or more unmodified internucleosides. In some embodiments, one or both flanking regions each independently comprise one or more phosphorothioate internucleoside linkages (e.g., phosphorothioate internucleoside linkages or other linkages) between at least two, at least three, at least four, at least five or more nucleotides. In some embodiments, the gap region and two flanking regions each independently comprise modified internucleoside linkages (e.g., phosphorothioate internucleoside linkages or other linkages) between at least two, at least three, at least four, at least five or more nucleotides.

[0223] A gapmer may be produced using appropriate methods. Representative U.S. patents, U.S. patent publications, and PCT publications that teach the preparation of gapmers include, but are not limited to, U.S. Pat. Nos. 5,013,830; 5,149,797; 5,220,007; 5,256,775; 5,366,878; 5,403,711; 5,491,133; 5,565,350; 5,623,065; 5,652,355; 5,652,356; 5,700,922; 5,898,031; 7,432,250; and 7,683,036; U.S. patent publication Nos. US20090286969, US20100197762, and US20110112170; and PCT publication Nos. WO2008049085 and WO2009090182, each of which is herein incorporated by reference in its entirety.

i. Mixmers

[0224] In some embodiments, an oligonucleotide described herein may be a mixmer or comprise a mixmer sequence pattern. In general, mixmers are oligonucleotides that comprise both naturally and non-naturally occurring nucleotides or comprise two different types of non-naturally occurring nucleotides typically in an alternating pattern. Mixmers generally have higher binding affinity than unmodified oligonucleotides and may be used to specifically bind a target molecule, e.g., to block a binding site on the target molecule. Generally, mixmers do not recruit an RNAse to the target molecule and thus do not promote cleavage of the target molecule. Such oligonucleotides that are incapable of recruiting RNAse H have been described, for example, see WO2007/112754 or WO2007/112753.

[0225] In some embodiments, the mixmer comprises or consists of a repeating pattern of nucleotide analogues and naturally occurring nucleotides, or one type of nucleotide analogue and a second type of nucleotide analogue. However, a mixmer need not comprise a repeating pattern and may instead comprise any arrangement of modified nucleotides and naturally occurring nucleotides or any arrangement of one type of modified nucleotide and a second type of modified nucleotide. The repeating pattern, may, for instance be every second or every third nucleotide is a modified nucleotide, such as LNA, and the remaining nucleotides are naturally occurring nucleotides, such as DNA, or are a 2' substituted nucleotide analogue such as 2'MOE or 2' fluoro analogues, or any other modified nucleotide described herein. It is recognized that the repeating pattern of modified nucleotide, such as LNA units, may be combined with modified nucleotide at fixed positions—e.g. at the 5' or 3' termini.

[0226] In some embodiments, a mixmer does not comprise a region of more than 5, more than 4, more than 3, or more than 2 consecutive naturally occurring nucleotides, such as DNA nucleotides. In some embodiments, the mixmer comprises at least a region consisting of at least two consecutive modified nucleotide, such as at least two consecutive LNAs. In some embodiments, the mixmer comprises at least a region consisting of at least three consecutive modified nucleotide units, such as at least three consecutive LNAs.

[0227] In some embodiments, the mixmer does not comprise a region of more than 7, more than 6, more than 5, more than 4, more than 3, or more than 2 consecutive nucleotide analogues, such as LNAs. In some embodiments, LNA units may be replaced with other nucleotide analogues, such as those referred to herein.

[0228] Mixmers may be designed to comprise a mixture of affinity enhancing modified nucleotides, such as in non-limiting example LNA nucleotides and 2'-O-methyl nucleotides. In some embodiments, a mixmer comprises modified internucleoside linkages (e.g., phosphorothioate internucleoside linkages or other linkages) between at least two, at least three, at least four, at least five or more nucleotides.

[0229] A mixmer may be produced using any suitable method. Representative U.S. patents, U.S. patent publications, and PCT publications that teach the preparation of mixmers include U.S. patent publication Nos. US20060128646, US20090209748, US20090298916, US20110077288, and US20120322851, and U.S. Pat. No. 7,687,617.

[0230] In some embodiments, a mixmer comprises one or more morpholino nucleotides. For example, in some embodiments, a mixmer may comprise morpholino nucleotides mixed (e.g., in an alternating manner) with one or more other nucleotides (e.g., DNA, RNA nucleotides) or modified nucleotides (e.g., LNA, 2'-O-Methyl nucleotides).

[0231] In some embodiments, mixmers are useful for splice correcting or exon skipping, for example, as reported in Touznik A., et al., LNA/DNA mixmer-based antisense oligonucleotides correct alternative splicing of the SMN2 gene and restore SMN protein expression in type 1 SMA fibroblasts Scientific Reports, volume 7, Article number: 3672 (2017), Chen S. et al., Synthesis of a Morpholino

Nucleic Acid (MNA)-Uridine Phosphoramidite, and Exon Skipping Using MNA/2'-O-Methyl Mixmer Antisense Oligonucleotide, Molecules 2016, 21, 1582, the contents of each which are incorporated herein by reference.

j. RNA Interference (RNAi)

[0232] In some embodiments, oligonucleotides provided herein may be in the form of small interfering RNAs (siRNA), also known as short interfering RNA or silencing RNA. SiRNA, is a class of double-stranded RNA molecules, typically about 20-25 base pairs in length that target nucleic acids (e.g., mRNAs) for degradation via the RNA interference (RNAi) pathway in cells. Specificity of siRNA molecules may be determined by the binding of the antisense strand of the molecule to its target RNA. Effective siRNA molecules are generally less than 30 to 35 base pairs in length to prevent the triggering of non-specific RNA interference pathways in the cell via the interferon response, although longer siRNA can also be effective.

[0233] Following selection of an appropriate target RNA sequence, siRNA molecules that comprise a nucleotide sequence complementary to all or a portion of the target sequence, i.e. an antisense sequence, can be designed and prepared using appropriate methods (see, e.g., PCT Publication Number WO 2004/016735; and U.S. Patent Publication Nos. 2004/0077574 and 2008/0081791).

[0234] The siRNA molecule can be double stranded (i.e. a dsRNA molecule comprising an antisense strand and a complementary sense strand) or single-stranded (i.e. a ssRNA molecule comprising just an antisense strand). The siRNA molecules can comprise a duplex, asymmetric duplex, hairpin or asymmetric hairpin secondary structure, having self-complementary sense and antisense strands.

[0235] Double-stranded siRNA may comprise RNA strands that are the same length or different lengths. Double-stranded siRNA molecules can also be assembled from a single oligonucleotide in a stem-loop structure, wherein self-complementary sense and antisense regions of the siRNA molecule are linked by means of a nucleic acid based or non-nucleic acid-based linker(s), as well as circular single-stranded RNA having two or more loop structures and a stem comprising self-complementary sense and antisense strands, wherein the circular RNA can be processed either in vivo or in vitro to generate an active siRNA molecule capable of mediating RNAi. Small hairpin RNA (shRNA) molecules thus are also contemplated herein. These molecules comprise a specific antisense sequence in addition to the reverse complement (sense) sequence, typically separated by a spacer or loop sequence. Cleavage of the spacer or loop provides a single-stranded RNA molecule and its reverse complement, such that they may anneal to form a dsRNA molecule (optionally with additional processing steps that may result in addition or removal of one, two, three or more nucleotides from the 3' end and/or the 5' end of either or both strands). A spacer can be of a sufficient length to permit the antisense and sense sequences to anneal and form a double-stranded structure (or stem) prior to cleavage of the spacer (and, optionally, subsequent processing steps that may result in addition or removal of one, two, three, four, or more nucleotides from the 3' end and/or the 5' end of either or both strands). A spacer sequence is may be an unrelated nucleotide sequence that is situated between two complementary nucleotide sequence regions which, when annealed into a double-stranded nucleic acid, comprise a shRNA.

[0236] The overall length of the siRNA molecules can vary from about 14 to about 100 nucleotides depending on the type of siRNA molecule being designed. Generally between about 14 and about 50 of these nucleotides are complementary to the RNA target sequence, i.e. constitute the specific antisense sequence of the siRNA molecule. For example, when the siRNA is a double- or single-stranded siRNA, the length can vary from about 14 to about 50 nucleotides, whereas when the siRNA is a shRNA or circular molecule, the length can vary from about 40 nucleotides to about 100 nucleotides.

[0237] An siRNA molecule may comprise a 3' overhang at one end of the molecule, The other end may be blunt-ended or have also an overhang (5' or 3'). When the siRNA molecule comprises an overhang at both ends of the molecule, the length of the overhangs may be the same or different. In one embodiment, the siRNA molecule of the present disclosure comprises 3' overhangs of about 1 to about 3 nucleotides on both ends of the molecule.

k. microRNA (miRNAs)

[0238] In some embodiments, an oligonucleotide may be a microRNA (miRNA).

[0239] MicroRNAs (referred to as "miRNAs") are small non-coding RNAs, belonging to a class of regulatory molecules that control gene expression by binding to complementary sites on a target RNA transcript. Typically, miRNAs are generated from large RNA precursors (termed pri-miRNAs) that are processed in the nucleus into approximately 70 nucleotide pre-miRNAs, which fold into imperfect stem-loop structures. These pre-miRNAs typically undergo an additional processing step within the cytoplasm where mature miRNAs of 18-25 nucleotides in length are excised from one side of the pre-miRNA hairpin by an RNase III enzyme, Dicer.

[0240] As used herein, miRNAs including pri-miRNA, pre-miRNA, mature miRNA or fragments of variants thereof that retain the biological activity of mature miRNA. In one embodiment, the size range of the miRNA can be from 21 nucleotides to 170 nucleotides. In one embodiment the size range of the miRNA is from 70 to 170 nucleotides in length. In another embodiment, mature miRNAs of from 21 to 25 nucleotides in length can be used.

l. Aptamers

[0241] In some embodiments, oligonucleotides provided herein may be in the form of aptamers. Generally, in the context of molecular payloads, aptamer is any nucleic acid that binds specifically to a target, such as a small molecule, protein, nucleic acid in a cell. In some embodiments, the aptamer is a DNA aptamer or an RNA aptamer. In some embodiments, a nucleic acid aptamer is a single-stranded DNA or RNA (ssDNA or ssRNA). It is to be understood that a single-stranded nucleic acid aptamer may form helices and/or loop structures. The nucleic acid that forms the nucleic acid aptamer may comprise naturally occurring nucleotides, modified nucleotides, naturally occurring nucleotides with hydrocarbon linkers (e.g., an alkylene) or a polyether linker (e.g., a PEG linker) inserted between one or more nucleotides, modified nucleotides with hydrocarbon or PEG linkers inserted between one or more nucleotides, or a combination of thereof. Exemplary publications and patents describing aptamers and method of producing aptamers include, e.g., Lorsch and Szostak, 1996; Jayasena, 1999; U.S. Pat. Nos. 5,270,163; 5,567,588; 5,650,275; 5,670,637; 5,683,867; 5,696,249; 5,789,157; 5,843,653; 5,864,026;

5,989,823; 6,569,630; 8,318,438 and PCT application WO 99/31275, each incorporated herein by reference.

m. Ribozymes

[0242] In some embodiments, oligonucleotides provided herein may be in the form of a ribozyme. A ribozyme (ribonucleic acid enzyme) is a molecule, typically an RNA molecule, that is capable of performing specific biochemical reactions, similar to the action of protein enzymes. Ribozymes are molecules with catalytic activities including the ability to cleave at specific phosphodiester linkages in RNA molecules to which they have hybridized, such as mRNAs, RNA-containing substrates, lncRNAs, and ribozymes, themselves.

[0243] Ribozymes may assume one of several physical structures, one of which is called a "hammerhead." A hammerhead ribozyme is composed of a catalytic core containing nine conserved bases, a double-stranded stem and loop structure (stem-loop II), and two regions complementary to the target RNA flanking regions the catalytic core. The flanking regions enable the ribozyme to bind to the target RNA specifically by forming double-stranded stems I and III. Cleavage occurs in cis (i.e., cleavage of the same RNA molecule that contains the hammerhead motif) or in trans (cleavage of an RNA substrate other than that containing the ribozyme) next to a specific ribonucleotide triplet by a transesterification reaction from a 3', 5'-phosphate diester to a 2', 3'-cyclic phosphate diester. Without wishing to be bound by theory, it is believed that this catalytic activity requires the presence of specific, highly conserved sequences in the catalytic region of the ribozyme.

[0244] Modifications in ribozyme structure have also included the substitution or replacement of various non-core portions of the molecule with non-nucleotidic molecules. For example, Benseler et al. (J. Am. Chem. Soc. (1993) 115:8483-8484) disclosed hammerhead-like molecules in which two of the base pairs of stem II, and all four of the nucleotides of loop II were replaced with non-nucleoside linkers based on hexaethylene glycol, propanediol, bis(triethylene glycol) phosphate, tris(propanediol)bisphosphate, or bis(propanediol) phosphate. Ma et al. (Biochem. (1993) 32:1751-1758; Nucleic Acids Res. (1993) 21:2585-2589) replaced the six nucleotide loop of the TAR ribozyme hairpin with non-nucleotidic, ethylene glycol-related linkers. Thomson et al. (Nucleic Acids Res. (1993) 21:5600-5603) replaced loop II with linear, non-nucleotidic linkers of 13, 17, and 19 atoms in length.

[0245] Ribozyme oligonucleotides can be prepared using well known methods (see, e.g., PCT Publications WO9118624; WO9413688; WO9201806; and WO 92/07065; and U.S. Pat. Nos. 5,436,143 and 5,650,502) or can be purchased from commercial sources (e.g., US Biochemicals) and, if desired, can incorporate nucleotide analogs to increase the resistance of the oligonucleotide to degradation by nucleases in a cell. The ribozyme may be synthesized in any known manner, e.g., by use of a commercially available synthesizer produced, e.g., by Applied Biosystems, Inc. or Milligen. The ribozyme may also be produced in recombinant vectors by conventional means. See, Molecular Cloning: A Laboratory Manual, Cold Spring Harbor Laboratory (Current edition). The ribozyme RNA sequences maybe synthesized conventionally, for example, by using RNA polymerases such as T7 or SP6.

n. Guide Nucleic Acids

[0246] In some embodiments, oligonucleotides are guide nucleic acid, e.g., guide RNA (gRNA) molecules. Generally, a guide RNA is a short synthetic RNA composed of (1) a scaffold sequence that binds to a nucleic acid programmable DNA binding protein (napDNAbp), such as Cas9, and (2) a nucleotide spacer portion that defines the DNA target sequence (e.g., genomic DNA target) to which the gRNA binds in order to bring the nucleic acid programmable DNA binding protein in proximity to the DNA target sequence. In some embodiments, the napDNAbp is a nucleic acid-programmable protein that forms a complex with (e.g., binds or associates with) one or more RNA(s) that targets the nucleic acid-programmable protein to a target DNA sequence (e.g., a target genomic DNA sequence). In some embodiments, a nucleic acid-programmable nuclease, when in a complex with an RNA, may be referred to as a nuclease:RNA complex. Guide RNAs can exist as a complex of two or more RNAs, or as a single RNA molecule.

[0247] Guide RNAs (gRNAs) that exist as a single RNA molecule may be referred to as single-guide RNAs (sgRNAs), though gRNA is also used to refer to guide RNAs that exist as either single molecules or as a complex of two or more molecules. Typically, gRNAs that exist as a single RNA species comprise two domains: (1) a domain that shares homology to a target nucleic acid (i.e., directs binding of a Cas9 complex to the target); and (2) a domain that binds a Cas9 protein. In some embodiments, domain (2) corresponds to a sequence known as a tracrRNA and comprises a stem-loop structure. In some embodiments, domain (2) is identical or homologous to a tracrRNA as provided in Jinek et al., Science 337:816-821 (2012), the entire contents of which is incorporated herein by reference.

[0248] In some embodiments, a gRNA comprises two or more of domains (1) and (2), and may be referred to as an extended gRNA. For example, an extended gRNA will bind two or more Cas9 proteins and bind a target nucleic acid at two or more distinct regions, as described herein. The gRNA comprises a nucleotide sequence that complements a target site, which mediates binding of the nuclease/RNA complex to said target site, providing the sequence specificity of the nuclease:RNA complex. In some embodiments, the RNA-programmable nuclease is the (CRISPR-associated system) Cas9 endonuclease, for example, Cas9 (Csn1) from *Streptococcus pyogenes* (see, e.g., "Complete genome sequence of an M1 strain of *Streptococcus pyogenes*." Ferretti J. J., McShan W. M., Ajdic D. J., Savic D. J., Savic G., Lyon K., Primeaux C., Sezate S., Suvorov A. N., Kenton S., Lai H. S., Lin S. P., Qian Y., Jia H. G., Najar F. Z., Ren Q., Zhu H., Song L., White J., Yuan X., Clifton S. W., Roe B. A., McLaughlin R. E., Proc. Natl. Acad. Sci. U.S.A. 98:4658-4663 (2001); "CRISPR RNA maturation by trans-encoded small RNA and host factor RNase III." Deltcheva E., Chylinski K., Sharma C. M., Gonzales K., Chao Y., Pirzada Z. A., Eckert M. R., Vogel J., Charpentier E., Nature 471:602-607 (2011); and "A programmable dual-RNA-guided DNA endonuclease in adaptive bacterial immunity." Jinek M., Chylinski K., Fonfara I., Hauer M., Doudna J. A., Charpentier E. Science 337:816-821 (2012), the entire contents of each of which are incorporated herein by reference.

o. Splice Altering Oligonucleotides

[0249] In some embodiments, a oligonucleotide (e.g., an antisense oligonucleotide including a morpholino) of the present disclosure target splicing. In some embodiments, the oligonucleotide targets splicing by inducing exon skipping and restoring the reading frame within a gene. As a non-limiting example, the oligonucleotide may induce skipping of an exon encoding a frameshift mutation and/or an exon that encodes a premature stop codon. In some embodiments, an oligonucleotide may induce exon skipping by blocking spliceosome recognition of a splice site. In some embodiments, exon skipping results in a truncated but functional protein compared to the reference protein (e.g., truncated but functional DMD protein as described below). In some embodiments, the oligonucleotide promotes inclusion of a particular exon (e.g., exon 7 of the SMN2 gene described below). In some embodiments, an oligonucleotide may induce inclusion of an exon by targeting a splice site inhibitory sequence. RNA splicing has been implicated in muscle diseases, including Duchenne muscular dystrophy (DMD) and spinal muscular atrophy (SMA).

[0250] Alterations (e.g., deletions, point mutations, and duplications) in the gene encoding dystrophin (DMD) cause DMD. These alterations can lead to frameshift mutations and/or nonsense mutations. In some embodiments, an oligonucleotide of the present disclosure promotes skipping of one or more DMD exons (e.g., exon 8, exon 43, exon 44, exon 45, exon 50, exon 51, exon 52, exon 53, and/or exon 55) and results in a functional truncated protein. See, e.g., U.S. Pat. No. 8,486,907 published on Jul. 16, 2013 and U.S. 20140275212 published on Sep. 18, 2014.

[0251] In SMA, there is loss of functional SMN1. Although the SMN2 gene is a paralog to SMN1, alternative splicing of the SMN2 gene predominantly leads to skipping of exon 7 and subsequent production of a truncated SMN protein that cannot compensate for SMN1 loss. In some embodiments, an oligonucleotide of the present disclosure promotes inclusion of SMN2 exon 7. In some embodiments, an oligonucleotide is an antisense oligonucleotide that targets SMN2 splice site inhibitory sequences (see, e.g., U.S. Pat. No. 7,838,657, which was published on Nov. 23, 2010).

p. Multimers

[0252] In some embodiments, molecular payloads may comprise multimers (e.g., concatemers) of 2 or more oligonucleotides connected by a linker. In this way, in some embodiments, the oligonucleotide loading of a complex/conjugate can be increased beyond the available linking sites on a targeting agent (e.g., available thiol sites on an antibody) or otherwise tuned to achieve a particular payload loading content. Oligonucleotides in a multimer can be the same or different (e.g., targeting different genes or different sites on the same gene or products thereof).

[0253] In some embodiments, multimers comprise 2 or more oligonucleotides linked together by a cleavable linker. However, in some embodiments, multimers comprise 2 or more oligonucleotides linked together by a non-cleavable linker. In some embodiments, a multimer comprises 2, 3, 4, 5, 6, 7, 8, 9, 10 or more oligonucleotides linked together. In some embodiments, a multimer comprises 2 to 5, 2 to 10 or 4 to 20 oligonucleotides linked together.

[0254] In some embodiments, a multimer comprises 2 or more oligonucleotides linked end-to-end (in a linear arrangement). In some embodiments, a multimer comprises 2 or more oligonucleotides linked end-to-end via a oligonucleotide based linker (e.g., poly-dT linker, an abasic linker). In some embodiments, a multimer comprises a 5' end of one oligonucleotide linked to a 3' end of another oligonucleotide. In some embodiments, a multimer comprises a 3' end of one oligonucleotide linked to a 3' end of another oligonucleotide. In some embodiments, a multimer comprises a 5' end of one oligonucleotide linked to a 5' end of another oligonucleotide. Still, in some embodiments, multimers can comprise a branched structure comprising multiple oligonucleotides linked together by a branching linker.

[0255] Further examples of multimers that may be used in the complexes provided herein are disclosed, for example, in US Patent Application Number 2015/0315588 A1, entitled Methods of delivering multiple targeting oligonucleotides to a cell using cleavable linkers, which was published on Nov. 5, 2015; US Patent Application Number 2015/0247141 A1, entitled Multimeric Oligonucleotide Compounds, which was published on Sep. 3, 2015, US Patent Application Number US 2011/0158937 A1, entitled Immunostimulatory Oligonucleotide Multimers, which was published on Jun. 30, 2011; and U.S. Pat. No. 5,693,773, entitled Triplex-Forming Antisense Oligonucleotides Having Abasic Linkers Targeting Nucleic Acids Comprising Mixed Sequences Of Purines And Pyrimidines, which issued on Dec. 2, 1997, the contents of each of which are incorporated herein by reference in their entireties.

ii. Small Molecules:

[0256] Any suitable small molecule may be used as a molecular payload, as described herein. Non-limiting examples are provided below for selected genes of Table 1.

DMPK/DM1

[0257] In some embodiments, e.g., for the treatment of DM, the small molecule is as described in US Patent Application Publication 2016052914A1, published on Feb. 25, 2016, entitled "Compounds And Methods For Myotonic Dystrophy Therapy". Further examples of small molecule payloads are provided in Lopez-Morato M, et al., Small Molecules Which Improve Pathogenesis of Myotonic Dystrophy Type 1, (Review) Front. Neurol., 18 May 2018. For example, in some embodiments, the small molecule is an MBNL1 upregulator such as phenylbuthazone, ketoprofen, ISOX, or vorinostat. In some embodiments, the small molecule is an H-Ras pathway inhibitor such as manumycin A. In some embodiments, the small molecule is a protein kinase modulator such as Ro-318220, C16, C51, Metformin, AICAR, lithium chloride, TDZD-8 or Bio. In some embodiments, the small molecule is a plant alkaloid such as harmine. In some embodiments, the small molecule is a transcription inhibitor such as pentamidine, propamidine, heptamidiine or actinomycin D. In some embodiments, the small molecule is an inhibitor of Glycogen synthase kinase 3 beta (GSK3B), for example, as disclosed in Jones K, et al., GSK3β mediates muscle pathology in myotonic dystrophy. J Clin Invest. 2012 December; 122(12):4461-72; and Wei C, et al., GSK30 is a new therapeutic target for myotonic dystrophy type 1. Rare Dis. 2013; 1: e26555; and Palomo V, et al., Subtly Modulating Glycogen Synthase Kinase 3 β: Allosteric Inhibitor Development and Their Potential for the Treatment of Chronic Diseases. J Med Chem. 2017 Jun. 22; 60(12):4983-5001, the contents of each of which are incorporated herein by reference in their entireties. In some embodiments, the small molecule is a substituted pyrido[2,3-d]pyrimidines and pentamidine-like compound, as disclosed in Gonzalez A L, et al., In silico discovery of substituted pyrido[2,3-d]pyrimidines and pentamidine-like compounds with biological activity in myotonic dystrophy models. PLoS One. 2017 Jun. 5; 12(6):e0178931, the contents of which are incorporated herein by reference in its entirety. In some embodiments, the small molecule is an MBNL1 modulator, for example, as disclosed in: Zhange F, et al., A flow cytometry-based screen identifies MBNL1 modulators that rescue splicing defects in myotonic dystrophy type I. Hum Mol Genet. 2017 Aug. 15; 26(16):3056-3068, the contents of which are incorporated herein by reference in its entirety.

DUX4/FSHD

[0258] In some embodiments, e.g., for the treatment of FSHD, the small molecule payload is as described in US Patent Application Publication 20170340606, published on Nov. 30, 2017, entitled "METHODS OF TREATING MUSCULAR DYSTROPHY" or as described in US Patent Application Publication 20180050043, published on Feb. 22, 2018, entitled "INHIBITION OF DUX4 EXPRESSION USING BROMODOMAIN AND EXTRA-TERMINAL DOMAIN PROTEIN INHIBITORS (BETi). Further examples of small molecule payloads are provided in Bosnakovski, D., et al., High-throughput screening identifies inhibitors of DUX4-induced myoblast toxicity, Skelet Muscle, February 2014, and Choi. S., et al., "Transcriptional Inhibitors Identified in a 160,000-Compound Small-Molecule DUX4 Viability Screen," Journal of Biomolecular Screening, 2016. For example, in some embodiments, the small molecule is a transcriptional inhibitor, such as SHC351, SHC540, SHC572. In some embodiments, the small molecule is STR00316 increases production or activity of another protein, such as integrin. In some embodiments, the small molecule is a bromodomain inhibitor (BETi), such as JQ1, PF1-1, I-BET-762, I-BET-151, RVX-208, or CPI-0610.

DNM/CNM

[0259] In some embodiments, e.g., for the treatment of CNM, the small molecule, for the treatment of CNM, is as described in US Patent Application Publication Number 20160264976, published on Sep. 15, 2016, entitled "DYNAMIN 2 INHIBITOR FOR TREATMENT OF CENTRONUCLEAR MYOPATHIES". For example, in some embodiments, the small molecule is selected from a group consisting of 3-Hydroxynaphthalene-2-carboxylic acid (3,4-dihydroxybenzylidene) hydrazide, 3-Hydroxy-N'-[(2,4,5-trihydroxyphenyl)methylidene]naphthalene-2-carbohydrazide. In some embodiments, the small molecule is as described in US Patent Application Publication Number 20180000762, published Jan. 4, 2018, entitled "COMPOSITION AND METHOD FOR MUSCLE REPAIR AND REGENERATION". In some embodiments, the small molecule is a retinoic receptor agonist, such as 4-[(E)-2-[5,6,7,8-Tetrahydro-5,5,8,8-tetramethyl-3-(1H-pyrazol-1-ylmethyl-)-2-naphthalenyl]-ethenyl]-benzoic acid. In some embodiments, the small molecule is as described in US Patent Application Publication Number 20170119748, published May 4, 2017, entitled "METHODS, COMPOUNDS, AND COMPOSITIONS FOR THE TREATMENT OF MUSCULOSKELETAL DISEASES." The contents of each of these publications listed above are incorporated herein in their entirety.

Pompe Disease

[0260] In some embodiments, e.g., for the treatment of Pompe disease, the small molecule is a 1-deoxynojirimycin (DNJ) derivative, such as N-butyl-DNJ, N-methyl-DNJ, or N-cyclopropylmethyl-DNJ as described in US Patent Application Publication Number 20160051528, published on Feb. 25, 2016, entitled "METHOD FOR TREATMENT OF POMPE DISEASE USING 1-DEOXYNOJIRIMYCIN DERIVATIVES". In some embodiments, the small molecule DNJ derivative is used as a molecular chaperone to increase the activity of a GAA. In some embodiments, the non-inhibitory acid alpha glucosidase chaperone ML247 small molecule is utilized as in Marugan, et al., "Discovery, SAR, and Biological Evaluation of a Non-Inhibitory Chaperone for Acid Alpha Glucosidase," published in Probe Reports from NIH Molecular Libraries in December 2011. For example, the small molecule chaperone ML247 is utilized to increase the activity of a PD-associated GAA allele or a wild-type GAA allele. The contents of each of these publications listed above are incorporated herein in their entirety.

FXN/Friedreich's Ataxia

[0261] In some embodiments, e.g., for the treatment of Friedreich's Ataxia, the small molecule is as described in Herman D. et al. "Histone deacetylase inhibitors reverse gene silencing in Friedreich's ataxia." Nat Chem Biol. 2006; 2:551-558. In some embodiments, the small molecule is as described in Rai, M. et al. "HDAC inhibitors correct frataxin deficiency in a Friedreich ataxia mouse model." PLoS One. 2008 Apr. 9; 3(4):e1958. Further examples of small molecule payloads are provided in Richardson, T. E. et al, "Therapeutic strategies in Friedreich's Ataxia", Brain Res. 2013 Jun. 13; 1514: 91-97; Zeier Z et al. "Bromodomain inhibitors regulate the C90RF72 locus in ALS" Exp Neurol. 2015 September; 271:241-50.; and Gottesfeld J. M. "Small molecules affecting transcription in Friedreich ataxia." Pharmacol Ther. 2007 November; 116(2):236-48. For example, in some embodiments, the small molecule is an inhibitor of a histone deacetylase, e.g., BML-210 and compound 106. In some embodiments, the small molecule is 170-Estradiol or methylene blue. In some embodiments, the small molecule targets, e.g., binds to, a disease-associated-repeat and/or R-loop. In some embodiments, the small molecule is as described in WO 2004/003565, published Jan. 8, 2004, "A screening method and compounds for treating friedreich ataxia". In some embodiments, the small molecule is a Glutathione peroxidase mimetic.

DMD/Dystrophinopathies

[0262] In some embodiments, the small molecule enhances exon skipping of an mRNA expression from a mutant DMD allele. In some embodiments, the small molecule is as described in US Patent Application Publication US20140080896A1, published Mar. 20, 2014, entitled "IDENTIFICATION OF SMALL MOLECULES THAT FACILITATE THERAPEUTIC EXON SKIPPING". Further examples of small molecule payloads are provided in U.S. Pat. No. 9,982,260, issued May 29, 2018, entitled "Identification of structurally similar small molecules that enhance therapeutic exon skipping". For example, in some embodiments, the small molecule is an enhancer of exon skipping such as perphenazine, flupentixol, zuclopenthixol or corynanthine. In some embodiments, a small molecule enhancer of exon skipping inhibits the ryanodine receptor or calmodulin. In some embodiments, the small molecule is an H-Ras pathway inhibitor such as manumycin A. In some embodiments, the small molecule is a suppressor of stop codons and desensitizes ribosomes to premature stop codons. In some embodiments, the small molecule is ataluren, as described in McElroy S. P. et al. "A Lack of Premature Termination Codon Read Through Efficacy of PTC124 (Ataluren) in a Diverse Array of Reporter Assays." PLOS Biology, published Jun. 25, 2013. In some embodiments, the small molecule is a corticosteroid, e.g., as described in Manzur, A. Y. et al. "Glucocorticoid corticosteroids for Duchenne muscular dystrophy". Cochrane Database Syst Rev. 2004;(2):CD003725. In some embodiments, the small molecule upregulates the expression and/or activity of genes that can replace the function of dystrophin, such as utrophin. In some embodiments, a utrophin modulator is as described in International Publication No. WO2007091106, published Aug. 16, 2007, entitled "TREATMENT OF DUCHENNE MUSCULAR DYSTROPHY" and/or International Publication No. WO/2017/168151, published Oct. 5, 2017, entitled "COMPOSITION FOR THE TREATMENT OF DUCHENNE MUSCULAR DYSTROPHY".

MYH7/Hypertrophic Cardiomyopathy

[0263] In some embodiments, the small molecule is a hypomethylating agent, such as 5-Azacytidine or 5-Aza-2'-Deoxycytidine, which modulates the expression of the MYH7 gene, such as in US Patent Application Publication 20160106771, published on Apr. 21, 2016, entitled Therapies for Cardiomyopathy; in some embodiments, the small molecule is a JAK-STAT inhibitor such as nifuroxazide, ketoprofen, sulfasalazine, 5,15-diphenylporphyrin, or AG490, such as in US Patent Application Publication 20180185478, published on Jul. 5, 2018, entitled Treatment for Myopathy; in some embodiments the small molecule is para-Nitroblebbistatin, which reduces the force of myosin contraction while not changing the dissociation of ADP, as in Tang, W., et al. "Modulating Beta-Cardiac Myosin Function at the Molecular and Tissue Levels," Front. Physiol. 2016 (7): 659, the contents of any of which are incorporated herein by reference in their entirety.

iii. Peptides/Proteins

[0264] Any suitable peptide or protein may be used as a molecular payload, as described herein. In some embodiments, a protein is an enzyme (e.g., an acid alpha-glucosidase, e.g., as encoded by the GAA gene). These peptides or proteins may be produced, synthesized, and/or derivatized using several methodologies, e.g. phage displayed peptide libraries, one-bead one-compound peptide libraries, or positional scanning synthetic peptide combinatorial libraries. Exemplary methodologies have been characterized in the art and are incorporated by reference (Gray, B. P. and Brown, K. C. "Combinatorial Peptide Libraries: Mining for Cell-Binding Peptides" Chem Rev. 2014, 114:2, 1020-1081.; Samoylova, T. I. and Smith, B. F. "Elucidation of muscle-binding peptides by phage display screening." Muscle Nerve, 1999, 22:4. 460-6.).

[0265] Non-limiting examples are provided below for selected genes of Table 1.

DMPK/DM1

[0266] A peptide or protein payload, e.g., for the treatment of DM1, may correspond to a sequence of a protein that preferentially binds to a nucleic acid, e.g. a disease-associated repeat, or a protein, e.g. MBNL1, found in muscle cells. In some embodiments, the peptide is as described in US Patent Application 2018/0021449, published on Jan. 25, 2018, "Antisense conjugates for decreasing expression of DMPK". In some embodiments, the peptide is as described in Garcia-Lopez et al., "In vivo discovery of a peptide that prevents CUG-RNA hairpin formation and reverses RNA toxicity in myotonic dystrophy models", PNAS Jul. 19, 2011. 108 (29) 11866-11871. In some embodiments, the peptide or protein may target, e.g., bind to, a disease-associated repeat, e.g. a RNA CUG repeat expansion.

[0267] In some embodiments, e.g., for the treatment of DM1, the peptide or protein comprises a fragment of an MBNL protein, e.g., MBNL1. In some embodiments, the peptide or protein comprises at least one zinc finger. In some embodiments, the peptide or protein may comprise about 2-25 amino acids, about 2-20 amino acids, about 2-15 amino acids, about 2-10 amino acids, or about 2-5 amino acids. The peptide or protein may comprise naturally-occurring amino acids, e.g. cysteine, alanine, or non-naturally-occurring or modified amino acids. Non-naturally occurring amino acids include j-amino acids, homo-amino acids, proline derivatives, 3-substituted alanine derivatives, linear core amino acids, N-methyl amino acids, and others known in the art. In some embodiments, the peptide may be linear; in other embodiments, the peptide may be cyclic, e.g. bicyclic.

DUX4/FSHD

[0268] In some embodiments, e.g., for the treatment of FSHD, the peptide or protein may bind a DME1 or DME2 enhancer to inhibit DUX4 expression, e.g., by blocking binding of an activator.

DNM2/CNM

[0269] In some embodiments, e.g., for the treatment of CNM, the peptide is a dynamin inhibitor peptide with amino acid sequence QVPSRPNRAP, as described in US Patent Application Publication Number 20160264976, published on Sep. 15, 2016, entitled "DYNAMIN 2 INHIBITOR FOR TREATMENT OF CENTRONUCLEAR MYOPATHIES".

Pompe Disease

[0270] In some embodiments, e.g., for the treatment of Pompe disease, the molecular payload is a protein or enzyme such as an acid alpha-glucosidase or wild-type GAA protein or an active fragment thereof as in US Patent Application Publication Number 20160346363, published on Dec. 1, 2016, entitled "METHODS AND ORAL FORMULATIONS FOR ENZYME REPLACEMENT THERAPY OF HUMAN LYSOSOMAL AND METABOLIC DISEASES," US Patent Application Publication Number 20160279254, published Sep. 29, 2016, entitled "METHODS AND MATERIALS FOR TREATMENT OF POMPE'S DISEASE", or US Patent Application Publication Number 20130243746, published on Sep. 19, 2013, entitled "METHODS AND MATERIALS FOR TREATMENT OF POMPE'S DISEASE". In some embodiments, the acid alpha-glucosidase or wild-type GAA protein increases the GAA activity of a subject. In some embodiments, the acid alpha-glucosidase or wild-type GAA protein is encoded by the GAA gene.

ACVR1/FOP

[0271] In some embodiments, e.g., for the treatment of FOP, the peptide or protein is a BMP inhibitor such as regulatory SMAD 6 and 7 or fragment thereof. Additional examples of peptides or proteins are included in Cappato, S. et al. "The Horizon of a Therapy for Rare Genetic Diseases: A "Druggable" Future for Fibrodysplasia Ossificans Progressiva" Int. J. Mol. Sci. 2018, 19(4), 989. The contents of each of the foregoing are incorporated herein by reference in their entireties.

FXN/Freidrich Ataxia

[0272] In some embodiments, e.g., for the treatment of Friedreich's Ataxia, the peptide is as described in U.S. Pat. No. 8,815,230, filed Aug. 30, 2010, "Methods for treating Friedreich's ataxia with interferon gamma". In some embodiments, the peptide is as described in Britti, E. et al. "Frataxin-deficient neurons and mice models of Friedreich ataxia are improved by TAT-MTScs-FXN treatment." J Cell Mol Med. 2018 Feb. 22(2):834-848. In some embodiments, the peptide is as described in Zhao, H. et al., "Peptide SS-31 upregulates frataxin expression and improves the quality of mitochondria: implications in the treatment of Friedreich ataxia", Sci Rep. 2017 Aug. 29; 7(1):9840. In some embodiments, the peptide is as described in Vyas, P. M. et al. "A TAT-frataxin fusion protein increases lifespan and cardiac function in a conditional Friedreich's ataxia mouse model", Hum Mol Genet. 2012 Mar. 15; 21(6):1230-47. In some embodiments, the peptide or protein may target, e.g., bind to, a disease-associated repeat, e.g. a GAA repeat expansion.

DMD/Dystrophinopathies

[0273] In some embodiments, e.g., for the treatment of dystrophinopathies, such as Duchenne muscular dystrophy, a peptide may facilitate exon skipping in an mRNA expressed from a mutant DMD allele. In some embodiments, a peptide may promote the expression of functional dystrophin and/or the expression of a protein capable of functioning in place of dystrophin. In some embodiments, payload is a protein that is a functional fragment of dystrophin, e.g. an amino acid segment of a functional dytrophin protein.

iv. Nucleic Acid Constructs

[0274] Any suitable gene expression construct may be used as a molecular payload, as described herein. In some embodiments, a gene expression construct may be a vector or a cDNA fragment. In some embodiments, a gene expression construct may be messenger RNA (mRNA). In some embodiments, a mRNA used herein may be a modified mRNA, e.g., as described in U.S. Pat. No. 8,710,200, issued on Apr. 24, 2014, entitled "Engineered nucleic acids encoding a modified erythropoietin and their expression". In some embodiments, a mRNA may comprise a 5' methyl cap. In some embodiments, a mRNA may comprise a polyA tail, optionally of up to 160 nucleotides in length. A gene expression construct may encode a sequence of a protein that is deficient in a muscle disease. In some embodiments, the gene expression construct may be expressed, e.g., overexpressed, within the nucleus of a muscle cell. In some embodiments, the gene expression construct encodes a gene that is deficient in a muscle disease. In some embodiments, the gene expression constructs encodes a protein that comprises at least one zinc finger. In some embodiments, the gene expression construct encodes a protein that binds to a gene in Table 1. In some embodiments, the gene expression construct encodes a protein that leads to a reduction in the expression of a protein (e.g., mutant protein) encoded by a gene in Table 1. In some embodiments, the gene expression construct encodes a gene editing enzyme. Additional examples of nucleic acid constructs that may be used as molecular payloads are provided in International Patent Application Publication WO2017152149A1, published on Sep. 19, 2017, entitled, "CLOSED-ENDED LINEAR DUPLEX DNA FOR NON-VIRAL GENE TRANSFER"; U.S. Pat. No. 8,853,377B2, issued on Oct. 7, 2014, entitled, "MRNA FOR USE IN TREATMENT OF HUMAN GENETIC DISEASES"; and US Patent U.S. Pat. No. 8,822, 663B2, issued on Sep. 2, 2014, ENGINEERED NUCLEIC ACIDS AND METHODS OF USE THEREOF," the contents of each of which are incorporated herein by reference in their entireties.

[0275] Further non-limiting examples are provided below for selected genes/disease of Table 1.

DMPK/DM1

[0276] In some embodiments, e.g., for the treatment of DM, the gene expression construct encodes a MBNL protein, e.g., MBNL1.

DUX4/FSHD

[0277] In some embodiments, e.g., for the treatment of FSHD, the gene expression construct encodes a oligonucleotide (e.g., an shRNA targeting DUX4) or a protein that downregulates the expression of DUX4 (e.g., a peptide or protein that binds to DME1 or DME2 enhancer to inhibit DUX4 expression, e.g., by blocking binding of an activator).

DNM2/CNM

[0278] In some embodiments, e.g., for the treatment of CNM1, a gene expression construct may encode a sequence of a protein that downregulates the expression of a mutant DNM2 protein, or which expresses wild-type DNM2. In some embodiments, a gene expression construct encodes an oligonucleotide (e.g., an shRNA) that inhibits expression of DNM2. However, in some embodiments, an expression construct encodes Spliceosome-Mediated RNA Trans-splicing components that may be used to reprogram mutated DNM2-mRNA, as disclosed in Trochet D., et al., Reprogramming the Dynamin 2 mRNA by Spliceosome-mediated RNA Trans-splicing Mol Ther Nucleic Acids. 2016 September; 5(9): e362, the contents of which are incorporated herein by reference.

Pompe Disease

[0279] In some embodiments, e.g., for the treatment of Pompe disease, the gene expression construct encodes a wild-type GAA protein. A gene expression construct may encode a sequence of a protein that leads to decreased expression of ACVR1 gene or decreased activity of GYS1 protein. In some embodiments, e.g., for the treatment of Pompe disease, the gene expression construct encodes and oligonucleotide (e.g., shRNA) that inhibits expression of GYS1.

ACVR1/FOP

[0280] A gene expression construct may encode a sequence of a protein that leads to decreased expression of ACVR1 gene or decreased activity of ACVR1 protein. In some embodiments, the gene expression construct encodes a protein that leads to a reduction in the expression of a epigenetic regulators that negatively regulate the expression of ACVR1, e.g. histone deactylases. In some embodiments, the gene expression construct encodes an oligonucleotide (e.g., shRNA) that inhibits expression of ACVR1.

FXN/Friedreich's Ataxia

[0281] A gene expression construct may encode a sequence of a protein that leads to increased expression of frataxin. In some embodiments, the gene expression construct may be expressed, e.g., overexpressed, within the nucleus of a muscle cell. In some embodiments, the gene expression construct encodes frataxin. In some embodiments, the gene expression constructs encodes a protein that inhibit the function of epigenetic regulators that negatively regulate the expression of FXN, e.g. histone deactylases. In some embodiments, the gene expression construct encodes a protein that binds to a disease-associated-repeat expansion of a GAA trinucleotide. In some embodiments, the gene expression construct encodes a protein that leads to a reduction in the expression of a epigenetic regulators that negatively regulate the expression of FXN, e.g. histone deactylases. In some embodiments, the gene expression construct encodes a gene editing enzyme. In some embodiments, the gene expression construct encodes erythropoietin (see, e.g. Miller, J. L. et al, "Erythropoietin and small molecule agonists of the tissue-protective erythropoietin receptor increase FXN expression in neuronal cells in vitro and in FXN-deficient KIKO mice in vivo", Neuropharmacology. 2017 Sep. 1; 123:34-45.). In some embodiments, the gene expression construct encodes interferon gamma (see, e.g. U.S. Pat. No. 8,815,230, filed Aug. 30, 2010, "Methods for treating Friedreich's ataxia with interferon gamma").

DMD/Dystrophinopathies

[0282] A gene expression construct may encode a sequence of a dystrophin protein, a dystrophin fragment, a mini-dystrophin, a utrophin protein, or any protein that shares a common function with dystrophin. In some embodiments, the gene expression construct may be expressed, e.g., overexpressed, within the nucleus of a muscle cell. In some embodiments, the gene expression constructs encodes a protein that comprises at least one zinc finger. In some embodiments, the gene expression construct encodes a protein that promotes the expression of dystrophin or a protein that shares function with dystrophin, e.g., utrophin. In some embodiments, the gene expression construct encodes a gene editing enzyme. In some embodiments, the gene expression construct is as described in U.S. Patent Application Publication US20170368198A1, published Dec. 28, 2017, entitled "Optimized mini-dystrophin genes and expression cassettes and their use"; Duan D. "Myodys, a full-length dystrophin plasmid vector for Duchenne and Becker muscular dystrophy gene therapy." Curr Opin Mol Ther 2008; 10:86-94; and expression cassettes disclosed in Tang, Y. et al., "AAV-directed muscular dystrophy gene therapy" Expert Opin Biol Ther. 2010 Mar. 10(3):395-408; the contents of each of which are incorporated herein by reference in their entireties.

C. Linkers

[0283] Complexes described herein generally comprise a linker that connects a muscle-targeting agent to a molecular payload. A linker comprises at least one covalent bond. In some embodiments, a linker may be a single bond, e.g., a disulfide bond or disulfide bridge, that connects a muscle-targeting agent to a molecular payload. However, in some embodiments, a linker may connect a muscle-targeting agent to a molecular payload through multiple covalent bonds. In some embodiments, a linker may be a cleavable linker. However, in some embodiments, a linker may be a non-cleavable linker. A linker is generally stable in vitro and in vivo, and may be stable in certain cellular environments. Additionally, generally a linker does not negatively impact the functional properties of either the muscle-targeting agent or the molecular payload. Examples and methods of synthesis of linkers are known in the art (see, e.g. Kline, T. et al. "Methods to Make Homogenous Antibody Drug Conjugates." Pharmaceutical Research, 2015, 32:11, 3480-3493.; Jain, N. et al. "Current ADC Linker Chemistry" Pharm Res. 2015, 32:11, 3526-3540.; McCombs, J. R. and Owen, S. C. "Antibody Drug Conjugates: Design and Selection of Linker, Payload and Conjugation Chemistry" AAPS J. 2015, 17:2, 339-351.).

[0284] A precursor to a linker typically will contain two different reactive species that allow for attachment to both the muscle-targeting agent and a molecular payload. In some embodiments, the two different reactive species may be a nucleophile and/or an electrophile. In some embodiments, a linker is connected to a muscle-targeting agent via conjugation to a lysine residue or a cysteine residue of the muscle-targeting agent. In some embodiments, a linker is connected to a cysteine residue of a muscle-targeting agent via a maleimide-containing linker, wherein optionally the maleimide-containing linker comprises a maleimidocaproyl or maleimidomethyl cyclohexane-1-carboxylate group. In some embodiments, a linker is connected to a cysteine residue of a muscle-targeting agent or thiol functionalized molecular payload via a 3-arylpropionitrile functional group. In some embodiments, a linker is connected to a muscle-targeting agent and/or a molecular payload via an amide bond, a hydrazide, a triazole, a thioether, or a disulfide bond.

i. Cleavable Linkers

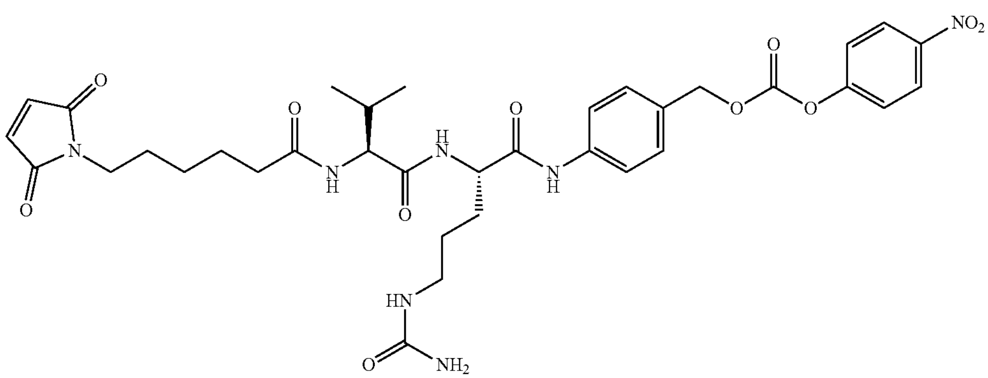
[0285] A cleavable linker may be a protease-sensitive linker, a pH-sensitive linker, or a glutathione-sensitive linker. These linkers are generally cleavable only intracellularly and are preferably stable in extracellular environments, e.g. extracellular to a muscle cell.

[0286] Protease-sensitive linkers are cleavable by protease enzymatic activity. These linkers typically comprise peptide sequences and may be 2-10 amino acids, about 2-5 amino acids, about 5-10 amino acids, about 10 amino acids, about 5 amino acids, about 3 amino acids, or about 2 amino acids in length. In some embodiments, a peptide sequence may comprise naturally-occurring amino acids, e.g. cysteine, alanine, or non-naturally-occurring or modified amino acids. Non-naturally occurring amino acids include j-amino acids, homo-amino acids, proline derivatives, 3-substituted alanine derivatives, linear core amino acids, N-methyl amino acids, and others known in the art. In some embodiments, a protease-sensitive linker comprises a valine-citrulline or alanine-citrulline dipeptide sequence. In some embodiments, a protease-sensitive linker can be cleaved by a lysosomal protease, e.g. cathepsin B, and/or an endosomal protease.

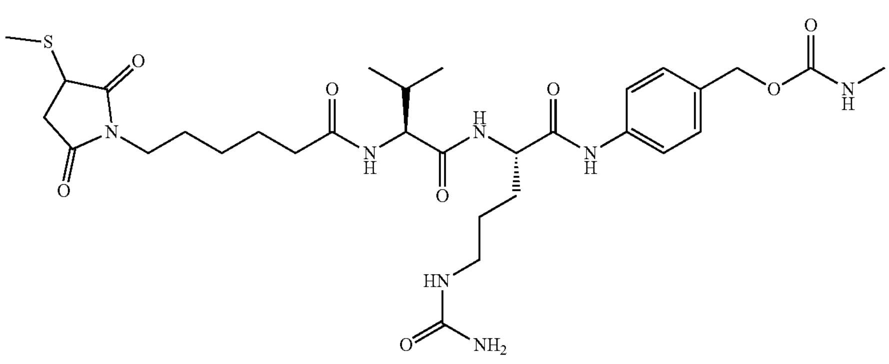
[0287] A pH-sensitive linker is a covalent linkage that readily degrades in high or low pH environments. In some embodiments, a pH-sensitive linker may be cleaved at a pH in a range of 4 to 6. In some embodiments, a pH-sensitive linker comprises a hydrazone or cyclic acetal. In some embodiments, a pH-sensitive linker is cleaved within an endosome or a lysosome.

[0288] In some embodiments, a glutathione-sensitive linker comprises a disulfide moiety. In some embodiments, a glutathione-sensitive linker is cleaved by an disulfide exchange reaction with a glutathione species inside a cell. In some embodiments, the disulfide moiety further comprises at least one amino acid, e.g. a cysteine residue.

[0289] In some embodiments, the linker is a Val-cit linker (e.g., as described in U.S. Pat. No. 6,214,345, incorporated herein by reference). In some embodiments, before conjugation, the val-cit linker has a structure of:



[0290] In some embodiments, after conjugation, the val-cit linker has a structure of:



ii. Non-Cleavable Linkers

[0291] In some embodiments, non-cleavable linkers may be used. Generally, a non-cleavable linker cannot be readily degraded in a cellular or physiological environment. In some embodiments, a non-cleavable linker comprises an optionally substituted alkyl group, wherein the substitutions may include halogens, hydroxyl groups, oxygen species, and other common substitutions. In some embodiments, a linker may comprise an optionally substituted alkyl, an optionally substituted alkylene, an optionally substituted arylene, a heteroarylene, a peptide sequence comprising at least one non-natural amino acid, a truncated glycan, a sugar or sugars that cannot be enzymatically degraded, an azide, an alkyne-azide, a peptide sequence comprising a LPXT sequence, a thioether, a biotin, a biphenyl, repeating units of polyethylene glycol or equivalent compounds, acid esters, acid amides, sulfamides, and/or an alkoxy-amine linker. In some embodiments, sortase-mediated ligation will be utilized to covalently link a muscle-targeting agent comprising a LPXT sequence to a molecular payload comprising a $(G)_n$ sequence (see, e.g. Proft T. Sortase-mediated protein ligation: an emerging biotechnology tool for protein modification and immobilization. Biotechnol Lett. 2010, 32(1):1-10.). In some embodiments, a linker comprises a LPXTG sequence (SEQ ID NO: 16), where X is any amino acid.

[0292] In some embodiments, a linker may comprise a substituted alkylene, an optionally substituted alkenylene, an optionally substituted alkynylene, an optionally substituted cycloalkylene, an optionally substituted cycloalkenylene, an optionally substituted arylene, an optionally substituted heteroarylene further comprising at least one heteroatom selected from N, O, and S; an optionally substituted heterocyclylene further comprising at least one heteroatom selected from N, O, and S; an imino, an optionally substituted nitrogen species, an optionally substituted oxygen species O, an optionally substituted sulfur species, or a poly(alkylene oxide), e.g. polyethylene oxide or polypropylene oxide.

iii. Linker Conjugation

[0293] In some embodiments, a linker is connected to a muscle-targeting agent and/or molecular payload via a phosphate, thioether, ether, carbon-carbon, or amide bond. In some embodiments, a linker is connected to an oligonucleotide through a phosphate or phosphorothioate group, e.g. a terminal phosphate of an oligonucleotide backbone. In some embodiments, a linker is connected to an muscle-targeting agent, e.g. an antibody, through a lysine or cysteine residue present on the muscle-targeting agent

[0294] In some embodiments, a linker is connected to a muscle-targeting agent and/or molecular payload by a cycloaddition reaction between an azide and an alkyne to form a triazole, wherein the azide and the alkyne may be located on the muscle-targeting agent, molecular payload, or the linker. In some embodiments, an alkyne may be a cyclic alkyne, e.g., a cyclooctyne. In some embodiments, an alkyne may be bicyclononyne (also known as bicyclo[6.1.0]nonyne or BCN) or substituted bicyclononyne. In some embodiments, a cyclooctane is as described in International Patent Application Publication WO2011136645, published on Nov. 3, 2011, entitled, "Fused Cyclooctyne Compounds And Their Use In Metal-free Click Reactions". In some embodiments, an azide may be a sugar or carbohydrate molecule that comprises an azide. In some embodiments, an azide may be 6-azido-6-deoxygalactose or 6-azido-N-acetylgalactosamine. In some embodiments, a sugar or carbohydrate molecule that comprises an azide is as described in International Patent Application Publication WO2016170186, published on Oct. 27, 2016, entitled, "Process For The Modification Of A Glycoprotein Using A Glycosyltransferase That Is Or Is Derived From A P(1,4)-N-Acetylgalactosaminyltransferase". In some embodiments, a cycloaddition reaction between an azide and an alkyne to form a triazole, wherein the azide and the alkyne may be located on the muscle-targeting agent, molecular payload, or the linker is as described in International Patent Application Publication WO2014065661, published on May 1, 2014, entitled, "Modified antibody, antibody-conjugate and process for the preparation thereof"; or International Patent Application Publication WO2016170186, published on Oct. 27, 2016, entitled, "Process For The Modification Of A Glycoprotein Using A Glycosyltransferase That Is Or Is Derived From A P(1,4)-N-Acetylgalactosaminyltransferase".

[0295] In some embodiments, a linker further comprises a spacer, e.g., a polyethylene glycol spacer or an acyl/carbomoyl sulfamide spacer, e.g., a HydraSpace™ spacer. In some embodiments, a spacer is as described in Verkade, J. M. M. et al., "A Polar Sulfamide Spacer Significantly Enhances the Manufacturability, Stability, and Therapeutic Index of Antibody-Drug Conjugates", Antibodies, 2018, 7, 12.

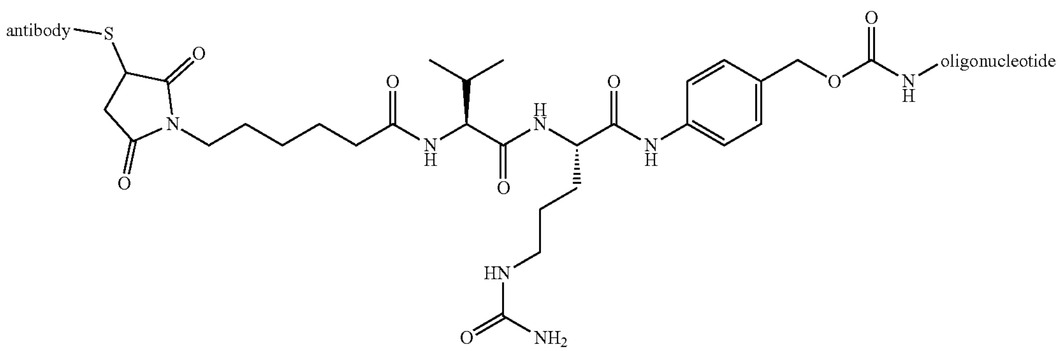
[0296] In some embodiments, a linker is connected to a muscle-targeting agent and/or molecular payload by the Diels-Alder reaction between a dienophile and a diene/hetero-diene, wherein the dienophile and the diene/hetero-diene may be located on the muscle-targeting agent, molecular payload, or the linker. In some embodiments a linker is connected to a muscle-targeting agent and/or molecular payload by other pericyclic reactions, e.g. ene reaction. In some embodiments, a linker is connected to a muscle-targeting agent and/or molecular payload by an amide, thioamide, or sulfonamide bond reaction. In some embodiments, a linker is connected to a muscle-targeting agent and/or molecular payload by a condensation reaction to form an oxime, hydrazone, or semicarbazide group existing between the linker and the muscle-targeting agent and/or molecular payload.

[0297] In some embodiments, a linker is connected to a muscle-targeting agent and/or molecular payload by a conjugate addition reactions between a nucleophile, e.g. an amine or a hydroxyl group, and an electrophile, e.g. a carboxylic acid or an aldehyde. In some embodiments, a nucleophile may exist on a linker and an electrophile may exist on a muscle-targeting agent or molecular payload prior to a reaction between a linker and a muscle-targeting agent or molecular payload. In some embodiments, an electrophile may exist on a linker and a nucleophile may exist on a muscle-targeting agent or molecular payload prior to a reaction between a linker and a muscle-targeting agent or molecular payload. In some embodiments, an electrophile may be an azide, a silicon centers, a carbonyl, a carboxylic acid, an anhydride, an isocyanate, a thioisocyanate, a succinimidyl ester, a sulfosuccinimidyl ester, a maleimide, an alkyl halide, an alkyl pseudohalide, an epoxide, an episulfide, an aziridine, an aryl, an activated phosphorus center, and/or an activated sulfur center. In some embodiments, a nucleophile may be an optionally substituted alkene, an optionally substituted alkyne, an optionally substituted aryl, an optionally substituted heterocyclyl, a hydroxyl group, an amino group, an alkylamino group, an anilido group, or a thiol group.

D. Examples of Antibody-Molecular Payload Complexes

[0298] Other aspects of the present disclosure provide complexes comprising any one the muscle targeting agent (e.g., a transferrin receptor antibodies) described herein covalently linked to any of the molecular payloads (e.g., an oligonucleotide) described herein. In some embodiments, the muscle targeting agent (e.g., a transferrin receptor antibody) is covalently linked to a molecular payload (e.g., an oligonucleotide) via a linker. Any of the linkers described herein may be used. In some embodiments, the linker is linked to the 5' end, the 3' end, or internally of the oligonucleotide. In some embodiments, the linker is linked to the antibody via a thiol-reactive linkage (e.g., via a cysteine in the antibody).

[0299] An exemplary structure of a complex comprising a transferrin receptor antibody covalently linked to an oligonucleotide via a Val-cit linker is provided below:



wherein the linker is linked to the 5' end, the 3' end, or internally of the oligonucleotide, and wherein the linker is linked to the antibody via a thiol-reactive linkage (e.g., via a cysteine in the antibody).

[0300] It should be appreciated that antibodies can be linked to oligonucleotides with different stochiometries, a property that may be referred to as a drug to antibody ratios (DAR) with the "drug" being the oligonucleotide. In some embodiments, one oligonucleotide is linked to an antibody (DAR=1). In some embodiments, two oligonucleotides are linked to an antibody (DAR=2). In some embodiments, three oligonucleotides are linked to an antibody (DAR=3). In some embodiments, four oligonucleotides are linked to an antibody (DAR=4). In some embodiments, a mixture of different complexes, each having a different DAR, is provided. In some embodiments, an average DAR of complexes in such a mixture may be in a range of 1 to 3, 1 to 4, 1 to 5 or more. DAR may be increased by conjugating oligonucleotides to different sites on an antibody and/or by conjugating multimers to one or more sites on antibody. For example, a DAR of 2 may be achieved by conjugating a single oligonucleotide to two different sites on an antibody or by conjugating a dimer oligonucleotide to a single site of an antibody.

[0301] In some embodiments, the complex described herein comprises a transferrin receptor antibody (e.g., an antibody or any variant thereof as described herein) covalently linked to an oligonucleotide. In some embodiments, the complex described herein comprises a transferrin receptor antibody (e.g., an antibody or any variant thereof as described herein) covalently linked to an oligonucleotide via a linker (e.g., a Val-cit linker). In some embodiments, the linker (e.g., a Val-cit linker) is linked to the 5' end, the 3' end, or internally of the oligonucleotide. In some embodiments, the linker (e.g., a Val-cit linker) is linked to the antibody (e.g., an antibody or any variant thereof as described herein) via a thiol-reactive linkage (e.g., via a cysteine in the antibody).

[0302] In some embodiments, the complex described herein comprises a transferrin receptor antibody covalently linked to an oligonucleotide, wherein the transferrin receptor antibody comprises a CDR-H1, a CDR-H2, and a CDR-H3 that are the same as the CDR-H1, CDR-H2, and CDR-H3 shown in Table 1.1; and a CDR-L1, a CDR-L2, and a CDR-L3 that are the same as the CDR-L1, CDR-L2, and CDR-L3 shown in Table 1.1.

[0303] In some embodiments, the complex described herein comprises a transferrin receptor antibody covalently linked to an oligonucleotide, wherein the transferrin receptor antibody comprises a VH having the amino acid sequence of SEQ ID NO: 33 and a VL having the amino acid sequence of SEQ ID NO: 34. In some embodiments, the complex described herein comprises a transferrin receptor antibody covalently linked to an oligonucleotide, wherein the transferrin receptor antibody comprises a VH having the amino acid sequence of SEQ ID NO: 35 and a VL having the amino acid sequence of SEQ ID NO: 36.

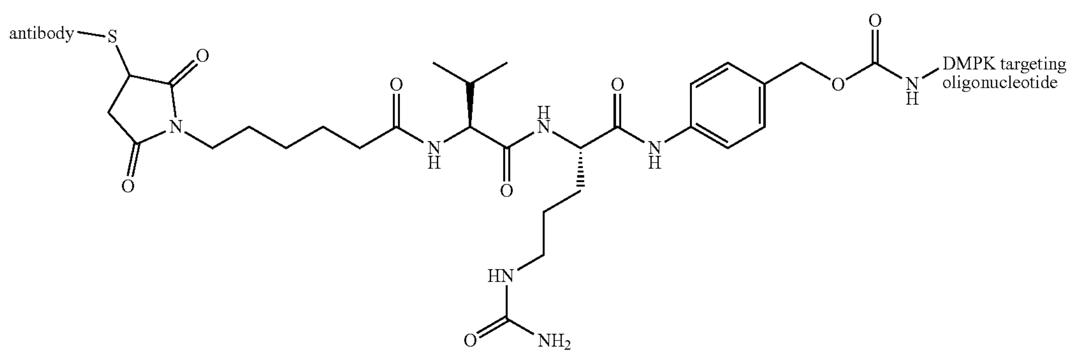
[0304] In some embodiments, the complex described herein comprises a transferrin receptor antibody covalently linked to an oligonucleotide, wherein the transferrin receptor antibody comprises a heavy chain having the amino acid sequence of SEQ ID NO: 39 and a light chain having the amino acid sequence of SEQ ID NO: 40. In some embodiments, the complex described herein comprises a transferrin receptor antibody covalently linked to an oligonucleotide, wherein the transferrin receptor antibody comprises a heavy chain having the amino acid sequence of SEQ ID NO: 41 and a light chain having the amino acid sequence of SEQ ID NO: 42.

[0305] In some embodiments, the complex described herein comprises a transferrin receptor antibody covalently linked to an oligonucleotide via a linker (e.g., a Val-cit linker), wherein the transferrin receptor antibody comprises a CDR-H1, a CDR-H2, and a CDR-H3 that are the same as the CDR-H1, CDR-H2, and CDR-H3 shown in Table 1.1; and a CDR-L1, a CDR-L2, and a CDR-L3 that are the same as the CDR-L1, CDR-L2, and CDR-L3 shown in Table 1.1.

[0306] In some embodiments, the complex described herein comprises a transferrin receptor antibody covalently linked to an oligonucleotide via a linker (e.g., a Val-cit linker), wherein the transferrin receptor antibody comprises a VH having the amino acid sequence of SEQ ID NO: 33 and a VL having the amino acid sequence of SEQ ID NO: 34. In some embodiments, the complex described herein comprises a transferrin receptor antibody covalently linked to an oligonucleotide via a linker (e.g., a Val-cit linker), wherein the transferrin receptor antibody comprises a VH having the amino acid sequence of SEQ ID NO: 35 and a VL having the amino acid sequence of SEQ ID NO: 36.

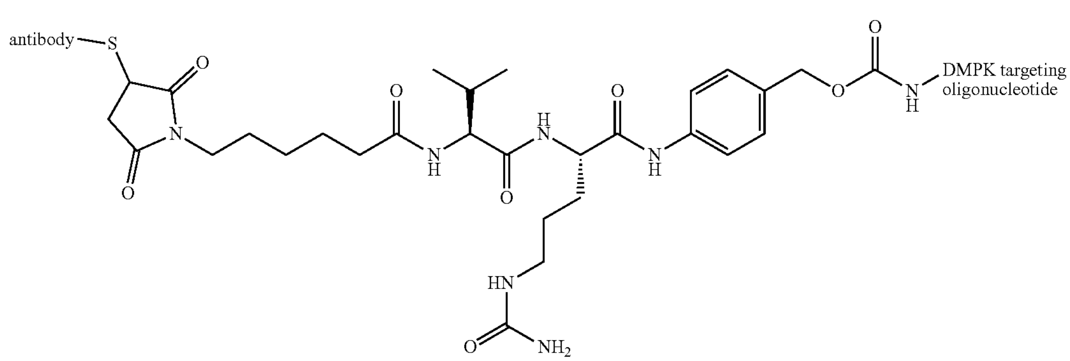
[0307] In some embodiments, the complex described herein comprises a transferrin receptor antibody covalently linked to an oligonucleotide via a linker (e.g., a Val-cit linker), wherein the transferrin receptor antibody comprises a heavy chain having the amino acid sequence of SEQ ID NO: 39 and a light chain having the amino acid sequence of SEQ ID NO: 40. In some embodiments, the complex described herein comprises a transferrin receptor antibody covalently linked to an oligonucleotide via a linker (e.g., a Val-cit linker), wherein the transferrin receptor antibody comprises a heavy chain having the amino acid sequence of SEQ ID NO: 41 and a light chain having the amino acid sequence of SEQ ID NO: 42.

[0308] In some embodiments, the complex described herein comprises a transferrin receptor antibody covalently linked to an oligonucleotide via a Val-cit linker, wherein the transferrin receptor antibody comprises a CDR-H1, a CDR-H2, and a CDR-H3 that are the same as the CDR-H1, CDR-H2, and CDR-H3 shown in Table 1.1; and a CDR-L1, a CDR-L2, and a CDR-L3 that are the same as the CDR-L1, CDR-L2, and CDR-L3 shown in Table 1.1, and wherein the complex comprises the structure of:



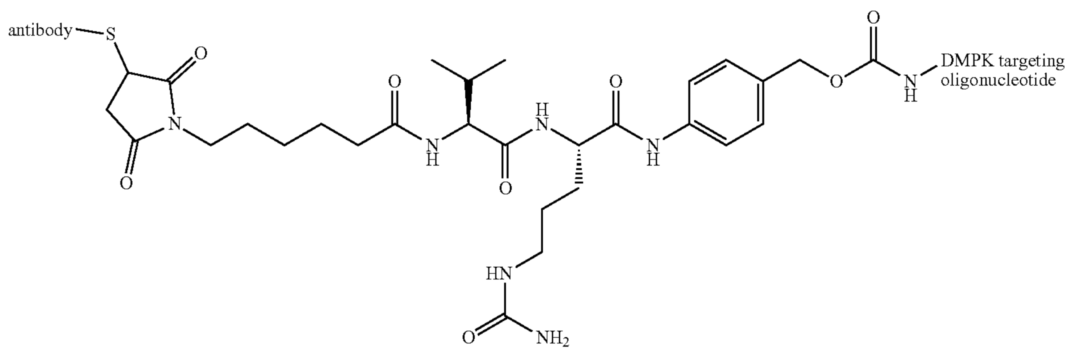
wherein the linker Val-cit linker is linked to the 5' end, the 3' end, or internally of the oligonucleotide, and wherein the Val-cit linker is linked to the antibody (e.g., an antibody or any variant thereof as described herein) via a thiol-reactive linkage (e.g., via a cysteine in the antibody).

[0309] In some embodiments, the complex described herein comprises a transferrin receptor antibody covalently linked to an oligonucleotide via a Val-cit linker, wherein the transferrin receptor antibody comprises a VH having the amino acid sequence of SEQ ID NO: 33 and a VL having the amino acid sequence of SEQ ID NO: 34, and wherein the complex comprises the structure of:



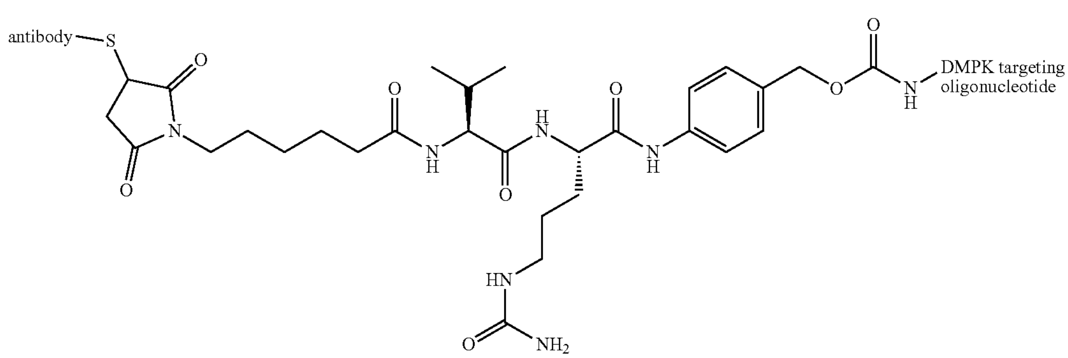
wherein the linker Val-cit linker is linked to the 5' end, the 3' end, or internally of the oligonucleotide, and wherein the Val-cit linker is linked to the antibody (e.g., an antibody or any variant thereof as described herein) via a thiol-reactive linkage (e.g., via a cysteine in the antibody).

[0310] In some embodiments, the complex described herein comprises a transferrin receptor antibody covalently linked to an oligonucleotide via a Val-cit linker, wherein the transferrin receptor antibody comprises a VH having the amino acid sequence of SEQ ID NO: 35 and a VL having the amino acid sequence of SEQ ID NO: 36, and wherein the complex comprises the structure of:



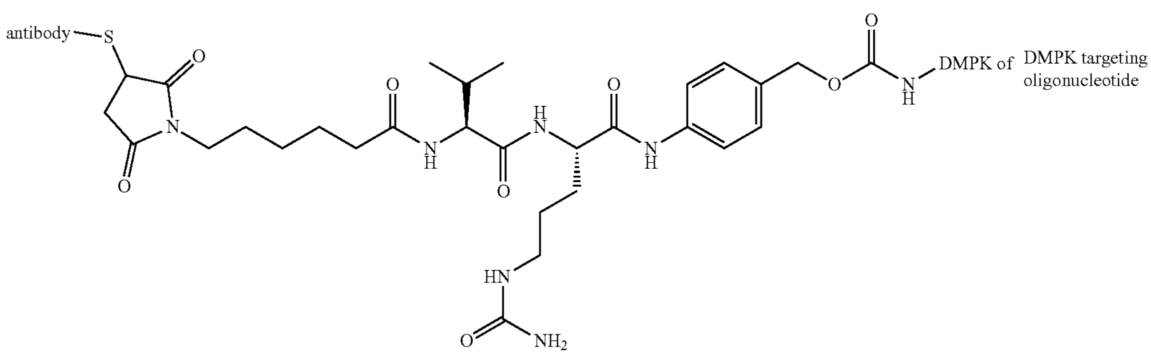
wherein the linker Val-cit linker is linked to the 5' end, the 3' end, or internally of the oligonucleotide, and wherein the Val-cit linker is linked to the antibody (e.g., an antibody or any variant thereof as described herein) via a thiol-reactive linkage (e.g., via a cysteine in the antibody).

[0311] In some embodiments, the complex described herein comprises a transferrin receptor antibody covalently linked to an oligonucleotide via a Val-cit linker, wherein the transferrin receptor antibody comprises a heavy chain having the amino acid sequence of SEQ ID NO: 39 and a light chain having the amino acid sequence of SEQ ID NO: 40, and wherein the complex comprises the structure of:



[0312] wherein the linker Val-cit linker is linked to the 5' end, the 3' end, or internally of an oligonucleotide, and wherein the Val-cit linker is linked to the antibody (e.g., an antibody or any variant thereof as described herein) via a thiol-reactive linkage (e.g., via a cysteine in the antibody).

[0313] In some embodiments, the complex described herein comprises a transferrin receptor antibody covalently linked to an oligonucleotide via a Val-cit linker, wherein the transferrin receptor antibody comprises a heavy chain having the amino acid sequence of SEQ ID NO: 41 and a light chain having the amino acid sequence of SEQ ID NO: 42, and wherein the complex comprises the structure of:



wherein the linker Val-cit linker is linked to the 5' end, the 3' end, or internally of an oligonucleotide, and wherein the Val-cit linker is linked to the antibody (e.g., an antibody or any variant thereof as described herein) via a thiol-reactive linkage (e.g., via a cysteine in the antibody).

[0314] Further examples of complexes and molecular payloads (e.g., oligonucleotides useful for targeting muscle genes) are provided in US Patent Application Publication US20210308272, published on Oct. 7, 2021, entitled, “MUSCLE TARGETING COMPLEXES AND USES THEREOF FOR TREATING MYOTONIC DYSTROPHY”; US Patent Application Publication US20210317226, published on Oct. 14, 2021, entitled, “MUSCLE TARGETING COMPLEXES AND USES THEREOF FOR TREATING POMPE DISEASE”; US Patent Application Publication US20210308274, published on Oct. 7, 2021, 2020, entitled, “MUSCLE TARGETING COMPLEXES AND USES THEREOF FOR TREATING FRIEDREICH'S ATAXIA”; US Patent Application Publication US20210261680, published on Aug. 26, 2021, entitled, “MUSCLE-TARGETING COMPLEXES AND USES THEREOF”; US Patent Application Publication US20220378934, published on Dec. 1, 2022, entitled, “MUSCLE-TARGETING COMPLEXES AND USES THEREOF IN TREATING MUSCLE ATROPHY”; US Patent Application Publication US20210308273, published on Oct. 7, 2021, entitled, “MUSCLE TARGETING COMPLEXES AND USES THEREOF FOR TREATING DYSTROPHINOPATHIES”; US Patent Application Publication US20210324101, published on Oct. 21, 2021, entitled, “MUSCLE TARGETING COMPLEXES AND USES THEREOF FOR TREATING HYPERTROPHIC CARDIOMYOPATHY”; US Patent Application Publication US20220193250, published on Jun. 23, 2022, entitled, “MUSCLE TARGETING COMPLEXES AND USES THEREOF FOR TREATING FACIOSCAPULOHUMERAL MUSCULAR DYSTROPHY”; the entire contents of each of which are incorporated herein by reference.

III. Formulations

[0315] Complexes provided herein may be formulated in any suitable manner. Generally, complexes provided herein are formulated in a manner suitable for pharmaceutical use. For example, complexes can be delivered to a subject using a formulation that minimizes degradation, facilitates delivery and/or uptake, or provides another beneficial property to the complexes in the formulation. In some embodiments, provided herein are compositions comprising complexes and pharmaceutically acceptable carriers. Such compositions can be suitably formulated such that when administered to a subject, either into the immediate environment of a target cell or systemically, a sufficient amount of the complexes enter target muscle cells. In some embodiments, complexes are formulated in buffer solutions such as phosphate-buffered saline solutions, liposomes, micellar structures, and capsids.

[0316] It should be appreciated that, in some embodiments, compositions may include separately one or more components of complexes provided herein (e.g., muscle-targeting agents, linkers, molecular payloads, or precursor molecules of any one of them).

[0317] In some embodiments, complexes are formulated in water or in an aqueous solution (e.g., water with pH adjustments). In some embodiments, complexes are formulated in basic buffered aqueous solutions (e.g., PBS). In some embodiments, formulations as disclosed herein comprise an excipient. In some embodiments, an excipient confers to a composition improved stability, improved absorption, improved solubility and/or therapeutic enhancement of the active ingredient. In some embodiments, an excipient is a buffering agent (e.g., sodium citrate, sodium phosphate, a tris base, or sodium hydroxide) or a vehicle (e.g., a buffered solution, petrolatum, dimethyl sulfoxide, or mineral oil).

[0318] In some embodiments, a complex or component thereof (e.g., oligonucleotide or antibody) is lyophilized for extending its shelf-life and then made into a solution before use (e.g., administration to a subject). Accordingly, an excipient in a composition comprising a complex, or component thereof, described herein may be a lyoprotectant (e.g., mannitol, lactose, polyethylene glycol, or polyvinyl pyrolidone), or a collapse temperature modifier (e.g., dextran, ficoll, or gelatin).

[0319] In some embodiments, a pharmaceutical composition is formulated to be compatible with its intended route of administration. Examples of routes of administration include parenteral, e.g., intravenous, intradermal, subcutaneous, administration. Typically, the route of administration is intravenous or subcutaneous. In some embodiments, the route of administration is extramuscular parenteral administration.

[0320] Pharmaceutical compositions suitable for injectable use include sterile aqueous solutions (where water soluble) or dispersions and sterile powders for the extemporaneous preparation of sterile injectable solutions or dispersions. The carrier can be a solvent or dispersion medium containing, for example, water, ethanol, polyol (for example, glycerol, propylene glycol, and liquid polyethylene glycol, and the like), and suitable mixtures thereof. In some embodiments, formulations include isotonic agents, for example, sugars, polyalcohols such as mannitol, sorbitol, and sodium chloride in the composition. Sterile injectable solutions can be prepared by incorporating the complexes in a required amount in a selected solvent with one or a combination of ingredients enumerated above, as required, followed by filtered sterilization.

[0321] In some embodiments, a composition may contain at least about 0.1% of the a complex, or component thereof, or more, although the percentage of the active ingredient(s) may be between about 1% and about 80% or more of the weight or volume of the total composition. Factors such as solubility, bioavailability, biological half-life, route of administration, product shelf life, as well as other pharmacological considerations will be contemplated by one skilled in the art of preparing such pharmaceutical formulations, and as such, a variety of dosages and treatment regimens may be desirable.

IV. Methods of Use/Treatment

[0322] Complexes comprising a muscle-targeting agent covalently to a molecular payload as described herein are effective in treating a muscle disease (e.g., a rare muscle disease). In some embodiments, complexes are effective in treating a muscle disease provided in Table 1. In some embodiments, a muscle disease is associated with a disease allele, for example, a disease allele for a particular muscle disease may comprise a genetic alteration in a corresponding gene listed in Table 1.

[0323] In some embodiments, a subject may be a human subject, a non-human primate subject, a rodent subject, or any suitable mammalian subject. In some embodiments, a subject may have a muscle disease provided in Table 1.

[0324] An aspect of the disclosure includes a methods involving administering to a subject an effective amount of a complex as described herein. In some embodiments, an effective amount of a pharmaceutical composition that comprises a complex comprising a muscle-targeting agent covalently to a molecular payload can be administered to a subject in need of treatment. In some embodiments, a pharmaceutical composition comprising a complex as described herein may be administered by a suitable route, which may include intravenous administration, e.g., as a bolus or by continuous infusion over a period of time. In some embodiments, intravenous administration may be performed by intramuscular, intraperitoneal, intracerebrospinal, subcutaneous, intra-articular, intrasynovial, or intrathecal routes. In some embodiments, a pharmaceutical composition may be in solid form, aqueous form, or a liquid form. In some embodiments, an aqueous or liquid form may be nebulized or lyophilized. In some embodiments, a nebulized or lyophilized form may be reconstituted with an aqueous or liquid solution.

[0325] Compositions for intravenous administration may contain various carriers such as vegetable oils, dimethylactamide, dimethyformamide, ethyl lactate, ethyl carbonate, isopropyl myristate, ethanol, and polyols (glycerol, propylene glycol, liquid polyethylene glycol, and the like). For intravenous injection, water soluble antibodies can be administered by the drip method, whereby a pharmaceutical formulation containing the antibody and a physiologically acceptable excipients is infused. Physiologically acceptable excipients may include, for example, 5% dextrose, 0.9% saline, Ringer's solution or other suitable excipients. Intramuscular preparations, e.g., a sterile formulation of a suitable soluble salt form of the antibody, can be dissolved and administered in a pharmaceutical excipient such as Water-for-Injection, 0.9% saline, or 5% glucose solution.

[0326] In some embodiments, a pharmaceutical composition that comprises a complex comprising a muscle-targeting agent covalently to a molecular payload is administered via site-specific or local delivery techniques. Examples of these techniques include implantable depot sources of the complex, local delivery catheters, site specific carriers, direct injection, or direct application.

[0327] In some embodiments, a pharmaceutical composition that comprises a complex comprising a muscle-targeting agent covalently to a molecular payload is administered at an effective concentration that confers therapeutic effect on a subject. Effective amounts vary, as recognized by those skilled in the art, depending on the severity of the disease, unique characteristics of the subject being treated, e.g. age, physical conditions, health, or weight, the duration of the treatment, the nature of any concurrent therapies, the route of administration and related factors. These related factors are known to those in the art and may be addressed with no more than routine experimentation. In some embodiments, an effective concentration is the maximum dose that is considered to be safe for the patient. In some embodiments, an effective concentration will be the lowest possible concentration that provides maximum efficacy.

[0328] Empirical considerations, e.g. the half-life of the complex in a subject, generally will contribute to determination of the concentration of pharmaceutical composition that is used for treatment. The frequency of administration may be empirically determined and adjusted to maximize the efficacy of the treatment.

[0329] Generally, for administration of any of the complexes described herein, an initial candidate dosage may be about 1 to 100 mg/kg, or more, depending on the factors described above, e.g. safety or efficacy. In some embodiments, a treatment will be administered once. In some embodiments, a treatment will be administered daily, biweekly, weekly, bimonthly, monthly, or at any time interval that provide maximum efficacy while minimizing safety risks to the subject. Generally, the efficacy and the treatment and safety risks may be monitored throughout the course of treatment

[0330] The efficacy of treatment may be assessed using any suitable methods. In some embodiments, the efficacy of treatment may be assessed by evaluation of observation of symptoms associated with a muscle disease.

[0331] In some embodiments, a pharmaceutical composition that comprises a complex comprising a muscle-targeting agent covalently to a molecular payload described herein is administered to a subject at an effective concentration sufficient to inhibit activity or expression of a target gene by at least 10%, at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90% or at least 95% relative to a control, e.g. baseline level of gene expression prior to treatment.

[0332] In some embodiments, a single dose or administration of a pharmaceutical composition that comprises a complex comprising a muscle-targeting agent covalently to a molecular payload described herein to a subject is sufficient to inhibit activity or expression of a target gene for at least 1-5, 1-10, 5-15, 10-20, 15-30, 20-40, 25-50, or more days. In some embodiments, a single dose or administration of a pharmaceutical composition that comprises a complex comprising a muscle-targeting agent covalently to a molecular payload described herein to a subject is sufficient to inhibit activity or expression of a target gene for at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, or 12 weeks. In some embodiments, a single dose or administration of a pharmaceutical composition that comprises a complex comprising a muscle-targeting agent covalently to a molecular payload described herein to a subject is sufficient to inhibit activity or expression of a target gene for at least 1, 2, 3, 4, 5, or 6 months.

[0333] In some embodiments, a pharmaceutical composition may comprises more than one complex comprising a muscle-targeting agent covalently to a molecular payload. In some embodiments, a pharmaceutical composition may further comprise any other suitable therapeutic agent for treatment of a subject, e.g. a human subject having a muscle disease (e.g., a muscle disease provided in Table 1). In some embodiments, the other therapeutic agents may enhance or supplement the effectiveness of the complexes described herein. In some embodiments, the other therapeutic agents may function to treat a different symptom or disease than the complexes described herein.

Examples

Example 1: Targeting DMPK with Transfected Antisense Oligonucleotides

[0334] A gapmer antisense oligonucleotide that targets both wild-type and mutant alleles of DMPK (DTX-P-060) was tested in vitro for its ability to reduce expression levels of DMPK in an immortalized cell line. Briefly, Hepa 1-6 cells were transfected with the DTX-P-060 (100 nM) formulated with lipofectamine 2000. DMPK expression levels were evaluated 72 hours following transfection. A control experiment was also performed in which vehicle (phosphate-buffered saline) was delivered to Hepa 1-6 cells in culture and the cells were maintained for 72 hours. As shown in FIG. **1**, it was found that the DTX-P-060 reduced DMPK expression levels by ~90% compared with controls.

Example 2: Targeting DMPK with a Muscle-Targeting Complex

[0335] A muscle-targeting complex was generated comprising the DMPK ASO used in Example 1 (DTX-P-060) covalently linked, via a cathepsin cleavable linker, to DTX-A-002 (RI7 217 (Fab)), an anti-transferrin receptor antibody.

[0336] Briefly, a maleimidocaproyl-L-valine-L-citrulline-p-aminobenzyl alcohol p-nitrophenyl carbonate (MC-Val-Cit-PABC-PNP) linker molecule was coupled to NH_2—C_6-DTX-P-060 using an amide coupling reaction. Excess linker and organic solvents were removed by gel permeation chromatography. The purified Val-Cit-linker-DTX-P-060 was then coupled to a thiol-reactive anti-transferrin receptor antibody (DTX-A-002).

[0337] The product of the antibody coupling reaction was subjected to hydrophobic interaction chromatography (HIC-HPLC). FIG. **2**A shows a resulting HIC-HPLC trace, in which fractions B7-C2 of the trace (denoted by vertical lines) contained ASO to antibody ratio of 1 or 2 as determined by SDS-PAGE. These fractions were pooled to arrive at the final muscle-targeting complex, referred to as DTX-C-008. Densitometry confirmed that DTX-C-008 had an average ASO to antibody ratio of 1.48, and SDS-PAGE revealed a purity of 86.4% (FIG. **2**B).

[0338] Using the same approach, a control complex was generated comprising the DMPK ASO used in Example 1 (DTX-P-060) covalently linked via a Val-Cit linker to an IgG2a (Fab) antibody (DTX-C-007).

[0339] The purified DTX-C-008 was then tested for cellular internalization and inhibition of DMPK. Hepa 1-6 cells, which have relatively high expression levels of transferrin receptor, were incubated in the presence of vehicle control, DTX-C-008 (100 nM), or DTX-C-007 (100 nM) for 72 hours. After the 72 hour incubation, the cells were isolated and assayed for expression levels of DMPK (FIG. **3**). Cells treated with the DTX-C-008 demonstrated a reduction in DMPK expression by ~65% relative to the cells treated with the vehicle control. Meanwhile, cells treated with the DTX-C-007 had DMPK expression levels comparable to the vehicle control (no reduction in DMPK expression). These data indicate that the anti-transferrin receptor antibody of the DTX-C-008 enabled cellular internalization of the complex, thereby allowing the DMPK ASO to inhibit expression of DMPK.

Example 3: Targeting DMPK in Mouse Muscle Tissues with a Muscle-Targeting Complex

[0340] The muscle-targeting complex described in Example 2, DTX-C-008, was tested for inhibition of DMPK in mouse tissues. C57BL/6 wild-type mice were intravenously injected with a single dose of a vehicle control, DMPK-1 (3 mg/kg of RNA), DTX-C-008 (3 mg/kg of RNA, corresponding to 20 mg/kg antibody conjugate), or DTX-C-007 (3 mg/kg of RNA, corresponding to 20 mg/kg antibody conjugate). DTX-P-060, the DMPK ASO as described in Example 1, was used as a control. Each experimental condition was replicated in three individual C57BL/6 wild-type mice. Following a seven-day period after injection, the mice were euthanized and segmented into isolated tissue types. Individual tissue samples were subsequently assayed for expression levels of DMPK (FIGS. **4**A-**4**E and **5**A-**5**B).

[0341] Mice treated with the DTX-C-008 complex demonstrated a reduction in DMPK expression in a variety of skeletal, cardiac, and smooth muscle tissues. For example, as shown in FIGS. **4**A-**4**E, DMPK expression levels were significantly reduced in gastrocnemius (50% reduction), heart (30% reduction), esophagus (45% reduction), tibialis anterior (47% reduction), and soleus (31% reduction) tissues, relative to the mice treated with the vehicle control. Meanwhile, mice treated with the DTX-C-007 complex had DMPK expression levels comparable to the vehicle control (no reduction in DMPK expression) for all assayed muscle tissue types.

[0342] Mice treated with the DTX-C-008 complex demonstrated no change in DMPK expression in non-muscle tissues such as spleen and brain tissues (FIGS. **5**A and **5**B).

[0343] These data indicate that the anti-transferrin receptor antibody of the DTX-C-008 enabled cellular internalization of the complex into muscle-specific tissues in an in vivo mouse model, thereby allowing the DMPK ASO to inhibit expression of DMPK. These data further demonstrate that the DTX-C-008 complex is capable of specifically targeting muscle tissues.

Example 4: Targeting DMPK in Mouse Muscle Tissues with a Muscle-Targeting Complex

[0344] The muscle-targeting complex described in Example 2, DTX-C-008, was tested for dose-dependent inhibition of DMPK in mouse tissues. C57BL/6 wild-type mice were intravenously injected with a single dose of a vehicle control (phosphate-buffered saline, PBS), DTX-P-060 (10 mg/kg of RNA), DTX-C-008 (3 mg/kg or 10 mg/kg of RNA, wherein 3 mg/kg corresponds to 20 mg/kg antibody conjugate), or DTX-C-007 (3 mg/kg or 10 mg/kg of RNA, wherein 3 mg/kg corresponds to 20 mg/kg antibody conjugate). DTX-P-060, the DMPK ASO as described in Example 1, was used as a control. Each experimental condition was replicated in five individual C57BL/6 wild-type mice. Following a seven-day period after injection, the mice were euthanized and segmented into isolated tissue types. Individual tissue samples were subsequently assayed for expression levels of DMPK (FIGS. **6**A-**6**F).

[0345] Mice treated with the DTX-C-008 complex demonstrated a reduction in DMPK expression in a variety of skeletal muscle tissues. As shown in FIGS. **6**A-**6**F, DMPK expression levels were significantly reduced in tibialis anterior (58% and 75% reduction for 3 mg/kg and 10 mg/kg DTX-C-008, respectively), soleus (55% and 66% reduction for 3 mg/kg and 10 mg/kg DTX-C-008, respectively), extensor digitorum longus (EDL) (52% and 72% reduction for 3 mg/kg and 10 mg/kg DTX-C-008, respectively), gastrocnemius (55% and 77% reduction for 3 mg/kg and 10 mg/kg DTX-C-008, respectively), heart (19% and 35% reduction for 3 mg/kg and 10 mg/kg DTX-C-008, respectively), and diaphragm (53% and 70% reduction for 3 mg/kg and 10 mg/kg DTX-C-008, respectively) tissues, relative to the mice treated with the vehicle control. Notably, all assayed muscle tissue types experienced dose-dependent inhibition of DMPK, with greater reduction in DMPK levels at 10 mg/kg antibody conjugate relative to 3 mg/kg antibody conjugate.

[0346] Meanwhile, mice treated with the control DTX-C-007 complex had DMPK expression levels comparable to the vehicle control (no reduction in DMPK expression) for all assayed muscle tissue types.

[0347] These data indicate that the anti-transferrin receptor antibody of the DTX-C-008 enabled cellular internalization of the complex into muscle-specific tissues in an in vivo mouse model, thereby allowing the DMPK ASO to inhibit expression of DMPK. These data further demonstrate that the DTX-C-008 complex is capable of specifically targeting muscle tissues for dose-dependent inhibition of DMPK.

Example 5: Targeting DMPK in Cynomolgus Monkey Muscle Tissues with a Muscle-Targeting Complex

[0348] A muscle-targeting complex comprising DTX-P-060 (DTX-C-012), was generated and purified using methods described in Example 2. DTX-C-012 is a complex comprising a human anti-transferrin antibody covalently linked, via a cathepsin cleavable Val-Cit linker, to DTX-P-060, an antisense oligonucleotide that targets DMPK. Following HIC-HPLC purification, densitometry confirmed that DTX-C-012 had an average ASO to antibody ratio of 1.32, and SDS-PAGE revealed a purity of 92.3%.

[0349] DTX-C-012 was tested for dose-dependent inhibition of DMPK in male cynomolgus monkey tissues. Male cynomolgus monkeys (19-31 months; 2-3 kg) were intravenously injected with a single dose of a saline control, DTX-P-060 (naked DMPK ASO) (10 mg/kg of RNA), or DTX-C-012 (10 mg/kg of RNA) on Day 0. Each experimental condition was replicated in three individual male cynomolgus monkeys. On Day 7 after injection, tissue biopsies (including muscle tissues) were collected. DMPK mRNA expression levels, ASO detection assays, serum clinical chemistries, tissue histology, clinical observations, and body weights were analyzed. The monkeys were euthanized on Day 14.

[0350] Significant knockdown (KD) of DMPK mRNA expression using DTX-C-012 was observed in soleus, deep flexor, and masseter muscles relative to saline control, with 39% KD, 62% KD, and 41% KD, respectively (FIGS. **7**A-**7**C). Robust knockdown of DMPK mRNA expression DTX-C-012 was further observed in gastrocnemius (62% KD; FIG. **7**D), EDL (29% KD; FIG. **7**E), tibialis anterior muscle (23% KD; FIG. **7**F), diaphragm (54% KD; FIG. **7**G), tongue (43% KD; FIG. **7**H), heart muscle (36% KD; FIG. **7**I), quadriceps (58% KD; FIG. **7**J), bicep (51% KD; FIG. **7**K), and deltoid muscles (47% KD; FIG. **7**L). Knockdown of DMPK mRNA expression DTX-C-012 in smooth muscle was also observed in the intestine, with 63% KD at jejunum-duodenum ends (FIG. **8**A) and 70% KD in ileum (FIG. **8**B). Notably, naked DMPK ASO (i.e., not linked to a muscle-targeting agent), DTX-P-060, had minimal effects on DMPK expression levels relative to the vehicle control (i.e., little or no reduction in DMPK expression) for all assayed muscle tissue types. Monkeys treated with the DTX-C-012 complex demonstrated no change in DMPK expression in non-muscle tissues, such as liver, kidney, brain, and spleen tissues (FIGS. **9**A-**9**D). Additional tissues were examined, as depicted in FIG. **10**, which shows normalized DMPK mRNA tissue expression levels across several tissue types in cynomolgus monkeys. (N=3 male cynomolgus monkeys)

[0351] Prior to euthanization, all monkeys were tested for reticulocyte levels, platelet levels, hemoglobin expression, alanine aminotransferase (ALT) expression, aspartate aminotransferase (AST) expression, and blood urea nitrogen (BUN) levels on days 2, 7, and 14 after dosing. As shown in FIG. **12**, monkeys dosed with antibody-oligonucleotide complex had normal reticulocyte levels, platelet levels, hemoglobin expression, alanine aminotransferase (ALT) expression, aspartate aminotransferase (AST) expression, and blood urea nitrogen (BUN) levels throughout the length of the experiment. These data show that a single dose of a complex comprising DTX-P-060 is safe and tolerated in cynomolgus monkeys.

[0352] These data demonstrate that the anti-transferrin receptor antibody of the DTX-C-012 complex enabled cellular internalization of the complex into muscle-specific tissues in an in vivo cynomolgus monkey model, thereby allowing the DMPK ASO (DTX-P-060) to inhibit expression of DMPK. These data further demonstrate that the DTX-C-012 complex is capable of specifically targeting muscle tissues for dose-dependent inhibition of DMPK without substantially impacting non-muscle tissues. This is direct contrast with the limited ability of DTX-P-060, a naked DMPK ASO (i.e., not linked to a muscle-targeting agent), to inhibit expression of DMPK in muscle tissues of an in vivo cynomolgus monkey model.

Example 6: Targeting DMPK in Mouse Muscle Tissues with a Muscle-Targeting Complex

[0353] The muscle-targeting complex described in Example 2, DTX-C-008, was tested for time-dependent inhibition of DMPK in mouse tissues. C57BL/6 wild-type mice were intravenously injected with a single dose of a vehicle control (saline), DTX-P-060 (10 mg/kg of RNA), or DTX-C-008 (10 mg/kg of RNA) and euthanized after a prescribed period of time, as described in Table 2. Following euthanization, the mice were segmented into isolated tissue types and tissue samples were subsequently assayed for expression levels of DMPK (FIGS. **11**A-**11**B).

TABLE 2

Experimental conditions			
Group	Dosage	Days after injection before euthanization	Number of mice
1	Vehicle (saline)	3 days	3
2	Vehicle (saline)	7 days	3
3	Vehicle (saline)	14 days	3
4	Vehicle (saline)	28 days	3
5	DTX-P-060	3 days	3
6	DTX-P-060	7 days	3
7	DTX-P-060	14 days	3
8	DTX-P-060	28 days	3
9	DTX-C-008	3 days	3
10	DTX-C-008	7 days	3
11	DTX-C-008	14 days	3
12	DTX-C-008	28 days	3

Mice treated with the DTX-C-008 complex demonstrated approximately 50% reduction in DMPK expression in gastrocnemius (FIG. **11**A) and tibialis anterior (FIG. **11**B) muscles for all of Groups 9-12 (3-28 days between injection and euthanization), relative to vehicle. Mice treated with the DTX-P-060 naked oligonucleotide did not demonstrate significant reduction in DMPK expression.

EQUIVALENTS AND TERMINOLOGY

[0354] The disclosure illustratively described herein suitably can be practiced in the absence of any element or elements, limitation or limitations that are not specifically disclosed herein. Thus, for example, in each instance herein any of the terms "comprising", "consisting essentially of", and "consisting of" may be replaced with either of the other two terms. The terms and expressions which have been employed are used as terms of description and not of limitation, and there is no intention that in the use of such terms and expressions of excluding any equivalents of the features shown and described or portions thereof, but it is recognized that various modifications are possible within the scope of the disclosure. Thus, it should be understood that although the present disclosure has been specifically disclosed by preferred embodiments, optional features, modification and variation of the concepts herein disclosed may be resorted to by those skilled in the art, and that such modifications and variations are considered to be within the scope of this disclosure.

[0355] In addition, where features or aspects of the disclosure are described in terms of Markush groups or other grouping of alternatives, those skilled in the art will recognize that the disclosure is also thereby described in terms of any individual member or subgroup of members of the Markush group or other group.

[0356] It should be appreciated that, in some embodiments, sequences presented in the sequence listing may be referred to in describing the structure of an oligonucleotide or other nucleic acid. In such embodiments, the actual oligonucleotide or other nucleic acid may have one or more alternative nucleotides (e.g., an RNA counterpart of a DNA nucleotide or a DNA counterpart of an RNA nucleotide) and/or one or more modified nucleotides and/or one or more modified internucleotide linkages and/or one or more other modification compared with the specified sequence while retaining essentially same or similar complementary properties as the specified sequence.

[0357] The use of the terms "a" and "an" and "the" and similar referents in the context of describing the invention (especially in the context of the following claims) are to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. The terms "comprising," "having," "including," and "containing" are to be construed as open-ended terms (i.e., meaning "including, but not limited to,") unless otherwise noted. Recitation of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value falling within the range, unless otherwise indicated herein, and each separate value is incorporated into the specification as if it were individually recited herein. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g., "such as") provided herein, is intended merely to better illuminate the invention and does not pose a limitation on the scope of the invention unless otherwise claimed. No language in the specification should be construed as indicating any non-claimed element as essential to the practice of the invention.

[0358] Embodiments of this invention are described herein. Variations of those embodiments may become apparent to those of ordinary skill in the art upon reading the foregoing description.

[0359] The inventors expect skilled artisans to employ such variations as appropriate, and the inventors intend for the invention to be practiced otherwise than as specifically described herein. Accordingly, this invention includes all modifications and equivalents of the subject matter recited in the claims appended hereto as permitted by applicable law. Moreover, any combination of the above-described elements in all possible variations thereof is encompassed by the invention unless otherwise indicated herein or otherwise clearly contradicted by context. Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific embodiments of the invention described herein. Such equivalents are intended to be encompassed by the following claims.

SEQUENCE LISTING

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YINLDKAVLG TSNFKVSASP LLYTLIEKTM QNVKHPVTGQ FLYQDSNWAS KVEKLTLDNA  540
AFPFLAYSGI PAVSFCFCED TDYPYLGTTM DTYKELIERI PELNKVARAA AEVAGQFVIK  600
LTHDVELNLD YERYNSQLLS FVRDLNQYRA DIKEMGLSLQ WLYSARGDFF RATSRLTTDF  660
GNAEKTDRFV MKKLNDRVMR VEYHFLSPYV SPKESPFRHV FWGSGSHTLP ALLENLKLRK  720
QNNGAFNETL FRNQLALATW TIQGAANALS GDVWDIDNEF                       760

SEQ ID NO: 2            moltype = AA  length = 760
FEATURE                 Location/Qualifiers
source                  1..760
                        mol_type = protein
                        organism = Macaca mulatta
SEQUENCE: 2
MMDQARSAFS NLFGGEPLSY TRFSLARQVD GDNSHVEMKL GVDEEENTDN NTKPNGTKPK  60
RCGGNICYGT IAVIIFFLIG FMIGYLGYCK GVEPKTECER LAGTESPARE EPEEDFPAAP  120
RLYWDDLKRK LSEKLDTTDF TSTIKLLNEN LYVPREAGSQ KDENLALYIE NQFREFKLSK  180
VWRDQHFVKI QVKDSAQNSV IIVDKNGGLV YLVENPGGYV AYSKAATVTG KLVHANFGTK  240
KDFEDLDSPV NGSIVIVRAG KITFAEKVAN AESLNAIGVL IYMDQTKFPI VKADLSFFGH  300
AHLGTGDPYT PGFPSFNHTQ FPPSQSSGLP NIPVQTISRA AAEKLFGNME GDCPSDWKTD  360
STCKMVTSEN KSVKLTVSNV LKETKILNIF GVIKGFVEPD HYVVVGAQRD AWGPGAAKSS  420
VGTALLLKLA QMFSDMVLKD GFQPSRSIIF ASWSAGDFGS VGATEWLEGY LSSLHLKAFT  480
YINLDKAVLG TSNFKVSASP LLYTLIEKTM QDVKHPVTGR SLYQDSNWAS KVEKLTLDNA  540
AFPFLAYSGI PAVSFCFCED TDYPYLGTTM DTYKELVERI PELNKVARAA AEVAGQFVIK  600
LTHDTELNLD YERYNSQLLL FLRDLNQYRA DVKEMGLSLQ WLYSARGDFF RATSRLTTDF  660
RNAEKRDKFV MKKLNDRVMR VEYYFLSPYV SPKESPFRHV FWGSGSHTLS ALLESLKLRR  720
QNNSAFNETL FRNQLALATW TIQGAANALS GDVWDIDNEF                       760

SEQ ID NO: 3            moltype = AA  length = 760
FEATURE                 Location/Qualifiers
source                  1..760
                        mol_type = protein
                        organism = Macaca fascicularis
SEQUENCE: 3
MMDQARSAFS NLFGGEPLSY TRFSLARQVD GDNSHVEMKL GVDEEENTDN NTKANGTKPK  60
RCGGNICYGT IAVIIFFLIG FMIGYLGYCK GVEPKTECER LAGTESPARE EPEEDFPAAP  120
RLYWDDLKRK LSEKLDTTDF TSTIKLLNEN LYVPREAGSQ KDENLALYIE NQFREFKLSK  180
VWRDQHFVKI QVKDSAQNSV IIVDKNGGLV YLVENPGGYV AYSKAATVTG KLVHANFGTK  240
KDFEDLDSPV NGSIVIVRAG KITFAEKVAN AESLNAIGVL IYMDQTKFPI VKADLSFFGH  300
AHLGTGDPYT PGFPSFNHTQ FPPSQSSGLP NIPVQTISRA AAEKLFGNME GDCPSDWKTD  360
STCKMVTSEN KSVKLTVSNV LKETKILNIF GVIKGFVEPD HYVVVGAQRD AWGPGAAKSS  420
VGTALLLKLA QMFSDMVLKD GFQPSRSIIF ASWSAGDFGS VGATEWLEGY LSSLHLKAFT  480
YINLDKAVLG TSNFKVSASP LLYTLIEKTM QDVKHPVTGR SLYQDSNWAS KVEKLTLDNA  540
AFPFLAYSGI PAVSFCFCED TDYPYLGTTM DTYKELVERI PELNKVARAA AEVAGQFVIK  600
LTHDTELNLD YERYNSQLLL FLRDLNQYRA DVKEMGLSLQ WLYSARGDFF RATSRLTTDF  660
RNAEKRDKFV MKKLNDRVMR VEYYFLSPYV SPKESPFRHV FWGSGSHTLS ALLESLKLRR  720
QNNSAFNETL FRNQLALATW TIQGAANALS GDVWDIDNEF                       760

SEQ ID NO: 4            moltype = AA  length = 763
FEATURE                 Location/Qualifiers
source                  1..763
                        mol_type = protein
                        organism = Mus musculus
SEQUENCE: 4
MMDQARSAFS NLFGGEPLSY TRFSLARQVD GDNSHVEMKL AADEEENADN NMKASVRKPK  60
RFNGRLCFAA IALVIFFLIG FMSGYLGYCK RVEQKEECVK LAETEETDKS ETMETEDVPT  120
SSRLYWADLK TLLSEKLNSI EFADTIKQLS QNTYTPREAG SQKDESLAYY IENQFHEFKF  180
SKVWRDEHYV KIQVKSSIGQ NMVTIVQSNG NLDPVESPEG YVAFSKPTEV SGKLVHANFG  240
TKKDFEELSY SVNGSLVIVR AGEITFAEKV ANAQSFNAIG VLIYMDKNKF PVVEADLALF  300
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GHAHLGTGDP YTPGFPSFNH TQFPPSQSSG LPNIPVQTIS RAAAEKLFGK MEGSCPARWN  360
IDSSCKLELS QNQNVKLIVK NVLKERRILN IFGVIKGYEE PDRYVVVGAQ RDALGAGVAA  420
KSSVGTGLLL KLAQVFSDMI SKDGFRPSRS IIFASWTAGD FGAVGATEWL EGYLSSLHLK  480
AFTYINLDKV VLGTSNFKVS ASPLLYTLMG KIMQDVKHPV DGKSLYRDSN WISKVEKLSF  540
DNAAYPFLAY SGIPAVSFCF CEDADYPYLG TRLDTYEALT QKVPQLNQMV RTAAEVAGQL  600
IIKLTHDVEL NLDYEMYNSK LLSFMKDLNQ FKTDIRDMGL SLQWLYSARG DYFRATSRLT  660
TDFHNAEKTN RFVMREINDR IMKVEYHFLS PYVSPRESPF RHIFWGSGSH TLSALVENLK  720
LRQKNITAFN ETLFRNQLAL ATWTIQGVAN ALSGDIWNID NEF                    763

SEQ ID NO: 5            moltype = AA  length = 197
FEATURE                 Location/Qualifiers
REGION                  1..197
                        note = Synthetic polypeptide
source                  1..197
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 5
FVKIQVKDSA QNSVIIVDKN GRLVYLVENP GGYVAYSKAA TVTGKLVHAN FGTKKDFEDL  60
YTPVNGSIVI VRAGKITFAE KVANAESLNA IGVLIYMDQT KFPIVNAELS FFGHAHLGTG  120
DPYTPGFPSF NHTQFPPSRS SGLPNIPVQT ISRAAAEKLF GNMEGDCPSD WKTDSTCRMV  180
TSESKNVKLT VSNVLKE                                                  197

SEQ ID NO: 6            moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Synthetic polypeptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 6
ASSLNIA                                                             7

SEQ ID NO: 7            moltype = AA  length = 12
FEATURE                 Location/Qualifiers
REGION                  1..12
                        note = Synthetic polypeptide
source                  1..12
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 7
SKTFNTHPQS TP                                                       12

SEQ ID NO: 8            moltype = AA  length = 12
FEATURE                 Location/Qualifiers
REGION                  1..12
                        note = Synthetic polypeptide
source                  1..12
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 8
TARGEHKEEE LI                                                       12

SEQ ID NO: 9            moltype = AA  length = 9
FEATURE                 Location/Qualifiers
REGION                  1..9
                        note = Synthetic polypeptide
source                  1..9
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 9
CQAQGQLVC                                                           9

SEQ ID NO: 10           moltype = AA  length = 9
FEATURE                 Location/Qualifiers
REGION                  1..9
                        note = Synthetic polypeptide
source                  1..9
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 10
CSERSMNFC                                                           9

SEQ ID NO: 11           moltype = AA  length = 9
FEATURE                 Location/Qualifiers
REGION                  1..9
                        note = Synthetic polypeptide
```

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```
source                  1..9
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 11
CPKTRRVPC                                                                   9

SEQ ID NO: 12           moltype = AA  length = 20
FEATURE                 Location/Qualifiers
REGION                  1..20
                        note = Synthetic polypeptide
source                  1..20
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 12
WLSEAGPVVT VRALRGTGSW                                                       20

SEQ ID NO: 13           moltype = AA  length = 9
FEATURE                 Location/Qualifiers
REGION                  1..9
                        note = Synthetic polypeptide
source                  1..9
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 13
CMQHSMRVC                                                                   9

SEQ ID NO: 14           moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Synthetic polypeptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 14
DDTRHWG                                                                     7

SEQ ID NO: 15           moltype =   length =
SEQUENCE: 15
000

SEQ ID NO: 16           moltype = AA  length = 5
FEATURE                 Location/Qualifiers
source                  1..5
                        mol_type = protein
                        organism = synthetic construct
VARIANT                 3
                        note = X can be any naturally occurring amino acid
SEQUENCE: 16
LPXTG                                                                       5

SEQ ID NO: 17           moltype = AA  length = 5
FEATURE                 Location/Qualifiers
REGION                  1..5
                        note = Synthetic polypeptide
source                  1..5
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 17
SYWMH                                                                       5

SEQ ID NO: 18           moltype = AA  length = 17
FEATURE                 Location/Qualifiers
REGION                  1..17
                        note = Synthetic polypeptide
source                  1..17
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 18
EINPTNGRTN YIEKFKS                                                          17

SEQ ID NO: 19           moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Synthetic polypeptide
source                  1..7
                        mol_type = protein
```

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```
                          organism = synthetic construct
SEQUENCE: 19
GTRAYHY                                                                   7

SEQ ID NO: 20             moltype = AA  length = 11
FEATURE                   Location/Qualifiers
REGION                    1..11
                          note = Synthetic polypeptide
source                    1..11
                          mol_type = protein
                          organism = synthetic construct
SEQUENCE: 20
RASDNLYSNL A                                                              11

SEQ ID NO: 21             moltype = AA  length = 7
FEATURE                   Location/Qualifiers
REGION                    1..7
                          note = Synthetic polypeptide
source                    1..7
                          mol_type = protein
                          organism = synthetic construct
SEQUENCE: 21
DATNLAD                                                                   7

SEQ ID NO: 22             moltype = AA  length = 9
FEATURE                   Location/Qualifiers
REGION                    1..9
                          note = Synthetic polypeptide
source                    1..9
                          mol_type = protein
                          organism = synthetic construct
SEQUENCE: 22
QHFWGTPLT                                                                 9

SEQ ID NO: 23             moltype = AA  length = 7
FEATURE                   Location/Qualifiers
REGION                    1..7
                          note = Synthetic polypeptide
source                    1..7
                          mol_type = protein
                          organism = synthetic construct
SEQUENCE: 23
GYTFTSY                                                                   7

SEQ ID NO: 24             moltype = AA  length = 6
FEATURE                   Location/Qualifiers
REGION                    1..6
                          note = Synthetic polypeptide
source                    1..6
                          mol_type = protein
                          organism = synthetic construct
SEQUENCE: 24
NPTNGR                                                                    6

SEQ ID NO: 25             moltype = AA  length = 6
FEATURE                   Location/Qualifiers
REGION                    1..6
                          note = Synthetic polypeptide
source                    1..6
                          mol_type = protein
                          organism = synthetic construct
SEQUENCE: 25
TSYWMH                                                                    6

SEQ ID NO: 26             moltype = AA  length = 13
FEATURE                   Location/Qualifiers
REGION                    1..13
                          note = Synthetic polypeptide
source                    1..13
                          mol_type = protein
                          organism = synthetic construct
SEQUENCE: 26
WIGEINPTNG RTN                                                            13

SEQ ID NO: 27             moltype = AA  length = 6
FEATURE                   Location/Qualifiers
```

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```
REGION                 1..6
                       note = Synthetic polypeptide
source                 1..6
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 27
ARGTRA                                                                    6

SEQ ID NO: 28          moltype = AA  length = 7
FEATURE                Location/Qualifiers
REGION                 1..7
                       note = Synthetic polypeptide
source                 1..7
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 28
YSNLAWY                                                                   7

SEQ ID NO: 29          moltype = AA  length = 10
FEATURE                Location/Qualifiers
REGION                 1..10
                       note = Synthetic polypeptide
source                 1..10
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 29
LLVYDATNLA                                                                10

SEQ ID NO: 30          moltype = AA  length = 8
FEATURE                Location/Qualifiers
REGION                 1..8
                       note = Synthetic polypeptide
source                 1..8
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 30
QHFWGTPL                                                                  8

SEQ ID NO: 31          moltype = AA  length = 9
FEATURE                Location/Qualifiers
REGION                 1..9
                       note = Synthetic polypeptide
source                 1..9
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 31
QHFAGTPLT                                                                 9

SEQ ID NO: 32          moltype = AA  length = 8
FEATURE                Location/Qualifiers
REGION                 1..8
                       note = Synthetic polypeptide
source                 1..8
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 32
QHFAGTPL                                                                  8

SEQ ID NO: 33          moltype = AA  length = 116
FEATURE                Location/Qualifiers
REGION                 1..116
                       note = Synthetic polypeptide
source                 1..116
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 33
QVQLQQPGAE LVKPGASVKL SCKASGYTFT SYWMHWVKQR PGQGLEWIGE INPTNGRTNY  60
IEKFKSKATL TVDKSSSTAY MQLSSLTSED SAVYYCARGT RAYHYWGQGT SVTVSS      116

SEQ ID NO: 34          moltype = AA  length = 107
FEATURE                Location/Qualifiers
REGION                 1..107
                       note = Synthetic polypeptide
source                 1..107
                       mol_type = protein
                       organism = synthetic construct
```

-continued

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SEQUENCE: 34
DIQMTQSPAS LSVSVGETVT ITCRASDNLY SNLAWYQQKQ GKSPQLLVYD ATNLADGVPS  60
RFSGSGSGTQ YSLKINSLQS EDFGTYYCQH FWGTPLTFGA GTKLELK                107

SEQ ID NO: 35           moltype = AA  length = 116
FEATURE                 Location/Qualifiers
REGION                  1..116
                        note = Synthetic polypeptide
source                  1..116
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 35
EVQLVQSGAE VKKPGASVKV SCKASGYTFT SYWMHWVRQA PGQRLEWIGE INPTNGRTNY  60
IEKFKSRATL TVDKSASTAY MELSSLRSED TAVYYCARGT RAYHYWGQGT MVTVSS      116

SEQ ID NO: 36           moltype = AA  length = 107
FEATURE                 Location/Qualifiers
REGION                  1..107
                        note = Synthetic polypeptide
source                  1..107
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 36
DIQMTQSPSS LSASVGDRVT ITCRASDNLY SNLAWYQQKP GKSPKLLVYD ATNLADGVPS  60
RFSGSGSGTD YSLKINSLQS EDFGTYYCQH FWGTPLTFGA GTKLELK                107

SEQ ID NO: 37           moltype = AA  length = 330
FEATURE                 Location/Qualifiers
REGION                  1..330
                        note = Synthetic polypeptide
source                  1..330
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 37
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS  60
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPELLGG  120
PSVFLFPPKP KDTLMISRTP EVTCVVVDVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYN  180
STYRVVSVLT VLHQDWLNGK EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSRDE  240
LTKNQVSLTC LVKGFYPSDI AVEWESNGQP ENNYKTTPPV LDSDGSFFLY SKLTVDKSRW  300
QQGNVFSCSV MHEALHNHYT QKSLSLSPGK                                   330

SEQ ID NO: 38           moltype = AA  length = 110
FEATURE                 Location/Qualifiers
REGION                  1..110
                        note = Synthetic polypeptide
source                  1..110
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 38
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS  60
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP             110

SEQ ID NO: 39           moltype = AA  length = 446
FEATURE                 Location/Qualifiers
REGION                  1..446
                        note = Synthetic polypeptide
source                  1..446
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 39
QVQLQQPGAE LVKPGASVKL SCKASGYTFT SYWMHWVKQR PGQGLEWIGE INPTNGRTNY  60
IEKFKSKATL TVDKSSSTAY MQLSSLTSED SAVYYCARGT RAYHYWGQGT SVTVSSASTK  120
GPSVFPLAPS SKSTSGGTAA LGCLVKDYFP EPVTVSWNSG ALTSGVHTFP AVLQSSGLYS  180
LSSVVTVPSS SLGTQTYICN VNHKPSNTKV DKKVEPKSCD KTHTCPPCPA PELLGGPSVF  240
LFPPKPKDTL MISRTPEVTC VVVDVSHEDP EVKFNWYVDG VEVHNAKTKP REEQYNSTYR  300
VVSVLTVLHQ DWLNGKEYKC KVSNKALPAP IEKTISKAKG QPREPQVYTL PPSRDELTKN  360
QVSLTCLVKG FYPSDIAVEW ESNGQPENNY KTTPPVLDSD GSFFLYSKLT VDKSRWQQGN  420
VFSCSVMHEA LHNHYTQKSL SLSPGK                                       446

SEQ ID NO: 40           moltype = AA  length = 226
FEATURE                 Location/Qualifiers
REGION                  1..226
                        note = Synthetic polypeptide
source                  1..226
                        mol_type = protein
                        organism = synthetic construct
```

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SEQUENCE: 40
QVQLQQPGAE LVKPGASVKL SCKASGYTFT SYWMHWVKQR PGQGLEWIGE INPTNGRTNY  60
IEKFKSKATL TVDKSSSTAY MQLSSLTSED SAVYYCARGT RAYHYWGQGT SVTVSSASTK  120
GPSVFPLAPS SKSTSGGTAA LGCLVKDYFP EPVTVSWNSG ALTSGVHTFP AVLQSSGLYS  180
LSSVVTVPSS SLGTQTYICN VNHKPSNTKV DKKVEPKSCD KTHTCP                 226

SEQ ID NO: 41          moltype = AA  length = 446
FEATURE                Location/Qualifiers
REGION                 1..446
                       note = Synthetic polypeptide
source                 1..446
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 41
EVQLVQSGAE VKKPGASVKV SCKASGYTFT SYWMHWVRQA PGQRLEWIGE INPTNGRTNY  60
IEKFKSRATL TVDKSASTAY MELSSLRSED TAVYYCARGT RAYHYWGQGT MVTVSSASTK  120
GPSVFPLAPS SKSTSGGTAA LGCLVKDYFP EPVTVSWNSG ALTSGVHTFP AVLQSSGLYS  180
LSSVVTVPSS SLGTQTYICN VNHKPSNTKV DKKVEPKSCD KTHTCPPCPA PELLGGPSVF  240
LFPPKPKDTL MISRTPEVTC VVVDVSHEDP EVKFNWYVDG VEVHNAKTKP REEQYNSTYR  300
VVSVLTVLHQ DWLNGKEYKC KVSNKALPAP IEKTISKAKG QPREPQVYTL PPSRDELTKN  360
QVSLTCLVKG FYPSDIAVEW ESNGQPENNY KTTPPVLDSD GSFFLYSKLT VDKSRWQQGN  420
VFSCSVMHEA LHNHYTQKSL SLSPGK                                       446

SEQ ID NO: 42          moltype = AA  length = 217
FEATURE                Location/Qualifiers
REGION                 1..217
                       note = Synthetic polypeptide
source                 1..217
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 42
DIQMTQSPSS LSASVGDRVT ITCRASDNLY SNLAWYQQKP GKSPKLLVYD ATNLADGVPS  60
RFSGSGSGTD YSLKINSLQS EDFGTYYCQH FWGTPLTFGA GTKLELKAST KGPSVFPLAP  120
SSKSTSGGTA ALGCLVKDYF PEPVTVSWNS GALTSGVHTF PAVLQSSGLY SLSSVVTVPS  180
SSLGTQTYIC NVNHKPSNTK VDKKVEPKSC DKTHTCP                           217

SEQ ID NO: 43          moltype = AA  length = 226
FEATURE                Location/Qualifiers
REGION                 1..226
                       note = Synthetic polypeptide
source                 1..226
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 43
QVQLQQPGAE LVKPGASVKL SCKASGYTFT SYWMHWVKQR PGQGLEWIGE INPTNGRTNY  60
IEKFKSKATL TVDKSSSTAY MQLSSLTSED SAVYYCARGT RAYHYWGQGT SVTVSSASTK  120
GPSVFPLAPS SKSTSGGTAA LGCLVKDYFP EPVTVSWNSG ALTSGVHTFP AVLQSSGLYS  180
LSSVVTVPSS SLGTQTYICN VNHKPSNTKV DKKVEPKSCD KTHTCP                 226

SEQ ID NO: 44          moltype = AA  length = 226
FEATURE                Location/Qualifiers
REGION                 1..226
                       note = Synthetic polypeptide
source                 1..226
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 44
EVQLVQSGAE VKKPGASVKV SCKASGYTFT SYWMHWVRQA PGQRLEWIGE INPTNGRTNY  60
IEKFKSRATL TVDKSASTAY MELSSLRSED TAVYYCARGT RAYHYWGQGT MVTVSSASTK  120
GPSVFPLAPS SKSTSGGTAA LGCLVKDYFP EPVTVSWNSG ALTSGVHTFP AVLQSSGLYS  180
LSSVVTVPSS SLGTQTYICN VNHKPSNTKV DKKVEPKSCD KTHTCP                 226

SEQ ID NO: 45          moltype = DNA  length = 2859
FEATURE                Location/Qualifiers
source                 1..2859
                       mol_type = genomic DNA
                       organism = Homo sapiens
SEQUENCE: 45
agggggctg gaccaagggg tggggagaag gggaggaggc ctcggccggc cgcagagaga  60
agtggccaga gaggcccagg ggacagccag ggacaggcag acatgcagcc agggctccag  120
ggcctggaca ggggctgcca ggccctgtga caggaggacc ccgagccccc ggcccgggga  180
ggggccatgg tgctgcctgt ccaacatgtc agccgaggtg cggctgaggc ggctccagca  240
gctggtgttg gacccgggct tcctggggct ggagcccctg ctcgaccttc tcctgggcgt  300
ccaccaggag ctgggcgcct ccgaactggc ccaggacaag tacgtggccg acttcttgca  360
gtgggcggag cccatcgtgg tgaggcttaa ggaggtccga ctgcagaggg acgacttcga  420
gattctgaag gtgatcggac gcggggcgtt cagcgaggta gcggtagtga agatgaagca  480
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gacgggccag gtgtatgcca tgaagatcat gaacaagtgg gacatgctga agaggggcga  540
ggtgtcgtgc ttccgtgagg agagggacgt gttggtgaat ggggaccggc ggtggatcac  600
gcagctgcac ttcgccttcc aggatgagaa ctacctgtac ctggtcatgg agtattacgt  660
gggcggggac ctgctgacac tgctgagcaa gtttggggag cggattccgg ccgagatggc  720
gcgcttctac ctggcggaga ttgtcatggc catagactcg gtgcaccggc ttggctacgt  780
gcacagggac atcaaacccg acaacatcct gctggaccgc tgtggccaca tccgcctggc  840
cgacttcggc tcttgcctca agctgcgggc agatggaacg gtgcggtcgc tggtggctgt  900
gggcacccca gactacctgt cccccgagat cctgcaggct gtgggcggtg ggcctgggac  960
aggcagctac gggcccgagt gtgactggtg ggcgctgggt gtattcgcct atgaaatgtt  1020
ctatgggcag acgcccttct acgcggattc cacggcggag acctatggca agatcgtcca  1080
ctacaaggag cacctctctc tgccgctggt ggacgaaggg gtccctgagg aggctcgaga  1140
cttcattcag cggttgctgt gtcccccgga gacacggctg ggccggggtg gagcaggcga  1200
cttccggaca catcccttct tctttggcct cgactgggat ggtctccggg acagcgtgcc  1260
cccctttaca ccggatttcg aaggtgccac cgacacatgc aacttcgact tggtggagga  1320
cgggctcact gccatggaga cactgtcgga cattcgggaa ggtgcgccgc taggggtcca  1380
cctgcctttt gtgggctact cctactcctg catggccctc agggacagtg aggtcccagg  1440
ccccacaccc atggaactgg aggccgagca gctgcttgag ccacacgtgc aagcgcccag  1500
cctggagccc tcggtgtccc cacaggatga aacagctgaa gtggcagttc cagcggctgt  1560
ccctgcggca gaggctgagg ccgaggtgac gctgcgggag ctccaggaag ccctggagga  1620
ggaggtgctc acccggcaga gcctgagccg ggagatggag gccatccgca cggacaacca  1680
gaacttcgcc agtcaactac gcgaggcaga ggctcggaac cgggacctag aggcacacgt  1740
ccggcagttg caggagcgga tggagttgct gcaggcagag ggagccacag ctgtcacggg  1800
ggtccccagt ccccgggcca cggatccacc ttcccatcta gatggccccc cggccgtggc  1860
tgtgggccag tgcccgctgg tggggccagg ccccatgcac cgccgccacc tgctgctccc  1920
tgccagggtc cctaggcctg gcctatcgga ggcgctttcc ctgctcctgt tcgccgttgt  1980
tctgtctcgt gccgccgccc tgggctgcat tgggttggtg gcccacgccg gccaactcac  2040
cgcagtctgg cgccgcccag gagccgcccg cgctccctga accctagaac tgtcttcgac  2100
tccggggccc cgttggaaga ctgagtgccc ggggcacggc acagaagccg cgcccaccgc  2160
ctgccagttc acaaccgctc cgagcgtggg tctccgccca gctccagtcc tgtgatccgg  2220
gcccgccccc tagcggccgg ggagggaggg gccgggtccg cggccggcga acggggctcg  2280
aagggtcctt gtagccggga atgctgctgc tgctgctgct gctgctgctg ctgctgctgc  2340
tgctgctgct gctgctgctg ctggggggat cacagaccat ttctttcttt cggccaggct  2400
gaggccctga cgtggatggg caaactgcag gcctgggaag gcagcaagcc gggccgtccg  2460
tgttccatcc tccacgcacc cccacctatc gttggttcgc aaagtgcaaa gctttcttgt  2520
gcatgacgcc ctgctctggg gagcgtctgg cgcgatctct gcctgcttac tcgggaaatt  2580
tgcttttgcc aaacccgctt tttcggggat cccgcgcccc cctcctcact tgcgctgctc  2640
tcggagcccc agccggctcc gcccgcttcg gcggtttgga tatttattga cctcgtcctc  2700
cgactcgctg acaggctaca ggacccccaa caaccccaat ccacgttttg gatgcactga  2760
gaccccgaca ttcctcggta tttattgtct gtccccacct aggaccccca cccccgaccc  2820
tcgcgaataa aaggccctcc atctgcccaa agctctgga                         2859

SEQ ID NO: 46           moltype = DNA  length = 2683
FEATURE                 Location/Qualifiers
source                  1..2683
                        mol_type = genomic DNA
                        organism = Mus musculus
SEQUENCE: 46
gaactggcca gagagaccca agggatagtc agggacgggc agacatgcag ctagggttct  60
ggggcctgga caggggcagc caggccctgt gacgggaaga ccccgagctc cggcccgggg  120
aggggccatg gtgttgcctg cccaacatgt cagccgaagt gcggctgagg cagctccagc  180
agctggtgct ggacccaggc ttcctgggac tggagcccct gctcgacctt ctcctgggcg  240
tccaccagga gctgggtgcc tctcacctag cccaggacaa gtatgtggcc gacttcttgc  300
agtgggtgga gcccattgca gcaaggctta aggaggtccg actgcagagg gatgattttg  360
agattttgaa ggtgatcggg cgtggggcgt tcagcgaggt agcggtggtg aagatgaaac  420
agacgggcca agtgtatgcc atgaagatta tgaataagtg ggacatgctg aagagaggcg  480
aggtgtcgtg cttccgggaa gaaagggatg tattagtgaa aggggaccgg cgctggatca  540
cacagctgca ctttgccttc caggatgaga actacctgta cctggtcatg gaatactacg  600
tgggcgggga cctgctaacg ctgctgagca agtttgggga gcggatcccc gccgagatgg  660
ctcgcttcta cctggccgag attgtcatgg ccatagactc cgtgcaccgg ctgggctacg  720
tgcacaggga catcaaacca gataacattc tgctggaccg atgtgggcac attcgcctgg  780
cagacttcgg ctcctgcctc aaactgcagc ctgatggaat ggtgaggtcg ctggtggctg  840
tgggcacccc ggactacctg tctcctgaga ttctgcaggc cgttggtgga gggcctgggg  900
caggcagcta cgggccagag tgtgactggt gggcactggg cgtgttcgcc tatgagatgt  960
tctatgggca gacccccttc tacgcggact ccacagccga gacatatgcc aagattgtgc  1020
actacaggga acacttgtcg ctgccgctgg cagacacagt tgtccccgag gaagctcagg  1080
acctcattcg tgggctgctg tgtcctgctg agataaggct aggtcgaggt ggggcagact  1140
tcgagggtgc cacggacaca tgcaatttcg atgtggtgga ggaccggctc actgccatgg  1200
tgagcggggg cggggagacg ctgtcagaca tgcaggaaga catgcccctt ggggtgcgcc  1260
tgcccttcgt gggctactcc tactgctgca tggccttcag agacaatcag gtcccggacc  1320
ccacccctat ggaactagag gccctgcagt tgcctgtgtc agacttgcaa gggcttgact  1380
tgcagccccc agtgtcccca ccggatcaag tggctgaaga ggctgaccta gtggctgtcc  1440
ctgcccctgt ggctgaggca gagaccacgg taacgctgca gcagctccag gaagccctgg  1500
aagaagaggt tctcacccgg cagagcctga gccgcgagct ggaggccatc cggaccgcca  1560
accagaactt ctccagccaa ctacaggagg ccgaggtccg aaaccgagac ctggaggcgc  1620
atgttcggca gctacaggaa cggatggaga tgctgcaggc cccaggagcc gcagccatca  1680
cgggggtccc cagtccccgg gccacggatc caccttccca tctagatggc cccccggccg  1740
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tggctgtggg ccagtgcccg ctggtggggc caggccccat gcaccgccgt cacctgctgc  1800
tccctgccag gatccctagg cctggcctat ccgaggcgcg ttgcctgctc ctgttcgccg  1860
ctgctctggc tgctgccgcc acactgggct gcactgggtt ggtggcctat accggcggtc  1920
tcaccccagt ctggtgtttc ccgggagcca ccttcgcccc ctgaacccta agactccaag  1980
ccatctttca tttaggcctc ctaggaaggt cgagcgacca gggagcgacc caaagcgtct  2040
ctgtgcccat cgcgcccccc cccccccccc accgctccgc tccacacttc tgtgagcctg  2100
ggtccccacc cagctccgct cctgtgatcc aggcctgcca cctggcggcc ggggagggag  2160
gaacagggct cgtgcccagc accccctggtt cctgcagagc tggtagccac cgctgctgca  2220
gcagctgggc attcgccgac cttgctttac tcagccccga cgtggatggg caaactgctc  2280
agctcatccg atttcacttt ttcactctcc cagccatcag ttacaagcca taagcatgag  2340
ccccctattt ccagggacat cccattccca tagtgatgga tcagcaagac ctctgccagc  2400
acacacggag tctttggctt cggacagcct cactcctggg ggttgctgca actccttccc  2460
cgtgtacacg tctgcactct aacaacggag ccacagctgc actccccccct cccccaaagc  2520
agtgtgggta tttattgatc ttgttatctg actcactgac agactccggg acccacgttt  2580
tagatgcatt gagactcgac attcctcggt atttattgtc tgtccccacc tacgacctcc  2640
actcccgacc cttgcgaata aaatacttct ggtctgccct aaa                    2683

SEQ ID NO: 47           moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 47
ggacggcccg gcttgctgcc                                              20

SEQ ID NO: 48           moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 48
gggcccggat cacaggactg                                              20

SEQ ID NO: 49           moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 49
caaacttgct cagcagtgtc                                              20

SEQ ID NO: 50           moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 50
aaacttgctc agcagtgtca                                              20

SEQ ID NO: 51           moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 51
cggatggcct ccatctcccg                                              20

SEQ ID NO: 52           moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 52
ctcggccgga atccgctccc                                              20
```

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SEQ ID NO: 53           moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 53
tctcggccgg aatccgctcc                                              20

SEQ ID NO: 54           moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 54
tgctcagcag tgtcagcagg                                              20

SEQ ID NO: 55           moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 55
ttgtcgggtt tgatgtccct                                              20

SEQ ID NO: 56           moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 56
gttgtcgggt ttgatgtccc                                              20

SEQ ID NO: 57           moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 57
tccgccaggt agaagcgcgc                                              20

SEQ ID NO: 58           moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 58
catggcatac acctggcccg                                              20

SEQ ID NO: 59           moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 59
aacttgctca gcagtgtcag                                              20

SEQ ID NO: 60           moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
```

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```
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 60
cagctgcgtg atccaccgcc                                                20

SEQ ID NO: 61           moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 61
cgaatgtccg acagtgtctc                                                20

SEQ ID NO: 62           moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 62
gaagtcggcc aggcggatgt                                                20

SEQ ID NO: 63           moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 63
tgtcgggttt gatgtccctg                                                20

SEQ ID NO: 64           moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 64
ggatggcctc catctcccgg                                                20

SEQ ID NO: 65           moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 65
aggatgttgt cgggtttgat                                                20

SEQ ID NO: 66           moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 66
gtcgggtttg atgtccctgt                                                20

SEQ ID NO: 67           moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 67
aatactccat gaccaggtac                                                20

SEQ ID NO: 68           moltype = RNA  length = 20
```

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```
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 68
cttgttcatg atcttcatgg                                                20

SEQ ID NO: 69          moltype = RNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 69
tcagtgcatc caaaacgtgg                                                20

SEQ ID NO: 70          moltype = RNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 70
ctgtcccgga gaccatccca                                                20

SEQ ID NO: 71          moltype = RNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 71
gggcctggga cctcactgtc                                                20

SEQ ID NO: 72          moltype = RNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 72
cccacgtaat actccatgac                                                20

SEQ ID NO: 73          moltype = RNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 73
ctctgccgca gggacagccg                                                20

SEQ ID NO: 74          moltype = RNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 74
ctgtgcacgt agccaagccg                                                20

SEQ ID NO: 75          moltype = RNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other RNA
                       organism = synthetic construct
```

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```
SEQUENCE: 75
tgcccatcca cgtcagggcc                                          20

SEQ ID NO: 76           moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 76
agcgcctccg ataggccagg                                          20

SEQ ID NO: 77           moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 77
tgtgcacgta gccaagccgg                                          20

SEQ ID NO: 78           moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 78
gaccaggtac aggtagttct                                          20

SEQ ID NO: 79           moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 79
ccatctcggc cggaatccgc                                          20

SEQ ID NO: 80           moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 80
catctcggcc ggaatccgct                                          20

SEQ ID NO: 81           moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 81
ttgccatagg tctccgccgt                                          20

SEQ ID NO: 82           moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 82
acagcggtcc agcaggatgt                                          20

SEQ ID NO: 83           moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
```

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```
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 83
aaagcgcctc cgataggcca                                      20

SEQ ID NO: 84           moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 84
gccaaagaag aagggatgtg                                      20

SEQ ID NO: 85           moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 85
cacgtaatac tccatgacca                                      20

SEQ ID NO: 86           moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 86
atctcggccg gaatccgctc                                      20

SEQ ID NO: 87           moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 87
gcttcatctt cactaccgct                                      20

SEQ ID NO: 88           moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 88
gccatctcgg ccggaatccg                                      20

SEQ ID NO: 89           moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 89
cagggacagc cgctggaact                                      20

SEQ ID NO: 90           moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 90
atgacaatct ccgccaggta                                      20
```

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```
SEQ ID NO: 91          moltype = RNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 91
ggccatgaca atctccgcca                                       20

SEQ ID NO: 92          moltype = RNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 92
atactccatg accaggtaca                                       20

SEQ ID NO: 93          moltype = RNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 93
gcctctgcct cgcgtagttg                                       20

SEQ ID NO: 94          moltype = RNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 94
gaatgtccga cagtgtctcc                                       20

SEQ ID NO: 95          moltype = RNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 95
cgttccatct gcccgcagct                                       20

SEQ ID NO: 96          moltype = RNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 96
ccttgtagtg gacgatcttg                                       20

SEQ ID NO: 97          moltype = RNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 97
atctccgcca ggtagaagcg                                       20

SEQ ID NO: 98          moltype = RNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
```

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```
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 98
ctcaggctct gccgggtgag                                           20

SEQ ID NO: 99           moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 99
tgcttcatct tcactaccgc                                           20

SEQ ID NO: 100          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 100
gcaggatgtt gtcgggtttg                                           20

SEQ ID NO: 101          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 101
ggcctcagcc tctgccgcag                                           20

SEQ ID NO: 102          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 102
tgttgtcggg tttgatgtcc                                           20

SEQ ID NO: 103          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 103
ccacgtaata ctccatgacc                                           20

SEQ ID NO: 104          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 104
ccgttccatc tgcccgcagc                                           20

SEQ ID NO: 105          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 105
ttcccgagta agcaggcaga                                           20

SEQ ID NO: 106          moltype = RNA  length = 20
```

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```
FEATURE               Location/Qualifiers
misc_feature          1..20
                      note = Synthetic Polynucleotide
source                1..20
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 106
tgatcttcat ggcatacacc                                              20

SEQ ID NO: 107        moltype = RNA  length = 20
FEATURE               Location/Qualifiers
misc_feature          1..20
                      note = Synthetic Polynucleotide
source                1..20
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 107
agggacagcc gctggaactg                                              20

SEQ ID NO: 108        moltype = RNA  length = 20
FEATURE               Location/Qualifiers
misc_feature          1..20
                      note = Synthetic Polynucleotide
source                1..20
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 108
gggtttgatg tccctgtgca                                              20

SEQ ID NO: 109        moltype = RNA  length = 20
FEATURE               Location/Qualifiers
misc_feature          1..20
                      note = Synthetic Polynucleotide
source                1..20
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 109
tgacaatctc cgccaggtag                                              20

SEQ ID NO: 110        moltype = RNA  length = 20
FEATURE               Location/Qualifiers
misc_feature          1..20
                      note = Synthetic Polynucleotide
source                1..20
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 110
cacagcggtc cagcaggatg                                              20

SEQ ID NO: 111        moltype = RNA  length = 20
FEATURE               Location/Qualifiers
misc_feature          1..20
                      note = Synthetic Polynucleotide
source                1..20
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 111
gcgtagaagg gcgtctgccc                                              20

SEQ ID NO: 112        moltype = RNA  length = 20
FEATURE               Location/Qualifiers
misc_feature          1..20
                      note = Synthetic Polynucleotide
source                1..20
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 112
ctcagcctct gccgcaggga                                              20

SEQ ID NO: 113        moltype = RNA  length = 20
FEATURE               Location/Qualifiers
misc_feature          1..20
                      note = Synthetic Polynucleotide
source                1..20
                      mol_type = other RNA
                      organism = synthetic construct
```

-continued

```
SEQUENCE: 113
gtctcagtgc atccaaaacg                                              20

SEQ ID NO: 114          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 114
ggacgatctt gccataggtc                                              20

SEQ ID NO: 115          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 115
tcagcagtgt cagcaggtcc                                              20

SEQ ID NO: 116          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 116
gctcctgggc ggcgccagac                                              20

SEQ ID NO: 117          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 117
agcaggatgt tgtcgggttt                                              20

SEQ ID NO: 118          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 118
atccgctcct gcaactgccg                                              20

SEQ ID NO: 119          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 119
aggagcaggg aaagcgcctc                                              20

SEQ ID NO: 120          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 120
acacctggcc cgtctgcttc                                              20

SEQ ID NO: 121          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
```

-continued

```
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 121
cccagcgccc accagtcaca                                                20

SEQ ID NO: 122          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 122
gctccctctg cctgcagcaa                                                20

SEQ ID NO: 123          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 123
gctcaggctc tgccgggtga                                                20

SEQ ID NO: 124          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 124
ttgatgtccc tgtgcacgta                                                20

SEQ ID NO: 125          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 125
gcctcagcct ctgccgcagg                                                20

SEQ ID NO: 126          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 126
ggtagttctc atcctggaag                                                20

SEQ ID NO: 127          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 127
cagcgcccac cagtcacact                                                20

SEQ ID NO: 128          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 128
cccaaacttg ctcagcagtg                                                20
```

-continued

```
SEQ ID NO: 129          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 129
cttgccatag gtctccgccg                                              20

SEQ ID NO: 130          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 130
tacacctggc ccgtctgctt                                              20

SEQ ID NO: 131          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 131
ccagcgccca ccagtcacac                                              20

SEQ ID NO: 132          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 132
ggcctcagcc tggccgaaag                                              20

SEQ ID NO: 133          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 133
aatctccgcc aggtagaagc                                              20

SEQ ID NO: 134          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 134
atggcataca cctggcccgt                                              20

SEQ ID NO: 135          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 135
ccatgacaat ctccgccagg                                              20

SEQ ID NO: 136          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
```

-continued

```
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 136
tccccaaact tgctcagcag                                                  20

SEQ ID NO: 137          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 137
gatgttgtcg ggtttgatgt                                                  20

SEQ ID NO: 138          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 138
gtttgcccat ccacgtcagg                                                  20

SEQ ID NO: 139          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 139
cggacggccc ggcttgctgc                                                  20

SEQ ID NO: 140          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 140
ctccgccagg tagaagcgcg                                                  20

SEQ ID NO: 141          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 141
gtacaggtag ttctcatcct                                                  20

SEQ ID NO: 142          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 142
agggcgtctg cccatagaac                                                  20

SEQ ID NO: 143          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 143
tggccacagc ggtccagcag                                                  20

SEQ ID NO: 144          moltype = RNA  length = 20
```

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```
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic Polynucleotide
source                   1..20
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 144
cgtagttgac tggcgaagtt                                              20

SEQ ID NO: 145           moltype = RNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic Polynucleotide
source                   1..20
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 145
tctgccgcag ggacagccgc                                              20

SEQ ID NO: 146           moltype = RNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic Polynucleotide
source                   1..20
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 146
aagcgcctcc gataggccag                                              20

SEQ ID NO: 147           moltype = RNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic Polynucleotide
source                   1..20
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 147
gacagaacaa cggcgaacag                                              20

SEQ ID NO: 148           moltype = RNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic Polynucleotide
source                   1..20
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 148
gctcagcagt gtcagcaggt                                              20

SEQ ID NO: 149           moltype = RNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic Polynucleotide
source                   1..20
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 149
atgatcttca tggcatacac                                              20

SEQ ID NO: 150           moltype = RNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic Polynucleotide
source                   1..20
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 150
tttgcccatc cacgtcaggg                                              20

SEQ ID NO: 151           moltype = RNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic Polynucleotide
source                   1..20
                         mol_type = other RNA
                         organism = synthetic construct
```

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```
SEQUENCE: 151
acttgctcag cagtgtcagc                                             20

SEQ ID NO: 152          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 152
tgatgtccct gtgcacgtag                                             20

SEQ ID NO: 153          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 153
aaataccgag gaatgtcggg                                             20

SEQ ID NO: 154          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 154
ggcgaataca cccagcgccc                                             20

SEQ ID NO: 155          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 155
agacaataaa taccgaggaa                                             20

SEQ ID NO: 156          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 156
cccgtctgct tcatcttcac                                             20

SEQ ID NO: 157          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 157
ctgcctgcag caactccatc                                             20

SEQ ID NO: 158          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 158
cctcagcctc tgccgcaggg                                             20

SEQ ID NO: 159          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
```

-continued

```
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 159
gtgtccggaa gtcgcctgct                                                20

SEQ ID NO: 160          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 160
tgcacgtgtg gctcaagcag                                                20

SEQ ID NO: 161          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 161
gacaataaat accgaggaat                                                20

SEQ ID NO: 162          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 162
gccatgacaa tctccgccag                                                20

SEQ ID NO: 163          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 163
gctgtcccgg agaccatccc                                                20

SEQ ID NO: 164          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 164
catgaccagg tacaggtagt                                                20

SEQ ID NO: 165          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 165
agcgcccacc agtcacactc                                                20

SEQ ID NO: 166          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 166
tctcagtgca tccaaaacgt                                                20
```

-continued

```
SEQ ID NO: 167          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 167
tttgggcaga tggagggcct                                              20

SEQ ID NO: 168          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 168
gatgtccctg tgcacgtagc                                              20

SEQ ID NO: 169          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 169
cagcagtgtc agcaggtccc                                              20

SEQ ID NO: 170          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 170
catgacaatc tccgccaggt                                              20

SEQ ID NO: 171          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 171
acttgttcat gatcttcatg                                              20

SEQ ID NO: 172          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 172
gtggaatccg cgtagaaggg                                              20

SEQ ID NO: 173          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 173
tggccatgac aatctccgcc                                              20

SEQ ID NO: 174          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
```

-continued

```
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 174
gggacagaca ataaataccg                                              20

SEQ ID NO: 175          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 175
ccgctcccca aacttgctca                                              20

SEQ ID NO: 176          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 176
cggctcaggc tctgccgggt                                              20

SEQ ID NO: 177          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 177
ggctcctggg cggcgccaga                                              20

SEQ ID NO: 178          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 178
tttcccgagt aagcaggcag                                              20

SEQ ID NO: 179          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 179
ggatgttgtc gggtttgatg                                              20

SEQ ID NO: 180          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 180
caggtagttc tcatcctgga                                              20

SEQ ID NO: 181          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 181
tgcccataga acatttcata                                              20

SEQ ID NO: 182          moltype = RNA  length = 20
```

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```
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 182
tagttctcat cctggaaggc                                              20

SEQ ID NO: 183         moltype = RNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 183
atgtccctgt gcacgtagcc                                              20

SEQ ID NO: 184         moltype = RNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 184
cgggcccgga tcacaggact                                              20

SEQ ID NO: 185         moltype = RNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 185
tggacgatct tgccataggt                                              20

SEQ ID NO: 186         moltype = RNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 186
gttggccggc gtgggccacc                                              20

SEQ ID NO: 187         moltype = RNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 187
ctcagtgcat ccaaaacgtg                                              20

SEQ ID NO: 188         moltype = RNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 188
tcgaagttgc atgtgtcggt                                              20

SEQ ID NO: 189         moltype = RNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other RNA
                       organism = synthetic construct
```

-continued

```
SEQUENCE: 189
tggaacacgg acggcccggc                                              20

SEQ ID NO: 190          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 190
ccgagagcag cgcaagtgag                                              20

SEQ ID NO: 191          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 191
tcctgcaact gccggacgtg                                              20

SEQ ID NO: 192          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 192
tcaccaacac gtccctctcc                                              20

SEQ ID NO: 193          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 193
tgcctgcagc aactccatcc                                              20

SEQ ID NO: 194          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 194
ttggccggcg tgggccacca                                              20

SEQ ID NO: 195          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 195
gagcctctgc ctcgcgtagt                                              20

SEQ ID NO: 196          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 196
aagggcgtct gcccatagaa                                              20

SEQ ID NO: 197          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
```

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```
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 197
acagacaata aataccgagg                                          20

SEQ ID NO: 198          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 198
ggacagacaa taaataccga                                          20

SEQ ID NO: 199          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 199
acgtgtgcct ctaggtcccg                                          20

SEQ ID NO: 200          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 200
ggcacgagac agaacaacgg                                          20

SEQ ID NO: 201          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 201
tgaccaggta caggtagttc                                          20

SEQ ID NO: 202          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 202
ctctgccggg tgagcacctc                                          20

SEQ ID NO: 203          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 203
gacaatctcc gccaggtaga                                          20

SEQ ID NO: 204          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 204
tctccgccag gtagaagcgc                                          20
```

-continued

```
SEQ ID NO: 205          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 205
ctctgcctcg cgtagttgac                                              20

SEQ ID NO: 206          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 206
ctttgggcag atggagggcc                                              20

SEQ ID NO: 207          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 207
acaggtagtt ctcatcctgg                                              20

SEQ ID NO: 208          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 208
ccaaacttgc tcagcagtgt                                              20

SEQ ID NO: 209          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 209
tcgggtttga tgtccctgtg                                              20

SEQ ID NO: 210          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 210
ggcttgctgc cttcccaggc                                              20

SEQ ID NO: 211          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 211
tacaggtagt tctcatcctg                                              20

SEQ ID NO: 212          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
```

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```
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 212
ttgcccatcc acgtcagggc                                          20

SEQ ID NO: 213          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 213
aggtacaggt agttctcatc                                          20

SEQ ID NO: 214          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 214
gacagacaat aaataccgag                                          20

SEQ ID NO: 215          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 215
tagaacattt cataggcgaa                                          20

SEQ ID NO: 216          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 216
agggcctttt attcgcgagg                                          20

SEQ ID NO: 217          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 217
gcctcgcgta gttgactggc                                          20

SEQ ID NO: 218          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 218
ccagcaggat gttgtcgggt                                          20

SEQ ID NO: 219          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 219
gtagttgact ggcgaagttc                                          20

SEQ ID NO: 220          moltype = RNA  length = 20
```

-continued

```
FEATURE               Location/Qualifiers
misc_feature          1..20
                      note = Synthetic Polynucleotide
source                1..20
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 220
tgcggatggc ctccatctcc                                              20

SEQ ID NO: 221        moltype = RNA  length = 20
FEATURE               Location/Qualifiers
misc_feature          1..20
                      note = Synthetic Polynucleotide
source                1..20
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 221
acaatctccg ccaggtagaa                                              20

SEQ ID NO: 222        moltype = RNA  length = 20
FEATURE               Location/Qualifiers
misc_feature          1..20
                      note = Synthetic Polynucleotide
source                1..20
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 222
gcgaatacac ccagcgccca                                              20

SEQ ID NO: 223        moltype = RNA  length = 20
FEATURE               Location/Qualifiers
misc_feature          1..20
                      note = Synthetic Polynucleotide
source                1..20
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 223
gtagttctca tcctggaagg                                              20

SEQ ID NO: 224        moltype = RNA  length = 20
FEATURE               Location/Qualifiers
misc_feature          1..20
                      note = Synthetic Polynucleotide
source                1..20
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 224
ggctcaggct ctgccgggtg                                              20

SEQ ID NO: 225        moltype = RNA  length = 20
FEATURE               Location/Qualifiers
misc_feature          1..20
                      note = Synthetic Polynucleotide
source                1..20
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 225
ccattcacca acacgtccct                                              20

SEQ ID NO: 226        moltype = RNA  length = 20
FEATURE               Location/Qualifiers
misc_feature          1..20
                      note = Synthetic Polynucleotide
source                1..20
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 226
accaggtaca ggtagttctc                                              20

SEQ ID NO: 227        moltype = RNA  length = 20
FEATURE               Location/Qualifiers
misc_feature          1..20
                      note = Synthetic Polynucleotide
source                1..20
                      mol_type = other RNA
                      organism = synthetic construct
```

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```
SEQUENCE: 227
ctgcagtttg cccatccacg                                              20

SEQ ID NO: 228          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 228
ttgttcatga tcttcatggc                                              20

SEQ ID NO: 229          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 229
tttgatgtcc ctgtgcacgt                                              20

SEQ ID NO: 230          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 230
gcggtccagc aggatgttgt                                              20

SEQ ID NO: 231          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 231
gtctatggcc atgacaatct                                              20

SEQ ID NO: 232          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 232
ggagcaggga aagcgcctcc                                              20

SEQ ID NO: 233          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 233
tgcctcgcgt agttgactgg                                              20

SEQ ID NO: 234          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 234
gcggatggcc tccatctccc                                              20

SEQ ID NO: 235          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
```

-continued

```
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 235
tttcataggc gaatacaccc                                              20

SEQ ID NO: 236          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 236
gcctgtcagc gagtcggagg                                              20

SEQ ID NO: 237          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 237
ccacttcagc tgtttcatcc                                              20

SEQ ID NO: 238          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 238
catccgctcc tgcaactgcc                                              20

SEQ ID NO: 239          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 239
tctagggttc agggagcgcg                                              20

SEQ ID NO: 240          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 240
caccaacacg tccctctcct                                              20

SEQ ID NO: 241          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 241
caggagcagg gaaagcgcct                                              20

SEQ ID NO: 242          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 242
caatctccgc caggtagaag                                              20
```

-continued

```
SEQ ID NO: 243          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 243
atgttgtcgg gtttgatgtc                                                20

SEQ ID NO: 244          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 244
ccatccgctc ctgcaactgc                                                20

SEQ ID NO: 245          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 245
gcgtcacctc ggcctcagcc                                                20

SEQ ID NO: 246          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 246
gagggccttt tattcgcgag                                                20

SEQ ID NO: 247          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 247
agcggcagag agaggtgctc                                                20

SEQ ID NO: 248          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 248
catccaaaac gtggattggg                                                20

SEQ ID NO: 249          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 249
ttgggcagat ggagggcctt                                                20

SEQ ID NO: 250          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
```

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```
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 250
cctctgcctc gcgtagttga                                              20

SEQ ID NO: 251          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 251
acagaacaac ggcgaacagg                                              20

SEQ ID NO: 252          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 252
caggatgttg tcgggtttga                                              20

SEQ ID NO: 253          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 253
cggcctcagc ctctgccgca                                              20

SEQ ID NO: 254          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 254
cagcaggatg ttgtcgggtt                                              20

SEQ ID NO: 255          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 255
gcagagagag gtgctccttg                                              20

SEQ ID NO: 256          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 256
tccagttcca tgggtgtggg                                              20

SEQ ID NO: 257          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 257
cctcagcctg gccgaaagaa                                              20

SEQ ID NO: 258          moltype = RNA  length = 20
```

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```
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 258
gggcctttta ttcgcgaggg                                        20

SEQ ID NO: 259         moltype = RNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 259
gtcggccagg cggatgtggc                                        20

SEQ ID NO: 260         moltype = RNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 260
gcttgctgcc ttcccaggcc                                        20

SEQ ID NO: 261         moltype = RNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 261
ggtccagcag gatgttgtcg                                        20

SEQ ID NO: 262         moltype = RNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 262
cggagaccat cccagtcgag                                        20

SEQ ID NO: 263         moltype = RNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 263
tctgcctcgc gtagttgact                                        20

SEQ ID NO: 264         moltype = RNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 264
aggtagttct catcctggaa                                        20

SEQ ID NO: 265         moltype = RNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other RNA
                       organism = synthetic construct
```

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```
SEQUENCE: 265
tccttgtagt ggacgatctt                                          20

SEQ ID NO: 266          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 266
gcatccaaaa cgtggattgg                                          20

SEQ ID NO: 267          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 267
gtccagcagg atgttgtcgg                                          20

SEQ ID NO: 268          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 268
agctcccgca gcgtcacctc                                          20

SEQ ID NO: 269          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 269
cgagagcagc gcaagtgagg                                          20

SEQ ID NO: 270          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 270
cagggaaagc gcctccgata                                          20

SEQ ID NO: 271          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 271
atttcatagg cgaatacacc                                          20

SEQ ID NO: 272          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 272
tcggccaggc ggatgtggcc                                          20

SEQ ID NO: 273          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
```

-continued

```
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 273
aagggatgtg tccggaagtc                                              20

SEQ ID NO: 274          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 274
cttgtagtgg acgatcttgc                                              20

SEQ ID NO: 275          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 275
agtcggccag gcggatgtgg                                              20

SEQ ID NO: 276          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 276
gcctcagcct ggccgaaaga                                              20

SEQ ID NO: 277          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 277
agcgtcacct cggcctcagc                                              20

SEQ ID NO: 278          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 278
cagcggcaga gagaggtgct                                              20

SEQ ID NO: 279          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 279
ccagcggcag agagaggtgc                                              20

SEQ ID NO: 280          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 280
ttgtagtgga cgatcttgcc                                              20
```

-continued

```
SEQ ID NO: 281          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 281
agggaaagcg cctccgatag                                              20

SEQ ID NO: 282          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 282
gggaaagcgc ctccgatagg                                              20

SEQ ID NO: 283          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 283
ggcagcaagc cgggccgtcc                                              20

SEQ ID NO: 284          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 284
cagtcctgtg atccgggccc                                              20

SEQ ID NO: 285          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 285
gacactgctg agcaagtttg                                              20

SEQ ID NO: 286          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 286
tgacactgct gagcaagttt                                              20

SEQ ID NO: 287          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 287
cgggagatgg aggccatccg                                              20

SEQ ID NO: 288          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
```

-continued

```
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 288
gggagcggat tccggccgag                                               20

SEQ ID NO: 289          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 289
ggagcggatt ccggccgaga                                               20

SEQ ID NO: 290          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 290
cctgctgaca ctgctgagca                                               20

SEQ ID NO: 291          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 291
agggacatca aacccgacaa                                               20

SEQ ID NO: 292          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 292
gggacatcaa acccgacaac                                               20

SEQ ID NO: 293          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 293
gcgcgcttct acctggcgga                                               20

SEQ ID NO: 294          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 294
cgggccaggt gtatgccatg                                               20

SEQ ID NO: 295          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 295
ctgacactgc tgagcaagtt                                               20

SEQ ID NO: 296          moltype = DNA  length = 20
```

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```
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 296
ggcggtggat cacgcagctg                                              20

SEQ ID NO: 297         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 297
gagacactgt cggacattcg                                              20

SEQ ID NO: 298         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 298
acatccgcct ggccgacttc                                              20

SEQ ID NO: 299         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 299
cagggacatc aaacccgaca                                              20

SEQ ID NO: 300         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 300
ccgggagatg gaggccatcc                                              20

SEQ ID NO: 301         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 301
atcaaacccg acaacatcct                                              20

SEQ ID NO: 302         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 302
acagggacat caaacccgac                                              20

SEQ ID NO: 303         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
```

-continued

```
SEQUENCE: 303
gtacctggtc atggagtatt                                         20

SEQ ID NO: 304          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 304
ccatgaagat catgaacaag                                         20

SEQ ID NO: 305          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 305
ccacgttttg gatgcactga                                         20

SEQ ID NO: 306          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 306
tgggatggtc tccgggacag                                         20

SEQ ID NO: 307          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 307
gacagtgagg tcccaggccc                                         20

SEQ ID NO: 308          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 308
gtcatggagt attacgtggg                                         20

SEQ ID NO: 309          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 309
cggctgtccc tgcggcagag                                         20

SEQ ID NO: 310          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 310
cggcttggct acgtgcacag                                         20

SEQ ID NO: 311          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
```

-continued

```
                         note = Synthetic Polynucleotide
source                   1..20
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 311
ggccctgacg tggatgggca                                                  20

SEQ ID NO: 312           moltype = DNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic Polynucleotide
source                   1..20
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 312
cctggcctat cggaggcgct                                                  20

SEQ ID NO: 313           moltype = DNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic Polynucleotide
source                   1..20
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 313
ccggcttggc tacgtgcaca                                                  20

SEQ ID NO: 314           moltype = DNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic Polynucleotide
source                   1..20
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 314
agaactacct gtacctggtc                                                  20

SEQ ID NO: 315           moltype = DNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic Polynucleotide
source                   1..20
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 315
gcggattccg gccgagatgg                                                  20

SEQ ID NO: 316           moltype = DNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic Polynucleotide
source                   1..20
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 316
agcggattcc ggccgagatg                                                  20

SEQ ID NO: 317           moltype = DNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic Polynucleotide
source                   1..20
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 317
acggcggaga cctatggcaa                                                  20

SEQ ID NO: 318           moltype = DNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic Polynucleotide
source                   1..20
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 318
acatcctgct ggaccgctgt                                                  20
```

-continued

```
SEQ ID NO: 319          moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 319
tggcctatcg gaggcgcttt                                        20

SEQ ID NO: 320          moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 320
cacatccctt cttctttggc                                        20

SEQ ID NO: 321          moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 321
tggtcatgga gtattacgtg                                        20

SEQ ID NO: 322          moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 322
gagcggattc cggccgagat                                        20

SEQ ID NO: 323          moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 323
agcggtagtg aagatgaagc                                        20

SEQ ID NO: 324          moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 324
cggattccgg ccgagatggc                                        20

SEQ ID NO: 325          moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 325
agttccagcg gctgtccctg                                        20

SEQ ID NO: 326          moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
```

-continued

```
                          mol_type = other DNA
                          organism = synthetic construct
SEQUENCE: 326
tacctggcgg agattgtcat                                              20

SEQ ID NO: 327            moltype = DNA  length = 20
FEATURE                   Location/Qualifiers
misc_feature              1..20
                          note = Synthetic Polynucleotide
source                    1..20
                          mol_type = other DNA
                          organism = synthetic construct
SEQUENCE: 327
tggcggagat tgtcatggcc                                              20

SEQ ID NO: 328            moltype = DNA  length = 20
FEATURE                   Location/Qualifiers
misc_feature              1..20
                          note = Synthetic Polynucleotide
source                    1..20
                          mol_type = other DNA
                          organism = synthetic construct
SEQUENCE: 328
tgtacctggt catggagtat                                              20

SEQ ID NO: 329            moltype = DNA  length = 20
FEATURE                   Location/Qualifiers
misc_feature              1..20
                          note = Synthetic Polynucleotide
source                    1..20
                          mol_type = other DNA
                          organism = synthetic construct
SEQUENCE: 329
caactacgcg aggcagaggc                                              20

SEQ ID NO: 330            moltype = DNA  length = 20
FEATURE                   Location/Qualifiers
misc_feature              1..20
                          note = Synthetic Polynucleotide
source                    1..20
                          mol_type = other DNA
                          organism = synthetic construct
SEQUENCE: 330
ggagacactg tcggacattc                                              20

SEQ ID NO: 331            moltype = DNA  length = 20
FEATURE                   Location/Qualifiers
misc_feature              1..20
                          note = Synthetic Polynucleotide
source                    1..20
                          mol_type = other DNA
                          organism = synthetic construct
SEQUENCE: 331
agctgcgggc agatggaacg                                              20

SEQ ID NO: 332            moltype = DNA  length = 20
FEATURE                   Location/Qualifiers
misc_feature              1..20
                          note = Synthetic Polynucleotide
source                    1..20
                          mol_type = other DNA
                          organism = synthetic construct
SEQUENCE: 332
caagatcgtc cactacaagg                                              20

SEQ ID NO: 333            moltype = DNA  length = 20
FEATURE                   Location/Qualifiers
misc_feature              1..20
                          note = Synthetic Polynucleotide
source                    1..20
                          mol_type = other DNA
                          organism = synthetic construct
SEQUENCE: 333
cgcttctacc tggcggagat                                              20

SEQ ID NO: 334            moltype = DNA  length = 20
```

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```
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 334
ctcacccggc agagcctgag                                              20

SEQ ID NO: 335          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 335
gcggtagtga agatgaagca                                              20

SEQ ID NO: 336          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 336
caaacccgac aacatcctgc                                              20

SEQ ID NO: 337          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 337
ctgcggcaga ggctgaggcc                                              20

SEQ ID NO: 338          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 338
ggacatcaaa cccgacaaca                                              20

SEQ ID NO: 339          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 339
ggtcatggag tattacgtgg                                              20

SEQ ID NO: 340          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 340
gctgcgggca gatggaacgg                                              20

SEQ ID NO: 341          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
```

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```
SEQUENCE: 341
tctgcctgct tactcgggaa                                            20

SEQ ID NO: 342          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 342
ggtgtatgcc atgaagatca                                            20

SEQ ID NO: 343          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 343
cagttccagc ggctgtccct                                            20

SEQ ID NO: 344          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 344
tgcacaggga catcaaaccc                                            20

SEQ ID NO: 345          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 345
ctacctggcg gagattgtca                                            20

SEQ ID NO: 346          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 346
catcctgctg gaccgctgtg                                            20

SEQ ID NO: 347          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 347
gggcagacgc ccttctacgc                                            20

SEQ ID NO: 348          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 348
tccctgcggc agaggctgag                                            20

SEQ ID NO: 349          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
```

-continued

```
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 349
cgttttggat gcactgagac                                                  20

SEQ ID NO: 350          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 350
gacctatggc aagatcgtcc                                                  20

SEQ ID NO: 351          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 351
ggacctgctg acactgctga                                                  20

SEQ ID NO: 352          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 352
gtctggcgcc gcccaggagc                                                  20

SEQ ID NO: 353          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 353
aaacccgaca acatcctgct                                                  20

SEQ ID NO: 354          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 354
cggcagttgc aggagcggat                                                  20

SEQ ID NO: 355          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 355
gaggcgcttt ccctgctcct                                                  20

SEQ ID NO: 356          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 356
gaagcagacg ggccaggtgt                                                  20
```

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```
SEQ ID NO: 357          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 357
tgtgactggt gggcgctggg                                                  20

SEQ ID NO: 358          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 358
ttgctgcagg cagagggagc                                                  20

SEQ ID NO: 359          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 359
tcacccggca gagcctgagc                                                  20

SEQ ID NO: 360          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 360
tacgtgcaca gggacatcaa                                                  20

SEQ ID NO: 361          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 361
cctgcggcag aggctgaggc                                                  20

SEQ ID NO: 362          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 362
cttccaggat gagaactacc                                                  20

SEQ ID NO: 363          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 363
agtgtgactg gtgggcgctg                                                  20

SEQ ID NO: 364          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
```

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```
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 364
cactgctgag caagtttggg                                                20

SEQ ID NO: 365          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 365
cggcggagac ctatggcaag                                                20

SEQ ID NO: 366          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 366
aagcagacgg gccaggtgta                                                20

SEQ ID NO: 367          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 367
gtgtgactgg tgggcgctgg                                                20

SEQ ID NO: 368          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 368
ctttcggcca ggctgaggcc                                                20

SEQ ID NO: 369          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 369
gcttctacct ggcggagatt                                                20

SEQ ID NO: 370          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 370
acgggccagg tgtatgccat                                                20

SEQ ID NO: 371          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 371
cctggcggag attgtcatgg                                                20

SEQ ID NO: 372          moltype = DNA  length = 20
```

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```
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 372
ctgctgagca agtttgggga                                      20

SEQ ID NO: 373          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 373
acatcaaacc cgacaacatc                                      20

SEQ ID NO: 374          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 374
cctgacgtgg atgggcaaac                                      20

SEQ ID NO: 375          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 375
gcagcaagcc gggccgtccg                                      20

SEQ ID NO: 376          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 376
cgcgcttcta cctggcggag                                      20

SEQ ID NO: 377          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 377
aggatgagaa ctacctgtac                                      20

SEQ ID NO: 378          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 378
gttctatggg cagacgccct                                      20

SEQ ID NO: 379          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
```

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```
SEQUENCE: 379
ctgctggacc gctgtggcca                                              20

SEQ ID NO: 380          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 380
aacttcgcca gtcaactacg                                              20

SEQ ID NO: 381          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 381
gcggctgtcc ctgcggcaga                                              20

SEQ ID NO: 382          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 382
ctggcctatc ggaggcgctt                                              20

SEQ ID NO: 383          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 383
ctgttcgccg ttgttctgtc                                              20

SEQ ID NO: 384          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 384
acctgctgac actgctgagc                                              20

SEQ ID NO: 385          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 385
gtgtatgcca tgaagatcat                                              20

SEQ ID NO: 386          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 386
ccctgacgtg gatgggcaaa                                              20

SEQ ID NO: 387          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
```

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```
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 387
gctgacactg ctgagcaagt                                                  20

SEQ ID NO: 388          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 388
ctacgtgcac agggacatca                                                  20

SEQ ID NO: 389          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 389
cccgacattc ctcggtattt                                                  20

SEQ ID NO: 390          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 390
gggcgctggg tgtattcgcc                                                  20

SEQ ID NO: 391          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 391
ttcctcggta tttattgtct                                                  20

SEQ ID NO: 392          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 392
gtgaagatga agcagacggg                                                  20

SEQ ID NO: 393          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 393
gatggagttg ctgcaggcag                                                  20

SEQ ID NO: 394          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 394
ccctgcggca gaggctgagg                                                  20
```

-continued

```
SEQ ID NO: 395          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 395
agcaggcgac ttccggacac                                              20

SEQ ID NO: 396          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 396
ctgcttgagc cacacgtgca                                              20

SEQ ID NO: 397          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 397
attcctcggt atttattgtc                                              20

SEQ ID NO: 398          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 398
ctggcggaga ttgtcatggc                                              20

SEQ ID NO: 399          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 399
gggatggtct ccgggacagc                                              20

SEQ ID NO: 400          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 400
actacctgta cctggtcatg                                              20

SEQ ID NO: 401          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 401
gagtgtgact ggtgggcgct                                              20

SEQ ID NO: 402          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
```

-continued

```
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 402
acgttttgga tgcactgaga                                              20

SEQ ID NO: 403          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 403
aggccctcca tctgcccaaa                                              20

SEQ ID NO: 404          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 404
gctacgtgca cagggacatc                                              20

SEQ ID NO: 405          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 405
gggacctgct gacactgctg                                              20

SEQ ID NO: 406          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 406
acctggcgga gattgtcatg                                              20

SEQ ID NO: 407          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 407
catgaagatc atgaacaagt                                              20

SEQ ID NO: 408          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 408
cccttctacg cggattccac                                              20

SEQ ID NO: 409          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 409
ggcggagatt gtcatggcca                                              20

SEQ ID NO: 410          moltype = DNA  length = 20
```

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```
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 410
cggtatttat tgtctgtccc                                            20

SEQ ID NO: 411          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 411
tgagcaagtt tggggagcgg                                            20

SEQ ID NO: 412          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 412
acccggcaga gcctgagccg                                            20

SEQ ID NO: 413          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 413
tctggcgccg cccaggagcc                                            20

SEQ ID NO: 414          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 414
ctgcctgctt actcgggaaa                                            20

SEQ ID NO: 415          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 415
catcaaaccc gacaacatcc                                            20

SEQ ID NO: 416          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 416
tccaggatga gaactacctg                                            20

SEQ ID NO: 417          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
```

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```
SEQUENCE: 417
tatgaaatgt tctatgggca                                          20

SEQ ID NO: 418          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 418
gccttccagg atgagaacta                                          20

SEQ ID NO: 419          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 419
ggctacgtgc acagggacat                                          20

SEQ ID NO: 420          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 420
agtcctgtga tccgggcccg                                          20

SEQ ID NO: 421          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 421
acctatggca agatcgtcca                                          20

SEQ ID NO: 422          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 422
ggtggcccac gccggccaac                                          20

SEQ ID NO: 423          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 423
cacgttttgg atgcactgag                                          20

SEQ ID NO: 424          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 424
accgacacat gcaacttcga                                          20

SEQ ID NO: 425          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
```

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```
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 425
gccgggccgt ccgtgttcca                                                20

SEQ ID NO: 426          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 426
ctcacttgcg ctgctctcgg                                                20

SEQ ID NO: 427          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 427
cacgtccggc agttgcagga                                                20

SEQ ID NO: 428          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 428
ggagagggac gtgttggtga                                                20

SEQ ID NO: 429          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 429
ggatggagtt gctgcaggca                                                20

SEQ ID NO: 430          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 430
tggtggccca cgccggccaa                                                20

SEQ ID NO: 431          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 431
actacgcgag gcagaggctc                                                20

SEQ ID NO: 432          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 432
ttctatgggc agacgccctt                                                20
```

-continued

```
SEQ ID NO: 433          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 433
cctcggtatt tattgtctgt                                              20

SEQ ID NO: 434          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 434
tcggtattta ttgtctgtcc                                              20

SEQ ID NO: 435          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 435
cgggacctag aggcacacgt                                              20

SEQ ID NO: 436          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 436
ccgttgttct gtctcgtgcc                                              20

SEQ ID NO: 437          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 437
gaactacctg tacctggtca                                              20

SEQ ID NO: 438          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 438
gaggtgctca cccggcagag                                              20

SEQ ID NO: 439          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 439
tctacctggc ggagattgtc                                              20

SEQ ID NO: 440          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
```

-continued

```
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 440
gcgcttctac ctggcggaga                                              20

SEQ ID NO: 441           moltype = DNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic Polynucleotide
source                   1..20
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 441
gtcaactacg cgaggcagag                                              20

SEQ ID NO: 442           moltype = DNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic Polynucleotide
source                   1..20
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 442
ggccctccat ctgcccaaag                                              20

SEQ ID NO: 443           moltype = DNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic Polynucleotide
source                   1..20
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 443
ccaggatgag aactacctgt                                              20

SEQ ID NO: 444           moltype = DNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic Polynucleotide
source                   1..20
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 444
acactgctga gcaagtttgg                                              20

SEQ ID NO: 445           moltype = DNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic Polynucleotide
source                   1..20
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 445
cacagggaca tcaaacccga                                              20

SEQ ID NO: 446           moltype = DNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic Polynucleotide
source                   1..20
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 446
gcctgggaag gcagcaagcc                                              20

SEQ ID NO: 447           moltype = DNA  length = 20
FEATURE                  Location/Qualifiers
misc_feature             1..20
                         note = Synthetic Polynucleotide
source                   1..20
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 447
caggatgaga actacctgta                                              20

SEQ ID NO: 448           moltype = DNA  length = 20
```

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```
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 448
gccctgacgt ggatgggcaa                                          20

SEQ ID NO: 449         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 449
gatgagaact acctgtacct                                          20

SEQ ID NO: 450         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 450
ctcggtattt attgtctgtc                                          20

SEQ ID NO: 451         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 451
ttcgcctatg aaatgttcta                                          20

SEQ ID NO: 452         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 452
cctcgcgaat aaaaggccct                                          20

SEQ ID NO: 453         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 453
gccagtcaac tacgcgaggc                                          20

SEQ ID NO: 454         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 454
acccgacaac atcctgctgg                                          20

SEQ ID NO: 455         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
```

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```
SEQUENCE: 455
gaacttcgcc agtcaactac                                              20

SEQ ID NO: 456         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 456
ggagatggag gccatccgca                                              20

SEQ ID NO: 457         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 457
ttctacctgg cggagattgt                                              20

SEQ ID NO: 458         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 458
tgggcgctgg gtgtattcgc                                              20

SEQ ID NO: 459         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 459
ccttccagga tgagaactac                                              20

SEQ ID NO: 460         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 460
cacccggcag agcctgagcc                                              20

SEQ ID NO: 461         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 461
agggacgtgt tggtgaatgg                                              20

SEQ ID NO: 462         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 462
gagaactacc tgtacctggt                                              20

SEQ ID NO: 463         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
```

-continued

```
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 463
cgtggatggg caaactgcag                                            20

SEQ ID NO: 464          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 464
gccatgaaga tcatgaacaa                                            20

SEQ ID NO: 465          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 465
acgtgcacag ggacatcaaa                                            20

SEQ ID NO: 466          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 466
acaacatcct gctggaccgc                                            20

SEQ ID NO: 467          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 467
agattgtcat ggccatagac                                            20

SEQ ID NO: 468          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 468
ggaggcgctt tccctgctcc                                            20

SEQ ID NO: 469          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 469
ccagtcaact acgcgaggca                                            20

SEQ ID NO: 470          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 470
gggagatgga ggccatccgc                                            20
```

-continued

```
SEQ ID NO: 471          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 471
gggtgtattc gcctatgaaa                                              20

SEQ ID NO: 472          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 472
cctccgactc gctgacaggc                                              20

SEQ ID NO: 473          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 473
ggatgaaaca gctgaagtgg                                              20

SEQ ID NO: 474          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 474
ggcagttgca ggagcggatg                                              20

SEQ ID NO: 475          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 475
cgcgctccct gaaccctaga                                              20

SEQ ID NO: 476          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 476
aggagaggga cgtgttggtg                                              20

SEQ ID NO: 477          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 477
aggcgctttc cctgctcctg                                              20

SEQ ID NO: 478          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
```

-continued

```
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 478
cttctacctg gcggagattg                                            20

SEQ ID NO: 479          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 479
gacatcaaac ccgacaacat                                            20

SEQ ID NO: 480          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 480
gcagttgcag gagcggatgg                                            20

SEQ ID NO: 481          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 481
ggctgaggcc gaggtgacgc                                            20

SEQ ID NO: 482          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 482
ctcgcgaata aaaggccctc                                            20

SEQ ID NO: 483          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 483
gagcacctct ctctgccgct                                            20

SEQ ID NO: 484          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 484
cccaatccac gttttggatg                                            20

SEQ ID NO: 485          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 485
aaggccctcc atctgcccaa                                            20

SEQ ID NO: 486          moltype = DNA  length = 20
```

-continued

```
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 486
tcaactacgc gaggcagagg                                          20

SEQ ID NO: 487         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 487
cctgttcgcc gttgttctgt                                          20

SEQ ID NO: 488         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 488
tcaaacccga caacatcctg                                          20

SEQ ID NO: 489         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 489
tgcggcagag gctgaggccg                                          20

SEQ ID NO: 490         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 490
aacccgacaa catcctgctg                                          20

SEQ ID NO: 491         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 491
caaggagcac ctctctctgc                                          20

SEQ ID NO: 492         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 492
cccacaccca tggaactgga                                          20

SEQ ID NO: 493         moltype = DNA  length = 20
FEATURE                Location/Qualifiers
misc_feature           1..20
                       note = Synthetic Polynucleotide
source                 1..20
                       mol_type = other DNA
                       organism = synthetic construct
```

-continued

```
SEQUENCE: 493
ttctttcggc caggctgagg                                          20

SEQ ID NO: 494          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 494
ccctcgcgaa taaaaggccc                                          20

SEQ ID NO: 495          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 495
gccacatccg cctggccgac                                          20

SEQ ID NO: 496          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 496
ggcctgggaa ggcagcaagc                                          20

SEQ ID NO: 497          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 497
cgacaacatc ctgctggacc                                          20

SEQ ID NO: 498          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 498
ctcgactggg atggtctccg                                          20

SEQ ID NO: 499          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 499
agtcaactac gcgaggcaga                                          20

SEQ ID NO: 500          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 500
ttccaggatg agaactacct                                          20

SEQ ID NO: 501          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
```

-continued

```
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 501
aagatcgtcc actacaagga                                                20

SEQ ID NO: 502          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 502
ccaatccacg ttttggatgc                                                20

SEQ ID NO: 503          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 503
ccgacaacat cctgctggac                                                20

SEQ ID NO: 504          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 504
gaggtgacgc tgcgggagct                                                20

SEQ ID NO: 505          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 505
cctcacttgc gctgctctcg                                                20

SEQ ID NO: 506          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 506
tatcggaggc gctttccctg                                                20

SEQ ID NO: 507          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 507
ggtgtattcg cctatgaaat                                                20

SEQ ID NO: 508          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 508
ggccacatcc gcctggccga                                                20
```

-continued

```
SEQ ID NO: 509          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 509
gacttccgga cacatccctt                                              20

SEQ ID NO: 510          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 510
gcaagatcgt ccactacaag                                              20

SEQ ID NO: 511          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 511
ccacatccgc ctggccgact                                              20

SEQ ID NO: 512          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 512
tctttcggcc aggctgaggc                                              20

SEQ ID NO: 513          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 513
gctgaggccg aggtgacgct                                              20

SEQ ID NO: 514          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 514
agcacctctc tctgccgctg                                              20

SEQ ID NO: 515          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 515
gcacctctct ctgccgctgg                                              20

SEQ ID NO: 516          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
```

-continued

```
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 516
ggcaagatcg tccactacaa                                            20

SEQ ID NO: 517          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 517
ctatcggagg cgctttccct                                            20

SEQ ID NO: 518          moltype = DNA  length = 20
FEATURE                 Location/Qualifiers
misc_feature            1..20
                        note = Synthetic Polynucleotide
source                  1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 518
cctatcggag gcgctttccc                                            20

SEQ ID NO: 519          moltype = DNA  length = 1275
FEATURE                 Location/Qualifiers
source                  1..1275
                        mol_type = genomic DNA
                        organism = Homo sapiens
SEQUENCE: 519
atggccctcc cgacaccctc ggacagcacc ctccccgcgg aagcccgggg acgaggacgg  60
cgacggagac tcgtttggac cccgagccaa agcgaggccc tgcgagcctg ctttgagcgg  120
aacccgtacc cgggcatcgc caccagagaa cggctggccc aggccatcgg cattccggag  180
cccagggtcc agatttggtt tcagaatgag aggtcacgcc agctgaggca gcaccggcgg  240
gaatctcggc cctggcccgg gagacgcggc ccgccagaag gccggcgaaa gcggaccgcc  300
gtcaccggat cccagaccgc cctgctcctc cgagcctttg agaaggatcg ctttccaggc  360
atcgccgccc gggaggagct ggccagagag acgggcctcc cggagtccag gattcagatc  420
tggtttcaga atcgaagggc caggcacccg ggacagggtg gcagggcgcc cgcgcaggca  480
ggcggcctgt gcagcgcggc ccccggcggg ggtcaccctg ctccctcgtg ggtcgccttc  540
gcccacaccg gcgcgtgggg aacggggctt cccgcacccc acgtgccctg cgcgcctggg  600
gctctcccac agggggcttt cgtgagccag gcagcgaggg ccgcccccgc gctgcagccc  660
agccaggccg cgccggcaga ggggatctcc caacctgccc cggcgcgcgg ggatttcgcc  720
tacgccgccc cggctcctcc ggacggggcg ctctcccacc ctcaggctcc tcggtggcct  780
ccgcacccgg gcaaaagccg ggaggaccgg gacccgcagc gcgacggcct gccgggcccc  840
tgcgcggtgg cacagcctgg gcccgctcaa gcggggccgc agggccaagg ggtgcttgcg  900
ccacccacgt cccaggggag tccgtggtgg ggctggggcc ggggtcccca ggtcgccggg  960
gcggcgtggg aaccccaagc cggggcagct ccacctcccc agcccgcgcc cccggacgcc  1020
tccgcctccg cgcggcaggg gcagatgcaa ggcatcccgg cgccctccca ggcgctccag  1080
gagccggcgc cctggtctgc actcccctgc ggcctgctgc tggatgagct cctggcgagc  1140
ccggagtttc tgcagcaggc gcaacctctc ctagaaacgg aggccccggg ggagctggag  1200
gcctcggaag aggccgcctc gctggaagca cccctcagcg aggaagaata ccgggctctg  1260
ctggaggagc tttag                                                 1275

SEQ ID NO: 520          moltype = DNA  length = 2024
FEATURE                 Location/Qualifiers
source                  1..2024
                        mol_type = genomic DNA
                        organism = Mus musculus
SEQUENCE: 520
atggcagaag ctggcagccc tgttggtggc agtggtgtgg cacgggaatc ccggcggcgc  60
aggaagacgg tttggcaggc ctggcaagag caggccctgc tatcaacttt caagaagaag  120
agatacctga gcttcaagga gaggaaggag ctggccaagc gaatgggggt ctcagattgc  180
cgcatccgcg tgtggtttca gaaccgcagg aatcgcagtg gagaggaggg gcatgcctca  240
aagaggtcca tcagaggctc caggcggcta gcctcgccac agctccagga agagcttgga  300
tccaggccac agggtagagg catgcgctca tctggcagaa ggcctcgcac tcgactcacc  360
tcgctacagc tcaggatcct agggcaagcc tttgagagga acccacgacc aggctttgct  420
accagggagg agctggcgcg tgacacaggg ttgcccgagg acacgatcca catatggttt  480
caaaaccgaa gagctcggcg gcgccacagg aggggcaggc ccacagctca agatcaagac  540
ttgctggcgt cacaagggtc ggatggggcc cctgcaggtc cggaaggcag agagcgtgaa  600
ggtgcccagg agaacttgtt gccacaggaa gaagcaggaa gtacgggcat ggatacctcg  660
agccctagcg acttgccctc cttctgcgga gagtcccagc ctttccaagt ggcacagccc  720
cgtggagcag gccaacaaga ggcccccact cgagcaggca acgcaggctc tctggaaccc  780
ctccttgatc agctgctgga tgaagtccaa gtagaagagc ctgctccagc ccctctgaat  840
ttggatggag accctggtgg cagggtgcat gaaggttccc aggagagctt ttggccacag  900
gaagaagcag gaagtacagg catggatact tctagcccca gcgactcaaa ctccttctgc  960
```

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agagagtccc agccttccca agtggcacag ccctgtggag cgggccaaga agatgcccgc  1020
actcaagcag acagcacagg ccctctggaa ctcctcctcc ttgatcaact gctggacgaa  1080
gtccaaaagg aagagcatgt gccagtccca ctggattggg gtagaaatcc tggcagcagg  1140
gagcatgaag gttcccagga cagcttactg cccctggagg aagcagtaaa ttcgggcatg  1200
gatacctcga tccctagcat ctggccaacc ttctgcagag aatcccagcc tccccaagtg  1260
gcacagccct ctggaccagg ccaagcacag gcccccactc aaggtgggaa cacggacccc  1320
ctggagctct tcctctatca actgttggat gaagtccaag tagaagagca tgctccagcc  1380
cctctgaatt gggatgtaga tcctggtggc agggtgcatg aaggttcgtg ggagagcttt  1440
tggccacagg aagaagcagg aagtacaggc ctggatactt caagccccag cgactcaaac  1500
tccttcttca gagagtccaa gccttcccaa gtggcacagc gccgtggagc gggccaagaa  1560
gatgcccgca ctcaagcaga cagcacaggc cctctggaac tcctcctctt tgatcaactg  1620
ctggacgaag tccaaaagga agagcatgtg ccagccccac tggattgggg tagaaatcct  1680
ggcagcatgg agcatgaagg ttcccaggac agcttactgc cctggaggga agcagcaaat  1740
tcgggcaggg atacctcgat ccctagcatc tggccagcct tctgcagaaa atcccagcct  1800
ccccaagtgg cacagccctc tggaccaggc caagcacagg cccccattca aggtgggaac  1860
acggaccccc tggagctctt ccttgatcaa ctgctgaccg aagtccaact tgaggagcag  1920
gggcctgccc ctgtgaatgt ggaggaaaca tgggagcaaa tggacacaac acctatctgc  1980
ctctcacttc agaagaatat cagactcttc tagatatgct ctga                    2024

SEQ ID NO: 521         moltype = DNA  length = 25
FEATURE                Location/Qualifiers
misc_feature           1..25
                       note = Synthetic Polynucleotide
source                 1..25
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 521
gggcatttta atatatctct gaact                                         25

SEQ ID NO: 522         moltype = DNA  length = 25
FEATURE                Location/Qualifiers
misc_feature           1..25
                       note = Synthetic Polynucleotide
source                 1..25
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 522
agttcagaga tatattaaaa tgccc                                         25

SEQ ID NO: 523         moltype = DNA  length = 25
FEATURE                Location/Qualifiers
misc_feature           1..25
                       note = Synthetic
source                 1..25
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 523
cctcttacct cagttacaat ttata                                         25

SEQ ID NO: 524         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = Synthetic
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
modified_base          1
                       mod_base = OTHER
                       note = modified by 2prime Fluoro
modified_base          2
                       mod_base = cm
modified_base          3
                       mod_base = OTHER
                       note = modified by 2prime Fluoro
modified_base          4
                       mod_base = um
modified_base          5
                       mod_base = OTHER
                       note = modified by 2prime Fluoro
modified_base          6
                       mod_base = um
modified_base          7
                       mod_base = OTHER
                       note = modified by 2prime Fluoro
modified_base          8
                       mod_base = OTHER
```

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```
                        note = modified by 2prime OMe
modified_base           9
                        mod_base = OTHER
                        note = modified by 2prime Fluoro
modified_base           10
                        mod_base = um
modified_base           11
                        mod_base = OTHER
                        note = modified by 2prime Fluoro
modified_base           12
                        mod_base = um
modified_base           13
                        mod_base = OTHER
                        note = modified by 2prime Fluoro
modified_base           14
                        mod_base = gm
modified_base           15
                        mod_base = OTHER
                        note = modified by 2prime Fluoro
modified_base           16
                        mod_base = um
modified_base           17
                        mod_base = OTHER
                        note = modified by 2prime Fluoro
modified_base           18
                        mod_base = um
modified_base           19
                        mod_base = OTHER
                        note = modified by 2prime Fluoro
modified_base           20
                        mod_base = OTHER
                        note = modified by 2prime OMe
modified_base           21
                        mod_base = OTHER
                        note = modified by 2prime Fluoro
SEQUENCE: 524
tcctatgact gtagatttta t                                          21

SEQ ID NO: 525          moltype = RNA  length = 23
FEATURE                 Location/Qualifiers
misc_feature            1..23
                        note = Synthetic
source                  1..23
                        mol_type = other RNA
                        organism = synthetic construct
modified_base           1
                        mod_base = OTHER
                        note = modified by 2prime OMe
modified_base           2
                        mod_base = OTHER
                        note = modified by 2prime Fluoro
modified_base           3
                        mod_base = OTHER
                        note = modified by 2prime OMe
modified_base           4
                        mod_base = OTHER
                        note = modified by 2prime Fluoro
modified_base           5
                        mod_base = OTHER
                        note = modified by 2prime OMe
modified_base           6
                        mod_base = OTHER
                        note = modified by 2prime Fluoro
modified_base           7
                        mod_base = um
modified_base           8
                        mod_base = OTHER
                        note = modified by 2prime Fluoro
modified_base           9
                        mod_base = um
modified_base           10
                        mod_base = OTHER
                        note = modified by 2prime Fluoro
modified_base           11
                        mod_base = cm
modified_base           12
```

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```
                      mod_base = OTHER
                      note = modified by 2prime Fluoro
modified_base         13
                      mod_base = gm
modified_base         14
                      mod_base = OTHER
                      note = modified by 2prime Fluoro
modified_base         15
                      mod_base = cm
modified_base         16
                      mod_base = OTHER
                      note = modified by 2prime Fluoro
modified_base         17
                      mod_base = um
modified_base         18
                      mod_base = OTHER
                      note = modified by 2prime Fluoro
modified_base         19
                      mod_base = gm
modified_base         20
                      mod_base = OTHER
                      note = modified by 2prime Fluoro
modified_base         21
                      mod_base = OTHER
                      note = modified by 2prime OMe
modified_base         21..23
                      mod_base = OTHER
                      note = phosphorothioate linkage
modified_base         22
                      mod_base = OTHER
                      note = modified by 2prime Fluoro
modified_base         23
                      mod_base = um
SEQUENCE: 525
ataaaatcta cagtcatagg aat                                          23

SEQ ID NO: 526        moltype = RNA  length = 21
FEATURE               Location/Qualifiers
misc_feature          1..21
                      note = Synthetic
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct
modified_base         1
                      mod_base = OTHER
                      note = modified by 2prime Fluoro
modified_base         2
                      mod_base = gm
modified_base         3
                      mod_base = OTHER
                      note = modified by 2prime Fluoro
modified_base         4
                      mod_base = OTHER
                      note = modified by 2prime OMe
modified_base         5
                      mod_base = OTHER
                      note = modified by 2prime Fluoro
modified_base         6
                      mod_base = um
modified_base         7
                      mod_base = OTHER
                      note = modified by 2prime Fluoro
modified_base         8
                      mod_base = OTHER
                      note = modified by 2prime OMe
modified_base         9
                      mod_base = OTHER
                      note = modified by 2prime Fluoro
modified_base         10
                      mod_base = cm
modified_base         11
                      mod_base = OTHER
                      note = modified by 2prime Fluoro
modified_base         12
                      mod_base = um
modified_base         13
```

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```
                         mod_base = OTHER
                         note = modified by 2prime Fluoro
modified_base            14
                         mod_base = um
modified_base            15
                         mod_base = OTHER
                         note = modified by 2prime Fluoro
modified_base            16
                         mod_base = um
modified_base            17
                         mod_base = OTHER
                         note = modified by 2prime Fluoro
modified_base            18
                         mod_base = cm
modified_base            19
                         mod_base = OTHER
                         note = modified by 2prime Fluoro
modified_base            20
                         mod_base = um
modified_base            21
                         mod_base = OTHER
                         note = modified by 2prime Fluoro
SEQUENCE: 526
tgtaataacc atatctacct t                                                21

SEQ ID NO: 527           moltype = RNA  length = 23
FEATURE                  Location/Qualifiers
misc_feature             1..23
                         note = Synthetic
source                   1..23
                         mol_type = other RNA
                         organism = synthetic construct
modified_base            1
                         mod_base = OTHER
                         note = modified by 2prime OMe
modified_base            2
                         mod_base = OTHER
                         note = modified by 2prime Fluoro
modified_base            3
                         mod_base = gm
modified_base            4
                         mod_base = OTHER
                         note = modified by 2prime Fluoro
modified_base            5
                         mod_base = um
modified_base            6
                         mod_base = OTHER
                         note = modified by 2prime Fluoro
modified_base            7
                         mod_base = gm
modified_base            8
                         mod_base = OTHER
                         note = modified by 2prime Fluoro
modified_base            9
                         mod_base = um
modified_base            10
                         mod_base = OTHER
                         note = modified by 2prime Fluoro
modified_base            11
                         mod_base = um
modified_base            12
                         mod_base = OTHER
                         note = modified by 2prime Fluoro
modified_base            13
                         mod_base = gm
modified_base            14
                         mod_base = OTHER
                         note = modified by 2prime Fluoro
modified_base            15
                         mod_base = um
modified_base            16
                         mod_base = OTHER
                         note = modified by 2prime Fluoro
modified_base            17
                         mod_base = um
modified_base            18
```

-continued

```
                         mod_base = OTHER
                         note = modified by 2prime Fluoro
modified_base            19
                         mod_base = OTHER
                         note = modified by 2prime OMe
modified_base            20
                         mod_base = OTHER
                         note = modified by 2prime Fluoro
modified_base            21
                         mod_base = OTHER
                         note = modified by 2prime OMe
modified_base            21..23
                         mod_base = OTHER
                         note = phosphorothioate linkage
modified_base            22
                         mod_base = OTHER
                         note = modified by 2prime Fluoro
modified_base            23
                         mod_base = OTHER
                         note = modified by 2prime OMe
SEQUENCE: 527
aaggtagata tggttattac aaa                                      23

SEQ ID NO: 528           moltype = DNA  length = 1710
FEATURE                  Location/Qualifiers
source                   1..1710
                         mol_type = genomic DNA
                         organism = Homo sapiens
SEQUENCE: 528
atggccctcc cgacaccctc ggacagcacc ctccccgcgg aagcccgggg acgaggacgg  60
cgacggagac tcgtttggac cccgagccaa agcgaggccc tgcgagcctg ctttgagcgg  120
aacccgtacc cgggcatcgc caccagagaa cggctggccc aggccatcgg cattccggag  180
cccagggtcc agatttggtt tcagaatgag aggtcacgcc agctgaggca gcaccggcgg  240
gaatctcggc cctggcccgg gagacgcggc ccgccagaag gccggcgaaa gcggaccgcc  300
gtcaccggat cccagaccgc cctgctcctc cgagcctttg agaaggatcg ctttccaggc  360
atcgccgccc gggaggagct ggccagagag acgggcctcc cggagtccag gattcagatc  420
tggtttcaga atcgaagggc caggcacccg ggacagggtg gcagggcgcc cgcgcaggca  480
ggcggcctgt gcagcgcggc ccccggcggg ggtcaccctg ctccctcgtg ggtcgccttc  540
gcccacaccg gcgcgtgggg aacggggctt cccgcacccc acgtgccctg cgcgcctggg  600
gctctcccac agggggcttt cgtgagccag gcagcgaggg ccgcccccgc gctgcagccc  660
agccaggccg cgccggcaga ggggatctcc caacctgccc cggcgcgcgg ggatttcgcc  720
tacgccgccc cggctcctcc ggacggggcg ctctcccacc ctcaggctcc tcgctggcct  780
ccgcacccgg gcaaaagccg ggaggaccgg gacccgcagc gcgacggcct gccgggcccc  840
tgcgcggtgg cacagcctgg gcccgctcaa gcggggccgc agggccaagg ggtgcttgcg  900
ccacccacgt cccaggggag tccgtggtgg ggctggggcc ggggtcccca ggtcgccggg  960
gcggcgtggg aaccccaagc cggggcagct ccacctcccc agcccgcgcc cccggacgcc  1020
tccgcctccg cgcggcaggg gcagatgcaa ggcatcccgg cgccctccca ggcgctccag  1080
gagccggcgc cctggtctgc actcccctgc ggcctgctgc tggatgagct cctggcgagc  1140
ccggagtttc tgcagcaggc gcaacctctc ctagaaacgg aggcccccgg ggagctggag  1200
gcctcggaag aggccgcctc gctggaagca cccctcagcg aggaagaata ccgggctctg  1260
ctggaggagc tttaggacgc ggggttggga cggggtcggg tggttcgggg cagggcggtg  1320
gcctctcttt cgcggggaac acctggctgg ctacggaggg gcgtgtctcc gccccgcccc  1380
ctccaccggg ctgaccggcc tgggattcct gccttctagg tctaggcccg gtgagagact  1440
ccacaccgcg gagaactgcc attctttcct gggcatcccg gggatcccag agccggccca  1500
ggtaccagca gacctgcgcg cagtgcgcac cccggctgac gtgcaaggga gctcgctggc  1560
ctctctgtgc ccttgttctt ccgtgaaatt ctggctgaat gtctcccccc accttccgac  1620
gctgtctagg caaacctgga ttagagttac atctcctgga tgattagttc agagatatat  1680
taaaatgccc cctccctgtg gatcctatag                               1710

SEQ ID NO: 529           moltype = AA  length = 248
FEATURE                  Location/Qualifiers
source                   1..248
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 529
DYKDIVMTQS HKFMSTSVGD RVSITCKASQ DVGTAVAWYQ QKPGQSPKLL IYWASTRHTG  60
VPDRFTGSGS GTDFTLTISN VQSEDLADYF CQQYSSYLTF GAGTKLELKR GGGGSGGGGS  120
GGGGSGGGGS EVMLVESGGD LVKPGGSLKL SCAASGFTFS SYGMSWVRQT PDKRLEWVAT  180
ISSGGSYTYY PDSVKGRFTI SRDNAKNTLY LQMSSLKSED TAMYYCALLR SDWYFDVWGA  240
GTTVTVSS                                                       248

SEQ ID NO: 530           moltype = AA  length = 245
FEATURE                  Location/Qualifiers
source                   1..245
                         mol_type = protein
                         organism = synthetic construct
```

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```
SEQUENCE: 530
DYKDIVITQS HKFMSTSVGD RVSITCKASQ DVSTAVAWYQ QKPGQSPKLL IYSASYRYTG  60
VPDRFTGSGS GTDFTFTISS VQAEDLAVYY CQQHYSTPYT FGGGTKLEIK RGGGGSGGGG  120
SGGGGSGGGG SQVQLQQSGA ELVRPGASVT LSCKASGYTF TDYEMHWVKQ TPVHGLEWIG  180
AIDPETGGTA YNQKFKGKAT LTADKSSSTA YMELRSLTSE DSAVYYCTGW GFDYWGQGTT  240
LTVSS                                                             245

SEQ ID NO: 531          moltype = AA  length = 247
FEATURE                 Location/Qualifiers
source                  1..247
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 531
DYKDIVMTQS HKFMSTSVGD RVSITCKASQ DVGTAVAWYQ QKPGQSPKLL IYWASTRHTG  60
VPDRFTGSGS GTDFTLTISN VQSEDLADYF CQQYSSYPYT FGGGTKLEIK RGGGGSGGGG  120
SGGGGSGGGG SQVQLQQPGS ELVRPGASVK LSCKASGYTF TSYWMHWVKQ RPGQGLEWIG  180
NIYPGSGSTN YDEKFKSKAT LTVDTSSSTA YMQLSSLTSE DSAVYYCTRG ATALDYWGQG  240
TTLTVSS                                                           247

SEQ ID NO: 532          moltype = AA  length = 131
FEATURE                 Location/Qualifiers
source                  1..131
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 532
QVQLQESGPG LVKPSETLSL TCTVSAGSIS STSTSYYWGW IRQSPGKGLE WIGSIYYSGR  60
TYYNPSLKSR VTISVDRPNN QFSLKVSSVT AADSAVYYCA RHRRVLLWIG ELLDDYDRDV  120
WGQGTMVTVS S                                                      131

SEQ ID NO: 533          moltype = AA  length = 5
FEATURE                 Location/Qualifiers
source                  1..5
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 533
DYAMH                                                             5

SEQ ID NO: 534          moltype = AA  length = 17
FEATURE                 Location/Qualifiers
source                  1..17
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 534
GISTYFGRTN YNQKFKG                                                17

SEQ ID NO: 535          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 535
GLSGNYVMDY                                                        10

SEQ ID NO: 536          moltype = AA  length = 15
FEATURE                 Location/Qualifiers
source                  1..15
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 536
RASESWDSYG NSFMH                                                  15

SEQ ID NO: 537          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 537
RASNLES                                                           7

SEQ ID NO: 538          moltype = AA  length = 9
FEATURE                 Location/Qualifiers
source                  1..9
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 538
QQSNEAPPT                                                         9
```

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```
SEQ ID NO: 539          moltype = AA  length = 60
FEATURE                 Location/Qualifiers
source                  1..60
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 539
QVQLQQSGPE LVRPGVSVKI SCKGSGYTFT DYAMHWVKQS HAKSLEWIGG ISTYFGRTNY  60

SEQ ID NO: 540          moltype = AA  length = 111
FEATURE                 Location/Qualifiers
source                  1..111
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 540
DIVLTQSPAS LAVSLGQRAT ISCRASESVD DYGNSFMHWY QQKPGQPPKL LIYRASNLES  60
GIPARFSGSG SRTDFTLTIN PVEADDVATY YCQQSNEAPP TFGGGTKLEI R           111

SEQ ID NO: 541          moltype = AA  length = 5
FEATURE                 Location/Qualifiers
source                  1..5
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 541
DYGMH                                                              5

SEQ ID NO: 542          moltype = AA  length = 17
FEATURE                 Location/Qualifiers
source                  1..17
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 542
VISPYSGRTN YNQNFKG                                                 17

SEQ ID NO: 543          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 543
GLSGNYVVDY                                                         10

SEQ ID NO: 544          moltype = AA  length = 15
FEATURE                 Location/Qualifiers
source                  1..15
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 544
RASESWDSYG NSFMH                                                   15

SEQ ID NO: 545          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 545
RASNLES                                                            7

SEQ ID NO: 546          moltype = AA  length = 9
FEATURE                 Location/Qualifiers
source                  1..9
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 546
QQSNEGPPT                                                          9

SEQ ID NO: 547          moltype = AA  length = 119
FEATURE                 Location/Qualifiers
source                  1..119
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 547
QVQLQQSGPE LVRPGVSVKI SCKGSGYTFT DYGMHWVKQS HAKSLEWIGV ISPYSGRTNY  60
NQNFKGKATM TVDKSSSTAY LELARLTSED SAIYYCARGL SGNYVVDYWG QGTSVTVSS   119

SEQ ID NO: 548          moltype = AA  length = 111
```

-continued

```
FEATURE                 Location/Qualifiers
source                  1..111
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 548
DIVLTQSPAS LAVSLGQRAT ISCRASESVD SYGNSFMHWY QQKPGQPPKL LIYRASNLES  60
GIPARFSGSG SRTDFTLTIN PVEADDVATY YCQQSNEGPP TFGGGTKLEI K           111

SEQ ID NO: 549          moltype = AA  length = 5
FEATURE                 Location/Qualifiers
source                  1..5
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 549
DYAMH                                                             5

SEQ ID NO: 550          moltype = AA  length = 17
FEATURE                 Location/Qualifiers
source                  1..17
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 550
VISFYSGKTN YNQKFMG                                                17

SEQ ID NO: 551          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 551
GLSGNYVMDY                                                        10

SEQ ID NO: 552          moltype = AA  length = 15
FEATURE                 Location/Qualifiers
source                  1..15
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 552
RASESWDSYG NSFMH                                                  15

SEQ ID NO: 553          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 553
RASNLES                                                           7

SEQ ID NO: 554          moltype = AA  length = 9
FEATURE                 Location/Qualifiers
source                  1..9
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 554
QQSNEGPPT                                                         9

SEQ ID NO: 555          moltype = AA  length = 119
FEATURE                 Location/Qualifiers
source                  1..119
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 555
QVQLQQSGPE LVRPGVSVKI SCKGSGYTVT DYAMHWVKQS HAKSLEWIGV ISFYSGKTNY  60
NQKFMGKATM TVDKSSSTAY MELARLTSED SAIYYCARGL SGNYVMDYWG QGTSVTVSS   119

SEQ ID NO: 556          moltype = AA  length = 111
FEATURE                 Location/Qualifiers
source                  1..111
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 556
DIVLTQSPAS LAVSLGQRAT ISCRASESVD SYGNSFMHWY QQKPGQPPKL LIYRASNLES  60
GIPARFSGSG SRTDFTLTIN PVEADDVATY YCQQSNEGPP TFGGGTKLEI K           111

SEQ ID NO: 557          moltype = AA  length = 5
FEATURE                 Location/Qualifiers
```

-continued

```
source                  1..5
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 557
DYGMH                                                         5

SEQ ID NO: 558          moltype = AA  length = 17
FEATURE                 Location/Qualifiers
source                  1..17
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 558
VISPYSGKTN YSQKFKG                                            17

SEQ ID NO: 559          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 559
GLSGNFVMDF                                                    10

SEQ ID NO: 560          moltype = AA  length = 15
FEATURE                 Location/Qualifiers
source                  1..15
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 560
RASESWDSYG NSFMH                                              15

SEQ ID NO: 561          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 561
RASNLES                                                       7

SEQ ID NO: 562          moltype = AA  length = 9
FEATURE                 Location/Qualifiers
source                  1..9
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 562
QHSNEDPPT                                                     9

SEQ ID NO: 563          moltype = AA  length = 119
FEATURE                 Location/Qualifiers
source                  1..119
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 563
QVQLQQSGPE LVRPGVAVKI SCKGSGYKFI DYGMHWVKQS HTKSLQWIGV ISPYSGKTNY  60
SQKFKGKATM TVDKSSSTAY MELARLTSED SAIYYCARGL SGNFVMDFWG QGTSVTVSS   119

SEQ ID NO: 564          moltype = AA  length = 111
FEATURE                 Location/Qualifiers
source                  1..111
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 564
DIVLTQSPAS LAVSLGQRAT ISCRASESVD SYGPSFMHWY QQKPGQPPKL LIYRASNLES  60
GIPARFSGSG SRTDFTLTIN PVEADDVATY YCQHSNEDPP TFGGGTRLEI K           111

SEQ ID NO: 565          moltype = AA  length = 5
FEATURE                 Location/Qualifiers
source                  1..5
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 565
DYAMH                                                         5

SEQ ID NO: 566          moltype = AA  length = 17
FEATURE                 Location/Qualifiers
source                  1..17
                        mol_type = protein
```

-continued

```
                       organism = synthetic construct
SEQUENCE: 566
GISTYFGRTN YNQKFKG                                              17

SEQ ID NO: 567         moltype = AA  length = 10
FEATURE                Location/Qualifiers
source                 1..10
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 567
GLSGNYVMDY                                                      10

SEQ ID NO: 568         moltype = AA  length = 15
FEATURE                Location/Qualifiers
source                 1..15
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 568
RASESWDSYG NSFMH                                                15

SEQ ID NO: 569         moltype = AA  length = 7
FEATURE                Location/Qualifiers
source                 1..7
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 569
RASNLES                                                         7

SEQ ID NO: 570         moltype = AA  length = 9
FEATURE                Location/Qualifiers
source                 1..9
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 570
QQSNEAPPT                                                       9

SEQ ID NO: 571         moltype = AA  length = 119
FEATURE                Location/Qualifiers
source                 1..119
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 571
QVQLQQSGPE LVRPGVSVKI SCKGSGYTFT DYAMHWVKQS HAKSLEWIGG ISTYFGRTNY  60
NQKFKGRATM TVDKSSSTAY MELARLTSED SALYYCARGL SGNYVMDYWG QGTSVTVSS   119

SEQ ID NO: 572         moltype = AA  length = 111
FEATURE                Location/Qualifiers
source                 1..111
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 572
DIVLTQSPAS LAVSLGQRAT ISCRASESVD DYGNSFMHWY QQKPGQPPKL LIYRASNLES  60
GIPARFSGSG SRTDFTLTIN PVEADDVATY YCQQSNEAPP TFGGGTKLEI R           111

SEQ ID NO: 573         moltype = AA  length = 17
FEATURE                Location/Qualifiers
source                 1..17
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 573
EINPTNGRTN YNENFKS                                              17

SEQ ID NO: 574         moltype = AA  length = 7
FEATURE                Location/Qualifiers
source                 1..7
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 574
GTRAYHF                                                         7

SEQ ID NO: 575         moltype = AA  length = 11
FEATURE                Location/Qualifiers
source                 1..11
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 575
```

-continued

```
RASENIYSNL A                                                     11

SEQ ID NO: 576          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 576
AATDLAD                                                          7

SEQ ID NO: 577          moltype = AA  length = 116
FEATURE                 Location/Qualifiers
source                  1..116
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 577
QVQLQQPGAE LVRPGAAVKL SCKASGYTFT SYWMHWVKQR PGQGLEWIGE INPTNGRTNY  60
NENFKSKATL TVDKSSSTAY MQLSSLTSED SAVYYCARGT RAYHFWGQGT SVTVSS      116

SEQ ID NO: 578          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 578
DIQLTQTPAS LSVSVGETVT ITCRASENIY SNLAWYQQKQ GKSPQLLVYA ATDLADGVPS  60
RFRGSGSGTQ YSLKINSLQS EDFGSYYCQH FWGTPLTFGA GTKLELI                107

SEQ ID NO: 579          moltype = AA  length = 17
FEATURE                 Location/Qualifiers
source                  1..17
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 579
EINPINGRTN YSEKFKK                                               17

SEQ ID NO: 580          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 580
RASDNIYSNL A                                                     11

SEQ ID NO: 581          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 581
AATNLAD                                                          7

SEQ ID NO: 582          moltype = AA  length = 9
FEATURE                 Location/Qualifiers
source                  1..9
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 582
QHFWGTPLM                                                        9

SEQ ID NO: 583          moltype = AA  length = 116
FEATURE                 Location/Qualifiers
source                  1..116
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 583
QVQLQQPGAE LVKPGASVKL SCKASGYTFA SYWMHWVKQR PGQGLEWIGE INPINGRTNY  60
SEKFKKKATL TVDKSSSTAY MQLSSLTSED SAVYYCARGT RAYHYWGQGT SVTVSS      116

SEQ ID NO: 584          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 584
DIQMTQSPAS LSVSVGETVT ITCRASDNIY SNLAWYQQKQ GKSPQLLVYA ATNLADGVPS  60
```

-continued

```
RFSGSGSGTQ YSLKINSLQS EDFGSYYCQH FWGTPLMFGS GTKLELK                  107

SEQ ID NO: 585          moltype = AA  length = 17
FEATURE                 Location/Qualifiers
source                  1..17
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 585
EINPSNGRTN YNETFKS                                                   17

SEQ ID NO: 586          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 586
RASDNIYSNL A                                                         11

SEQ ID NO: 587          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 587
AVTNLAD                                                              7

SEQ ID NO: 588          moltype = AA  length = 116
FEATURE                 Location/Qualifiers
source                  1..116
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 588
QVQLQQPGAE LVKPGASVKL SCKASGYTFT SYWMHWVKQR PGQGLEWIGE INPSNGRTNY  60
NETFKSKATL TVDKSSSTAY MQLSSLTSED SAVYYCARGT RAYHYWGQGT SVTVSS      116

SEQ ID NO: 589          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 589
DIQMTQSPAS LSVSVGETVT ITCRASDNIY SNLAWYQQKQ GKSPQLLVYA VTNLADGVPS  60
RFSGSGSGTQ YSLKINSLQS EDFGSYYCQH FWGTPLTFGA GTKLELK                  107

SEQ ID NO: 590          moltype = AA  length = 6
FEATURE                 Location/Qualifiers
source                  1..6
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 590
SEYAWN                                                               6

SEQ ID NO: 591          moltype = AA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 591
YISYSGTTSY NPSLKS                                                    16

SEQ ID NO: 592          moltype = AA  length = 13
FEATURE                 Location/Qualifiers
source                  1..13
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 592
YGYGNPATRY FDV                                                       13

SEQ ID NO: 593          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 593
RASKSISKYL A                                                         11
```

-continued

```
SEQ ID NO: 594          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 594
SGSTLQS                                                           7

SEQ ID NO: 595          moltype = AA  length = 9
FEATURE                 Location/Qualifiers
source                  1..9
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 595
QQHNEYPWT                                                         9

SEQ ID NO: 596          moltype = AA  length = 124
FEATURE                 Location/Qualifiers
source                  1..124
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 596
VHDVQLQESG PGLVKPSQSL SLTCTVTGNS ITSEYAWNWI RQFPGNKLEW MGYISYSGTT  60
SYNPSLKSRI SITRDTSKNQ LFLQLNSVTT EDTATYFCAR YGYGNPATRY FDVWGAGTTV  120
TVSS                                                              124

SEQ ID NO: 597          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
source                  1..107
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 597
DVQITQSPSY LTASPGETIT INCRASKSIS KYLAWYQEKP GKTNKLLIYS GSTLQSGIPS  60
RFSGSGSGTD FTLTISNLEP EDFAMYYCQQ HNEYPWTFGG GTKLEIK               107

SEQ ID NO: 598          moltype = AA  length = 17
FEATURE                 Location/Qualifiers
source                  1..17
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 598
NIYPGSGSTK YDERFKS                                                17

SEQ ID NO: 599          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 599
GGYDSRAWFA Y                                                      11

SEQ ID NO: 600          moltype = AA  length = 15
FEATURE                 Location/Qualifiers
source                  1..15
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 600
RARQSVSTSS YSFMH                                                  15

SEQ ID NO: 601          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 601
YASIQES                                                           7

SEQ ID NO: 602          moltype = AA  length = 9
FEATURE                 Location/Qualifiers
source                  1..9
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 602
QHTWEIPFT                                                         9

SEQ ID NO: 603          moltype = AA  length = 120
```

-continued

FEATURE Location/Qualifiers
source 1..120
mol_type = protein
organism = synthetic construct
SEQUENCE: 603
QVQLQQPGSE LVRPGASVKL SCKASGYTFT SYWMHWVKQR HGQGLEWIGN IYPGSGSTKY 60
DERFKSKGTL TVDTSSSTAY MHLSSLTSED SAVYYCTRGG YDSRAWFAYW GQGTLVTVSA 120

SEQ ID NO: 604 moltype = AA length = 111
FEATURE Location/Qualifiers
source 1..111
mol_type = protein
organism = synthetic construct
SEQUENCE: 604
DIVLTQSPAS LAVSLGQRAT ISCRARQSVS TSSYSFMHWY RQKAGQPPKL LIKYASIQES 60
GVPARFSGSG SGTDFTLNIL PVEEEDTATY YCQHTWEIPF TFGSGTKLEI K 111

SEQ ID NO: 605 moltype = AA length = 17
FEATURE Location/Qualifiers
source 1..17
mol_type = protein
organism = synthetic construct
SEQUENCE: 605
NIYPGSGSTK YDEKRKS 17

SEQ ID NO: 606 moltype = AA length = 11
FEATURE Location/Qualifiers
source 1..11
mol_type = protein
organism = synthetic construct
SEQUENCE: 606
GGYDSRAWFA H 11

SEQ ID NO: 607 moltype = AA length = 15
FEATURE Location/Qualifiers
source 1..15
mol_type = protein
organism = synthetic construct
SEQUENCE: 607
RAROSVSTSS YSFMH 15

SEQ ID NO: 608 moltype = AA length = 7
FEATURE Location/Qualifiers
source 1..7
mol_type = protein
organism = synthetic construct
SEQUENCE: 608
YASIQES 7

SEQ ID NO: 609 moltype = AA length = 9
FEATURE Location/Qualifiers
source 1..9
mol_type = protein
organism = synthetic construct
SEQUENCE: 609
QHTWEIPFT 9

SEQ ID NO: 610 moltype = AA length = 120
FEATURE Location/Qualifiers
source 1..120
mol_type = protein
organism = synthetic construct
SEQUENCE: 610
QVQLQQPGSE LVRPGASVKL SCKASGYTFT SYWMHWVKQR HGQGLEWIGN IYPGSGSTKY 60
DEKFKSKGTL TVDTSSSTAY MHLSSLTSED SAVYYCTRGG YDSRAWFAHW GQGTLVTVSA 120

SEQ ID NO: 611 moltype = AA length = 111
FEATURE Location/Qualifiers
source 1..111
mol_type = protein
organism = synthetic construct
SEQUENCE: 611
DIVLTQSPAS LAVSLGQRAT ISCRARQSVS TSSYSFMHWY QQKPGQPPKL LIKYASIQES 60
GVPARFSGSG SGTDFTLNIL PVEEEDTATY YCQHTWEIPF TFGSGTNLEI K 111

SEQ ID NO: 612 moltype = AA length = 5

-continued

```
FEATURE               Location/Qualifiers
source                1..5
                      mol_type = protein
                      organism = synthetic construct
SEQUENCE: 612
DYYMY                                                        5

SEQ ID NO: 613        moltype = AA  length = 17
FEATURE               Location/Qualifiers
source                1..17
                      mol_type = protein
                      organism = synthetic construct
SEQUENCE: 613
SISNGGDNTY YPDTVKG                                           17

SEQ ID NO: 614        moltype = AA  length = 15
FEATURE               Location/Qualifiers
source                1..15
                      mol_type = protein
                      organism = synthetic construct
SEQUENCE: 614
QGALYDGYYR GAMDY                                             15

SEQ ID NO: 615        moltype = AA  length = 12
FEATURE               Location/Qualifiers
source                1..12
                      mol_type = protein
                      organism = synthetic construct
SEQUENCE: 615
TTSSSWPSSY FH                                                12

SEQ ID NO: 616        moltype = AA  length = 7
FEATURE               Location/Qualifiers
source                1..7
                      mol_type = protein
                      organism = synthetic construct
SEQUENCE: 616
STSNLAS                                                      7

SEQ ID NO: 617        moltype = AA  length = 9
FEATURE               Location/Qualifiers
source                1..9
                      mol_type = protein
                      organism = synthetic construct
SEQUENCE: 617
HQYHRSPFT                                                    9

SEQ ID NO: 618        moltype = AA  length = 124
FEATURE               Location/Qualifiers
source                1..124
                      mol_type = protein
                      organism = synthetic construct
SEQUENCE: 618
EVKLVESGGG LVQPGGFLKL SCATSGFTFS DYYMYWVRQT PEKRLEWVAS ISNGGDNTYY  60
PDTVKGRFTI SRDNAKNTLY LQMSRLKSED TAMYYCARQG ALYDGYYRGA MDYWGQGTSV  120
TVSS                                                         124

SEQ ID NO: 619        moltype = AA  length = 108
FEATURE               Location/Qualifiers
source                1..108
                      mol_type = protein
                      organism = synthetic construct
SEQUENCE: 619
QIVLTQSPAI MSASLGGRVT MTCTTSSSVP SSYFHWYQQK PGSSPKLWIY STSNLASGVP  60
ARFSGSGSGT SYSLTISSME AEDAATYYCH QYHRSPFTFG SGTKLEIK          108

SEQ ID NO: 620        moltype = AA  length = 5
FEATURE               Location/Qualifiers
source                1..5
                      mol_type = protein
                      organism = synthetic construct
SEQUENCE: 620
NYWIE                                                        5

SEQ ID NO: 621        moltype = AA  length = 17
FEATURE               Location/Qualifiers
```

-continued

```
source                 1..17
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 621
EILPGSGSTK YNEKFKG                                          17

SEQ ID NO: 622         moltype = AA  length = 12
FEATURE                Location/Qualifiers
source                 1..12
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 622
RGGYGYDGEF AY                                               12

SEQ ID NO: 623         moltype = AA  length = 11
FEATURE                Location/Qualifiers
source                 1..11
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 623
RAGQDITNYL N                                                11

SEQ ID NO: 624         moltype = AA  length = 7
FEATURE                Location/Qualifiers
source                 1..7
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 624
YTSRLHS                                                     7

SEQ ID NO: 625         moltype = AA  length = 9
FEATURE                Location/Qualifiers
source                 1..9
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 625
QQANTLPYT                                                   9

SEQ ID NO: 626         moltype = AA  length = 121
FEATURE                Location/Qualifiers
source                 1..121
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 626
QVQLQQSGAE LMKPGASVKI SCKAAGYTFS NYWIEWVKQR PGHGLEWIGE ILPGSGSTKY  60
NEKFKGKATF TADTSSNTAY MQLSSLTSED SAVYYCARRG GYGYDGEFAY WGQGTLVTVS  120
A                                                           121

SEQ ID NO: 627         moltype = AA  length = 107
FEATURE                Location/Qualifiers
source                 1..107
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 627
DIQMTQTTSS LSASLGDRVT INCRAGQDIT NYLNWFQQKP DGTVKLLIYY TSRLHSGVPS  60
RFSGSGSGTD YSLTITNLEQ EDIATYFCQQ ANTLPYTFGG GTKLEIK           107

SEQ ID NO: 628         moltype = AA  length = 8
FEATURE                Location/Qualifiers
source                 1..8
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 628
GYTFTNYW                                                    8

SEQ ID NO: 629         moltype = AA  length = 8
FEATURE                Location/Qualifiers
source                 1..8
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 629
INPINGRS                                                    8

SEQ ID NO: 630         moltype = AA  length = 9
FEATURE                Location/Qualifiers
source                 1..9
```

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```
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 630
ARGTRAMHY                                                      9

SEQ ID NO: 631           moltype = AA  length = 6
FEATURE                  Location/Qualifiers
source                   1..6
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 631
ENIYNN                                                         6

SEQ ID NO: 632           moltype = AA  length = 5
FEATURE                  Location/Qualifiers
source                   1..5
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 632
NYWMH                                                          5

SEQ ID NO: 633           moltype = AA  length = 17
FEATURE                  Location/Qualifiers
source                   1..17
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 633
EINPINGRSN YGERFKT                                             17

SEQ ID NO: 634           moltype = AA  length = 7
FEATURE                  Location/Qualifiers
source                   1..7
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 634
GTRAMHY                                                        7

SEQ ID NO: 635           moltype = AA  length = 11
FEATURE                  Location/Qualifiers
source                   1..11
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 635
RTSENIYNNL A                                                   11

SEQ ID NO: 636           moltype = AA  length = 7
FEATURE                  Location/Qualifiers
source                   1..7
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 636
AATNLAD                                                        7

SEQ ID NO: 637           moltype = AA  length = 7
FEATURE                  Location/Qualifiers
source                   1..7
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 637
GYTFTNY                                                        7

SEQ ID NO: 638           moltype = AA  length = 5
FEATURE                  Location/Qualifiers
source                   1..5
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 638
TRAMH                                                          5

SEQ ID NO: 639           moltype = AA  length = 7
FEATURE                  Location/Qualifiers
source                   1..7
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 639
SENIYNN                                                        7
```

-continued

```
SEQ ID NO: 640          moltype = AA  length = 6
FEATURE                Location/Qualifiers
source                 1..6
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 640
FWGTPL                                                             6

SEQ ID NO: 641          moltype = AA  length = 116
FEATURE                Location/Qualifiers
source                 1..116
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 641
QVQLQQPGAE LVKPGASVKL SCKASGYTFT NYWMHWVKQR PGQGLEWIGE INPINGRSNY  60
GERFKTKATL TVDKSSSTAY MQLSSLTSED SAVYYCARGT RAMHYWGQGT SVTVSS      116

SEQ ID NO: 642          moltype = AA  length = 107
FEATURE                Location/Qualifiers
source                 1..107
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 642
DIQMTQSPAS LSVSVGETVT ITCRTSENIY NNLAWYQQKQ GKSPQLLVYA ATNLADGVPS  60
RFSGSGSGTQ YSLKINSLQS EDFGNYYCQH FWGTPLTFGA GTKLELK                107

SEQ ID NO: 643          moltype = AA  length = 10
FEATURE                Location/Qualifiers
source                 1..10
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 643
GFTFSNYGMH                                                         10

SEQ ID NO: 644          moltype = AA  length = 17
FEATURE                Location/Qualifiers
source                 1..17
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 644
MIYYDSSKMN YADTVKG                                                 17

SEQ ID NO: 645          moltype = AA  length = 9
FEATURE                Location/Qualifiers
source                 1..9
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 645
PTSHYVVDV                                                          9

SEQ ID NO: 646          moltype = AA  length = 11
FEATURE                Location/Qualifiers
source                 1..11
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 646
QASQDIGNWL A                                                       11

SEQ ID NO: 647          moltype = AA  length = 7
FEATURE                Location/Qualifiers
source                 1..7
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 647
GATSLAD                                                            7

SEQ ID NO: 648          moltype = AA  length = 9
FEATURE                Location/Qualifiers
source                 1..9
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 648
LQAYNTPWT                                                          9

SEQ ID NO: 649          moltype = AA  length = 117
```

-continued

```
FEATURE                Location/Qualifiers
source                 1..117
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 649
EVQLVWSGGG LVQPQNSLTL SCVASGFTFS NYGMHWRQAP KKGLEWIAMI YYDSSKMNYA  60
DTVKGRFTIS RDNSKNTLYL EMNSLRSEDT AMYYCAVPTS HYVVDVWGQG VSVTVSS     117

SEQ ID NO: 650         moltype = AA  length = 108
FEATURE                Location/Qualifiers
source                 1..108
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 650
DIQMTQSPAS LSASLEEIVT ITCQASQDIG NWLAWYQQKP GKSPQLLIYG ATSLADGVPS  60
RFSGSRSGTQ FSLKISRVQV EDIGIYYCLQ AYNTPWTFGG GTKLELKR               108

SEQ ID NO: 651         moltype = AA  length = 60
FEATURE                Location/Qualifiers
source                 1..60
                       mol_type = protein
                       organism = Mus musculus
SEQUENCE: 651
MEFGLSWLFL VAILKGVQCE VQLQQSGTVL ARPGASVKMS CKASGYSFTI YWIHWVKQRP  60

SEQ ID NO: 652         moltype = AA  length = 130
FEATURE                Location/Qualifiers
source                 1..130
                       mol_type = protein
                       organism = Mus musculus
SEQUENCE: 652
MDMRVPAQLL GLLLLWLPGA RCDVQITQSP SYLAASPGET IIINCRASKS ISKYLAWYQE  60
KPGKTNKLLI YSGSTLQSGI PSRFSGSGSG TDFTLTISSL EPQDFAMYYC QQHNEYPWTF  120
GGGTKLEIKR                                                        130

SEQ ID NO: 653         moltype = RNA  length = 19
FEATURE                Location/Qualifiers
source                 1..19
                       mol_type = other RNA
                       organism = synthetic construct
modified_base          3
                       mod_base = OTHER
                       note = may be thymine
modified_base          5
                       mod_base = OTHER
                       note = may be thymine
modified_base          14
                       mod_base = OTHER
                       note = may be thymine
SEQUENCE: 653
ggtctaggcc cggtgagag                                              19

SEQ ID NO: 654         moltype = RNA  length = 19
FEATURE                Location/Qualifiers
source                 1..19
                       mol_type = other RNA
                       organism = synthetic construct
modified_base          3
                       mod_base = OTHER
                       note = may be thymine
modified_base          7..8
                       mod_base = OTHER
                       note = may be thymine
modified_base          13..14
                       mod_base = OTHER
                       note = may be thymine
modified_base          18
                       mod_base = OTHER
                       note = may be thymine
SEQUENCE: 654
cctggattag agttacatc                                              19

SEQ ID NO: 655         moltype = DNA  length = 25
FEATURE                Location/Qualifiers
source                 1..25
                       mol_type = other DNA
```

-continued

```
                        organism = synthetic construct
SEQUENCE: 655
cttctaggtc taggcccggt gagag                                     25

SEQ ID NO: 656          moltype = RNA  length = 25
FEATURE                 Location/Qualifiers
source                  1..25
                        mol_type = other RNA
                        organism = synthetic construct
modified_base           2
                        mod_base = OTHER
                        note = may be thymine
modified_base           4
                        mod_base = OTHER
                        note = may be thymine
modified_base           14
                        mod_base = OTHER
                        note = may be thymine
modified_base           20
                        mod_base = OTHER
                        note = may be thymine
SEQUENCE: 656
ctctcaccgg gcctagacct agaag                                     25

SEQ ID NO: 657          moltype = RNA  length = 35
FEATURE                 Location/Qualifiers
source                  1..35
                        mol_type = other RNA
                        organism = synthetic construct
modified_base           14
                        mod_base = OTHER
                        note = may be thymine
modified_base           18
                        mod_base = OTHER
                        note = may be thymine
modified_base           22
                        mod_base = OTHER
                        note = may be thymine
modified_base           34
                        mod_base = OTHER
                        note = may be thymine
SEQUENCE: 657
cgcggggaac acctggctgg ctacggaggg gcgtg                          35

SEQ ID NO: 658          moltype = RNA  length = 35
FEATURE                 Location/Qualifiers
source                  1..35
                        mol_type = other RNA
                        organism = synthetic construct
modified_base           4..5
                        mod_base = OTHER
                        note = may be thymine
modified_base           7
                        mod_base = OTHER
                        note = may be thymine
modified_base           11
                        mod_base = OTHER
                        note = may be thymine
modified_base           13
                        mod_base = OTHER
                        note = may be thymine
modified_base           22
                        mod_base = OTHER
                        note = may be thymine
modified_base           30
                        mod_base = OTHER
                        note = may be thymine
SEQUENCE: 658
gccttctagg tctaggcccg gtgagagact ccaca                          35

SEQ ID NO: 659          moltype = DNA  length = 38
FEATURE                 Location/Qualifiers
source                  1..38
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 659
```

-continued

gctccacgcc ctctgcaagg gacctgttgc tcgcgtgt 38

SEQ ID NO: 660 moltype = DNA length = 40
FEATURE Location/Qualifiers
source 1..40
mol_type = other DNA
organism = synthetic construct
SEQUENCE: 660
ctgggattcc tgccttctag gtctaggccc ggtgagagac 40

SEQ ID NO: 661 moltype = DNA length = 60
FEATURE Location/Qualifiers
source 1..60
mol_type = other DNA
organism = synthetic construct
SEQUENCE: 661
cgggttccac gctccttcgc cctctgcaag ggacctgtt gctcgcgtgt ctcccgcccc 60

SEQ ID NO: 662 moltype = DNA length = 60
FEATURE Location/Qualifiers
source 1..60
mol_type = other DNA
organism = synthetic construct
SEQUENCE: 662
agtcaagaca gcggcttcca gtttccatag aattactgga gaacctcaga gagccagccc 60

SEQ ID NO: 663 moltype = DNA length = 59
FEATURE Location/Qualifiers
source 1..59
mol_type = other DNA
organism = synthetic construct
SEQUENCE: 663
gctgaccggc ctgggattcc tgccactagg tctaggcccg gtgagagact ccacaccgc 59

SEQ ID NO: 664 moltype = DNA length = 3635
FEATURE Location/Qualifiers
source 1..3635
mol_type = genomic DNA
organism = Homo sapiens
SEQUENCE: 664
tcctggcggc tgcgaggttt cactgcaggg gcgccagtgg gctcagtgac gctgcggcct 60
ccttctgcct aggtcccaac gcttcggggc aggggtgcgg tcttgcaata ggaagccgag 120
cgtcttgcaa gcttcccgtc gggcaccagc tactcggccc cgcaccctac ctggtgcatt 180
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caaaggccaa gctgtgcgca aacagctttg ggacacggcc aacacggtga aggaaaagtt 1440
cgggaggaag ctttatgaat ccttactggt tgggagcctt cccgacatga acaagatgct 1500
ggataaggaa gacttcacta tgatgaagag agccatcttt gcaacgcagc ggcagtcttt 1560
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ggatgattcc tgctcgcagc tcacctcctt cctctacagt ttctgtcagc agagccggcg 1980
gcagcgtatc atccagcgga accgcacgga gcgcctctcc gaccttctgg actggaaata 2040
cctaggccgg tactatatgt ctgcgcgcca catggcgctg tccaaggcct ttccagagca 2100
cttcacctac gagcccaacg aggcggatgc ggcccagggg taccgctacc cacggccagc 2160

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ctcggtgcca ccgtcgccct cgctgtcacg acactccagc ccgcaccaga gtgaggacga  2220
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ggccgccaag gaccggcgca acatccgtgc accagagtgg ccgcgccgag cgtcctgcac  2340
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caccccgagc gagcccctca gccccaccag ctccctgggc gaggagcgta actaagtccg  2460
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acaaattcta cagcatacag agcaggccac gctcaggccc ggcatgcggg ccaccaagtt  2880
ctggaaacca cgtggtgtcc ctgcgaatgg ggcgatcaag tccagagccg gggcactttc  2940
agagtttgaa ggtaactgag agcagatggt cctccatttc aactccagaa gtggggctct  3000
gggagggatg ttctagccct ccctggcatg tcagagccag gctctgcctg gaggatccct  3060
ccatccggct cctgtcatcc cctacacttt ggccaagcaa gaggtggtag aaccacttgg  3120
ctgctcattc cttctggagg acacacagtc tcagtccaga tgccttcctg tctttctggc  3180
cctttctgga ccagatccta ctcttccttt ctaaatctga gatctccctc cagggaatcc  3240
gcctgcagag gacagagctg gctgtcttcc cccaccccta acctggctta ttcccaactg  3300
ctctgcccac tgtgaaacca ctaggttcta ggtcctggct tctagatctg gaaccttacc  3360
acgttactgc atactgatcc ctttcccatg atccagaact gaggtcactg ggttctagaa  3420
cccccacatt tacctcgagg ctcttccatc cccaaactgt gccctgcctt cagctttggt  3480
gaaagggagg gcccctcatg tgtgctgtgc tgtgtctgca ccgcttggtt tgcagttgag  3540
aggggagggc aggaggggtg tgattggagt gtgtccggag atgagatgaa aaaaatacat  3600
ctatatttaa gaatcccaaa aaaaaaaaaa aaaaa                             3635

SEQ ID NO: 665          moltype = DNA  length = 3681
FEATURE                 Location/Qualifiers
source                  1..3681
                        mol_type = genomic DNA
                        organism = Mus musculus
SEQUENCE: 665
actgcagctg cccgcccgat tcagtgtctc agctcaccct acctgagtcg gagcgctctg  60
gggcgggggt gcggtcgtgc aataggaagc ggagcgcctt gcaagcttcc cctgggacac  120
ccgctaactc taccggtcac caagtctgct gcgttcccag ccgatctctc tggtttccag  180
ttttggtgct cgaagtcccc tgcccgcagt agccatgcct ctcagccgca gtctctctgt  240
gtcctcgctt ccaggattgg aagactggga ggatgaattc gaccccgaga acgcagtgct  300
tttcgaggtg gcctgggagg tggccaacaa ggtgggtggc atctacactg tgctgcagac  360
gaaggcgaag gtgacagggg atgaatgggg tgacaactac tatctggtgg gaccatacac  420
ggagcagggt gtgaggacgc aggtagagct cctggagccc ccaactccgg aactgaagag  480
gactttggat tccatgaaca gcaagggttg taaggtgtat tttgggcgtt ggctgatcga  540
ggggggaccc ctagtggtgc tcctggatgt aggagcctca gcttgggccc tggagcgctg  600
gaagggtgag ctttgggaca cctgcaacat cggggtaccc tggtacgacc gcgaggccaa  660
tgacgctgtc ctgttcggct tcctcaccac ctggttcctg ggtgagttcc tggcccagaa  720
cgaagagaag ccgtatgtgg ttgcccactt ccacgaatgg ttggctggcg ttggtctgtg  780
tctgtgccgt gcccggcgct tgccggtggc aaccatcttc accactcatg ccacgctgct  840
ggggcgctac ctgtgtgctg gcgctgtgga cttctacaac aacctggaga atttcaatgt  900
agacaaggaa gcaggagaga ggcagatcta tcaccggtac tgcatggagc gtgcagcagc  960
tcactgtgcc catgtcttca ctaccgtatc ccagatcacc gcaatcgagg ctcaacacct  1020
ccttaagaga aaaccagata ttgtgacccc caacgggctg aatgtgaaga agttctctgc  1080
tatgcacgaa ttccagaacc ttcatgctca gagcaaagca cgaatccagg aatttgtgcg  1140
tggccatttt tatgggcacc tggacttcaa cctagacaag actttgtatt tctttatcgc  1200
tggccgctat gagttttcca acaagggagc tgatgtgttc ctggaggcat tggcccggct  1260
caactatctg ctcagagtga atggcagtga gcaaacagtt gtcgcattct tcatcatgcc  1320
ggcccggacc aataatttca acgtggaaac cctgaagggc caagccgtgc gcaaacaact  1380
atgggacaca gccaatacag tcaaggagaa atttgggagg aagctctacg aatccctttt  1440
agtggggagc ctcccggaca tgaacaagat gctggacaag gaggacttca ctatgatgaa  1500
gagagccatc tttgccactc agcggcagtc tttcccacca gtgtgcaccc acaacatgct  1560
ggacgactcc tcagacccca tcttgaccac catccgccga attggccttt tcaacagcag  1620
tgccgaccgt gtgaaggtga tttttcaccc agaattcctt tcttccacaa gccctctcct  1680
ccccgtggat tatgaggaat ttgtccgcgg ctgtcacctt ggggtcttcc cctcctacta  1740
tgagccctgg ggctacacac cagcggagtg cactgtcatg ggcatcccca gcatctccac  1800
caacctctcc ggctttggct gctttatgga ggaacacatc gcagatccct cagcttacgg  1860
catttacatt ctggatcgga ggttccgcag cctggatgat tcatgctcac agctcacctc  1920
cttcctgtac agcttctgcc agcagagccg gcgacagcgc atcatccagc ggaaccgcac  1980
agaacggttg tcggacttgc tagattggaa gtacctgggc cggtactaca tgtctgcgcg  2040
ccacatggct ctggccaagg cctttccaga ccacttcacc tatgaacccc atgaggtaga  2100
tgcgacccag gggtaccggt acccacgacc agcctccgtc ccgccgtcgc cctcactgtc  2160
tcgacactcc agcccacacc agagtgagga tgaggaagag ccacgggatg gacccctggg  2220
ggaagacagt gagcgttatg atgaggaaga ggaggctgcc aaggaccgcc gcaacatccg  2280
ggcacctgag tggccacgca gggcctcctg ttcctcctcc acaggtggca gcaagagaag  2340
caactcggtg gacactgggc cctccagctc actcagcaca cccactgagc ccctgagtcc  2400
taccagttcc ctgggtgagg agcgcaacta agctcccacc cccatcccat tccctgcctg  2460
tccagtgctc ctctcgcaga gggcctatgc agatgggagg gtgcctgaac ccccactccag  2520
actcttgagt gggacccct a cccagtgtgg tccatagcct aacctctgtt tcagacactc  2580
cagcccttga gctccaatct tggagttccc gcactccacg ccgccgtgcc tttcttggat  2640
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tgcaggatgc attctttgtg cactgatctg gagtctccag gcttagactg ggtcccagag  2700
gccaggcatc tgccattgtt tttcaatgcc agaggtttta ggacacctgg tttattggct  2760
tccaggctgt ggcttcttcg tttgatccta taatcataca gagtatgctt tgctcaggcc  2820
tgcctctggg accacctcat gttggattct gtgtggcttc ccgaatcagc caagttcaga  2880
gttaggacat ttcagggatt aacataattg aaaatcagcc tgcaaggtag ctcagtagct  2940
ctgtcgacag attgcttgtc tagcatgccc gaagccctgg gatctaactc tagaacctca  3000
taaacctggt gcggtgatac acatctgtaa tcccagcact cggtaggtag aggtagacgg  3060
atcaagagtt aaaggccatc atcctctgct acatagggag ttcaaggcca aactgggcaa  3120
catgagacac tgtctcaaaa gcaaagtaaa ggtggtggaa tgctcacggt cctccatttc  3180
aacccacgac tgcgatgctg ggacatgctg caaggttggc ctccctgggt gtgttcttca  3240
aaggagcatg cggagttgga ccagacacct ttctgccttt tttctggacc agaccttctt  3300
ttccttggtc cagtgtcccc tctagggaat gcctccattg agggcagaat gtctgtcaac  3360
cccacaagtg ctcagcccac tgtgaaacca ctgggttctg ggtcccagtg gctgaatcag  3420
gagtcttttg tcactgtgct gcaccccggt cccctttcct gatacaaaac cgagcccacc  3480
ggcttcttga agccccacat gtacctcgag gcctttctgc ctgcaagctt cagtgaatgg  3540
gcgggcccct cctcacgtgt gctgtgtctg gcccagtgcc tttggtttgc atttgggagg  3600
gggagggcag aaggtgtgtg attggagtgt gtctagagat gaaaaaaaaa aaaagaaaat  3660
acacctgtat ttaagaatgc c                                            3681

SEQ ID NO: 666          moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
source                  1..19
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 666
cttcctggat ggcttcaat                                               19

SEQ ID NO: 667          moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
source                  1..19
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 667
gtacattaag atggacttc                                               19

SEQ ID NO: 668          moltype = RNA  length = 30
FEATURE                 Location/Qualifiers
source                  1..30
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 668
tatctggata ggtggtatca agatctgtaa                                   30

SEQ ID NO: 669          moltype = RNA  length = 25
FEATURE                 Location/Qualifiers
source                  1..25
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 669
atgtaactga aaatgttctt cttta                                        25

SEQ ID NO: 670          moltype = RNA  length = 30
FEATURE                 Location/Qualifiers
source                  1..30
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 670
tggataggtg gtatcaacat ctgtaagcac                                   30

SEQ ID NO: 671          moltype = RNA  length = 22
FEATURE                 Location/Qualifiers
source                  1..22
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 671
gataggtggt atcaacatct gt                                           22

SEQ ID NO: 672          moltype = RNA  length = 30
FEATURE                 Location/Qualifiers
source                  1..30
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 672
tatctggata ggtggtatca acatctgtaa                                   30

SEQ ID NO: 673          moltype = RNA  length = 25
```

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```
FEATURE                 Location/Qualifiers
source                  1..25
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 673
aaacttggaa gagtgatgtg atgta                                    25

SEQ ID NO: 674          moltype = RNA  length = 27
FEATURE                 Location/Qualifiers
source                  1..27
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 674
gctcacttgt tgaggcaaaa cttggaa                                  27

SEQ ID NO: 675          moltype = RNA  length = 25
FEATURE                 Location/Qualifiers
source                  1..25
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 675
gccttggcaa catttccact tcctg                                    25

SEQ ID NO: 676          moltype = RNA  length = 25
FEATURE                 Location/Qualifiers
source                  1..25
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 676
tacacacttt acctgttgag aatag                                    25

SEQ ID NO: 677          moltype = RNA  length = 24
FEATURE                 Location/Qualifiers
source                  1..24
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 677
gataggtggt atcaacatct gtaa                                     24

SEQ ID NO: 678          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 678
gataggtggt atcaacatct g                                        21

SEQ ID NO: 679          moltype = RNA  length = 25
FEATURE                 Location/Qualifiers
source                  1..25
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 679
gataggtggt atcaacatct gtaag                                    25

SEQ ID NO: 680          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 680
ggtggtatca acatctgtaa                                          20

SEQ ID NO: 681          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 681
gtatcaacat ctgtaagcac                                          20

SEQ ID NO: 682          moltype = RNA  length = 30
FEATURE                 Location/Qualifiers
source                  1..30
                        mol_type = other RNA
                        organism = synthetic construct
```

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```
misc_difference        27
                       note = n can be any nucleotide
SEQUENCE: 682
cggctaattt cagagggcgc tttcttngac                                30

SEQ ID NO: 683         moltype = RNA  length = 26
FEATURE                Location/Qualifiers
source                 1..26
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 683
acagtggtgc tgagatagta taggcc                                    26

SEQ ID NO: 684         moltype = RNA  length = 22
FEATURE                Location/Qualifiers
source                 1..22
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 684
taggccactt tgttgctctt gc                                        22

SEQ ID NO: 685         moltype = RNA  length = 19
FEATURE                Location/Qualifiers
source                 1..19
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 685
ttcagagggc gctttcttc                                            19

SEQ ID NO: 686         moltype = RNA  length = 30
FEATURE                Location/Qualifiers
source                 1..30
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 686
tcttcaggtg caccttctgt ttctcaatct                                30

SEQ ID NO: 687         moltype = RNA  length = 30
FEATURE                Location/Qualifiers
source                 1..30
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 687
tctgtgatac tcttcaggtg caccttctgt                                30

SEQ ID NO: 688         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 688
tcttctgctc gggaggtgac a                                         21

SEQ ID NO: 689         moltype = RNA  length = 20
FEATURE                Location/Qualifiers
source                 1..20
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 689
ccagttacta ttcagaagac                                           20

SEQ ID NO: 690         moltype = RNA  length = 20
FEATURE                Location/Qualifiers
source                 1..20
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 690
tcttcaggtg caccttctgt                                           20

SEQ ID NO: 691         moltype = RNA  length = 18
FEATURE                Location/Qualifiers
source                 1..18
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 691
tgctgctgtc ttcttgct                                             18
```

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SEQ ID NO: 692          moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
source                  1..19
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 692
ttgttaactt tttcccatt                                              19

SEQ ID NO: 693          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 693
tgttaacttt ttcccattgg                                             20

SEQ ID NO: 694          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 694
cattttgtta actttttccc                                             20

SEQ ID NO: 695          moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
source                  1..19
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 695
ctgtagcttc accctttcc                                              19

SEQ ID NO: 696          moltype = RNA  length = 26
FEATURE                 Location/Qualifiers
source                  1..26
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 696
gagagcttcc tgtagcttca cccttt                                      26

SEQ ID NO: 697          moltype = RNA  length = 28
FEATURE                 Location/Qualifiers
source                  1..28
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 697
tcctgtagct tcaccctttc cacaggcg                                    28

SEQ ID NO: 698          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 698
tgtgttacct acccttgtcg                                             20

SEQ ID NO: 699          moltype = RNA  length = 27
FEATURE                 Location/Qualifiers
source                  1..27
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 699
tagactatct tttatattct gtaatat                                     27

SEQ ID NO: 700          moltype = RNA  length = 29
FEATURE                 Location/Qualifiers
source                  1..29
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 700
gagagcttcc tgtagcttca ccctttcca                                   29

SEQ ID NO: 701          moltype = RNA  length = 31
FEATURE                 Location/Qualifiers
source                  1..31
```

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```
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 701
ttcctgtagc ttcacccttt ccacaggcgt t                                      31

SEQ ID NO: 702           moltype = RNA  length = 23
FEATURE                  Location/Qualifiers
source                   1..23
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 702
agcttcctgt agcttcaccc ttt                                               23

SEQ ID NO: 703           moltype = RNA  length = 29
FEATURE                  Location/Qualifiers
source                   1..29
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 703
ggagagagct tcctgtagct tcacccttt                                         29

SEQ ID NO: 704           moltype = RNA  length = 23
FEATURE                  Location/Qualifiers
source                   1..23
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 704
gagagcttcc tgtagcttca ccc                                               23

SEQ ID NO: 705           moltype = RNA  length = 25
FEATURE                  Location/Qualifiers
source                   1..25
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 705
tatgtgttac ctacccttgt cggtc                                             25

SEQ ID NO: 706           moltype = RNA  length = 20
FEATURE                  Location/Qualifiers
source                   1..20
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 706
ggagagagct tcctgtagct                                                   20

SEQ ID NO: 707           moltype = RNA  length = 23
FEATURE                  Location/Qualifiers
source                   1..23
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 707
tcaccctttc cacaggcgtt gca                                               23

SEQ ID NO: 708           moltype = RNA  length = 28
FEATURE                  Location/Qualifiers
source                   1..28
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 708
gctgggagag agcttcctgt agcttcac                                          28

SEQ ID NO: 709           moltype = RNA  length = 28
FEATURE                  Location/Qualifiers
source                   1..28
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 709
tgttacctac ccttgtcggt ccttgtac                                          28

SEQ ID NO: 710           moltype = RNA  length = 28
FEATURE                  Location/Qualifiers
source                   1..28
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 710
ctgctgtctt cttgctatga ataatgtc                                          28
```

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SEQ ID NO: 711          moltype = RNA  length = 28
FEATURE                 Location/Qualifiers
source                  1..28
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 711
ggcgttgcac tttgcaatgc tgctgtct                                      28

SEQ ID NO: 712          moltype = RNA  length = 28
FEATURE                 Location/Qualifiers
source                  1..28
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 712
ttggaaatca agctgggaga gagcttcc                                      28

SEQ ID NO: 713          moltype = RNA  length = 28
FEATURE                 Location/Qualifiers
source                  1..28
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 713
ctacccttgt cggtccttgt acattttg                                      28

SEQ ID NO: 714          moltype = RNA  length = 28
FEATURE                 Location/Qualifiers
source                  1..28
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 714
gtcaatccga cctgagcttt gttgtaga                                      28

SEQ ID NO: 715          moltype = RNA  length = 27
FEATURE                 Location/Qualifiers
source                  1..27
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 715
cttgctatga ataatgtcaa tccgacc                                       27

SEQ ID NO: 716          moltype = RNA  length = 28
FEATURE                 Location/Qualifiers
source                  1..28
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 716
tatatgtgtt acctaccctt gtcggtcc                                      28

SEQ ID NO: 717          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 717
tttgtgtctt tctgagaaac                                               20

SEQ ID NO: 718          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 718
aaagacttac cttaagatac                                               20

SEQ ID NO: 719          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 719
atctgtcaaa tcgcctgcag                                               20

SEQ ID NO: 720          moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
source                  1..19
```

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```
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 720
cgccgccatt tctcaacag                                            19

SEQ ID NO: 721           moltype = RNA  length = 20
FEATURE                  Location/Qualifiers
source                   1..20
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 721
tttgtattta gcatgttccc                                           20

SEQ ID NO: 722           moltype = RNA  length = 17
FEATURE                  Location/Qualifiers
source                   1..17
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 722
ccgccatttc tcaacag                                              17

SEQ ID NO: 723           moltype = RNA  length = 21
FEATURE                  Location/Qualifiers
source                   1..21
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 723
ttctcaggaa tttgtgtctt t                                         21

SEQ ID NO: 724           moltype = RNA  length = 11
FEATURE                  Location/Qualifiers
source                   1..11
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 724
gacaactctt t                                                    11

SEQ ID NO: 725           moltype = RNA  length = 20
FEATURE                  Location/Qualifiers
source                   1..20
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 725
tcagcttctg ttagccactg                                           20

SEQ ID NO: 726           moltype = RNA  length = 24
FEATURE                  Location/Qualifiers
source                   1..24
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 726
tgttcagctt ctgttagcca ctga                                      24

SEQ ID NO: 727           moltype = RNA  length = 27
FEATURE                  Location/Qualifiers
source                   1..27
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 727
ctgttcagct tctgttagcc actgatt                                   27

SEQ ID NO: 728           moltype = RNA  length = 30
FEATURE                  Location/Qualifiers
source                   1..30
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 728
ttctcaacag atctgtcaaa tcgcctgcag                                30

SEQ ID NO: 729           moltype = RNA  length = 25
FEATURE                  Location/Qualifiers
source                   1..25
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 729
gccactgatt aaatatcttt atatc                                     25
```

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```
SEQ ID NO: 730          moltype = RNA  length = 30
FEATURE                Location/Qualifiers
source                 1..30
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 730
tctgttagcc actgattaaa tatctttata                                   30

SEQ ID NO: 731          moltype = RNA  length = 31
FEATURE                Location/Qualifiers
source                 1..31
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 731
gagaaactgt tcagcttctg ttagccactg a                                 31

SEQ ID NO: 732          moltype = RNA  length = 31
FEATURE                Location/Qualifiers
source                 1..31
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 732
tctttctgag aaactgttca gcttctgtta g                                 31

SEQ ID NO: 733          moltype = RNA  length = 26
FEATURE                Location/Qualifiers
source                 1..26
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 733
cagatctgtc aaatcgcctg caggta                                       26

SEQ ID NO: 734          moltype = RNA  length = 26
FEATURE                Location/Qualifiers
source                 1..26
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 734
caacagatct gtcaaatcgc ctgcag                                       26

SEQ ID NO: 735          moltype = RNA  length = 33
FEATURE                Location/Qualifiers
source                 1..33
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 735
aaactgttca gcttctgtta gccactgatt aaa                               33

SEQ ID NO: 736          moltype = RNA  length = 31
FEATURE                Location/Qualifiers
source                 1..31
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 736
gaaactgttc agcttctgtt agccactgat t                                 31

SEQ ID NO: 737          moltype = RNA  length = 28
FEATURE                Location/Qualifiers
source                 1..28
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 737
aaactgttca gcttctgtta gccactga                                     28

SEQ ID NO: 738          moltype = RNA  length = 28
FEATURE                Location/Qualifiers
source                 1..28
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 738
tgagaaactg ttcagcttct gttagcca                                     28

SEQ ID NO: 739          moltype = RNA  length = 32
FEATURE                Location/Qualifiers
source                 1..32
```

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```
                          mol_type = other RNA
                          organism = synthetic construct
SEQUENCE: 739
ttctgagaaa ctgttcagct tctgttagcc ac                                  32

SEQ ID NO: 740            moltype = RNA  length = 26
FEATURE                   Location/Qualifiers
source                    1..26
                          mol_type = other RNA
                          organism = synthetic construct
SEQUENCE: 740
ttctgagaaa ctgttcagct tctgtt                                         26

SEQ ID NO: 741            moltype = RNA  length = 25
FEATURE                   Location/Qualifiers
source                    1..25
                          mol_type = other RNA
                          organism = synthetic construct
SEQUENCE: 741
gatctgtcaa atcgcctgca ggtaa                                          25

SEQ ID NO: 742            moltype = RNA  length = 25
FEATURE                   Location/Qualifiers
source                    1..25
                          mol_type = other RNA
                          organism = synthetic construct
SEQUENCE: 742
ataatgaaaa cgccgccatt tctca                                          25

SEQ ID NO: 743            moltype = RNA  length = 25
FEATURE                   Location/Qualifiers
source                    1..25
                          mol_type = other RNA
                          organism = synthetic construct
SEQUENCE: 743
aaactgttca gcttctgtta gccac                                          25

SEQ ID NO: 744            moltype = RNA  length = 25
FEATURE                   Location/Qualifiers
source                    1..25
                          mol_type = other RNA
                          organism = synthetic construct
SEQUENCE: 744
ttgtgtcttt ctgagaaact gttca                                          25

SEQ ID NO: 745            moltype = RNA  length = 25
FEATURE                   Location/Qualifiers
source                    1..25
                          mol_type = other RNA
                          organism = synthetic construct
SEQUENCE: 745
ccaattctca ggaatttgtg tcttt                                          25

SEQ ID NO: 746            moltype = RNA  length = 25
FEATURE                   Location/Qualifiers
source                    1..25
                          mol_type = other RNA
                          organism = synthetic construct
SEQUENCE: 746
atcgcctgca ggtaaaagca tatgg                                          25

SEQ ID NO: 747            moltype = RNA  length = 30
FEATURE                   Location/Qualifiers
source                    1..30
                          mol_type = other RNA
                          organism = synthetic construct
SEQUENCE: 747
tgaaaacgcc gccatttctc aacagatctg                                     30

SEQ ID NO: 748            moltype = RNA  length = 30
FEATURE                   Location/Qualifiers
source                    1..30
                          mol_type = other RNA
                          organism = synthetic construct
SEQUENCE: 748
cataatgaaa acgccgccat ttctcaacag                                     30
```

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```
SEQ ID NO: 749          moltype = RNA  length = 30
FEATURE                Location/Qualifiers
source                 1..30
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 749
tgttcagctt ctgttagcca ctgattaaat                                    30

SEQ ID NO: 750          moltype = RNA  length = 24
FEATURE                Location/Qualifiers
source                 1..24
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 750
cagatctgtc aaatcgcctg cagg                                          24

SEQ ID NO: 751          moltype = RNA  length = 27
FEATURE                Location/Qualifiers
source                 1..27
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 751
caacagatct gtcaaatcgc ctgcagg                                       27

SEQ ID NO: 752          moltype = RNA  length = 29
FEATURE                Location/Qualifiers
source                 1..29
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 752
ctcaacagat ctgtcaaatc gcctgcagg                                     29

SEQ ID NO: 753          moltype = RNA  length = 23
FEATURE                Location/Qualifiers
source                 1..23
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 753
gatctgtcaa atcgcctgca ggt                                           23

SEQ ID NO: 754          moltype = RNA  length = 22
FEATURE                Location/Qualifiers
source                 1..22
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 754
gatctgtcaa atcgcctgca gg                                            22

SEQ ID NO: 755          moltype = RNA  length = 21
FEATURE                Location/Qualifiers
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 755
gatctgtcaa atcgcctgca g                                             21

SEQ ID NO: 756          moltype = RNA  length = 25
FEATURE                Location/Qualifiers
source                 1..25
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 756
cagatctgtc aaatcgcctg caggt                                         25

SEQ ID NO: 757          moltype = RNA  length = 23
FEATURE                Location/Qualifiers
source                 1..23
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 757
cagatctgtc aaatcgcctg cag                                           23

SEQ ID NO: 758          moltype = RNA  length = 25
FEATURE                Location/Qualifiers
source                 1..25
```

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```
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 758
gtgtctttct gagaaactgt tcagc                                          25

SEQ ID NO: 759          moltype = RNA  length = 28
FEATURE                 Location/Qualifiers
source                  1..28
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 759
gagaaactgt tcagcttctg ttagccac                                       28

SEQ ID NO: 760          moltype = RNA  length = 28
FEATURE                 Location/Qualifiers
source                  1..28
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 760
gaaactgttc agcttctgtt agccactg                                       28

SEQ ID NO: 761          moltype = RNA  length = 24
FEATURE                 Location/Qualifiers
source                  1..24
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 761
ctgttcagct tctgttagcc actg                                           24

SEQ ID NO: 762          moltype = RNA  length = 27
FEATURE                 Location/Qualifiers
source                  1..27
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 762
atctgtcaaa tcgcctgcag gtaaaag                                        27

SEQ ID NO: 763          moltype = RNA  length = 29
FEATURE                 Location/Qualifiers
source                  1..29
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 763
gatctgtcaa atcgcctgca ggtaaaagc                                      29

SEQ ID NO: 764          moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
source                  1..19
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 764
gctgaattat ttcttcccc                                                 19

SEQ ID NO: 765          moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
source                  1..19
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 765
tttttctgtc tgacagctg                                                 19

SEQ ID NO: 766          moltype = RNA  length = 18
FEATURE                 Location/Qualifiers
source                  1..18
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 766
tctgtttttg aggattgc                                                  18

SEQ ID NO: 767          moltype = RNA  length = 17
FEATURE                 Location/Qualifiers
source                  1..17
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 767
ccaccgcaga ttcaggc                                                   17
```

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```
SEQ ID NO: 768          moltype = RNA  length = 17
FEATURE                 Location/Qualifiers
source                  1..17
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 768
gcccaatgcc atcctgg                                                   17

SEQ ID NO: 769          moltype = RNA  length = 17
FEATURE                 Location/Qualifiers
source                  1..17
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 769
tttgcagacc tcctgcc                                                   17

SEQ ID NO: 770          moltype = RNA  length = 17
FEATURE                 Location/Qualifiers
source                  1..17
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 770
cagtttgccg ctgccca                                                   17

SEQ ID NO: 771          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 771
gttgcattca atgttctgac                                                20

SEQ ID NO: 772          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 772
atttttcctg tagaatactg g                                              21

SEQ ID NO: 773          moltype = RNA  length = 34
FEATURE                 Location/Qualifiers
source                  1..34
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 773
gctgcccaat gcgatcctgg agttcctgta agat                                34

SEQ ID NO: 774          moltype = RNA  length = 28
FEATURE                 Location/Qualifiers
source                  1..28
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 774
gctgcccaat gccatcctgg agttcctg                                       28

SEQ ID NO: 775          moltype = RNA  length = 31
FEATURE                 Location/Qualifiers
source                  1..31
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 775
gctgcccaat gccatcctgg agttcctgta a                                   31

SEQ ID NO: 776          moltype = RNA  length = 31
FEATURE                 Location/Qualifiers
source                  1..31
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 776
caatgccatc ctggagttcc tgtaagatac c                                   31

SEQ ID NO: 777          moltype = RNA  length = 32
FEATURE                 Location/Qualifiers
source                  1..32
```

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```
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 777
gctgcccaat gccatcctgg agttcctgta ag                          32

SEQ ID NO: 778           moltype = RNA  length = 30
FEATURE                  Location/Qualifiers
source                   1..30
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 778
ccaatgccat cctggagttc ctgtaagata                             30

SEQ ID NO: 779           moltype = RNA  length = 39
FEATURE                  Location/Qualifiers
source                   1..39
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 779
ttgccgctgc ccaatgccat cctggagttc ctgtaagat                   39

SEQ ID NO: 780           moltype = RNA  length = 34
FEATURE                  Location/Qualifiers
source                   1..34
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 780
gctgcccaat gccatcctgg agttcctgta agat                        34

SEQ ID NO: 781           moltype = RNA  length = 27
FEATURE                  Location/Qualifiers
source                   1..27
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 781
caatgccatc ctggagttcc tgtaaga                                27

SEQ ID NO: 782           moltype = RNA  length = 26
FEATURE                  Location/Qualifiers
source                   1..26
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 782
cagtttgccg ctgcccaatg ccatcc                                 26

SEQ ID NO: 783           moltype = RNA  length = 24
FEATURE                  Location/Qualifiers
source                   1..24
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 783
cttccccagt tgcattcaat gttc                                   24

SEQ ID NO: 784           moltype = RNA  length = 23
FEATURE                  Location/Qualifiers
source                   1..23
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 784
ctggcatctg tttttgagga ttg                                    23

SEQ ID NO: 785           moltype = RNA  length = 20
FEATURE                  Location/Qualifiers
source                   1..20
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 785
ttagatctgt cgccctacct                                        20

SEQ ID NO: 786           moltype = RNA  length = 39
FEATURE                  Location/Qualifiers
source                   1..39
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 786
gctgcccaat gccatcctgg agttcctgta agataccaa                   39
```

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```
SEQ ID NO: 787          moltype = RNA  length = 34
FEATURE                 Location/Qualifiers
source                  1..34
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 787
gcccaatgcc atcctggagt tcctgtaaga tacc                              34

SEQ ID NO: 788          moltype = RNA  length = 25
FEATURE                 Location/Qualifiers
source                  1..25
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 788
catcctggag ttcctgtaag atacc                                        25

SEQ ID NO: 789          moltype = RNA  length = 28
FEATURE                 Location/Qualifiers
source                  1..28
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 789
tgccatcctg gagttcctgt aagatacc                                     28

SEQ ID NO: 790          moltype = RNA  length = 25
FEATURE                 Location/Qualifiers
source                  1..25
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 790
tgccatcctg gagttcctgt aagat                                        25

SEQ ID NO: 791          moltype = RNA  length = 28
FEATURE                 Location/Qualifiers
source                  1..28
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 791
caatgccatc ctggagttcc tgtaagat                                     28

SEQ ID NO: 792          moltype = RNA  length = 31
FEATURE                 Location/Qualifiers
source                  1..31
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 792
gcccaatgcc atcctggagt tcctgtaaga t                                 31

SEQ ID NO: 793          moltype = RNA  length = 28
FEATURE                 Location/Qualifiers
source                  1..28
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 793
gcccaatgcc atcctggagt tcctgtaa                                     28

SEQ ID NO: 794          moltype = RNA  length = 34
FEATURE                 Location/Qualifiers
source                  1..34
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 794
gccgctgccc aatgacatcc tggagttcct gtaa                              34

SEQ ID NO: 795          moltype = RNA  length = 25
FEATURE                 Location/Qualifiers
source                  1..25
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 795
gccatcctgg agttcctgta agata                                        25

SEQ ID NO: 796          moltype = RNA  length = 25
FEATURE                 Location/Qualifiers
source                  1..25
```

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```
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 796
ccaatgccat cctggagttc ctgta                                          25

SEQ ID NO: 797           moltype = RNA  length = 25
FEATURE                  Location/Qualifiers
source                   1..25
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 797
ctgacaacag tttgccgctg cccaa                                          25

SEQ ID NO: 798           moltype = RNA  length = 25
FEATURE                  Location/Qualifiers
source                   1..25
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 798
tttgaggatt gctgaattat ttctt                                          25

SEQ ID NO: 799           moltype = RNA  length = 30
FEATURE                  Location/Qualifiers
source                   1..30
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 799
cagtttgccg ctgcccaatg ccatcctgga                                     30

SEQ ID NO: 800           moltype = RNA  length = 30
FEATURE                  Location/Qualifiers
source                   1..30
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 800
ttgccgctgc ccaatgccat cctggagttc                                     30

SEQ ID NO: 801           moltype = RNA  length = 25
FEATURE                  Location/Qualifiers
source                   1..25
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 801
tttgccgctg cccaatgcca tcctg                                          25

SEQ ID NO: 802           moltype = RNA  length = 22
FEATURE                  Location/Qualifiers
source                   1..22
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 802
ccaatgccat cctggagttc ct                                             22

SEQ ID NO: 803           moltype = RNA  length = 29
FEATURE                  Location/Qualifiers
source                   1..29
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 803
cccaatgcca tcctggagtt cctgtaaga                                      29

SEQ ID NO: 804           moltype = RNA  length = 28
FEATURE                  Location/Qualifiers
source                   1..28
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 804
ccgctgccca atgccatcct ggagttcc                                       28

SEQ ID NO: 805           moltype = RNA  length = 30
FEATURE                  Location/Qualifiers
source                   1..30
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 805
cccaatgcca tcctggagtt cctgtaagat                                     30
```

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```
SEQ ID NO: 806          moltype = RNA  length = 30
FEATURE                 Location/Qualifiers
source                  1..30
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 806
ccgctgccca atgccatcct ggagttcctg                                30

SEQ ID NO: 807          moltype = RNA  length = 30
FEATURE                 Location/Qualifiers
source                  1..30
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 807
tgcccaatgc catcctggag ttcctgtaag                                30

SEQ ID NO: 808          moltype = RNA  length = 28
FEATURE                 Location/Qualifiers
source                  1..28
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 808
cccaatgcca tcctggagtt cctgtaag                                  28

SEQ ID NO: 809          moltype = RNA  length = 28
FEATURE                 Location/Qualifiers
source                  1..28
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 809
tgcccaatgc catcctggag ttcctgta                                  28

SEQ ID NO: 810          moltype = RNA  length = 22
FEATURE                 Location/Qualifiers
source                  1..22
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 810
caatgccatc ctggagttcc tg                                        22

SEQ ID NO: 811          moltype = RNA  length = 29
FEATURE                 Location/Qualifiers
source                  1..29
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 811
agcctctcgc tcactcaccc tgcaaagga                                 29

SEQ ID NO: 812          moltype = RNA  length = 29
FEATURE                 Location/Qualifiers
source                  1..29
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 812
ccactcagag ctcagatctt ctaacttcc                                 29

SEQ ID NO: 813          moltype = RNA  length = 27
FEATURE                 Location/Qualifiers
source                  1..27
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 813
cttccactca gagctcagat cttctaa                                   27

SEQ ID NO: 814          moltype = RNA  length = 25
FEATURE                 Location/Qualifiers
source                  1..25
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 814
gggatccagt atacttacag gctcc                                     25

SEQ ID NO: 815          moltype = RNA  length = 17
FEATURE                 Location/Qualifiers
source                  1..17
```

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```
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 815
ctcagagctc agatctt                                                17

SEQ ID NO: 816          moltype = RNA  length = 15
FEATURE                 Location/Qualifiers
source                  1..15
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 816
ggctgctttg ccctc                                                  15

SEQ ID NO: 817          moltype = RNA  length = 27
FEATURE                 Location/Qualifiers
source                  1..27
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 817
ctcagatctt ctaacttcct ctttaac                                     27

SEQ ID NO: 818          moltype = RNA  length = 29
FEATURE                 Location/Qualifiers
source                  1..29
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 818
ctcagagctc agatcttcta acttcctct                                   29

SEQ ID NO: 819          moltype = RNA  length = 27
FEATURE                 Location/Qualifiers
source                  1..27
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 819
cgccttccac tcagagctca gatcttc                                     27

SEQ ID NO: 820          moltype = RNA  length = 27
FEATURE                 Location/Qualifiers
source                  1..27
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 820
tcagctcttg aagtaaacgg tttaccg                                     27

SEQ ID NO: 821          moltype = RNA  length = 27
FEATURE                 Location/Qualifiers
source                  1..27
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 821
tttgccctca gctcttgaag taaacgg                                     27

SEQ ID NO: 822          moltype = RNA  length = 27
FEATURE                 Location/Qualifiers
source                  1..27
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 822
ggctgctttg ccctcagctc ttgaagt                                     27

SEQ ID NO: 823          moltype = RNA  length = 26
FEATURE                 Location/Qualifiers
source                  1..26
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 823
caggagctag gtcaggctgc tttgcc                                      26

SEQ ID NO: 824          moltype = RNA  length = 25
FEATURE                 Location/Qualifiers
source                  1..25
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 824
tccaatagtg gtcagtccag gagct                                       25
```

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```
SEQ ID NO: 825          moltype = RNA  length = 27
FEATURE                Location/Qualifiers
source                 1..27
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 825
aaagagaatg ggatccagta tacttac                                     27

SEQ ID NO: 826          moltype = RNA  length = 27
FEATURE                Location/Qualifiers
source                 1..27
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 826
aaatagctag agccaaagag aatggga                                     27

SEQ ID NO: 827          moltype = RNA  length = 33
FEATURE                Location/Qualifiers
source                 1..33
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 827
ggctgctttg ccctcagctc ttgaagtaaa cgg                              33

SEQ ID NO: 828          moltype = RNA  length = 30
FEATURE                Location/Qualifiers
source                 1..30
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 828
aggctgcttt gccctcagct cttgaagtaa                                  30

SEQ ID NO: 829          moltype = RNA  length = 30
FEATURE                Location/Qualifiers
source                 1..30
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 829
gtcaggctgc tttgccctca gctcttgaag                                  30

SEQ ID NO: 830          moltype = RNA  length = 30
FEATURE                Location/Qualifiers
source                 1..30
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 830
aggtcaggct gctttgccct cagctcttga                                  30

SEQ ID NO: 831          moltype = RNA  length = 25
FEATURE                Location/Qualifiers
source                 1..25
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 831
cagagctcag atcttctaac ttcct                                       25

SEQ ID NO: 832          moltype = RNA  length = 25
FEATURE                Location/Qualifiers
source                 1..25
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 832
cttacaggct ccaatagtgg tcagt                                       25

SEQ ID NO: 833          moltype = RNA  length = 25
FEATURE                Location/Qualifiers
source                 1..25
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 833
atgggatcca gtatacttac aggct                                       25

SEQ ID NO: 834          moltype = RNA  length = 25
FEATURE                Location/Qualifiers
source                 1..25
```

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```
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 834
agagaatggg atccagtata cttac                                             25

SEQ ID NO: 835          moltype = RNA  length = 27
FEATURE                 Location/Qualifiers
source                  1..27
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 835
aacttcctct ttaacagaaa agcatac                                           27

SEQ ID NO: 836          moltype = RNA  length = 23
FEATURE                 Location/Qualifiers
source                  1..23
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 836
ctcatacctt ctgcttgatg atc                                               23

SEQ ID NO: 837          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 837
tcaaggaaga tggcatttct                                                   20

SEQ ID NO: 838          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 838
gaaagccagt cggtaagttc                                                   20

SEQ ID NO: 839          moltype = RNA  length = 15
FEATURE                 Location/Qualifiers
source                  1..15
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 839
cacccaccat caccc                                                        15

SEQ ID NO: 840          moltype = RNA  length = 23
FEATURE                 Location/Qualifiers
source                  1..23
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 840
cctctgtgat tttataactt gat                                               23

SEQ ID NO: 841          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 841
tgatatcctc aaggtcaccc                                                   20

SEQ ID NO: 842          moltype = RNA  length = 30
FEATURE                 Location/Qualifiers
source                  1..30
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 842
ggtacctcca acatcaagga agatggcatt                                        30

SEQ ID NO: 843          moltype = RNA  length = 26
FEATURE                 Location/Qualifiers
source                  1..26
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 843
atttctagtt tggagatggc agtttc                                            26
```

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```
SEQ ID NO: 844          moltype = RNA  length = 26
FEATURE                Location/Qualifiers
source                 1..26
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 844
catcaaggaa gatggcattt ctagtt                                        26

SEQ ID NO: 845          moltype = RNA  length = 26
FEATURE                Location/Qualifiers
source                 1..26
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 845
gagcaggtac ctccaacatc aaggaa                                        26

SEQ ID NO: 846          moltype = RNA  length = 30
FEATURE                Location/Qualifiers
source                 1..30
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 846
ctccaacatc aaggaagatg gcatttctag                                    30

SEQ ID NO: 847          moltype = RNA  length = 25
FEATURE                Location/Qualifiers
source                 1..25
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 847
accagagtaa cagtctgagt aggag                                         25

SEQ ID NO: 848          moltype = RNA  length = 25
FEATURE                Location/Qualifiers
source                 1..25
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 848
caccagagta acagtctgag tagga                                         25

SEQ ID NO: 849          moltype = RNA  length = 25
FEATURE                Location/Qualifiers
source                 1..25
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 849
tcaccagagt aacagtctga gtagg                                         25

SEQ ID NO: 850          moltype = RNA  length = 25
FEATURE                Location/Qualifiers
source                 1..25
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 850
gtcaccagag taacagtctg agtag                                         25

SEQ ID NO: 851          moltype = RNA  length = 26
FEATURE                Location/Qualifiers
source                 1..26
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 851
accagagtaa cagtctgagt aggagc                                        26

SEQ ID NO: 852          moltype = RNA  length = 24
FEATURE                Location/Qualifiers
source                 1..24
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 852
ttctgtccaa gcccggttga aatc                                          24

SEQ ID NO: 853          moltype = RNA  length = 30
FEATURE                Location/Qualifiers
source                 1..30
```

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```
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 853
acatcaagga agatggcatt tctagtttgg                                    30

SEQ ID NO: 854          moltype = RNA  length = 25
FEATURE                 Location/Qualifiers
source                  1..25
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 854
acatcaagga agatggcatt tctag                                         25

SEQ ID NO: 855          moltype = RNA  length = 25
FEATURE                 Location/Qualifiers
source                  1..25
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 855
atcatttttt ctcatacctt ctgct                                         25

SEQ ID NO: 856          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 856
cacccaccat caccctctgt g                                             21

SEQ ID NO: 857          moltype = RNA  length = 22
FEATURE                 Location/Qualifiers
source                  1..22
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 857
atcatctcgt tgatatcctc aa                                            22

SEQ ID NO: 858          moltype = RNA  length = 28
FEATURE                 Location/Qualifiers
source                  1..28
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 858
ctccaacatc aaggaagatg gcatttct                                      28

SEQ ID NO: 859          moltype = RNA  length = 25
FEATURE                 Location/Qualifiers
source                  1..25
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 859
catcaaggaa gatggcattt ctagt                                         25

SEQ ID NO: 860          moltype = RNA  length = 18
FEATURE                 Location/Qualifiers
source                  1..18
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 860
ttgctggtct tgtttttc                                                 18

SEQ ID NO: 861          moltype = RNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 861
ccgtaatgat tgttct                                                   16

SEQ ID NO: 862          moltype = RNA  length = 18
FEATURE                 Location/Qualifiers
source                  1..18
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 862
gctggtcttg tttttcaa                                                 18
```

-continued

```
SEQ ID NO: 863          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 863
tggtcttgtt tttcaaattt                                              20

SEQ ID NO: 864          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 864
gtcttgtttt tcaaattttg                                              20

SEQ ID NO: 865          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 865
cttgtttttc aaattttggg                                              20

SEQ ID NO: 866          moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
source                  1..19
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 866
tgtttttcaa attttgggc                                               19

SEQ ID NO: 867          moltype = RNA  length = 33
FEATURE                 Location/Qualifiers
source                  1..33
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 867
tccaactggg gacgcctctg ttccaaatcc tgc                               33

SEQ ID NO: 868          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 868
tcctgcattg ttgcctgtaa g                                            21

SEQ ID NO: 869          moltype = RNA  length = 30
FEATURE                 Location/Qualifiers
source                  1..30
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 869
tccaactggg gacgcctctg ttccaaatcc                                   30

SEQ ID NO: 870          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 870
actggggacg cctctgttcc a                                            21

SEQ ID NO: 871          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 871
ccgtaatgat tgttctagcc                                              20

SEQ ID NO: 872          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
source                  1..20
```

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```
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 872
tgttaaaaaa cttacttcga                                              20

SEQ ID NO: 873           moltype = RNA  length = 18
FEATURE                  Location/Qualifiers
source                   1..18
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 873
ctgttgcctc cggttctg                                                18

SEQ ID NO: 874           moltype = RNA  length = 18
FEATURE                  Location/Qualifiers
source                   1..18
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 874
ttggctctgg cctgtcct                                                18

SEQ ID NO: 875           moltype = RNA  length = 33
FEATURE                  Location/Qualifiers
source                   1..33
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 875
ttcaactgtt gcctccggtt ctgaaggtgt tct                               33

SEQ ID NO: 876           moltype = RNA  length = 25
FEATURE                  Location/Qualifiers
source                   1..25
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 876
tacttcatcc cactgattct gaatt                                        25

SEQ ID NO: 877           moltype = RNA  length = 25
FEATURE                  Location/Qualifiers
source                   1..25
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 877
ctgaaggtgt tcttgtactt catcc                                        25

SEQ ID NO: 878           moltype = RNA  length = 25
FEATURE                  Location/Qualifiers
source                   1..25
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 878
ctgttgcctc cggttctgaa ggtgt                                        25

SEQ ID NO: 879           moltype = RNA  length = 30
FEATURE                  Location/Qualifiers
source                   1..30
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 879
ctgttgcctc cggttctgaa ggtgttcttg                                   30

SEQ ID NO: 880           moltype = RNA  length = 30
FEATURE                  Location/Qualifiers
source                   1..30
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 880
caactgttgc ctccggttct gaaggtgttc                                   30

SEQ ID NO: 881           moltype = RNA  length = 30
FEATURE                  Location/Qualifiers
source                   1..30
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 881
ttgcctccgg ttctgaaggt gttcttgtac                                   30
```

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```
SEQ ID NO: 882          moltype = RNA  length = 25
FEATURE                 Location/Qualifiers
source                  1..25
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 882
gttgcctccg gttctgaagg tgttc                                          25

SEQ ID NO: 883          moltype = RNA  length = 23
FEATURE                 Location/Qualifiers
source                  1..23
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 883
ctccggttct gaaggtgttc ttg                                            23

SEQ ID NO: 884          moltype = RNA  length = 22
FEATURE                 Location/Qualifiers
source                  1..22
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 884
ctccggttct gaaggtgttc tt                                             22

SEQ ID NO: 885          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 885
ctccggttct gaaggtgttc t                                              21

SEQ ID NO: 886          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 886
ctccggttct gaaggtgttc                                                20

SEQ ID NO: 887          moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
source                  1..19
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 887
ctccggttct gaaggtgtt                                                 19

SEQ ID NO: 888          moltype = RNA  length = 25
FEATURE                 Location/Qualifiers
source                  1..25
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 888
cattcaactg ttgcctccgg ttctg                                          25

SEQ ID NO: 889          moltype = RNA  length = 24
FEATURE                 Location/Qualifiers
source                  1..24
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 889
ctgttgcctc cggttctgaa ggtg                                           24

SEQ ID NO: 890          moltype = RNA  length = 31
FEATURE                 Location/Qualifiers
source                  1..31
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 890
cattcaactg ttgcctccgg ttctgaaggt g                                   31

SEQ ID NO: 891          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
```

-continued

```
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 891
tactaacctt ggtttctgtg a                                        21

SEQ ID NO: 892          moltype = RNA  length = 27
FEATURE                 Location/Qualifiers
source                  1..27
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 892
tgtataggga ccctccttcc atgactc                                  27

SEQ ID NO: 893          moltype = RNA  length = 25
FEATURE                 Location/Qualifiers
source                  1..25
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 893
ctaaccttgg tttctgtgat tttct                                    25

SEQ ID NO: 894          moltype = RNA  length = 27
FEATURE                 Location/Qualifiers
source                  1..27
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 894
ggtatctttg atactaacct tggtttc                                  27

SEQ ID NO: 895          moltype = RNA  length = 22
FEATURE                 Location/Qualifiers
source                  1..22
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 895
attctttcaa ctagaataaa ag                                       22

SEQ ID NO: 896          moltype = RNA  length = 25
FEATURE                 Location/Qualifiers
source                  1..25
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 896
gattctgaat tctttcaact agaat                                    25

SEQ ID NO: 897          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 897
atcccactga ttctgaattc                                          20

SEQ ID NO: 898          moltype = RNA  length = 22
FEATURE                 Location/Qualifiers
source                  1..22
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 898
aaccgagacc ggacaggatt ct                                       22

SEQ ID NO: 899          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 899
ctgttgcagt aatctatgag                                          20

SEQ ID NO: 900          moltype = RNA  length = 23
FEATURE                 Location/Qualifiers
source                  1..23
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 900
tgccattgtt tcatcagctc ttt                                      23
```

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```
SEQ ID NO: 901          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 901
tgcagtaatc tatgagtttc                                                20

SEQ ID NO: 902          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 902
tcctgtagga cattggcagt                                                20

SEQ ID NO: 903          moltype = RNA  length = 18
FEATURE                 Location/Qualifiers
source                  1..18
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 903
gagtcttcta ggagcctt                                                  18

SEQ ID NO: 904          moltype = RNA  length = 25
FEATURE                 Location/Qualifiers
source                  1..25
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 904
ggccaaacct cggcttacct gaaat                                          25

SEQ ID NO: 905          moltype = RNA  length = 20
FEATURE                 Location/Qualifiers
source                  1..20
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 905
ggccaaacct cggcttacct                                                20

SEQ ID NO: 906          moltype = RNA  length = 29
FEATURE                 Location/Qualifiers
source                  1..29
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 906
aatcagctgg gagagagctt cctgtagct                                      29

SEQ ID NO: 907          moltype = RNA  length = 26
FEATURE                 Location/Qualifiers
source                  1..26
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 907
tcgttcttct gtcgtcgtaa cgtttc                                         26

SEQ ID NO: 908          moltype = RNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 908
caccgattgt cttcga                                                    16

SEQ ID NO: 909          moltype = RNA  length = 17
FEATURE                 Location/Qualifiers
source                  1..17
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 909
cccttgtacg atttatg                                                   17

SEQ ID NO: 910          moltype = RNA  length = 18
FEATURE                 Location/Qualifiers
source                  1..18
```

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```
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 910
tctgtgttta aggactct                                              18

SEQ ID NO: 911          moltype = RNA  length = 31
FEATURE                 Location/Qualifiers
source                  1..31
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 911
gccgctgccc aatgccatcc tggagttcct g                               31

SEQ ID NO: 912          moltype = RNA  length = 29
FEATURE                 Location/Qualifiers
source                  1..29
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 912
attagatctg tcgccctacc tctttttc                                   29

SEQ ID NO: 913          moltype = RNA  length = 27
FEATURE                 Location/Qualifiers
source                  1..27
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 913
tgtcgcccta cctcttttt ctgtctg                                     27

SEQ ID NO: 914          moltype = RNA  length = 25
FEATURE                 Location/Qualifiers
source                  1..25
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 914
gcccaatgcc atcctggagt tcctg                                      25

SEQ ID NO: 915          moltype = RNA  length = 30
FEATURE                 Location/Qualifiers
source                  1..30
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 915
gagcctctcg ctcactcacc ctgcaaagga                                 30

SEQ ID NO: 916          moltype = RNA  length = 36
FEATURE                 Location/Qualifiers
source                  1..36
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 916
atcatttttt ctcatacctt ctgctaggag ctaaaa                          36

SEQ ID NO: 917          moltype = RNA  length = 26
FEATURE                 Location/Qualifiers
source                  1..26
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 917
ggaagctaag gaagaagctg agcagg                                     26

SEQ ID NO: 918          moltype = DNA  length = 6054
FEATURE                 Location/Qualifiers
source                  1..6054
                        mol_type = genomic DNA
                        organism = Homo sapiens
SEQUENCE: 918
cagcccctga gaccaggtct ggctccacag ctctgtcctg ctctgtgtct ttccctgctg  60
ctctcaggtc ccctgcaggc cttggcccct ttcctcatct gtagacacac ttgagtagcc  120
caggcacagc catgggagat tcggagatgg cagtctttgg ggctgccgcc ccctacctgc  180
gcaagtcaga gaaggagcgg ctagaagcgc agaccaggcc ttttgacctc aagaaggatg  240
tcttcgtgcc tgatgacaaa caggagtttg tcaaggccaa gatcgtgtct cgagagggtg  300
gcaaagtcac tgccgagacc gagtatggca agacagtgac cgtgaaggag gaccaggtga  360
tgcagcagaa cccacccaag ttcgacaaaa tcgaggacat ggccatgctg accttcctgc  420
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aggtggtggc tgcctaccgg ggcaagaaga ggagcgaggc cccgccccac atcttctcca  600
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ccggagaatc cggagcaggg aagacagtca acaccaagag ggtcatccag tactttgctg  720
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source                  1..6313
                        mol_type = genomic DNA
                        organism = Mus musculus
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cttaaagcca aaa                                                     6313
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source                  1..16
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                        organism = synthetic construct
SEQUENCE: 920
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SEQ ID NO: 921          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 921
ctctcctcac ggaagc                                                  16

SEQ ID NO: 922          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 922
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SEQ ID NO: 923          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 923
aaacttgctc agcagt                                                16

SEQ ID NO: 924          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 924
caaacttgct cagcag                                                16

SEQ ID NO: 925          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 925
ccaaacttgc tcagca                                                16

SEQ ID NO: 926          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 926
cccaaacttg ctcagc                                                16

SEQ ID NO: 927          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 927
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SEQ ID NO: 928          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
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SEQUENCE: 928
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SEQ ID NO: 929          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
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SEQUENCE: 929
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SEQ ID NO: 930          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 930
ggtttgatgt ccctgt                                                16

SEQ ID NO: 931          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 931
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SEQ ID NO: 932          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
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source                1..16
                      mol_type = other DNA
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SEQ ID NO: 933        moltype = DNA  length = 16
FEATURE               Location/Qualifiers
source                1..16
                      mol_type = other DNA
                      organism = synthetic construct
SEQUENCE: 933
cacagcctgc aggatc                                                    16

SEQ ID NO: 934        moltype = DNA  length = 16
FEATURE               Location/Qualifiers
source                1..16
                      mol_type = other DNA
                      organism = synthetic construct
SEQUENCE: 934
ccacagcctg caggat                                                    16

SEQ ID NO: 935        moltype = DNA  length = 16
FEATURE               Location/Qualifiers
source                1..16
                      mol_type = other DNA
                      organism = synthetic construct
SEQUENCE: 935
cccacagcct gcagga                                                    16

SEQ ID NO: 936        moltype = DNA  length = 16
FEATURE               Location/Qualifiers
source                1..16
                      mol_type = other DNA
                      organism = synthetic construct
SEQUENCE: 936
gcccacagcc tgcagg                                                    16

SEQ ID NO: 937        moltype = DNA  length = 16
FEATURE               Location/Qualifiers
source                1..16
                      mol_type = other DNA
                      organism = synthetic construct
SEQUENCE: 937
cgcccacagc ctgcag                                                    16

SEQ ID NO: 938        moltype = DNA  length = 16
FEATURE               Location/Qualifiers
source                1..16
                      mol_type = other DNA
                      organism = synthetic construct
SEQUENCE: 938
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SEQ ID NO: 939        moltype = DNA  length = 16
FEATURE               Location/Qualifiers
source                1..16
                      mol_type = other DNA
                      organism = synthetic construct
SEQUENCE: 939
accgcccaca gcctgc                                                    16

SEQ ID NO: 940        moltype = DNA  length = 16
FEATURE               Location/Qualifiers
source                1..16
                      mol_type = other DNA
                      organism = synthetic construct
SEQUENCE: 940
caccgcccac agcctg                                                    16

SEQ ID NO: 941        moltype = DNA  length = 16
FEATURE               Location/Qualifiers
source                1..16
                      mol_type = other DNA
                      organism = synthetic construct
SEQUENCE: 941
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ccaccgccca cagcct                                                 16

SEQ ID NO: 942          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 942
cccaccgccc acagcc                                                 16

SEQ ID NO: 943          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 943
gcccaccgcc cacagc                                                 16

SEQ ID NO: 944          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 944
ccaggcccac cgccca                                                 16

SEQ ID NO: 945          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 945
cccaggccca ccgccc                                                 16

SEQ ID NO: 946          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 946
tcccaggccc accgcc                                                 16

SEQ ID NO: 947          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 947
tgcctgtccc aggccc                                                 16

SEQ ID NO: 948          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 948
ctgcctgtcc caggcc                                                 16

SEQ ID NO: 949          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
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                        organism = synthetic construct
SEQUENCE: 949
gctgcctgtc ccaggc                                                 16

SEQ ID NO: 950          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
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                        organism = synthetic construct
SEQUENCE: 950
ggtggcacct tcgaaa                                                 16

SEQ ID NO: 951          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
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source                  1..16
                        mol_type = other DNA
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SEQ ID NO: 952          moltype = DNA  length = 16
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source                  1..16
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                        organism = synthetic construct
SEQUENCE: 952
tcggtggcac cttcga                                                     16

SEQ ID NO: 953          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 953
agtgagcccg tcctcc                                                     16

SEQ ID NO: 954          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 954
cagtgagccc gtcctc                                                     16

SEQ ID NO: 955          moltype = DNA  length = 16
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source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 955
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SEQ ID NO: 956          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 956
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SEQ ID NO: 957          moltype = DNA  length = 16
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source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 957
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SEQ ID NO: 958          moltype = DNA  length = 16
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                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 958
cttcccgaat gtccga                                                     16

SEQ ID NO: 959          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 959
ccttcccgaa tgtccg                                                     16

SEQ ID NO: 960          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 960
```

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```
acсttccсga atgtcc                                                 16

SEQ ID NO: 961         moltype = DNA  length = 16
FEATURE                Location/Qualifiers
source                 1..16
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 961
caccttcccg aatgtc                                                 16

SEQ ID NO: 962         moltype = DNA  length = 16
FEATURE                Location/Qualifiers
source                 1..16
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 962
gcaccttccc gaatgt                                                 16

SEQ ID NO: 963         moltype = DNA  length = 16
FEATURE                Location/Qualifiers
source                 1..16
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 963
cgcaccttcc cgaatg                                                 16

SEQ ID NO: 964         moltype = DNA  length = 16
FEATURE                Location/Qualifiers
source                 1..16
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 964
atccgctcct gcaact                                                 16

SEQ ID NO: 965         moltype = DNA  length = 16
FEATURE                Location/Qualifiers
source                 1..16
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 965
catccgctcc tgcaac                                                 16

SEQ ID NO: 966         moltype = DNA  length = 16
FEATURE                Location/Qualifiers
source                 1..16
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 966
ccatccgctc ctgcaa                                                 16

SEQ ID NO: 967         moltype = DNA  length = 16
FEATURE                Location/Qualifiers
source                 1..16
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 967
gctccctctg cctgca                                                 16

SEQ ID NO: 968         moltype = DNA  length = 16
FEATURE                Location/Qualifiers
source                 1..16
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 968
aggtggatcc gtggcc                                                 16

SEQ ID NO: 969         moltype = DNA  length = 16
FEATURE                Location/Qualifiers
source                 1..16
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 969
gggaaggtgg atccgt                                                 16

SEQ ID NO: 970         moltype = DNA  length = 16
FEATURE                Location/Qualifiers
```

-continued

```
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 970
acaggagcag ggaaag                                                    16

SEQ ID NO: 971          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 971
cagactgcgg tgagtt                                                    16

SEQ ID NO: 972          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 972
ggctcctggg cggcgc                                                    16

SEQ ID NO: 973          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 973
ggcggctcct gggcgg                                                    16

SEQ ID NO: 974          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 974
cgcgggcggc tcctgg                                                    16

SEQ ID NO: 975          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 975
gagcgcgggc ggctcc                                                    16

SEQ ID NO: 976          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 976
ggttcaggga gcgcgg                                                    16

SEQ ID NO: 977          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 977
agttctaggg ttcagg                                                    16

SEQ ID NO: 978          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 978
cagttctagg gttcag                                                    16

SEQ ID NO: 979          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 979
```

-continued

```
acagttctag ggttca                                                16

SEQ ID NO: 980          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 980
gacagttcta gggttc                                                16

SEQ ID NO: 981          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 981
agacagttct agggtt                                                16

SEQ ID NO: 982          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 982
aagacagttc tagggt                                                16

SEQ ID NO: 983          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 983
gaagacagtt ctaggg                                                16

SEQ ID NO: 984          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 984
cgaagacagt tctagg                                                16

SEQ ID NO: 985          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 985
tcgaagacag ttctag                                                16

SEQ ID NO: 986          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 986
gtcgaagaca gttcta                                                16

SEQ ID NO: 987          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 987
agtcgaagac agttct                                                16

SEQ ID NO: 988          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
source                  1..16
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 988
gagtcgaaga cagttc                                                16

SEQ ID NO: 989          moltype = DNA  length = 16
FEATURE                 Location/Qualifiers
```

-continued

```
source                1..16
                      mol_type = other DNA
                      organism = synthetic construct
SEQUENCE: 989
ggagtcgaag acagtt                                              16

SEQ ID NO: 990        moltype = DNA  length = 16
FEATURE               Location/Qualifiers
source                1..16
                      mol_type = other DNA
                      organism = synthetic construct
SEQUENCE: 990
cggagtcgaa gacagt                                              16

SEQ ID NO: 991        moltype = DNA  length = 16
FEATURE               Location/Qualifiers
source                1..16
                      mol_type = other DNA
                      organism = synthetic construct
SEQUENCE: 991
ccggagtcga agacag                                              16

SEQ ID NO: 992        moltype = DNA  length = 16
FEATURE               Location/Qualifiers
source                1..16
                      mol_type = other DNA
                      organism = synthetic construct
SEQUENCE: 992
cccggagtcg aagaca                                              16

SEQ ID NO: 993        moltype = DNA  length = 16
FEATURE               Location/Qualifiers
source                1..16
                      mol_type = other DNA
                      organism = synthetic construct
SEQUENCE: 993
ccccggagtc gaagac                                              16

SEQ ID NO: 994        moltype = DNA  length = 16
FEATURE               Location/Qualifiers
source                1..16
                      mol_type = other DNA
                      organism = synthetic construct
SEQUENCE: 994
gccccggagt cgaaga                                              16

SEQ ID NO: 995        moltype = DNA  length = 16
FEATURE               Location/Qualifiers
source                1..16
                      mol_type = other DNA
                      organism = synthetic construct
SEQUENCE: 995
ggccccggag tcgaag                                              16

SEQ ID NO: 996        moltype = DNA  length = 16
FEATURE               Location/Qualifiers
source                1..16
                      mol_type = other DNA
                      organism = synthetic construct
SEQUENCE: 996
gggccccgga gtcgaa                                              16

SEQ ID NO: 997        moltype = DNA  length = 16
FEATURE               Location/Qualifiers
source                1..16
                      mol_type = other DNA
                      organism = synthetic construct
SEQUENCE: 997
acaataaata ccgagg                                              16
```

1.-91. (canceled)

92. A complex comprising an anti-transferrin receptor antibody covalently linked to a 5' end or a 3' end of an oligonucleotide,

wherein the anti-transferrin receptor antibody comprises a heavy chain variable region (VH) and a light chain variable region (VL), wherein:

the VH comprises a heavy chain complementarity determining region 1 (CDR-H1), a heavy chain complementarity determining region 2 (CDR-H2), and a heavy chain complementarity determining region 3 (CDR-H3) of a heavy chain variable region comprising the amino acid sequence of SEQ ID NO: 641, and comprises humanized framework regions,

the VL comprises a light chain complementarity determining region 1 (CDR-L1), a light chain complementarity determining region 2 (CDR-L2), and a light chain complementarity determining region 3 (CDR-L3) of a light chain variable region comprising the amino acid sequence of SEQ ID NO: 642, and comprises humanized framework regions,

wherein the oligonucleotide comprises one or more modifications and comprises an antisense strand comprising a region of complementarity of at least 15 nucleotides in length to an RNA encoded by a gene associated with a muscle disease, wherein the oligonucleotide is in the range of 15-30 nucleotides in length;

wherein the one or more modifications comprise a 2'-modified nucleoside selected from the group consisting of: a 2'-O-methyl nucleoside, a 2'-fluoro nucleoside, and combinations thereof, and/or comprise a modified backbone selected from a backbone comprising one or more phosphorothioate linkages and a phosphorodiamidate morpholino backbone.

93. The complex of claim **92**, wherein the antisense strand is 20-27 nucleotides in length.

94. The complex of claim **93**, wherein the region of complementarity is at least 18 nucleotides in length.

95. The complex of claim **93**, wherein the region of complementarity is at least 19 nucleotides in length.

96. The complex of claim **95**, wherein the oligonucleotide is double-stranded molecule and comprises the antisense strand hybridized to a sense strand.

97. The complex of claim **95**, wherein the oligonucleotide is a single stranded antisense oligonucleotide.

98. The complex of claim **96**, wherein each nucleoside of the oligonucleotide is selected from a 2'-O-methyl nucleoside and a 2'-fluoro nucleoside.

99. The complex of claim **98**, wherein the oligonucleotide comprises a modified backbone comprising one or more phosphorothioate linkages.

100. The complex of claim **97**, wherein the oligonucleotide is a phosphorodiamidate morpholino oligomer (PMO).

101. The complex of claim **92**, wherein the CDR-H1, CDR-H2, CDR-H3, CDR-L1, CDR-L2, and CDR-L3 are according to the Chothia numbering system.

102. The complex of claim **101**, wherein the anti-transferrin receptor antibody is in the form of a full-length IgG comprising a human IgG1 heavy chain constant region, or a functional variant thereof, and a human kappa light chain constant region.

103. The complex of claim **101**, wherein the anti-transferrin receptor antibody comprises a human kappa light chain constant region and a human IgG1 heavy chain constant region comprising at least one amino acid substitution relative to a wild-type human IgG1 heavy chain constant region.

104. The complex of claim **102**, wherein the full-length IgG comprises a heavy chain constant region comprising at least one amino acid substitution that reduces Fc receptor binding of the anti-transferrin receptor antibody.

105. The complex of claim **102**, wherein the full-length IgG comprises two or more amino acid substitutions in a CH2 domain and two or more amino acid substitutions in a CH3 domain, relative to a full-length IgG comprising an IgG1 constant region having an amino acid sequence of SEQ ID NO: 37.

106. The complex of claim **103**, wherein the anti-transferrin receptor antibody further comprises one or more sugar or carbohydrate molecules.

107. The complex of claim **106**, wherein the one or more sugar or carbohydrate molecules comprise a fucose unit.

108. The complex of claim **103**, wherein the complex is formable by a process comprising reacting a first electrophile of a linker precursor compound and a nucleophile of the anti-transferrin receptor antibody.

109. The complex of claim **108**, wherein the nucleophile of the anti-transferrin receptor antibody is a thiol group of a cysteine residue of the anti-transferrin receptor antibody.

110. The complex of claim **108**, wherein the first electrophile of the linker precursor compound is a maleimide moiety, and wherein the nucleophile of the anti-transferrin receptor antibody is a thiol-group of a cysteine residue of the anti-transferrin receptor antibody.

111. The complex of claim **110**, wherein the maleimide moiety is present in a (maleimidomethyl)cyclohexane-1-carboxylate group of the linker precursor compound.

112. The complex of claim **110**, wherein the complex is formable by a process comprising reacting a second electrophile of the linker precursor compound and a nucleophile covalently attached to the oligonucleotide.

113. The complex of claim **112**, wherein the nucleophile covalently attached to the oligonucleotide comprises an aminoalkyl group.

114. The complex of claim **113**, wherein the aminoalkyl group is a NH2-C6 group.

115. The complex of claim **112**, wherein the nucleophile covalently attached to the oligonucleotide is covalently attached to a terminal phosphate group of the oligonucleotide.

116. The complex of claim **92**, wherein the gene associated with a muscle disease is DUX4.

117. A method of delivering an oligonucleotide to a subject, the method comprising intravenously administering to the subject the complex of claim **92**.

118. The method of claim **117**, wherein the oligonucleotide is delivered to a skeletal muscle cell, a cardiac muscle cell, or a smooth muscle cell of the subject.

119. The method of claim **117**, wherein the subject is human.

120. The method of claim **119**, wherein the gene associated with a muscle disease is DUX4.

121. The method of claim **120**, wherein the subject has Facioscapulohumeral Muscular Dystrophy (FSHD).

* * * * *